Ridgetop Group Inc

ENGINEERING INNOVATION



ARULE

Software User Guide

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1. Introduction

The Adaptive Remaining Useful Life Estimator™ (ARULE™) is Ridgetop Group, Inc.'s prognostic-analysis technology for estimating the future health and remaining service life of systems, subsystems, components, and other defined analysis nodes from degradation-related Feature Data (FD).

ARULE is more than a single graphical software application. The technology comprises a prognostic methodology, the Adaptive Predictive Kernel (APK), signature-processing and adaptive-estimation algorithms, configurable System Definition (SDEF) and Node Definition (NDEF) structures, and software interfaces that allow the same underlying prognostic capability to be used through the standalone ARULE application or integrated into a larger Prognostics and Health Management (PHM) system.

At a high level, ARULE accepts application-specific Feature Data that characterize the progression of a known or hypothesized degradation mechanism. Each Feature Data observation associates a degradation-related Feature Data value with an ordered analysis coordinate that identifies where the observation occurs within the monitored degradation or usage history.

ARULE processes the resulting ordered Feature Data signature through a sequence of signature transformations and adaptive prognostic methods to generate:

- **Remaining Useful Life (RUL)** — the estimated remaining distance along the defined analysis coordinate from the current observation to the configured functional-failure condition;
- **State of Health (SoH)** — an ARULE health estimate associated with progression through the configured degradation-to-functional-failure interval; and
- **Prognostic Horizon (PH)** — the applicable prognostic-horizon quantity defined by the ARULE methodology.

The engineering interpretation of these outputs depends on the Feature Data definition, the ordered coordinate used to describe progression, and the configured functional-failure condition.

ARULE is not intended to convert arbitrary raw sensor measurements directly into remaining-life predictions. Condition-Based Data (CBD) must first be evaluated and, where appropriate, processed into a Feature Data representation that has a meaningful relationship to degradation and functional failure. Feature definition, feature quality, nominal behavior, degradation direction, coordinate basis, and failure mapping are therefore fundamental to the interpretation of ARULE results.

A distinguishing characteristic of ARULE is its adaptive, sequential processing methodology. Rather than relying only on a fixed beginning-of-life estimate, ARULE

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processes observations in sequence and updates its degradation representation and prognostic assessment as new information becomes available. The underlying methodology combines signature conditioning and transformation with adaptive-estimation techniques, including Extended Kalman Filtering (EKF), geometry-based calculations, linearization methods, and other supporting computational techniques.

Historical Development and Technology Evolution

ARULE is the result of a long-running Ridgetop Group research and development effort in Prognostics and Health Management (PHM), Condition-Based Maintenance (CBM), and Integrated Vehicle Health Management (IVHM).

Published ARULE research predates the current standalone software application. Early work documented in 2009 describes adaptive prognostic technology developed in connection with the NAVAIR Joint Strike Fighter program and demonstrates prognostic processing using multiple types of degradation information. Battery prognostics were also among the early published applications, including evaluation using NASA lithium-ion battery-health information.

These early implementations established several concepts that continue to define the ARULE methodology:

- use of application-specific degradation signatures rather than dependence on a single physical health quantity;
- explicit definition of a functional-failure condition toward which degradation progression is evaluated;
- sequential processing as new observations become available;
- adaptation of the prognostic model to observed degradation behavior; and
- evaluation of RUL and other prognostic information relative to a defined failure endpoint.

The modern ARULE software lineage subsequently formalized and productized these concepts through several stages of development. Research-oriented ARULE implementations provided an analytical environment for developing and evaluating the feature-based degradation methodology, transformation pipeline, and adaptive-estimation concepts that form the basis of the APK.

The ARULE graphical application represented an important transition from research-oriented analytical workflows to a configurable, user-facing engineering tool. The GUI introduced structured workflows for selecting input data, creating SDEF and NDEF configurations, executing the prognostic engine, and reviewing degradation-signature and prognostic outputs.

Subsequent development extended ARULE into a more modular and deployable prognostic capability suitable for integration within larger PHM platforms. In this

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architecture, the APK provides the core prognostic computation capability, while surrounding applications may provide data acquisition, Feature Data preparation, configuration management, visualization, advisory logic, and health-management functions.

This evolution supports two complementary forms of ARULE use:

1. **Engineering analysis and configuration development** — historical, experimental, or simulated data are evaluated using the standalone ARULE application to develop and review Feature Data definitions, SDEF/NDEF configurations, functional-failure mappings, signature transformations, and prognostic behavior.
2. **Integrated prognostic execution** — an external PHM application supplies qualified Feature Data to the ARULE prognostic engine and uses the returned SoH, RUL, PH, and status information as part of a larger health-management workflow.

The underlying prognostic methodology remains common to both forms of use, while surrounding data handling, automation, advisory logic, and operational decision functions depend on the host application.

Within **CellNostix™**, ARULE provides the prognostic-analysis capability within a broader battery PHM digital thread that includes data acquisition, battery modeling, model-to-test comparison, Feature Data preparation and qualification, prognostic assessment, advisory generation, and health-management functions. Section 8 describes this integrated role and provides a worked battery example.

Ridgetop Group, Inc., founded in 2000, develops PHM, CBM, reliability-engineering, modeling, and system-health technologies for government, commercial, research, and industrial applications. ARULE represents one component of this broader technology portfolio and supports engineering applications requiring advance understanding of degradation progression and future functional failure.

This User Guide is organized as follows:

1. Introduction
2. ARULE: Description and Operation
3. Prognostic Information: Evaluation of Accuracy
4. ARULE™ Installation
5. ARULE™ User Interface
6. Real-World Examples
7. How to Analyze Your Own Data with ARULE™
8. PHM Integration within CellNostix™
9. Appendix

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1.1 ARULE™ Overview

ARULE is an application-independent prognostic framework and adaptive-estimation capability designed to operate on degradation-related Feature Data from a wide range of monitored systems and applications.

The term *application-independent* does not mean that arbitrary data can be supplied to ARULE and automatically produce meaningful prognostic information. Instead, it means that the same underlying prognostic methodology can be applied across different asset classes and degradation mechanisms once the application-specific engineering problem has been appropriately defined.

That definition includes, at minimum:

- the system, subsystem, component, or analysis node being evaluated;
- the degradation mechanism, condition, or failure mode of interest;
- the source Condition-Based Data;
- the Feature Data extracted, calculated, or otherwise derived from that information;
- the expected direction and characteristic behavior of the degradation signature;
- the nominal or healthy Feature Data region;
- the ordered analysis coordinate used to represent progression; and
- the functional-failure condition toward which remaining life is estimated.

An ARULE application therefore defines both what degradation-related quantity is being evaluated and how successive observations are ordered.

The Feature Data value represents the degradation-related quantity of interest. The associated analysis coordinate establishes the progression basis used by the prognostic estimator and may represent elapsed time, operating cycles, accumulated usage, distance, mission count, sequential observations, or another ordered application-specific quantity.

The analysis coordinate should not be confused with the engineering domain from which the Feature Data were derived. A Feature Data value may originate from time-domain measurements, frequency-domain analysis, a physics-based model, statistical processing, or a model-to-test residual while still being evaluated by ARULE against a different progression coordinate.

This separation between application-specific feature engineering and application-independent prognostic processing is fundamental to the ARULE architecture.

ARULE may therefore operate on measured, derived, or model-informed Feature Data. Candidate Feature Data may originate from quantities such as vibration, impedance, resistance, capacity, temperature behavior, frequency response, model residuals, or other validated indicators. Physics-based, empirical, statistical, or data-driven models

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may also contribute to the upstream generation or qualification of the Feature Data supplied to ARULE.

Ridgetop's broader PHM framework is illustrated in Figure 1.

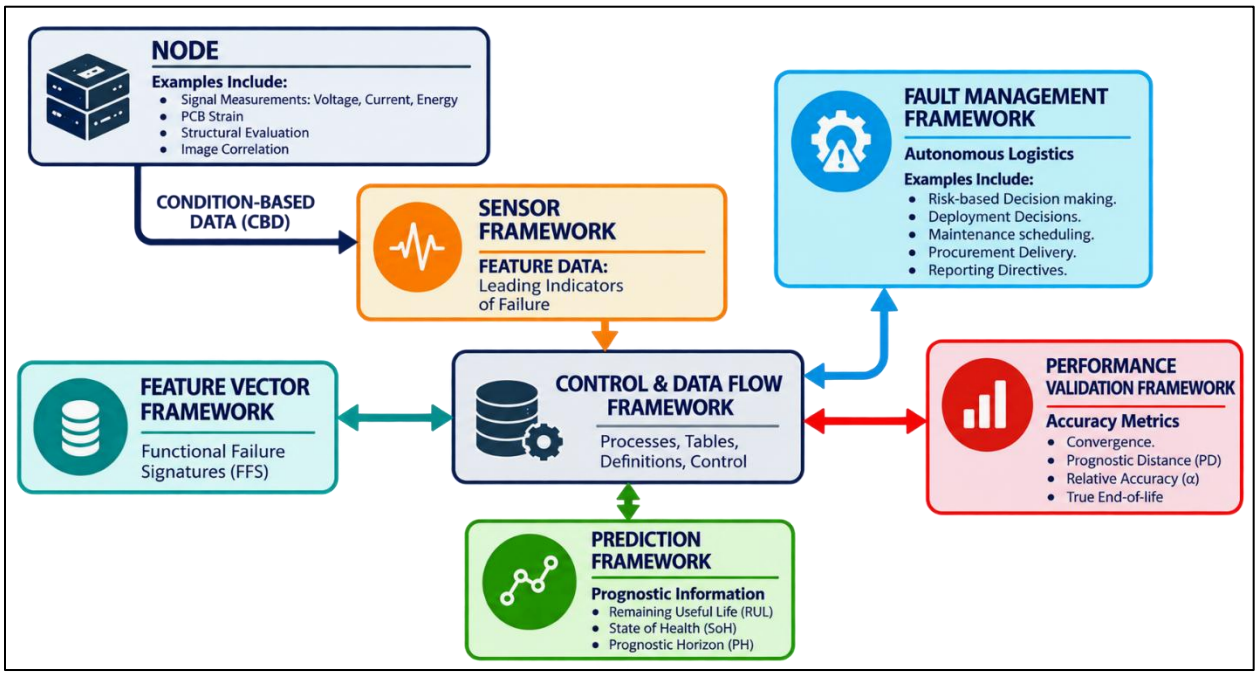


Figure 1 – Framework diagram for a PHM system

Within this framework, ARULE primarily provides the signature-processing and prognostic-estimation functions. Upstream PHM functions acquire Condition-Based Data and transform or qualify those measurements into degradation-related Feature Data. ARULE processes the resulting Feature Data and returns prognostic information. Downstream applications may combine those results with additional status, persistence, severity, mission, maintenance, safety, or operational criteria.

This separation of responsibilities is important:

ARULE produces prognostic information; it does not independently define the complete health-management decision process.

For example, ARULE does not inherently determine which sensor measurement should be selected as a degradation indicator, establish an application-specific alert policy, assign maintenance severity, or determine the operational action that should follow a prediction. Those responsibilities belong to the applicable upstream engineering process or downstream host application.

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The relationship between sensing, Feature Extraction, and ARULE prognostics can be further simplified using the three-stage PHM architecture shown in Figure 2:

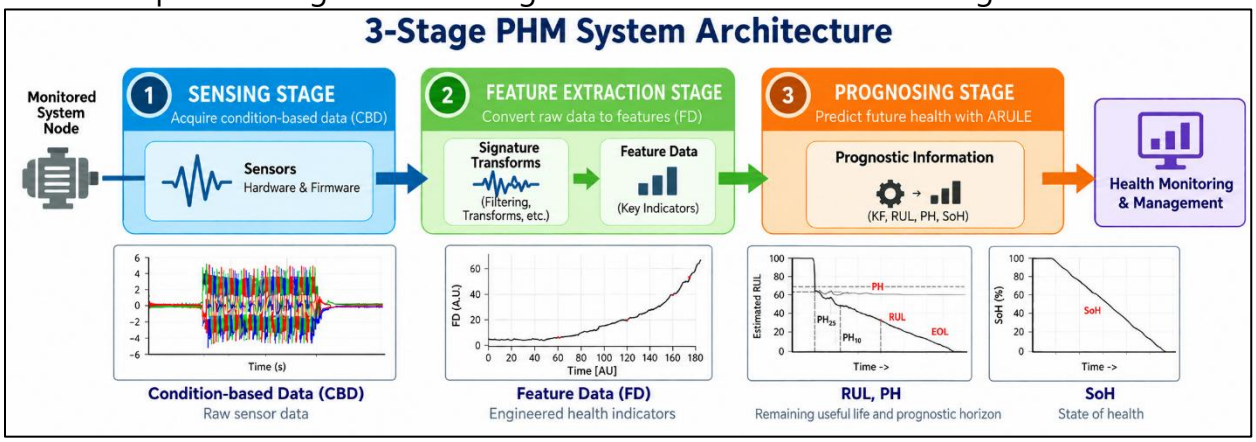


Figure 2 - Diagram of a 3-stage PHM system: (1) SENSING STAGE, (2) FEATURE EXTRACTION STAGE, and (3) PROGNOSING STAGE to Produce Prognostic Information with ARULE

This architecture provides the foundation for the remainder of Section 1.

1.2 Three-Stage Prognostic Health Management (PHM) System Architecture

The three-stage PHM architecture shown in Figure 2 provides a practical representation of the progression from measurements of a monitored asset to health and remaining-life information.

The three stages are:

1. **Sensing Stage** — acquires Condition-Based Data describing the state and operation of the monitored system or analysis node.
2. **Feature Extraction Stage** — applies application-specific processing to transform the acquired CBD into one or more Feature Data signatures associated with degradation, aging, damage, or another condition relevant to functional failure.
3. **Prognosing Stage** — processes the prepared Feature Data and estimates progression toward future functional failure, producing prognostic information such as RUL, SoH, and PH.

The architecture therefore represents the fundamental data progression:

- **Condition-Based Data → Feature Data → Prognostic Information**

The Sensing and Feature Extraction stages are inherently application-specific. They establish what is measured, how the measurements are conditioned or transformed, and

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which resulting quantity is sufficiently related to degradation to support prognostic analysis.

ARULE operates primarily in the Prognosing Stage.

The three-stage architecture is aligned with the functional organization described by the IEEE 1856-2017 Standard Framework for Prognostics and Health Management of Electronic Systems. The relationship between the Ridgetop three-stage architecture and the broader IEEE functional architecture is illustrated in Figure 3.

Conceptually:

- the **Sensing Stage** corresponds primarily to sensing and data-acquisition functions;
- the **Feature Extraction Stage** spans data-acquisition and data-manipulation functions;
- the **Prognosing Stage** encompasses applicable state-detection, health-assessment, and prognostic-assessment functions; and
- downstream advisory and health-management functions use the resulting prognostic information to support decisions and actions.

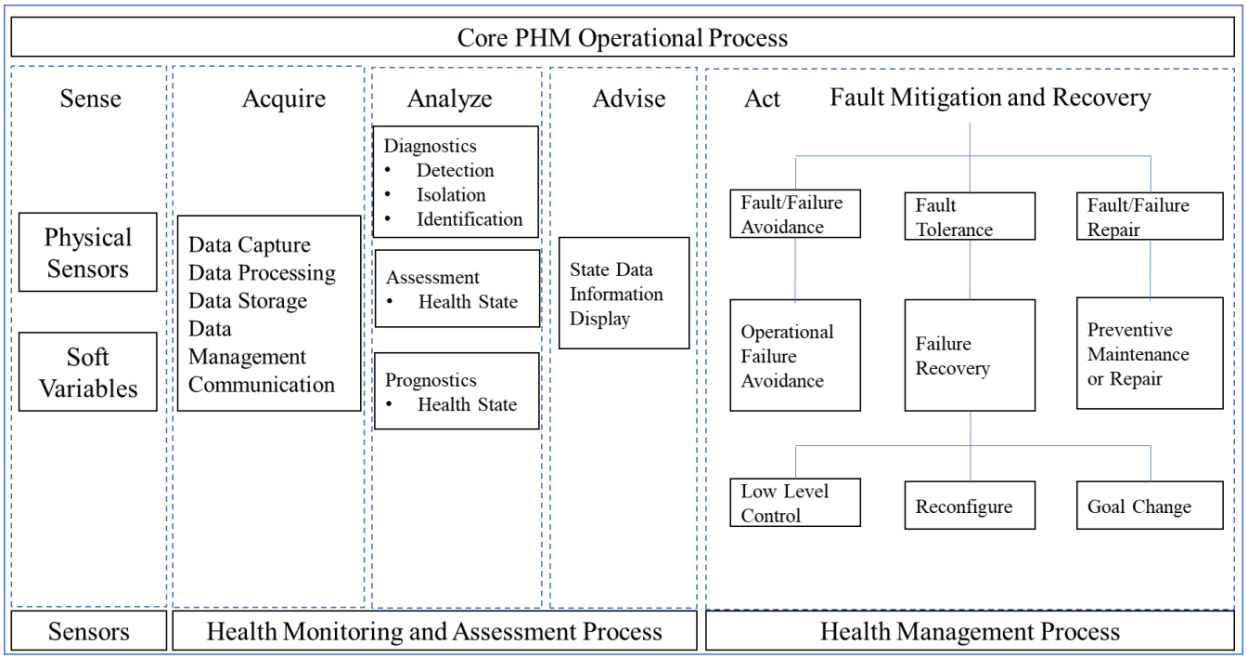


Figure 3 - General operational view of how the sensors, health monitoring, and assessment functions inherent within the electronic system enable the PHM capability operational processes [IEEE 1856-2017]

This layered view reinforces an important architectural principle: measurement, feature engineering, prognostic estimation, and health-management decisions are related but distinct functions.

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1.3 Sensing Stage

The Sensing Stage monitors one or more physical system nodes and acquires Condition-Based Data (CBD) associated with the state, operation, or environment of the monitored asset.

Depending on the application, sensors may measure electrical, mechanical, thermal, optical, chemical, acoustic, or other physical quantities.

Examples of CBD include:

- vibration;
- shock;
- temperature;
- voltage;
- current;
- pressure;
- strain;
- acoustic signals; and
- other directly measured system responses.

The output of the Sensing Stage may consist of raw, filtered, windowed, or otherwise conditioned measurement data.

Condition-Based Data may also be represented in different engineering domains. For example, sensor information may initially exist as a time-domain waveform, a periodically sampled measurement, a burst-mode acquisition, or another representation appropriate to the monitored system.

The domain in which CBD are acquired should not automatically be interpreted as the progression coordinate ultimately used by ARULE. The Sensing Stage establishes the source information from which degradation-related Feature Data may subsequently be extracted.

Important Sensing Stage considerations include:

- noisy data;
- sampling frequency;
- sampling interval;
- observation duration;
- periodic versus continuous acquisition;
- burst-mode sampling; and
- the relationship between the measurement process and the degradation mechanism being monitored.

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1.3.1 Noisy Data

CBD commonly contain information that is unrelated to the degradation mechanism of interest together with measurement and environmental noise.
Possible sources include:

- switching noise;
- thermal variation;
- signal distortion;
- changes in loading;
- sensor uncertainty;
- operating-condition variation; and
- other disturbances unrelated to degradation.

Figure 4 shows examples of noisy Condition-Based Data.

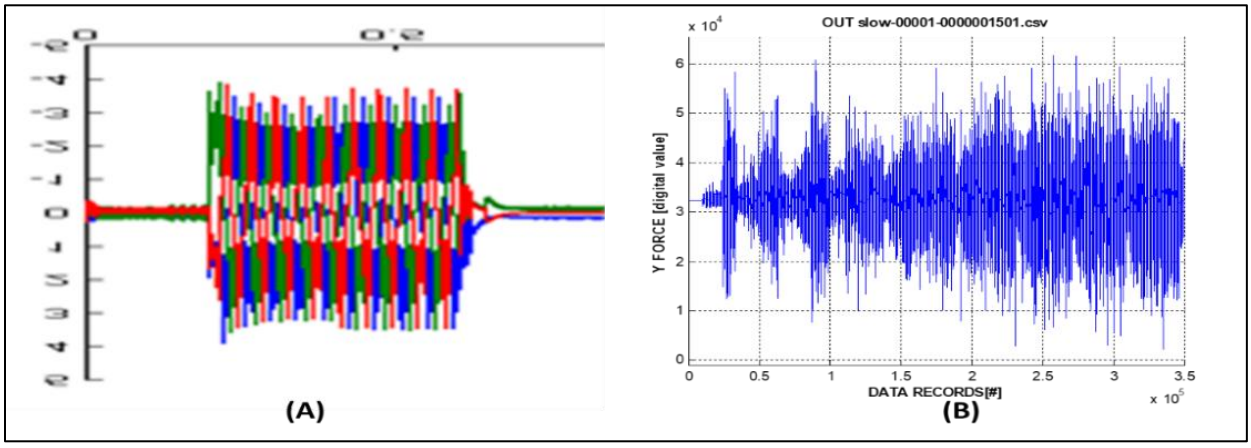


Figure 4 - Examples of noisy CBD: (A) phase currents and (B) y-direction accelerometer forces

Because useful degradation features may be obscured by noise or mixed with multiple system responses, further processing is often required. This processing may occur within a smart sensor, in downstream software, or across multiple stages of the PHM architecture.

Typical processing may include filtering, averaging, normalization, data fusion, domain transformation, model comparison, or extraction of one or more Condition Indicators or degradation precursors.

1.3.2 Continual and Periodic Sampling

The sampling strategy affects what information can be extracted from the measured system response.

As illustrated in Figure 5, acquisition may be:

- **continual**, at a sampling rate sufficient to capture the signal behavior of interest; or

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- **periodic**, where measurements are acquired at defined intervals appropriate to the expected degradation rate or monitoring objective.

Sampling strategy can also be used to reduce the amount of data that must be acquired, communicated, processed, and stored.

The correct strategy depends on both the dynamics of the source measurement and the degradation mechanism being monitored. A high-frequency sensor phenomenon may require high-rate acquisition even when the resulting degradation feature is evaluated only periodically.

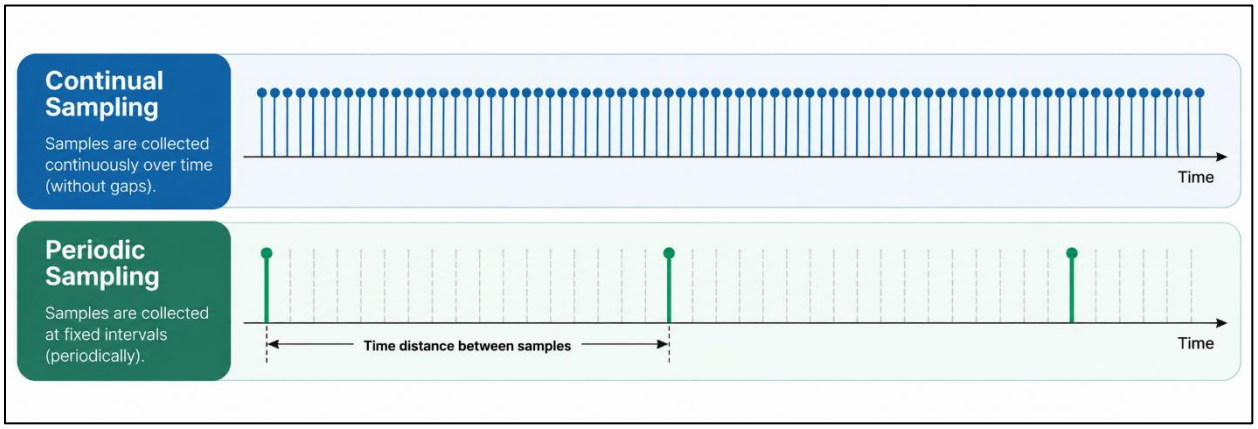


Figure 5 - Continual (top) and periodic sampling (bottom)

1.3.3 Periodic Burst-Mode Sampling with Averaging

A prognostic target may also be monitored using periodic high-rate burst acquisition. For example, a system may be evaluated once per hour while each observation consists of a short burst sampled at a much higher frequency. Processing performed within each burst can then produce a single Feature Data value associated with that observation.

Figure 6 illustrates this concept using a relatively slow observation period combined with higher-rate burst-mode sampling.

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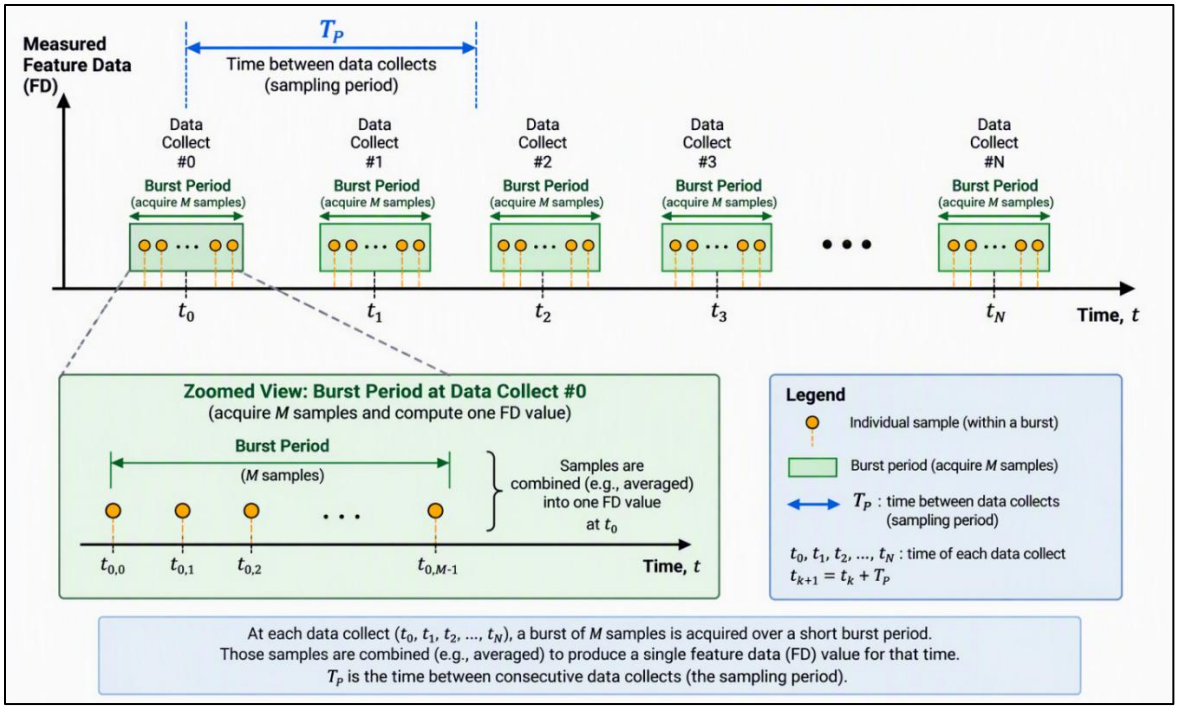


Figure 6 - Data sampling at 1/TP rate with burst mode at (s=7)/TB rate

This approach can reduce data-storage and communication requirements while preserving information associated with higher-frequency system behavior.

The distinction between the within-burst sampling domain and the between-observation progression coordinate is important.

For example, a high-frequency vibration waveform may be collected during each burst and processed in the frequency domain to generate one spectral Feature Data value. Those successive Feature Data values may then be ordered by operating hours, inspection number, or another progression coordinate for ARULE analysis.

1.4 Feature Extraction Stage: Application-Specific Data Processing

The Feature Extraction Stage accepts Condition-Based Data from one or more monitored system nodes and applies application-specific processing to produce Feature Data (FD) associated with degradation or damage.

Feature Extraction is one of the most application-dependent portions of the PHM process. The objective is not simply to reduce the amount of data, but to identify or calculate a quantity whose evolution provides useful information about progression toward the functional-failure condition of interest.

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Feature Extraction may involve processing in one or more engineering domains. Examples include:

- filtering and statistical processing of time-domain measurements;
- transforming time-domain measurements into frequency-domain representations;
- combining voltage and current measurements to derive impedance or resistance;
- compensating measurements for temperature or other operating conditions;
- applying physics-of-failure or empirical models;
- calculating statistical or distance-based Condition Indicators;
- comparing measurements with model-predicted behavior; or
- calculating residuals between measured and modeled quantities.

The domain in which a feature is derived is distinct from the coordinate used to track its degradation progression.

For example, a vibration Feature Data value could be the amplitude of a selected spectral component obtained from frequency-domain analysis. Successive values of that feature might then be ordered by operating hours or sequential inspections. In that case:

- the Feature Data amplitude is derived from the frequency domain; while
- the ARULE analysis coordinate represents accumulated operation or observation sequence.

Similarly, a battery Feature Data value could represent capacity, resistance growth, impedance, or a model-to-test residual, while its progression coordinate could represent charge/discharge cycles or elapsed test time.

Feature Extraction therefore defines both the degradation-related quantity to be evaluated and the context required to associate successive observations with an ordered degradation history.

Significant Feature Extraction topics include:

- Data-Processing Methods;
- Extracting FD;
- Modeling FD; and
- FD Signatures.

1.4.1 Data-Processing Methods

Feature Data can be extracted from Condition-Based Data using many application-specific analytical methods.

These may include:

- statistical methods;

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- reliability models;
- physics-of-failure models;
- signal-processing algorithms;
- data fusion;
- distance calculations such as Euclidean or Mahalanobis distance;
- pattern-recognition methods;
- machine-learning methods;
- empirical degradation models; and
- model-to-test comparison techniques.

For example, k-nearest-neighbor methods may be used to identify characteristic system behavior, distance metrics may quantify deviations from a healthy reference, and physics-based models may identify parameter changes that correlate with degradation.

The objective is to produce Feature Data whose change in value is sufficiently related to increasing damage or degradation to support health assessment and prognostic processing.

1.4.2 Extracting FD

Extracting useful Feature Data from CBD may require several levels of analysis.

Consider the switched-mode power supply (SMPS) example shown in Figure 7. The monitored output contains multiple candidate characteristics that could potentially respond to degradation of the output-filter components.

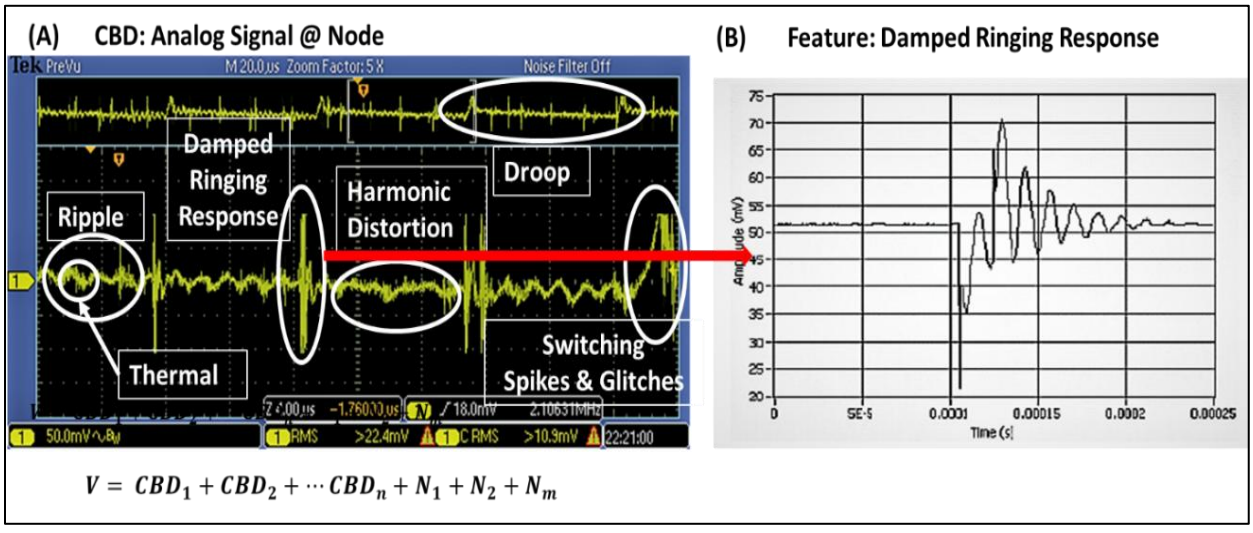


Figure 7 - SMPS output: six extractable features (A) and an extracted damped-ringing response (B)

After evaluating the measured output and the relevant failure mechanism, the damped-ringing response can be isolated as a candidate degradation-related response.

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Figure 8 illustrates additional features that can be extracted from the damped-ringing behavior.

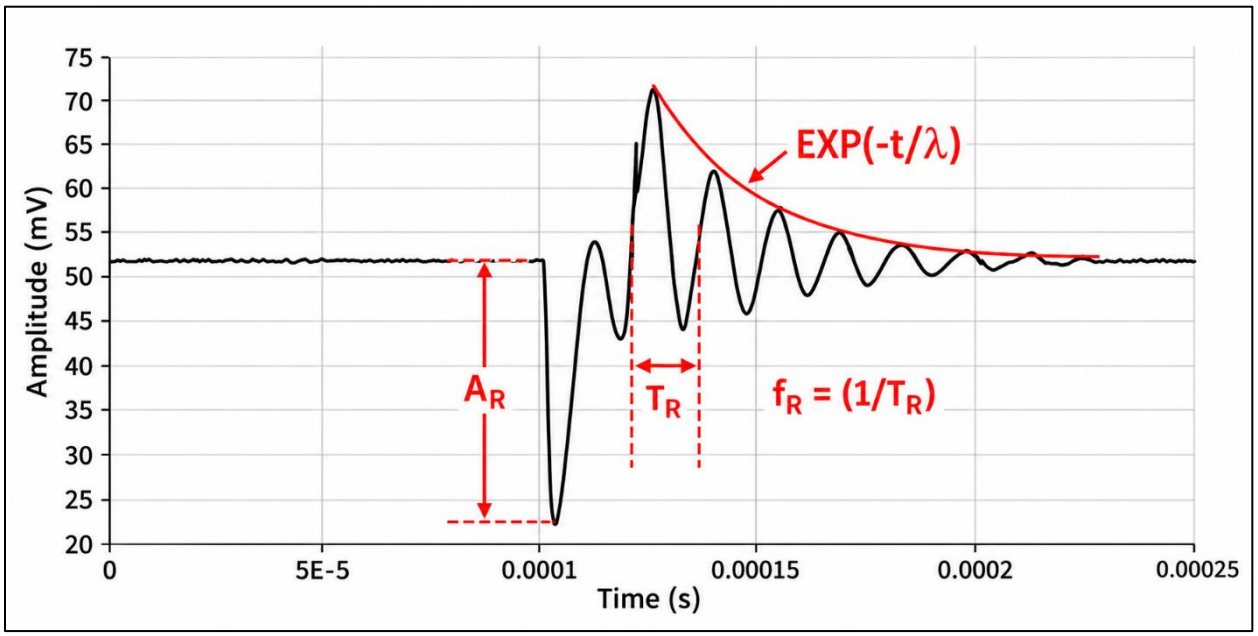


Figure 8 - Idealized damped-ringing response showing other extractable features

Possible features include:

- ringing amplitude;
- resonant frequency;
- ringing period;
- exponential decay or lifetime behavior; and
- other parameters related to the electrical characteristics of the filter.

The selected Feature Data should not be chosen solely because it is measurable. Its relationship to the degradation mechanism and functional-failure condition should be established using the available engineering evidence.

1.4.3 Modeling FD

Physics-of-Failure (PoF), empirical, or other engineering models may be used to identify which measurable quantity is most appropriate for degradation monitoring and to establish the relationship between that quantity and the underlying failure mechanism.

In the switched-mode power supply (SMPS) example, resonant frequency is selected as the Feature Data because the applicable analysis indicates that degradation-related changes in

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filtering capacitance produce a corresponding change in the resonant behavior of the output-filter circuit.

A measured system response may contain multiple physical contributions together with measurement and environmental noise. Conceptually, the measured response can be represented as:

$$V = CBD_1 + CBD_2 + \cdots CBD_n + N_1 + N_2 + \ldots + N_m \quad (1)$$

where:

- **CBD_1 ... CBD_n** represent Condition-Based Data contributions associated with the measured system response; and
- **N_1 ... N_m** represent noise or other unwanted contributions.

For the applicable SMPS example, the damped-ringing response can be represented as:

- Damped-ringing response (a CBD feature)

$$V_0 = V_{DC} + A_R\{exp(-t/\lambda)\}\{cos(\omega t + \phi)\} \quad (2)$$

where:

- **V_0** is the measured damped-ringing response;
- **V_DC** is the DC component;
- **A_R** is the ringing amplitude;
- **t** is time within the measured response;
- **λ** is the exponential decay or lifetime parameter;
- **ω** is angular frequency; and
- **φ** is phase.

The natural resonant frequency is related to the electrical characteristics of the circuit by:

$$\omega_0 = \sqrt{A_R + 1}\ (1/\sqrt{LC}) \quad (3)$$

Circuit quality can be expressed as:

$$Q = \sqrt{A_R + 1}\ (1/R)(\sqrt{C/L}) \quad (4)$$

-
- and the measurable angular frequency can be represented as:

$$\omega = \omega_0\sqrt{1 - 1/(4\,Q^2)} \quad (5)$$

For the assumptions applicable to this example, including a relatively high value of **Q** and comparatively small changes in **A_R**, **R**, and **L**, the relationship can be simplified to emphasize the effect of capacitance degradation:

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$$\omega \approx \omega_0 \sqrt{(C0/(C0 - dC))} \qquad C0 \text{ is the base value of } C \qquad (6)$$

where:

- C_0 is the nominal or baseline capacitance; and
- **dC** represents the degradation-related change in capacitance.

The corresponding Feature Data can then be defined as the measurable resonant frequency:

$$FD = \omega/2\pi \qquad (7)$$

This example illustrates how physics-based analysis can be used to identify a measurable quantity whose evolution is related to degradation.

It also illustrates an important distinction between the Feature Data amplitude and the ARULE progression coordinate. In this example, the Feature Data amplitude is resonant frequency and is therefore expressed in frequency units. Successive frequency observations could nevertheless be ordered by elapsed operating time, operating cycles, inspection number, or another application-specific progression coordinate.

For example, the resulting Feature Data signature could conceptually be expressed as:

- resonant frequency = f(operating time)

rather than interpreting frequency itself as the progression coordinate.

The selected Feature Data should therefore be evaluated with respect to both:

1. the physical or engineering relationship between the Feature Data amplitude and the degradation mechanism; and
2. the ordered coordinate over which successive Feature Data observations describe progression toward functional failure.

1.4.4 FD Signatures

A Feature Data signature represents the evolution of a degradation-related Feature Data value over an ordered application-specific coordinate.

Conceptually:

- FD=f(x)

where:

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- (x) is the **ordered analysis or progression coordinate**; and
- (FD) is the corresponding **Feature Data amplitude**.

The analysis coordinate functions as the independent variable, or x-axis, of the Feature Data signature.

Depending on the application, it may represent:

- elapsed time;
- operating hours;
- charge/discharge cycles;
- accumulated usage;
- mission count;
- distance;
- revolutions;
- sequential sample or inspection number; or
- another application-defined quantity that preserves the order of observations.

The Feature Data amplitude forms the dependent variable, or y-axis. Its units and physical meaning depend on the selected degradation feature and may represent frequency, impedance, resistance, capacity, vibration amplitude, temperature-derived metrics, model residuals, or other engineered quantities.

Accordingly, a Feature Data signature should be interpreted as:

- a degradation-related quantity progressing as a function of an ordered usage or observation coordinate

rather than assuming that the Feature Data source domain and the prognostic coordinate are the same.

Referring to Figure 9, a Feature Data signature may remain relatively stable during an initial no-degradation region and then change as degradation progresses.

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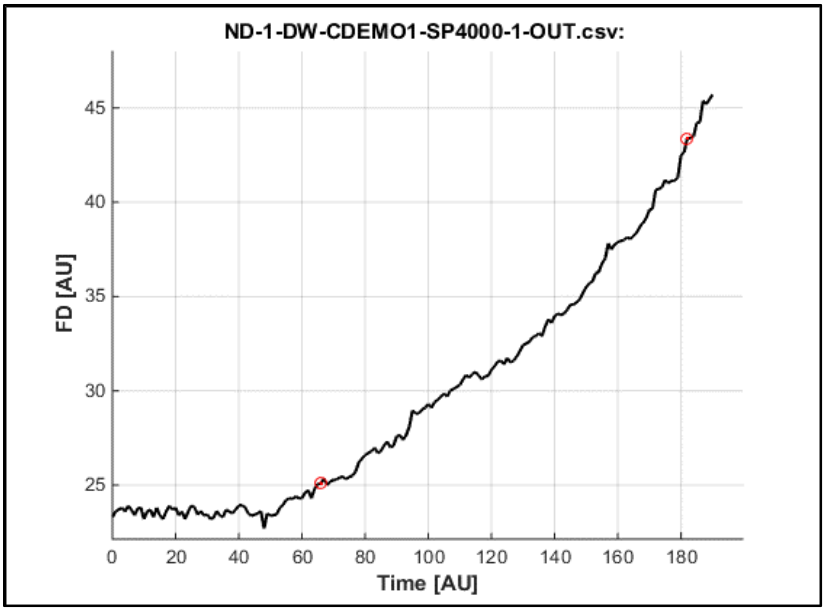


Figure 9 - Example of an FD signature (resonant frequency)

The signature may contain both degradation behavior and multiple sources of variation or noise. Figure 10 identifies representative regions and noise effects that may be present in an FD signature.

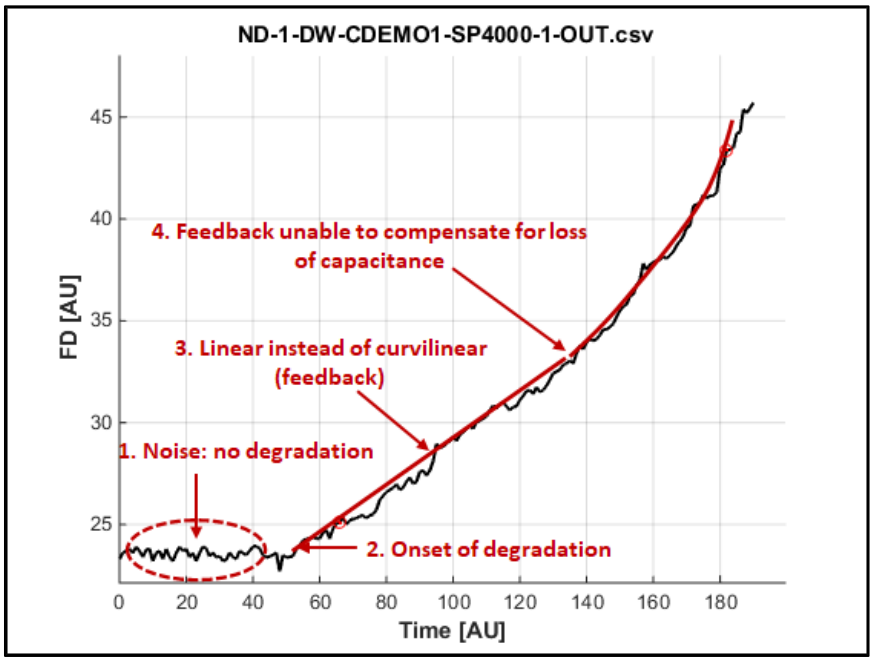


Figure 10 - Annotated FD signature: noise, degradation, linear degradation (feedback), and curvilinear degradation (feedback no longer effective)

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The objective of Feature Extraction is not necessarily to force the Feature Data into a perfectly linear or noise-free form. Rather, the resulting signature should contain sufficient degradation information for subsequent ARULE signature processing and prognostic estimation.

1.5 Prognosing Stage

The third stage of the PHM architecture is the **Prognosing Stage**, where ARULE processes the prepared Feature Data and estimates future progression toward the defined functional-failure condition.

The ARULE Adaptive Predictive Kernel comprises two major functional subsystems:

1. **Signatures Subsystem**
2. **Prognostics Subsystem**

The Signatures Subsystem conditions and transforms the application-specific Feature Data into degradation-signature representations suitable for prognostic estimation.

The Prognostics Subsystem applies adaptive estimation and prediction methods to estimate the future functional-failure condition and produce prognostic information such as RUL, SoH, and PH.

Because ARULE estimates future functional failure along the ordered progression of the Feature Data signature, the RUL coordinate basis is inherited from the analysis coordinate supplied with the input data**.**

For example:

- if Feature Data are ordered by operating hours, RUL is interpreted in operating hours;
- if they are ordered by charge/discharge cycles, RUL is interpreted in cycles;
- if they are ordered by mission count, RUL is interpreted in missions; and
- if they are ordered only by sequential sample number or another arbitrary progression coordinate, RUL must be interpreted in that same coordinate basis unless an external conversion has been established.

The physical units of the Feature Data amplitude do not determine the units of RUL.

For example, resonant frequency may be expressed in hertz while the ARULE progression coordinate is operating hours. In that case, the degradation signature represents frequency as a function of operating hours, and the resulting RUL is interpreted in operating hours.

This distinction is fundamental when interpreting ARULE results and comparing results across datasets or applications.

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1.5.1 Signatures Subsystem

The primary function of the ARULE Signatures Subsystem is to condition and transform Feature Data signatures into normalized degradation representations used by the prognostic estimator.

The applicable processing sequence is:

Feature Data (FD) → Fault-to-Failure Progression (FFP) → Degradation Progression Signature (DPS) → Functional Failure Signature (FFS)

The transforms are performed sequentially as observations are processed.

Conceptually:

- **FD** represents the application-specific degradation feature;
- **FFP** represents progression from the nominal Feature Data region toward the configured functional-failure condition;
- **DPS** provides a transformed degradation-progression representation suitable for subsequent modeling; and
- **FFS** provides the failure-signature representation used by the Prognostics Subsystem.

The Signatures Subsystem may also apply configured conditioning operations such as averaging, noise margins, normalization, and transformation parameters.

Processing the observations sequentially allows ARULE to update its degradation representation as new information becomes available and supports near-real-time or replay-based prognostic workflows.

1.5.2 Prognostics Subsystem

The Prognostics Subsystem accepts the processed Functional Failure Signature and applies adaptive prediction and estimation methods to generate prognostic information.

The prognostic process evaluates the evolving degradation signature relative to the configured functional-failure condition and produces estimates including:

- Remaining Useful Life (RUL);
- State of Health (SoH);
- Prognostic Horizon (PH); and
- applicable execution or estimator status information.

The estimator updates its assessment as each new observation is processed. Before sufficient degradation evidence is available, prognostic behavior may be influenced substantially by initialization parameters and prior assumptions. As additional Feature Data are processed and degradation becomes observable, the adaptive estimation

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process can update the fitted degradation behavior and the associated prognostic estimates.

Functional failure is defined relative to the monitored item's ability to continue operating within the applicable performance or specification criterion used for the analysis. It is therefore not necessarily equivalent to catastrophic physical failure.

ARULE predicts progression toward the configured functional-failure condition. The validity of the resulting RUL, SoH, and PH estimates therefore depends on the validity of the upstream Feature Data, coordinate definition, degradation mapping, configuration parameters, and functional-failure criterion. Together, the Sensing, Feature Extraction, and Prognosing stages provide a structured pathway from measured system behavior to application-specific prognostic information.

2. ARULE: Description and Operation

Referring to Figure 11, ARULE receives qualified Feature Data (FD) from an external application and processes the observations sequentially through the Adaptive Predictive Kernel (APK). For each ordered Feature Data observation, ARULE updates the applicable degradation-signature representation and prognostic state before returning prognostic information such as Remaining Useful Life (RUL), State of Health (SoH), Prognostic Horizon (PH), and associated status information.

The external application is responsible for supplying the applicable Feature Data, analysis coordinate, configuration information, and other required inputs. ARULE performs the signature-processing and adaptive prognostic-estimation functions described in this section.

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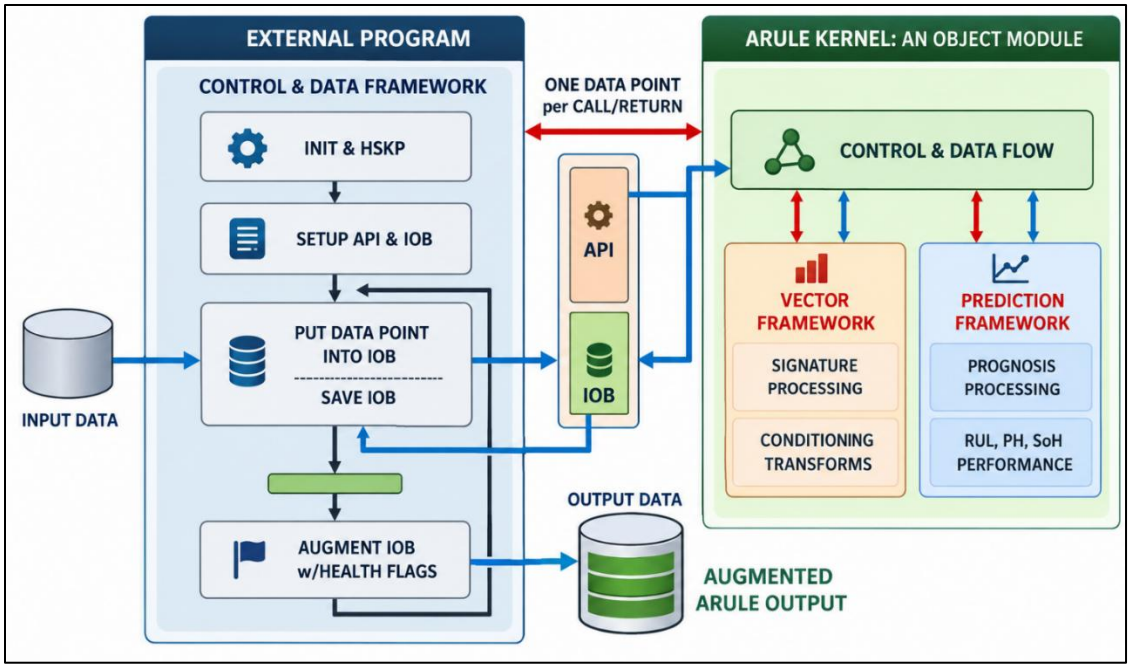


Figure 11 - Operational diagram: external program using ARULE through its API

ARULE is designed to support a wide range of electrical, mechanical, electromechanical, battery, and other monitored systems when suitable Condition-Based Data can be processed into Feature Data that have an established relationship to degradation and a defined functional-failure condition.

The same underlying prognostic methodology can therefore be applied across different applications without requiring all applications to use the same physical measurement or degradation variable. Depending on the application, the Feature Data may represent a directly measured quantity, a derived Condition Indicator, a frequency-domain feature, a model parameter, a model-to-test residual, or another validated degradation-related quantity.

ARULE processes the Feature Data sequentially and adapts its prognostic assessment as new degradation evidence becomes available. The rate at which the prognostic estimates stabilize or converge depends on several application-specific factors, including:

- Feature Data noise and variability;
- sampling or observation spacing;
- degradation-signature behavior;
- the difference between the configured initial time-to-functional-failure assumption and the subsequent degradation-supported estimate;
- the selected signature-processing and prognostic-model parameters; and

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- the quality of the Feature Data and functional-failure mapping.

Benchmark evaluations using selected ARULE reference datasets have demonstrated rapid convergence under favorable conditions. In low-noise reference cases with relatively small initialization error, estimates have converged to within a 10% accuracy margin after only a few degradation observations. Reference cases with higher noise or larger initialization error have required additional observations before reaching the same margin.

These benchmark results should be interpreted as examples of demonstrated behavior under the tested datasets and configurations, rather than as a universal guarantee of convergence rate or prognostic accuracy. Actual performance depends on the application, Feature Data quality, analysis coordinate, degradation behavior, functional-failure definition, initialization assumptions, and ARULE configuration.

The remainder of this section describes the ARULE software architecture, definition files, signature-processing sequence, and adaptive prognostic-estimation methods used to transform application-specific Feature Data into prognostic information.

2.1 Graphical User Interface Diagram

The ARULE Graphical User Interface (GUI) was designed and implemented as example of the “External Program” referenced in Figure 11. The underlying Adaptive Predictive Kernel (APK) within ARULE is a command line interface (CLI) program that can be embedded or called by an “External Program” such as the ARULE™ GUI. The CLI utilizes a flexible API, and Ridgetop offers engineering services to create customized installations for real–time system health monitoring. This User Guide covers the ARULE™ GUI implementation that links each of the PHM frameworks shown in Figure 12.

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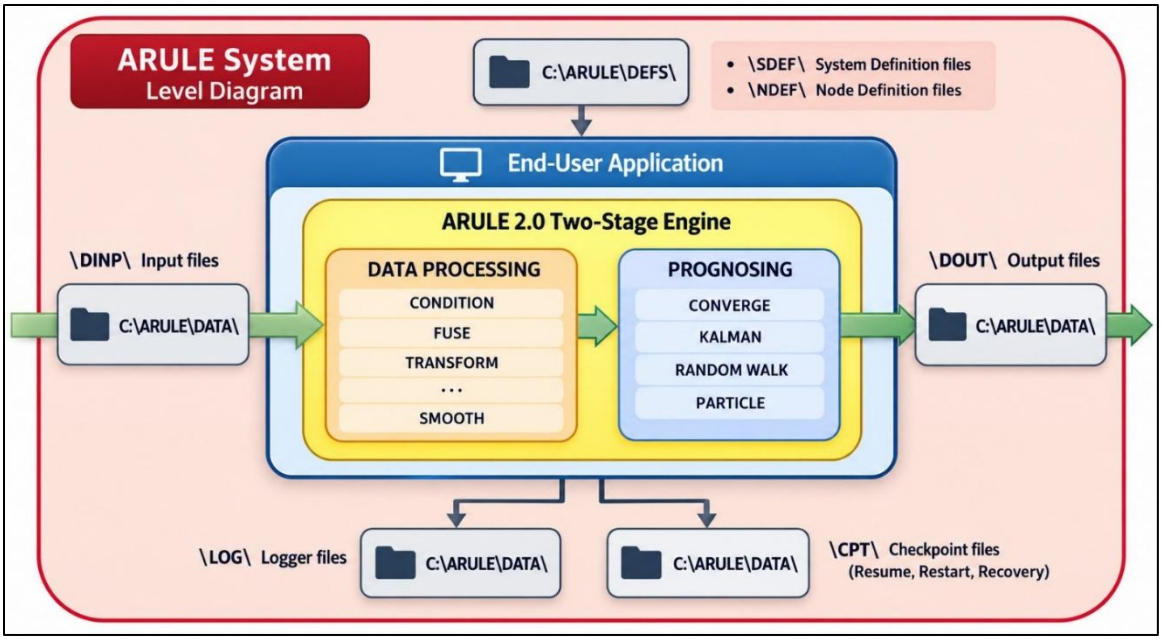


Figure 12 – Framework for a GUI–based ARULE™

2.2 Definition Files

The processing and control of ARULE™, including the ARULE™ Adaptive Prediction Kernel (APK), is driven by tables created from definition files: those tables are specified as editable files instead of programmed tables; they are called definition files and are described as follows:

- **System Definition (SDEF) Files** – Define the number of nodes in a system, and points to a corresponding node definition file.
- **Node Definition (NDEF) Files** – Defines the path for the input sensor data file, along with the user defined parameters and values required to process the input data with the ARULE™ APK. **Table 1** shows the complete list of NDEF parameters.

When an SDEF file and the corresponding NDEF files are created, the program can process the data input to generate key prognostic estimates for SoH, RUL, and PH. The generated data output files, log files, and system files support checkpoint/restart and recovery from unexpected end–of program events. The ARULE™ GUI also supports creating, editing, and visualization of the definition files, as well as plotting capabilities to preview the data output results.

Table 1. List of Node Definition (NDEF) Parameters

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ID NAME	DESCRIPTION	VALUE: TYPE, RANGE, and DEFAULT
1. FDZ	Specifies a nominal FD value to use when transforming FD into FFP signature data: FFP = [FD – FDC*(1 + FDNM/100)]/FDZ	• Any positive value • 0 means use FDC value
2. FDC	Specifies a nominal FD starting value to use in the absence of degradation/damage and a moving average is used to calculate FDC based on FDCPTS.	• 0.0 to any positive value • 1.0 (default)
3. FDCPTS	Specifies the number of pts to calculate the moving average of FDC.	• 0 to 25.0 • 10.0 (default)
4. FDPTS	Specifies number of data points as integer to average for FD.	• 1 to 5 • 5 (default)
5. FDNM	Specifies a noise margin; percent of the nominal value of FDC to use as a margin to mitigate noise	• 0 to 25.0 • 5.0 (default)
6. FDNV	Specifies the n or lambda value for transforming FFP signature data to DPS signature data	• Any non-zero positive value • 1.0 (default)
7. FFPFAIL	Specifies a percentage of FFP to use as a threshold value for functional failure: FF = FFP/(FFPFAIL/100)	• Any non-zero positive value • 70.0 (default)
8. PITTFF	Specifies an initial value to use for time-to-functional failure (TTFF): becomes the RUL value in the absence of degradation	• Any non-zero positive value • 200.0 (default)
9. PIFFSMOD	Specifies the expected shape of an FFS curve for EKF modeling: 1: convex, 2: linear, 3: concave, 4: convex-concave, 5: concave-convex;	• 1 to 5 • linear (default)

2.3 Directory Structure: Description and Contents

ARULE™ GUI files are installed in the default windows directory as outlined below:

C:\Program Files\ARULE **Executable subdirectory**

1. DEPENDENCIES\ ARULE™ GUI Dependencies
2. DOCS\ User Guide Document & Product Brief
3. ARULE.exe ARULE Executable Program

Other ARULE™ program files such as the input data files, output data files, and the system and node definition files are kept in a user specified directory selected when the software is first installed. There are two (2) main directories and seven (7) subdirectories

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that are installed in the user specified location. An example of how these subdirectories are organized in the default installation (**C:\Users*username*\My Documents\ARULE**) is shown below.

...\ARULE\DATA	**Data set of subdirectories**
1. CPT\	Checkpoint Files
2. DINP\	Data Input Files
3. DOUT\	Data Output Files
4. LOG\	Logger Files
5. SAVE\	Default Save/Download Directory

...\ARULE\DEFS	**Definition set of subdirectories**
6. NDEF\	Node Definition Files
7. SDEF\	System Definition Files

2.4 BNF for Definition Files

The ARULE™ GUI offers intuitive user forms to edit definition files, but it is important to note for freeform editing purposes, ARULE™ uses a simple BNF (Backus–Naur Form or Backus–Normal Form) for creating definition files:

- Input records are a maximum of 80 characters in length: including the new line character that ends the record.
- A percent sign (%) character starts a comment: it and all other characters that follow are ignored.
- Continuation of one record to the next is not supported.
- Data format is operand operator operand:
  - The first operand is a parameter keyword (such as NDNUMID).
  - There is only one operator – the equal sign (=) character.
  - The second operand is a keyword value to be assigned to the parameter keyword.
- Apostrophe (‘) characters are used to delimit the beginning and ending of string type of keyword values and are required when a string includes white space

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type of characters. To avoid errors in usage, the ARULE™ GUI automatically inserts apostrophes for you – do not enter them yourself.

- A semicolon (;) character to delimit a keyword value is automatically inserted by the ARULE™ GUI – so do not enter a semicolon yourself.

2.5 ARULE: Signature Processing

Referring back to Figure 1, ARULE comprises data algorithms and methods that have been incorporated into a Feature Vector Framework that processes signature data and a Prediction Framework that produces prognostic information. For each input data, ARULE produces a set of prognostic information before another data point is processed: ARULE uses data models, signature transforms, Extended Kalman Filtering, geometry-based computations, and various linearization and machine learning to produce fast, efficient, and accurate estimates of RUL, PH, and SoH. To illustrate and describe ARULE, the definition values for FDZ and FDNV shown in Figure 14 were changed respectively from 24.0 to 0.0 and from 2 to 1.275.

```
%*****************************************************************
% NODE1FULL: Full NDEF definition equivalent to CNODE1
%*****************************************************************
%**Feature Data: FD = FDZ*(dP/P)^FDNV + DC + NOISE
FDC       = 24.0;   % Feature Data (FD), DC Value
FDZ       = 24.0;   % Nominal AC value, when 0 use FDC value
FDNM      = 5.0;    % Noise margin - percent of the value of FDC
FDCPTS    = 10;     % data points to average for FDC: up to 25
FDNV      = 2;     % Degradation power n
%**Prognostic Modeling
FFPFAIL   = 70.0;   % Failure margin - percent above nominal
PITTFF    = 200.0;  % Default RUL = TTFF value
PIFFSMOD  = 2;      % model 1=Convex, 2=Linear, 3=Concave,
%                    4=convex-concave, 5=concave-convex, 6=convex-concave
%**       File Dependent Parameters
INFILE    = 'SP4000_1';  % Input file name, _OUT  appended for output
INTYPE    = '.txt' ;    % also .csv Input file type
OUTTYPE   = '.csv' ;    % also .txt Output file type
%*****************************************************************
ENDDEF    = -9;     % end of node definition
```

Figure 13 - An example instantiation of defined set of API keywords and values to use the ARULE kernel

2.5.1 Transforming a FD Signature into a Fault-to-Failure (FFP) Signature

ARULE conditions and transforms input FD signature data into Fault-to-Failure Progression (FFP) signature data:

$$\boldsymbol{FD_i = DC_i + N_i + FD_0(f(dP, P))} \quad \textbf{(8)}$$

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And ideal set of FFP is defined as,

$$\boldsymbol{FFP_i = FD_i / FD_0} \quad \textbf{(9)}$$

Referring back to Figure 13, specify the following.

$FDC \geq \max(\mathrm{DC_i})$, $\quad (FDNM/100) \geq \max(N_i)/100$, $\quad FDZ = \mathrm{FD_0}$

$FFP_i = (-FDC - NM)/FDZ + f(dP_i, P)$ [**from (8) and (9)]**

$FFP_i = f(dP, P) - c$ and when $c \ll max(f(dP_i, P))$

$$\boldsymbol{FFP_i \leq f(dP_i, P)} \quad \textbf{(10)}$$

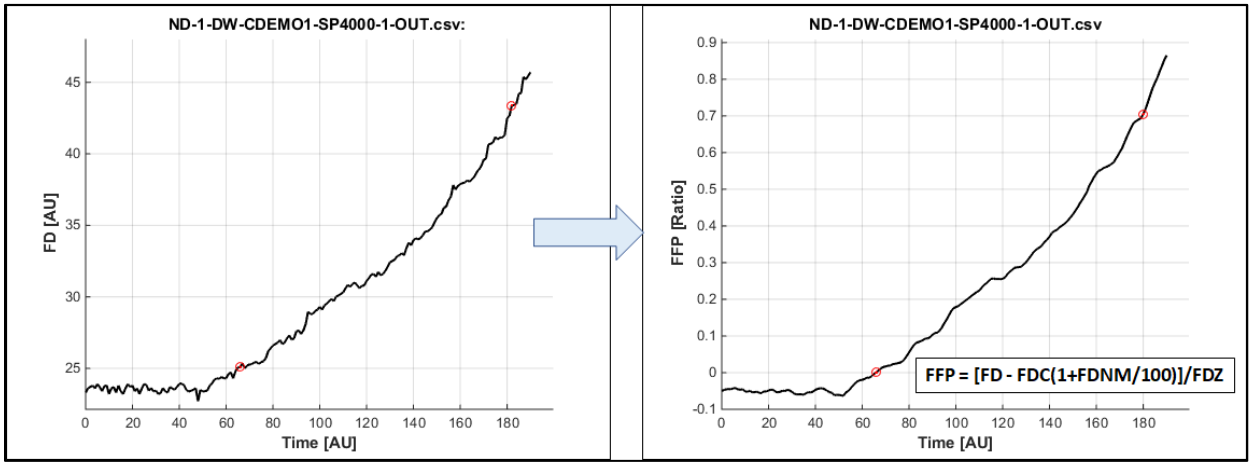


Figure 14 - FD signature data transformed into an FFP signature data

Note the following about the FD to FFP transform shown in Figure 14 :

1. The data has been shifted from a non-zero, application-specific reference of about 24 to a generic zero reference:
   - A nominal FD value was determined by averaging input data in the absence of degradation using values specified by FDC, FDCPTS, and FDCVAR.
   - A noise-margin value determined by a percent value of FDC as specified by FDNM
2. The data has been smoothed to allow specification of a smaller FDNM value to reduce the magnitude of offset distortion introduced by a noise-margin shift of the data.
3. Division of the data by the value of FDZ transforms the Y-axis values from units-of-measure to generic ratios, which reduces modeling complexity.

2.5.2 Transform FFP Signature Data into DPS Data

Modeling complexity and use is reduced in ARULE by use of a single model for DPS data based on the inherent inaccuracies associated with processing CBD that is noisy, distorted, digitized, and discretely sampled.

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$$DPS = (dP_i/P) \quad \textbf{where P represents a parameter that degrades} \quad \textbf{(11)}$$

$$f(dP_i, P) = (dP_i/P)^n \quad \textbf{power of n function} \quad \textbf{(12)}$$

Then by (11) and (12),

$$DPS = (FD_i/FD_o)^{1/n} = (FFP_i)^{1/n}$$

(13)

2.5.3 Significance of the FFP-to-DPS Transform

Because (dP_i/P) is a derivation (the rate of change of P to P), transforming FFP signature data into DPS data linearizes a curvilinear signature (Figure 15).

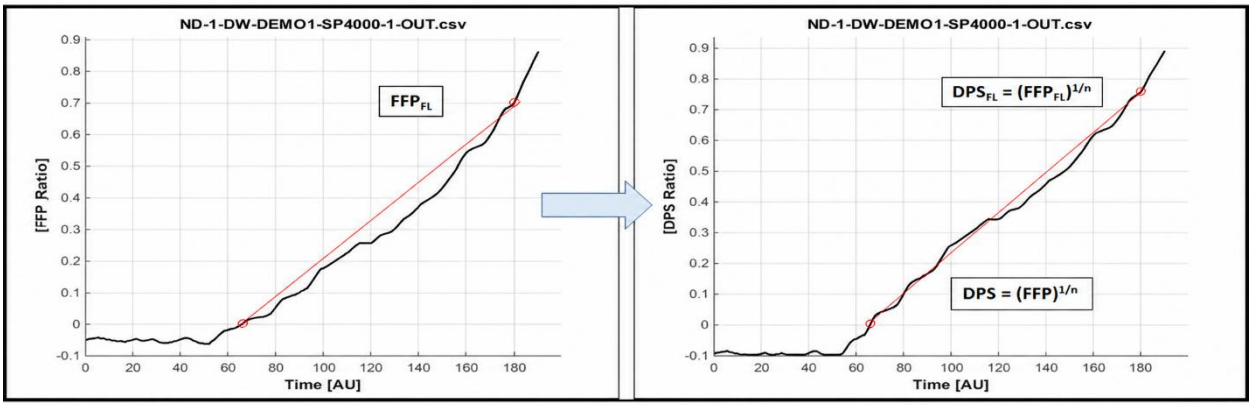


Figure 15 - FFP signature data transformed into DPS data

Referring back to Figure 15, note the following about the FFP to DPS transform:

- Negative FFP values are transformed to zero (0): no detectable degradation.
- The curvilinear FFP data has been linearized,

2.5.4 The FFP-to-DPS Model

Figure 16 contains ideal FFP plots for selected values of n (left) and the corresponding DPS plots: an ideal straight line.

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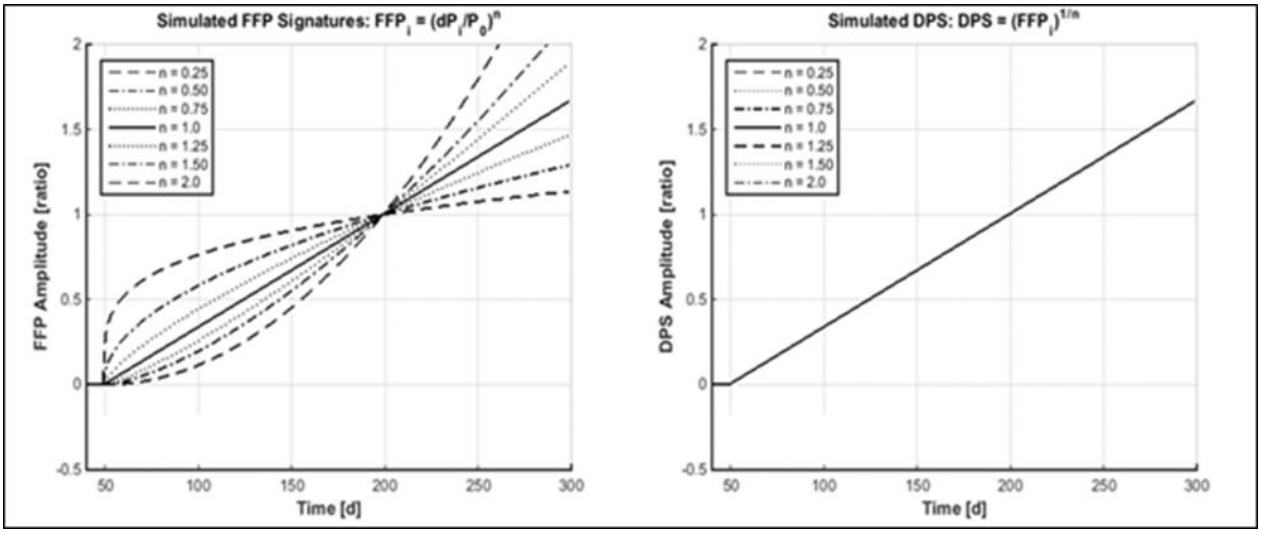


Figure 16 - Power Function: FFP on the left, DPS on the right

2.5.5 Transform FFP to DPS Example: Modeling and Specification

The FFP to DPS transform comprises the following steps: (1) obtain experimental and/or simulation data for the failure mode to be prognosticated; (2) select a beginning and ending point on the curve – above the noise level and below a level of functional failure; (3) determine the characteristic-signature curve of the failure mode – convex, linear, or concave; (4) and select the power value (see Figure 16 and Figure 17); (5) use the FDNV keyword to specify the value; (6) process the input FD; and (7) verify, validate, and change the specified FDNV to obtain a desired/reasonable level of DPS linearity (see Figure 18).

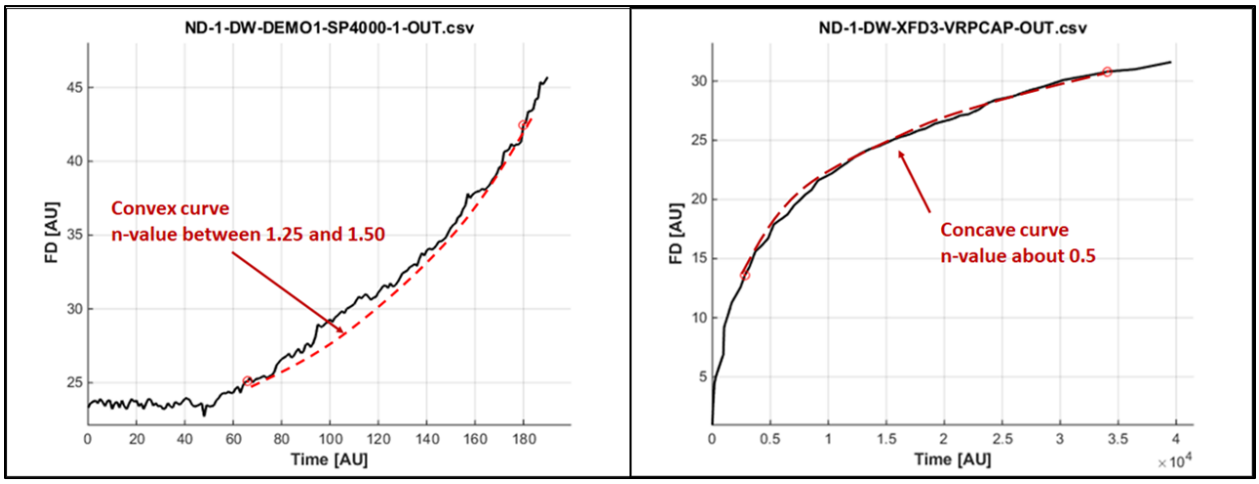


Figure 17 - Examples of FD input, their characteristic curves, and initial evaluation of probable n values

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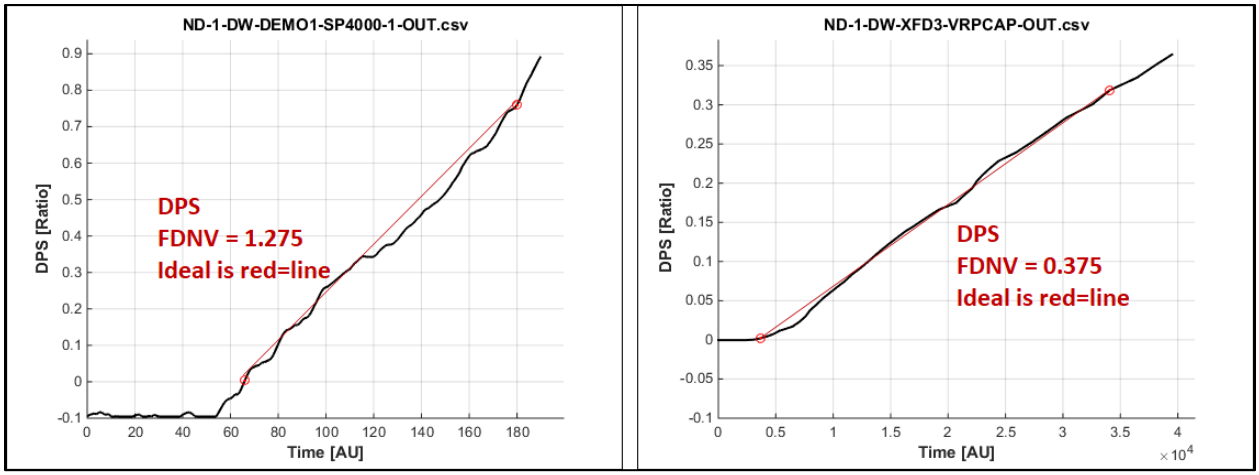


Figure 18 - Examples of DPS transforms for specified FDNV values for the FD input in Figure 17

2.5.6 Transforming DPS Data into Functional-Failure-Signature Data

A failure, as it relates to prognostics, is often ill-defined. For example, does failure occur when a drive shaft shears, or when the friction between the shaft and a roller bearing reaches a level where torque on the shaft exceeds design specifications, or when the shaft becomes warped and the unit begins to vibrate?

ARULE predictions are based on functional failure rather than simply failure or physical failure. Functional failure occurs when a prognostic target (a device, component, subassembly, and so on) is no longer capable of operating within specifications. Functional failure, therefore, occurs before physical failure or operational failure: much like a tire that is thread bare but still fully inflated.

The FFPFAIL keyword defines the FFP threshold at which functional failure occurs. ARULE autonomously transforms that value into its equivalent DPS value which transforms that value into an FFS value of 1, which is then transformed into a value of 100 percent (Figure 19):

$$\boldsymbol{FFS_i = 100\ (DPS_i / FL_{DPS})} \qquad \textbf{(14)}$$

Significant results of the transform are the following:

1. The DPS y-ratio values are transformed into FFS y-percent values.
2. The FFS y-values mean the following:
   a. FFS value = 0 -> degradation is not detectable.
   b. FFS value >= 100 -> functional failure has occurred
   c. FFS value between 0 and 100 -> degradation is detected

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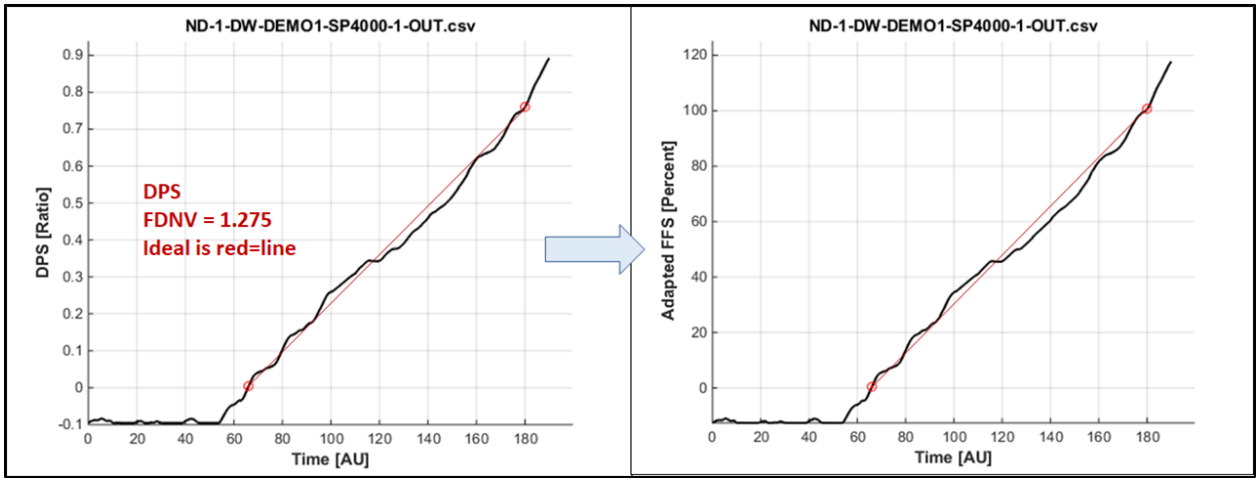


Figure 19 - DPS data transformed into FFS data – ideal transforms are red-colored lines

2.6 ARULE: Prognostic Processing and Filtering Methods and Results

ARULE employs Extended Kalman Filtering (EKF) in that input FFS data is compared to two sets of models. A data point is compared to a first, SoH model, then the model is adapted to the data point, and then the data point is adapted to the model with the adaptation biased in favor of a pessimistic level of SoH: the adaptation supports both gradual changes in health caused by increasing degradation and rapid changes in health caused by abrupt changes in health. The adapted data point is then compared to a second, RUL model, the model is adapted to the data point, then the adapted RUL model is used to project a future time of failure (Figure 20).

1. As though data points are particles that exhibit properties of momentum and inertia.
2. Data points travel in random-walk fashion through three data spaces as illustrated in Figure 21.

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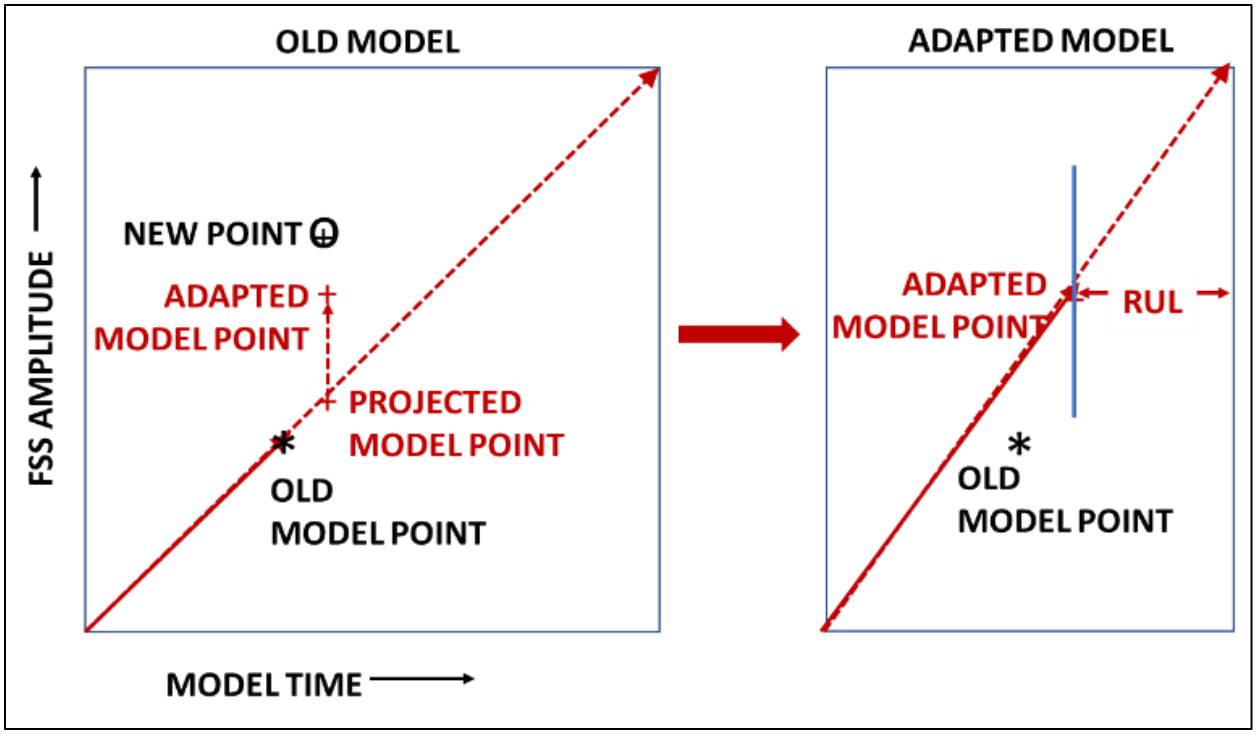


Figure 20 - EKF: adapt data point to 1st model, adapt 2nd model to adapted data, then use adapted 2nd model to predict RUL, PH, and SoH.

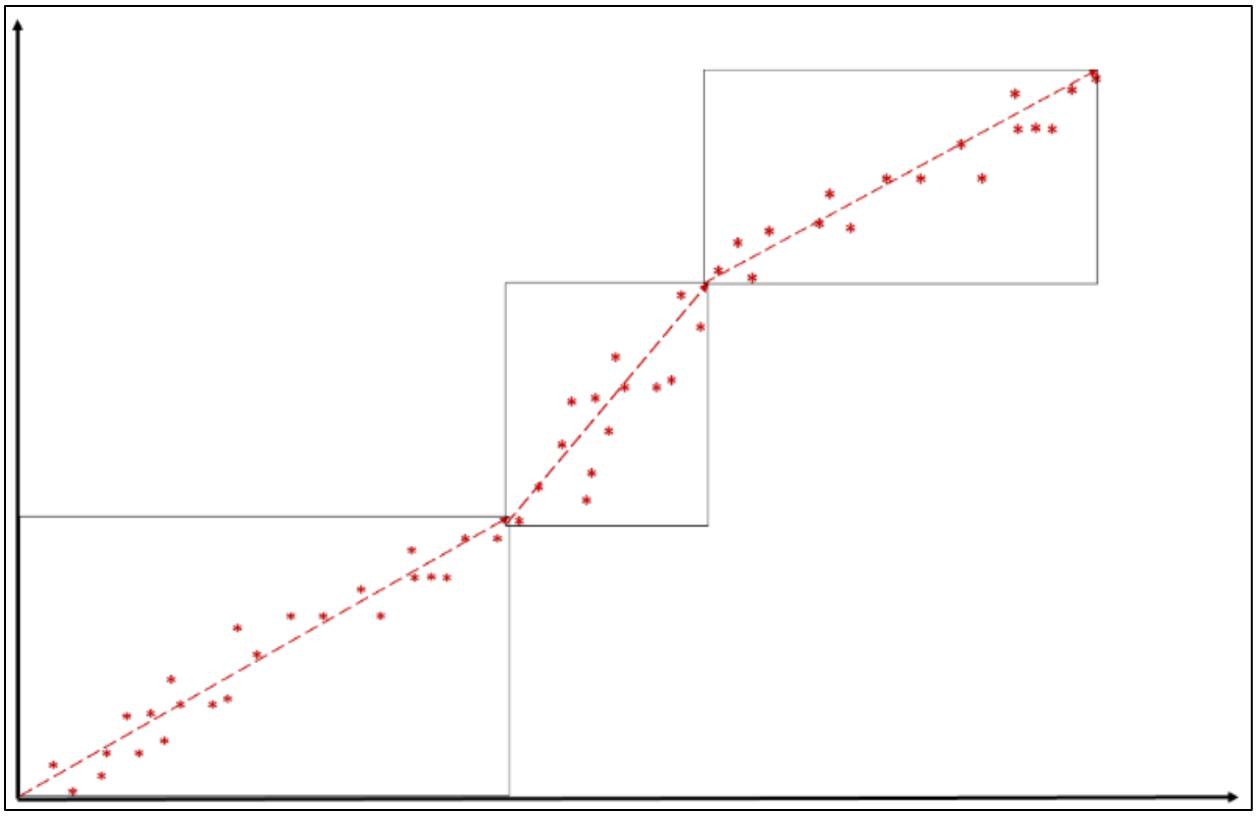


Figure 21 - Piecewise-linear 2nd EKF model: random-walk of data through multiple data spaces

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2.6.1 Second Model for EKF Processing

ARULE supports a PIFFSMOD keyword (**Table 1**, page 24) to specify one of five different models to be used as the final model for EKF processing. After conditioning and transforming, FFS data will exhibit a characteristic shape resembling one of those models: specify that model.

2.6.2 Estimated Prognostic Information

ARULE conditions and then partially adapts a first data model to FFS data, then adapts FFS data to the partially adapted first data model; then adapts a final model to the adapted FFS data and uses the adapted second model to estimate a future time of failure to produce prognostic information: RUL, PH, and SoH (Figure 24). Notice the very linear and accurate estimates of SoH.

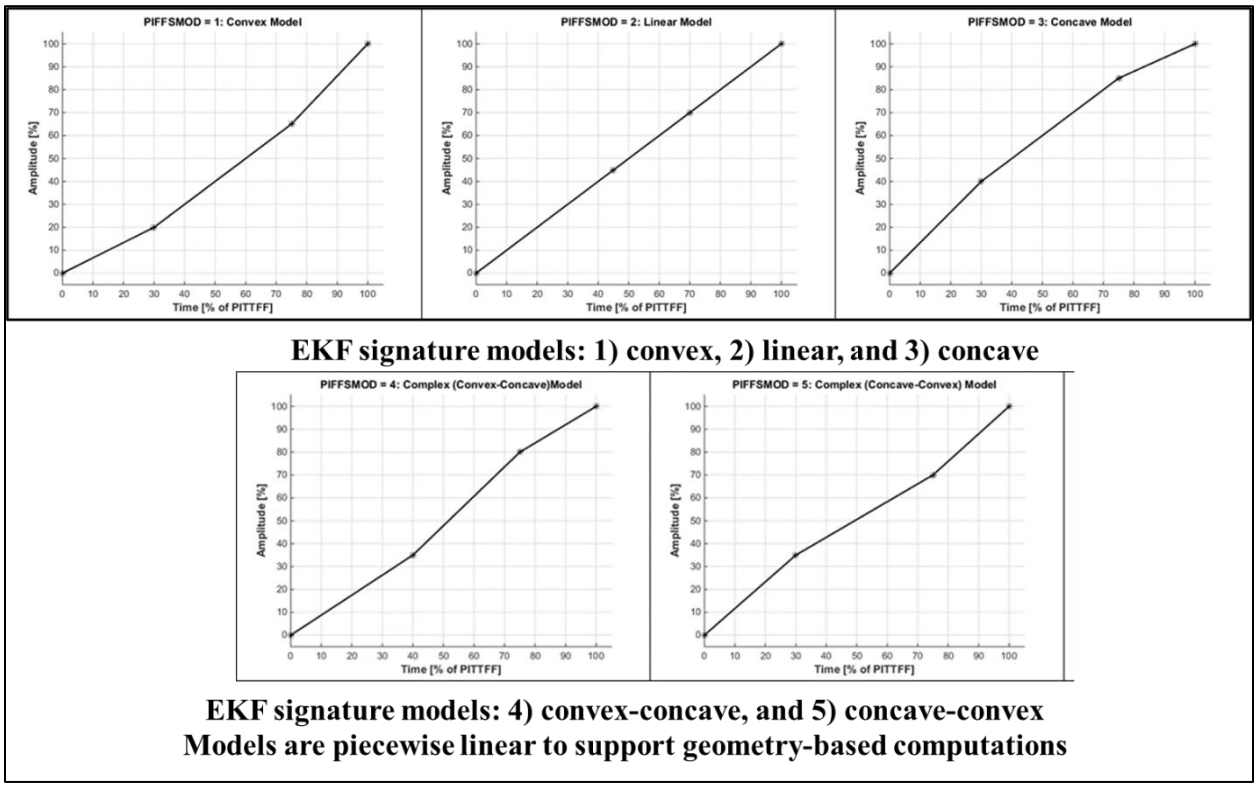


Figure 22 - Plots of the final signature models specified by the PIFFSMOD input parameter for ARULE to perform EKF processing of FFS data

2.6.3 Examples of Selecting and Specifying the 2nd EKF Model

The PIFFSMOD = 2 specifies final EKF processing is to use a linear model (Figure 22, top row, middle) and the final results are shown in Figure 23 (right-hand plots).

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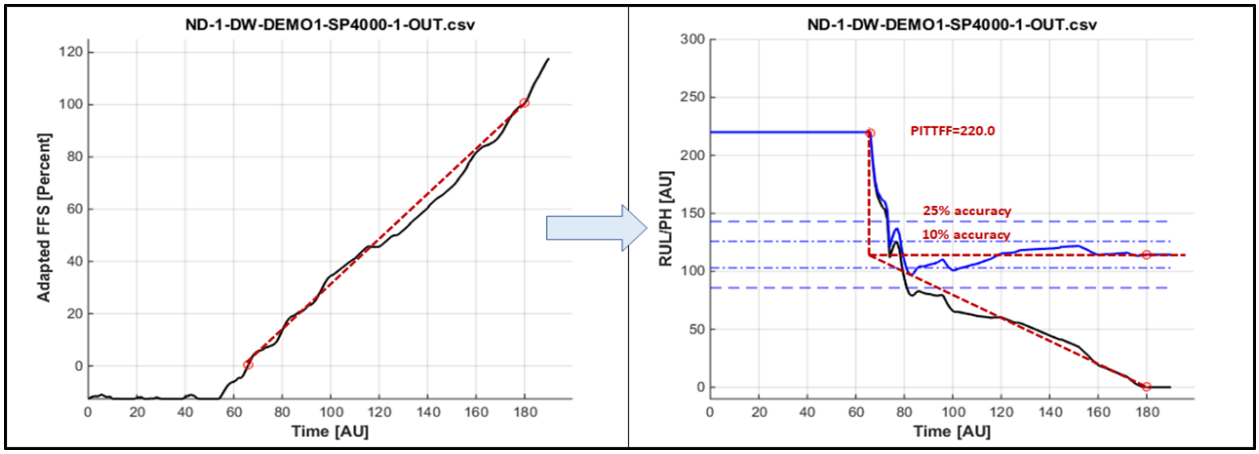


Figure 23 - Adjusted FFS plot (left) of the transformed input to final prediction stage (Extended Kalman Filtering) and plots of the resulting RUL/PH estimates.

2.6.4 Example Plots: Prognostic Estimates

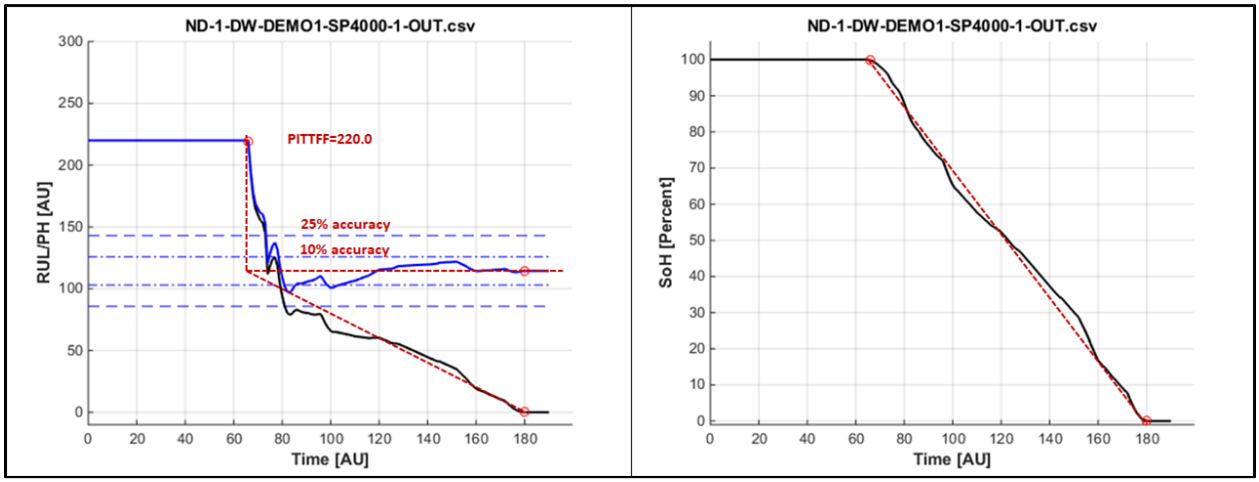


Figure 24 - Plots of RUL/PH estimates (left) and SoH estimates (right): 25% &10% margins of accuracy (blue dot-dashed lines) and ideal estimates are included (red-dashed lines).

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3. Prognostic Information: Evaluation of Accuracy

The purpose of prognostics is to detect degradation and create prognostic information such as estimates of state-of-health (SOH) and remaining useful life (RUL) of systems for the following benefits: (1) reduce operation costs, (2) provide advance warning of failures; (3) minimize unscheduled maintenance; (4) predict the time to perform preventive replacement; (5) increase maintenance cycles and operational readiness; (6) reduce sustainment costs by decreasing inspection, inventory, and down-time costs; and (7) increase reliability by improving the design and logistic support of existing systems.

3.1 Ideal Prognostic Information

Ideally, every instantiation of a prognostic target will fail in the same amount of time between the onset of degradation and the time when functional failure occurs; and there will be zero noise in the system; and the prognostic monitoring system will detect degradation exactly when it first occurs and will detect functional failure exactly when it occurs; and a prognostic engineer species the exact value to use as an estimated RUL value prior to the detection of degradation, enabling the PHM system to produce ideal RUL and PH values such as those plotted in Figure 25.

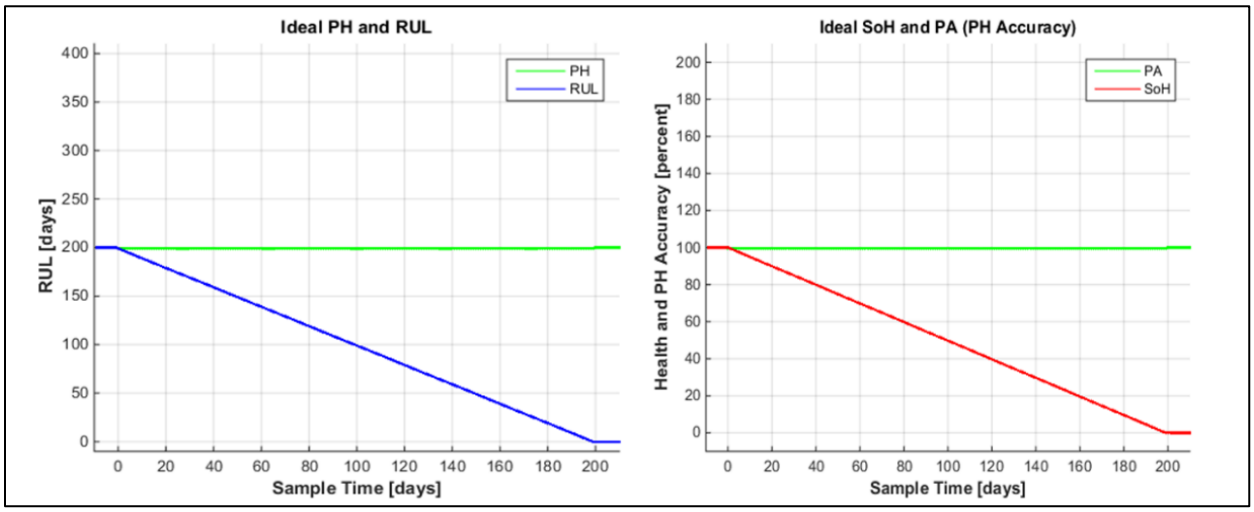


Figure 25 - Ideal plots of RUL, PH, and SOH values with 100% prognostic accuracy

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3.2 Idealistic, But Not Exact, Prognostic Information

Realistically, ideal plots cannot occur because, if nothing else, prognostic targets have a failure distribution. Referring to Figure 26, some targets have a fast rate of failure, some have a slow rate of failure, and the rest fail between the fastest and slowest. A temporizing approach is to specify some mean value, such as MTTF (mean-time-to-failure), as a Predicted Initial Time-to-Functional-Failure (PITTFF) value.

Given a PITTFF value that is 50% higher or lower in value compared to a true time of failure, then in the absence of noise in the data (anything not directly related to a failure mode of interest), ARULE produces prognostic information such as the idealized plots shown in Figure 27. Those plots illustrate the following: (1) fast convergence from initial-estimate values to true time of failure and (2) ability to converge from both high- and low-value initial estimates to very accurate estimates. But there are other inaccuracies that occur because of noisy data. Plots of RUL and PH are more likely to resemble one of the two non-ideal plots shown in Figure 28 rather than Figure 27.

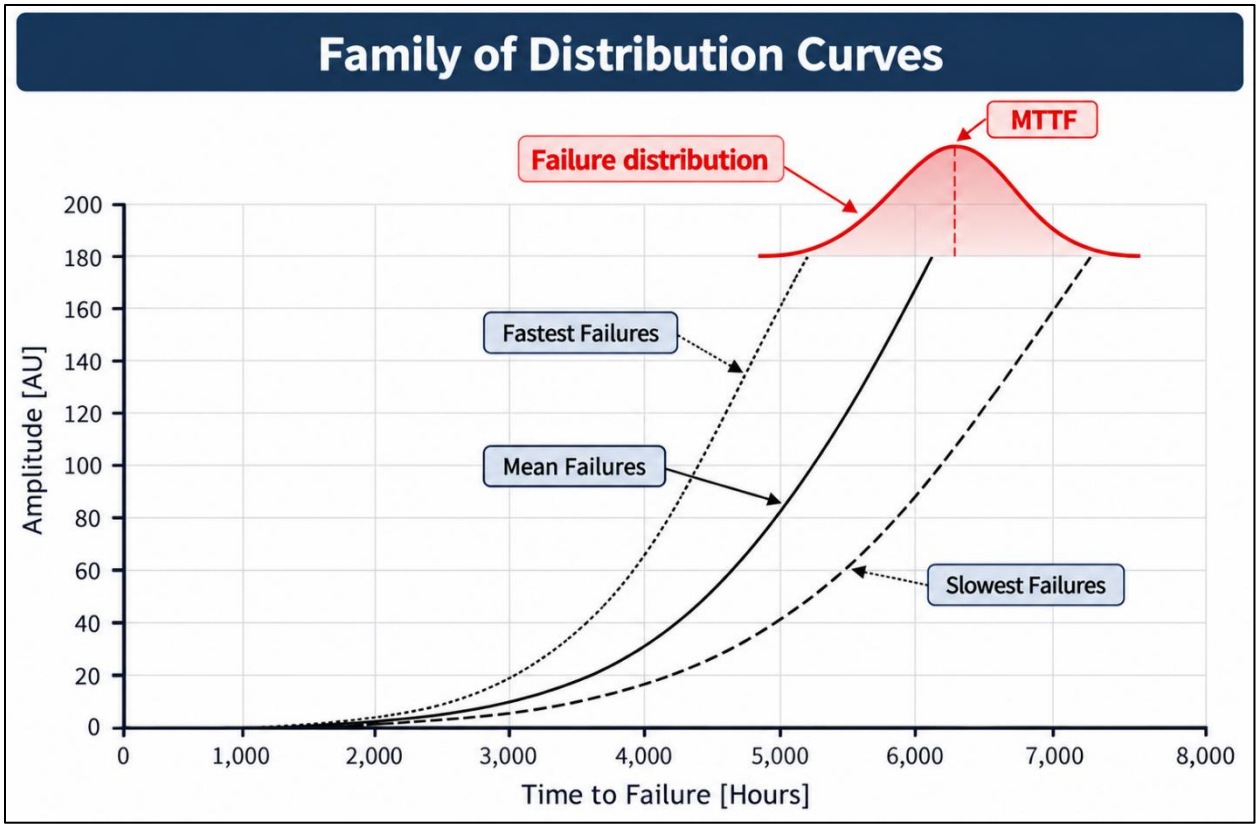


Figure 26 - Example of a failure distribution

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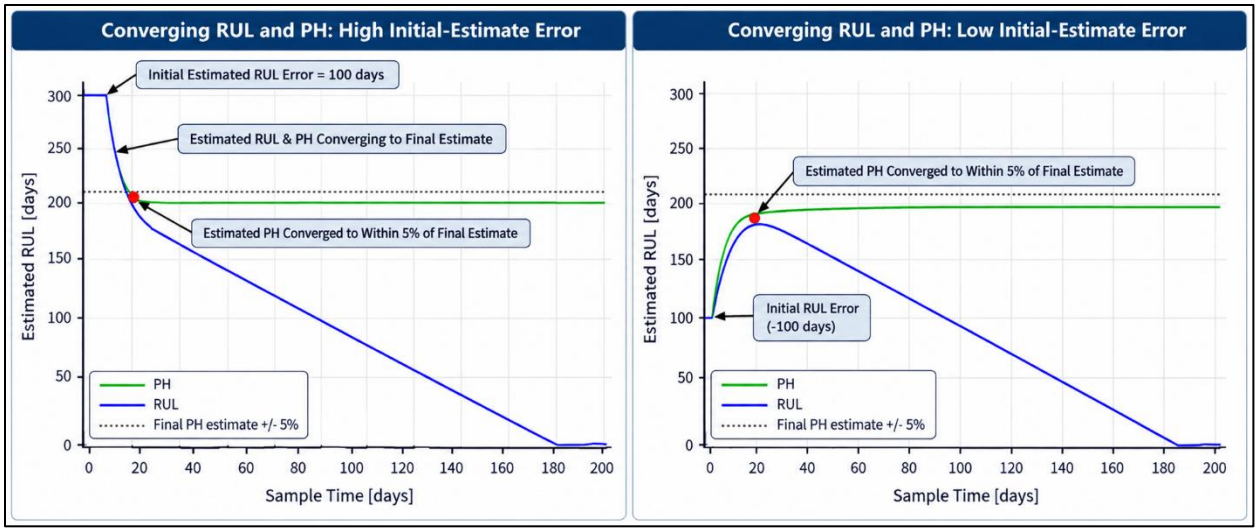


Figure 27 - Idealized plots of RUL and PH estimates given initial values having +/- 50% variation

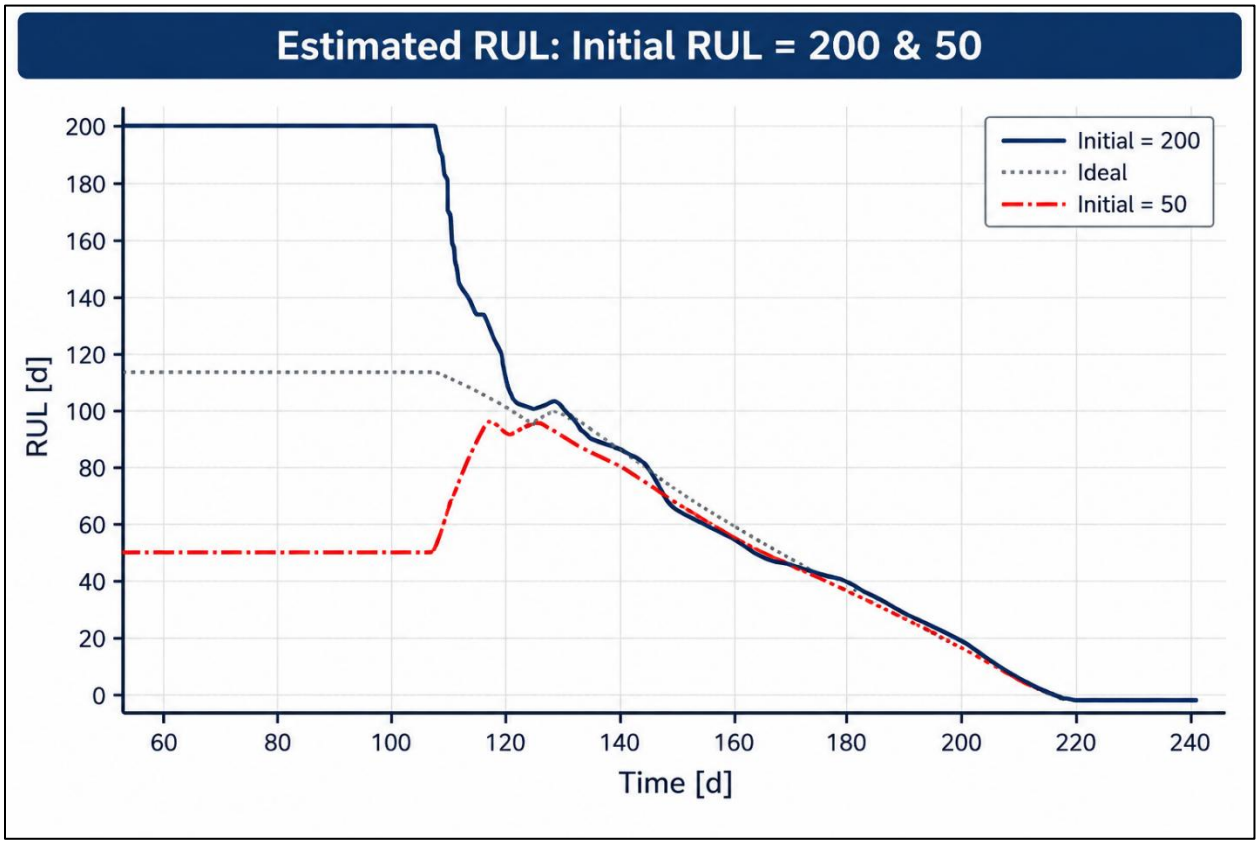


Figure 28 - Example plots of RUL given three values of initial estimates: 200, 115; and 50 days

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3.3 Accuracy

The plots in Figure 27 are annotated with the phrase "Estimated PH Converged to Within 5% of Final Estimate:" the '5%' is a value called alpha (α) that is used as a margin of accuracy and, in practice, is not reasonable: 25% and 10% are realistic values for noisy and distorted data. Ideally, accuracy would be relative to true values such as PH_0 and Prognostic Distance (PD_0). For example,

$$\boldsymbol{PD_0 = EOL_0 - BD_0} \quad \textbf{(15)}$$

Except for well-controlled experiments where faults/degradation are injected at specific, known times, true values cannot be exactly known or exactly detected – there will always be a finite, time-to-process-data delay: therefore, ARULE relates accuracy to PD_{MAX}, which can be exactly calculated as follows:

$$\boldsymbol{PD_{MAX} = EOL - BD} \quad \textbf{(16)}$$

But EOL, the time when functional failure occurs, becomes known only after functional failure occurs, a data sample is taken, and a failure condition is detected.

3.3.1 Near-real Time Estimates of Accuracy

For usability, ARULE provides near-real time estimates of the accuracy of prognostic information as each data point is sampled and processed. For each sample at time TS_i:

$$\boldsymbol{EOL_{EST(i)} = EOL_i \pm \Delta t} \quad \textbf{(17)}$$

Where Δt is algorithmically determined by using trending methods and dynamically adapting a signature model to input data. Then,

$$\boldsymbol{EOL_{EST(i)} \neq EOL_i} \quad \textbf{(18)}$$

$$\boldsymbol{PD_{MAX(i)} = EOL_{EST(i)} - BD} \quad \textbf{(19)}$$

$$\boldsymbol{RUL_i = EOL_i - TS_i} \quad \textbf{(20)}$$

$$\boldsymbol{PH_i = TS_i - BD + RUL_i = TS_i - BD + EOL_i - TS_i} \quad \textbf{(21)}$$

$$\boldsymbol{PH_i = EOL_i - BD} \quad \textbf{(22)}$$

In general, $\mathrm{PH_i} \neq \mathrm{PD_{MAX}}$. Initial calculations of PH_i will be higher than or lower than $\mathrm{PD_{MAX}}$ and accuracy can be calculated as:

$$\boldsymbol{accuracy = 100(PH_i/PD_{MAX})} \quad \textbf{(23)}$$

For example, suppose $\mathrm{PH_i} = 0.95 PD_{\mathrm{MAX}}$ (95% of maximum PD), then

$$accuracy = 100\,[(0.95)/1] = 95\ \%$$

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3.3.2 Alpha (α), Margin of Accuracy

Alpha (α), a margin of accuracy, is defined as the difference (in percent) between 100% and the calculated accuracy:

$$\boldsymbol{\alpha = 100 - accuracy = 100(1 - PH_i/PD_{MAX})} \quad \textbf{(24)}$$

3.3.3 Convergence of Accuracy

Experimentation reveals that convergence of initial estimate errors to actual time of failure is faster when the initial estimate is higher than the actual time of failure. The recommendation is to specify a value for Predicted Initial Time-to-Functional-Failure (PITTFF) in ARULE that is 50% to 100% higher than expected as shown in Figure 29 where ideal input data and PITTFF is 50% higher than the actual PH.

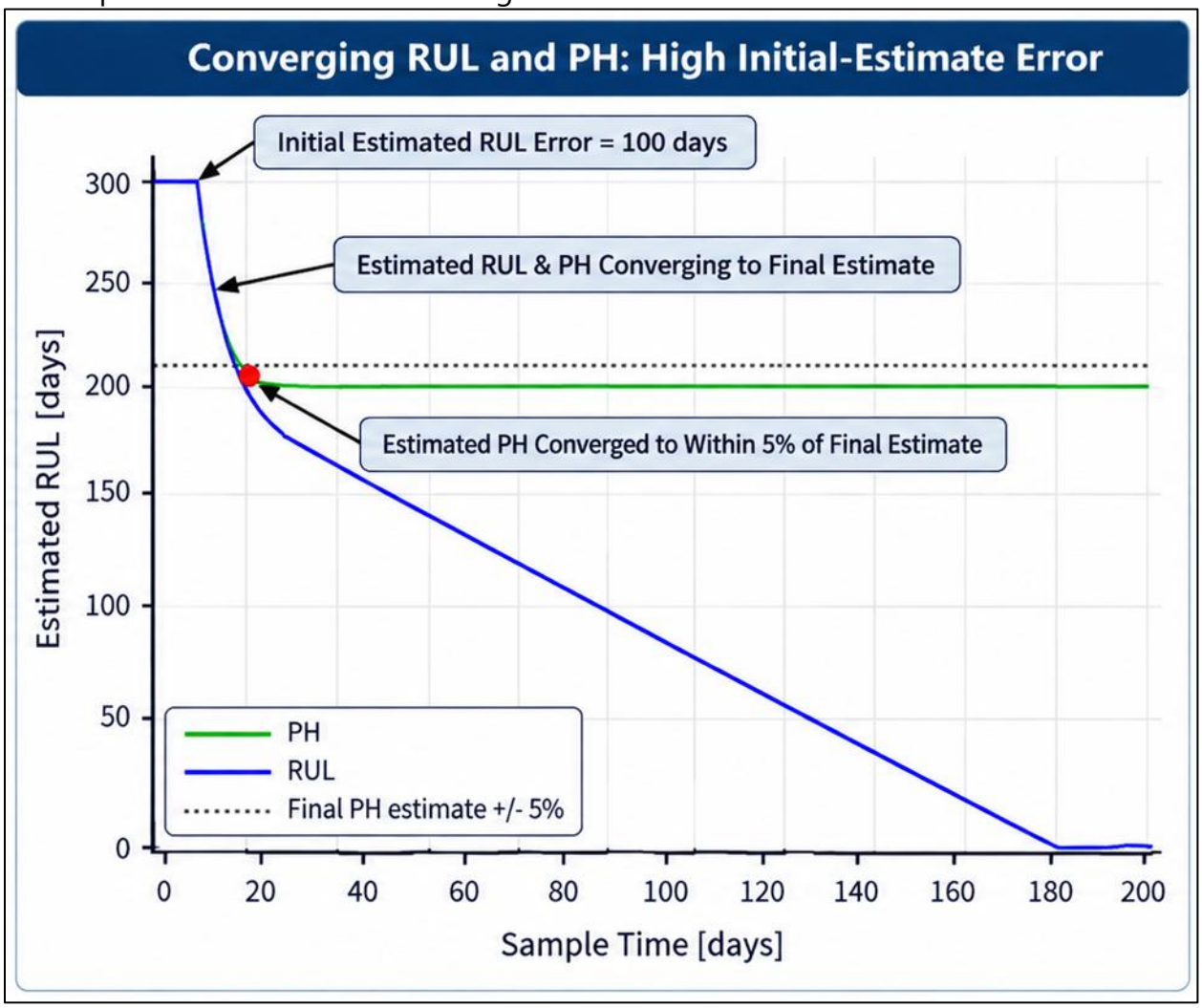


Figure 29 – Example of pseudo–ideal prognostic information. Ideal input FD signature data, non–ideal (300 days) estimate of time–to–failure (200 days) – an initial estimate error of 50 percent

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In Ridgetop's recently published IEEE paper, "Reducing Signature Models for Extended Kalman Filtering for Adaptive Prognostic Estimation", Ridgetop engineers demonstrated how accuracy (a) could be evaluated according to Prognostic Horizon (PH) rather than to remaining lifetime (l). By evaluating accuracy with this method, Ridgetop is able to overcome the issue where remaining lifetime is a quantized value that approaches zero, but can never actually reach zero because of sampling intervals, digitization, aperture jitter, measurement uncertainty, and so on. Figure 30 show how accuracy can be evaluated using the RUL and PH Plots, and Table 2 shows the accuracy results for all 20 data sets that utilized in the different real-world examples in Section 6.

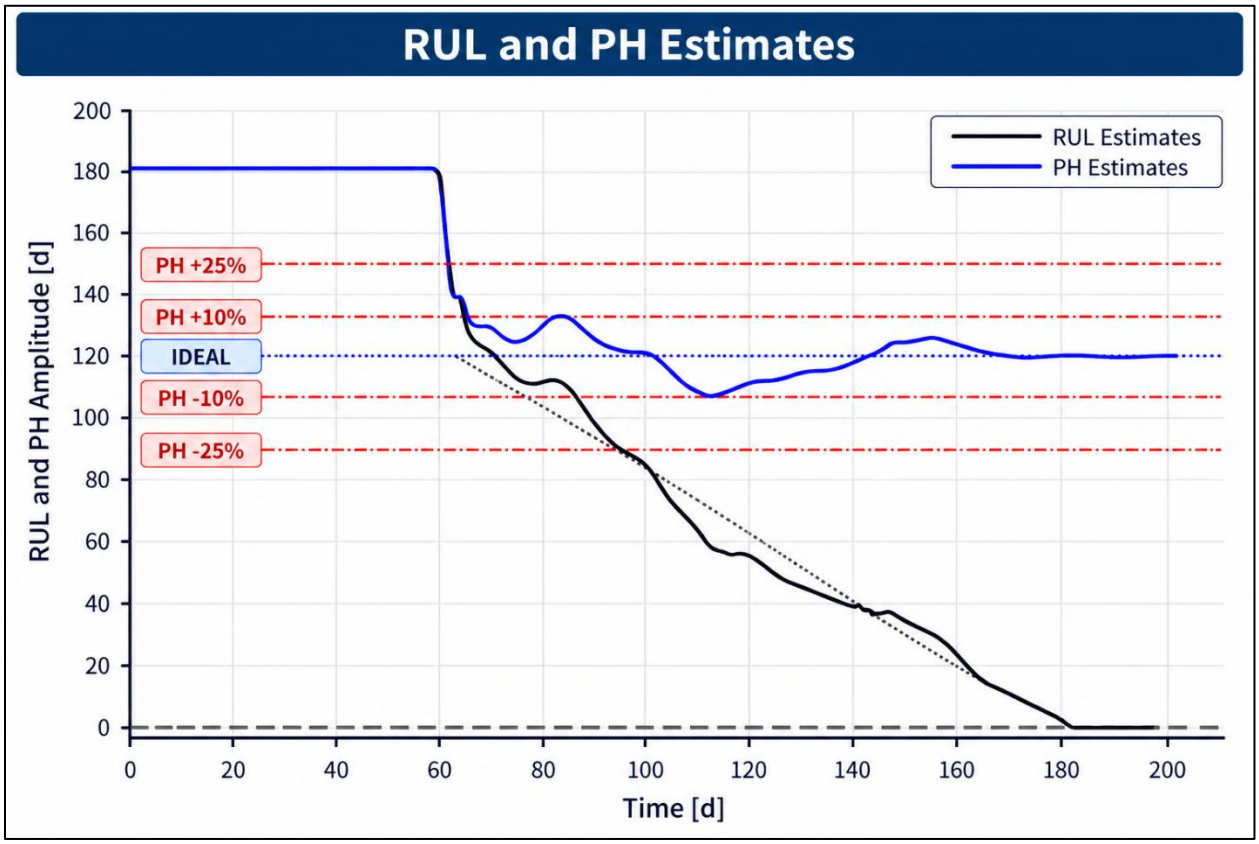


Figure 30 - Plot of RUL and PH estimates where the horizontal dashed lines represent 25% and 10% margins of accuracy

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Table 2. Accuracy Results: 25% and 10% Margins of PH Accuracy Relative to SoH

Input File * variation	PH (maximum)	SoH when PH estimates within accuracy percent	
		25%	10%
SP-4000-1	114.4	96.2	62.2
SP-4000-1*	109.8	94.2	62.6
SP-4000-2*	117.6	96.2	60.5
BATTIR	245.9	97.2	93.7
BHCHG	41.9	95.2	80.3
CS-CHGWC	32.9	93.1	93.1
CS-CHGBC	47.9	95.5	93.9
VR2400	127.2	88.2	59.1
VR2400*	123.0	88.0	65.2
EMARSPHT	14.4	96.7	96.7
EMASTPHA	16.0	96.4	69.0
Failures	16.6	96.2	75.1
ROTOGB2	53.4	75.0	58.3
VRPCAP	33,324.0	83.7	68.4
BPS03	15.5	97.8	87.6
CLEAK	22,124.0	93.5	78.3
CLEAK*	20,727.0	96.0	93.0
SUBRFA	12.5	91.5	59.4
SUBRFA*	13.0	71.9	71.9
TPWR	28.3	93.8	76.2

3.4 Contributing Factors Related to Inaccuracy

3.4.1 Signature Processing

Signature processing is a source of inaccuracy. Noise margin when trying to mitigate noise:

- A value too low might result in false detection of the onset of degradation.
- A value too high might result in excessive offset in the detection of degradation.

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Failure level:

- When set too low, detection of functional failure will occur much sooner than the actual time of failure.
- When set too high, detection of functional failure will occur much later than the actual time of functional failure, and might occur later than actual failure.

3.4.2 External Factors

Inaccuracies could be the result of factors outside of the ARULE Adaptive Prediction Kernel (APK).

- Actual Sampling period (in terms of hours and days rather than microseconds)
- Burst mode sampling at a much higher frequency than the sampling frequency
- Digitization errors due to (analog-to-digital - ADC) and (digital-to-analog - DAC) data conversions
- Rounding errors related to averaging, internal modeling, aperture jitter, and so on
- Time differences between the actual sampling and the system-recorded sampling moments
- Fluctuations in data values due to variations in, for example, the operating environment

3.5 Prognostic Information: Review of RUL, PH, and SOH

A first step in an evaluation process for a prognosing stage is to understand the types of and causes of inaccuracies in prognostic information. The plots in Figure 31 are used to illustrate, for review purposes, selected prognostic terminology: Table 3 on page 48 is a list of the names of prognostic terms plus a summarized description.

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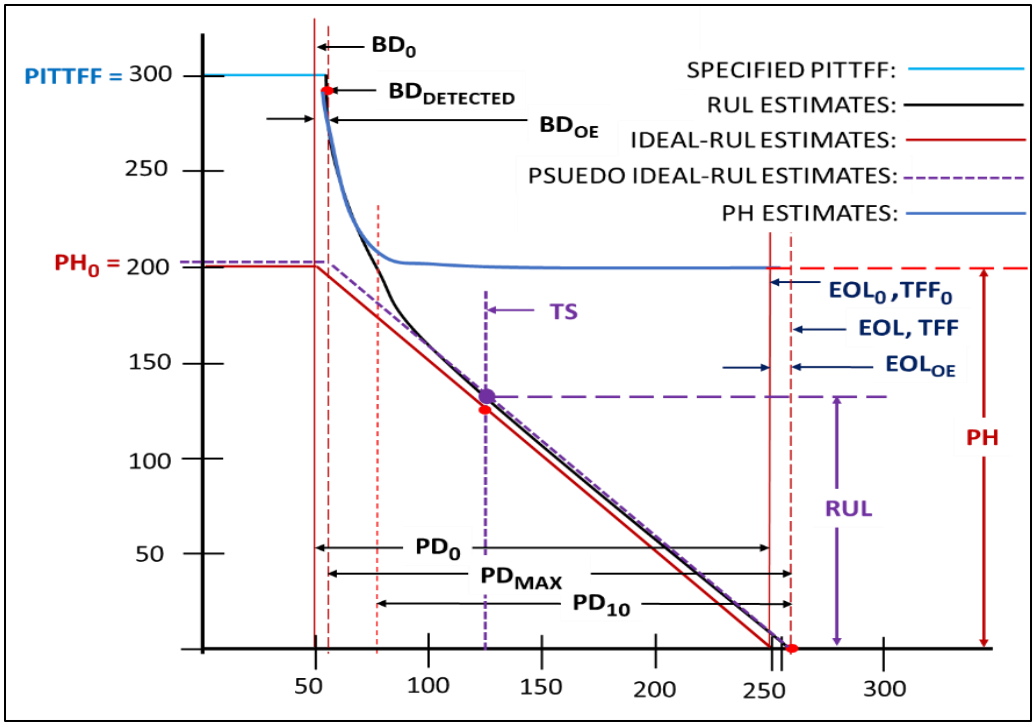


Figure 31 - Ideal and non-ideal plots for terminology and metrics explanation.

3.5.1 RUL

RUL is the estimated remaining distance along the defined analysis coordinate between the current observation and the estimated functional-failure endpoint. When the analysis coordinate represents time, RUL is expressed as a remaining time interval. $RUL_i = EOL_i - TS_i$ **(25)**

RUL is very important and useful in prognostics. It is used, for example, to schedule maintenance and repair to avoid an unplanned outage of a system. Another important use is to evaluate the likelihood that a degraded piece of equipment will last long enough to complete a mission within a specified period of time.

3.5.2 PH

One classical definition of PH (prognostic horizon) is the following: PH is the advance distance in time, to a specified level of accuracy, a failure can be predicted.

$$PH = time(1 \pm da) \quad \textbf{(26)}$$

where da is a value, such as 0.05, which defines the required accuracy of the value of PH. The primary issue with that definition is it is relative to a decreasing value. ARULE uses the following:

$$PD_{MAX} = EOL_{EST(i)} - BD \quad \textbf{(27)}$$

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$$RUL_\alpha = EOL_\alpha - TS_a \quad (28)$$

$$PH_\alpha = TS_\alpha - BD + RUL_\alpha \quad (29)$$

3.5.3 SOH

SoH information are values ranging from 100 % (full health) to 0 % (no health) and is calculated for every data sample,

$$SoH_i = 100(\, RUL_i/(EOL_i - BD)) \quad (30)$$

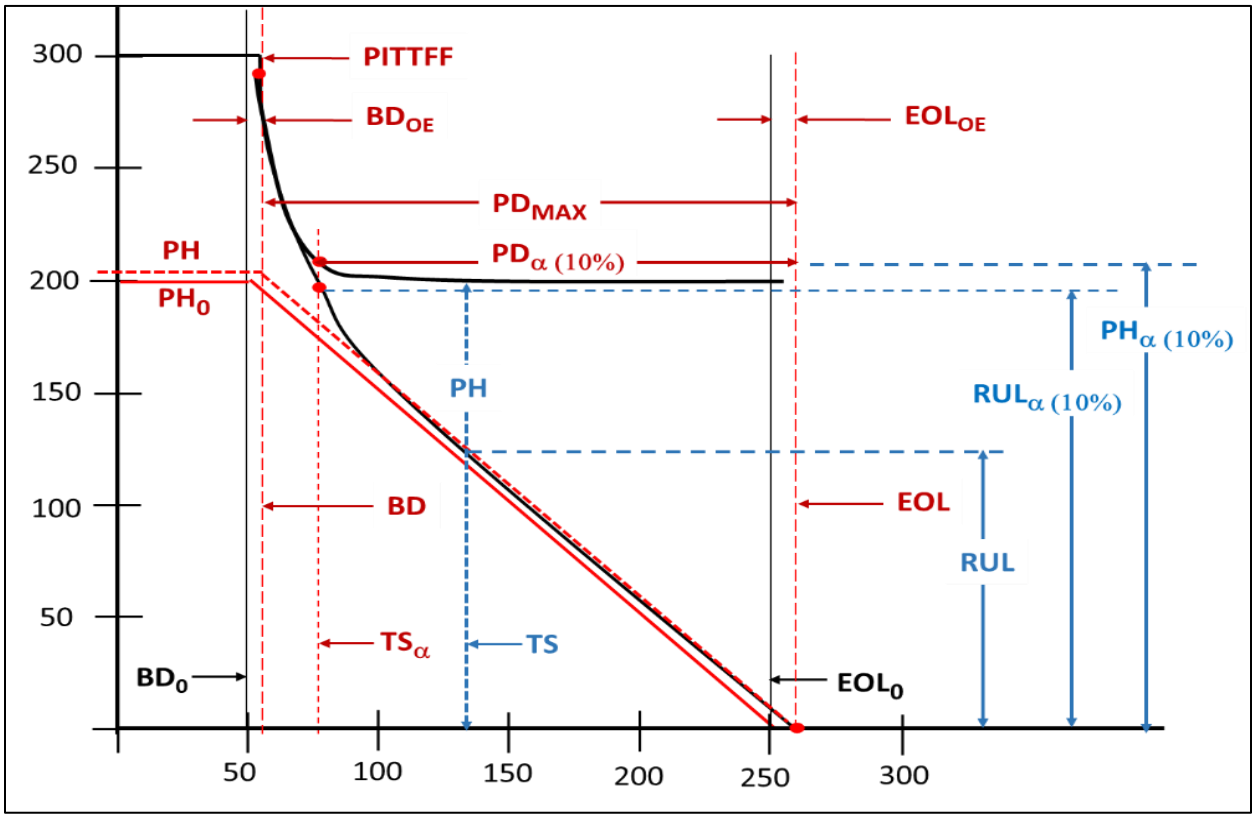


Figure 32 - Example plots to illustrate prediction information (except for SOH)

3.6 Prognostic Events

In prognostics, there are two major events of particular interest: a beginning event and an ending event – the beginning (onset) of degradation and the end of useful life when functional failure occurs.

3.6.1 Beginning of Degradation

- **True Beginning of Degradation** - Is a point in time, BD_0, when degradation of a prognostic target begins. In practice, except by lucky chance, it is not possible to

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truly detect the exact point in time because of, for example, noise, discrete sampling, and/or digitization of data. Sampling points are much more likely to happen before or after rather than exactly when degradation begins.

- **Detected Beginning of Degradation** - In practice, degradation is detected by processing at a point in time, $BD_{DETECTED}$ (also BD), a sampled data point (top-most red point in Figure 32) that is situated after degradation has begun.

3.6.2 End of Life – Functional Failure

- True End-of-Life (also TTF_0, Time of Functional Failure) - The point in time, EOL_0 or TTF_0, when End of Life (Functional Failure) occurs. In practice, except by lucky chance, it is not possible to detect this point in time because of, for example, noise, discrete sampling, digitization of data, and signature smoothing (lowers signal level).
- **End-of-Life** - In practice, End-of-Life is detected by processing at a point in time, EOL (also, $EOL_{DETECTED}$ or TFF), a sampled data point after the time corresponding to the failure level, as illustrated by the bottom-most, red-colored data point Figure 32. In one important respect, this is 'undesirable' because failure is reported after, instead of before, the event. One way to 'mitigate' that is to specify a lower failure level.

3.7 Prognostic Distance (PD)

Classically, PD is the time needed to make a failure prediction and take action, such as repair or shutdown, which can be restated as follows:

$PD = minimum\ required\ RUL$

That classical definition is qualitative and furthermore, it includes an indefinite, non-quantitative event (take action) that is not related to the accuracy of a prognosing stage and its prediction algorithms. A quantifiable definition of PD is needed.

3.7.1 True Prognostic Distance

True PD is the distance in time between the true onset of degradation and the true time of functional failure.

$$\boldsymbol{PD_0 = EOL_0 - BD_0} \quad (28)$$

In practice, PD_0 can only be measurably determined by using a well-controlled experiment in which the onset of degradation is forced at a known time and functional failure occurs as the result of an injected fault at a known time. Determining PD values, such PD_0, in a controlled, well-design experiment is useful for evaluation purposes of (1) the effects of sensor sampling rates, noise filtering, windowing and feature extraction, and digitization; (2) the effects of data conditioning to process input CBD and produce output FD signature

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data, with or without additional noise filtration and/or mitigation, to a prognosing stage; and (3) the efficacy and accuracy of a prognosing stage, such as the ARULE APK, to perform specified signature transforms and noise mitigation to provide prognostic information.

3.7.2 Maximum Prognostic Distance

The maximum PD is the distance in time between detection of degradation and the detection of functional failure. The value of PD_{MAX}, when compared to the value of PD_0, can be used as a primary performance metric of the sensing stage. Optimally, the difference between the two values should be no larger than one or two sampling periods and, at the same time, be within the desired or specified margin of accuracy.

Unacceptable differences in value are not due the deficiencies in the design and operation of ARULE: instead, corrective actions need to be applied to the sensing and data processing stages: (1) sensing stage - primarily noise filtering, sampling methods, and sampling rates; (2) data processing – primarily data conditioning and extraction, domain transforms, fusion of data with data and data with models, and input specifications to a prognosing stage.

3.7.3 Beginning of Degradation, Offset

Another inaccuracy is caused by an offset between the true time of the onset of degradation and the time when degradation is detected: BD_{OE}.

The difference in time between BD_0 and $BD_{DETECTED}$ is an offset. The magnitude is dependent on the following: the magnitude of unfiltered noise in the data – the higher the magnitude, the higher the value of Noise Margin (NM) required to mitigate noise and the larger offset; the time period of periodic sampling – higher sampling rates have smaller time periods, which lowers the offset; and measurement uncertainty, including inaccuracies due to quantization and rounding during data conversions.

3.7.4 End of Life, Offset

Another source of prognostic inaccuracy is the offset between the **true time of functional failure** and the time at which functional failure is detected from the sampled and processed data.

The End-of-Life offset, **EOL_OE**, is defined as:

$EOL_OE = EOL_DETECTED - EOL_0$

where:

- **EOL_0** is the true point at which functional failure occurs; and
- **EOL_DETECTED** is the point at which functional failure is detected through processing of the sampled data.

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The magnitude of this offset depends on several factors, including the sampling interval, measurement uncertainty, data conditioning, and the magnitude of noise present in the Feature Data. For example, a larger noise margin may be required when the Feature Data contain greater variability or unfiltered noise. Increasing the noise margin can delay detection of the configured functional-failure condition and therefore increase the End-of-Life offset.

Because the data are sampled discretely, the detected End-of-Life point will not necessarily coincide exactly with the true functional-failure point. The resulting offset should therefore be considered when evaluating the accuracy and timing of prognostic outputs.

3.7.5 Methods to Mitigate Offset Error

The following methods may be used to mitigate offset errors: (1) use a sensor and/or extract feature method(s) that filters more noise; (2) reduce the value of noise margin; (3) increase the sampling rate; (4) use data converters that have a higher effective number of bits (ENOB); (5) use measurement devices having a higher resolution; and (6) use adaptive application-specific calibration.

3.8 Summary of Prognostic Terminology and Performance Metrics

Table 3 is a summary list of prognostic terms, their names, and a brief description or explanation. More detailed information was previously presented in this chapter.

Table 3. List of Prognostic Terms, Names, and Description

Term	Name – Example Figure	Description/Explanation
$\boldsymbol{\alpha}$	Alpha (accuracy)	• Used with PD, PH, RUL, and SoH. • When estimates are within α of actual
$\mathbf{BD_0}$	True BD	• Actual time when degradation begins
$\mathbf{BD}$ $\mathbf{BD_{DETECTED}}$	Estimated beginning of degradation	• Estimated because of measurement uncertainty, noise margin, and sampling

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Term	Name – Example Figure	Description/Explanation
$\mathbf{BD_{OE}}$	BD offset $BD_{OE} = BD_{DETECTED} - BD_0$	• The inaccuracy (time) between actual and estimated beginning of degradation
Convergence	Convergence of estimates to within a specified margin (actual +/- margin)	• Transition from highly inaccurate values of RUL to within a specified accuracy
$\mathbf{EOL_0}$ $\mathbf{TFF_0}$	True EOL Time of Functional Failure	• Actual time when end of life (functional failure) occurs
EOL $\mathbf{EOL_{DETECTED}}$ TFF	Estimated time of End of Life Detected time of End of Life Estimated Time of Functional Failure	• Estimated because of measurement uncertainty, noise margin, sampling, level shifting due to conditioning/transforms
$\mathbf{EOL_{MAX(i)}}$	Estimated EOL at time after the time of a sample (TS_i)	• Used to calculate an estimated value for PD_{MAX} to produce estimates of accuracy
$\mathbf{EOL_{OE}}$	EOL offset $EOL_{OE} = EOL_{DETECTED} - EOL_0$	• Time between actual end of life and time when functional failure is detected
FF	Functional Failure	• State in which a prognostic target no longer operates within specifications
NM	Noise Margin	• Shift signature downward by NM value to mitigate noise and measurement uncertainty
PD	Estimated prognostic distance $PD = EOL - TS$	• Estimated time between functional failure and a time of a sample
$\mathbf{PD_0}$	True Maximum Prognostic Distance $PD_0 = EOL_0 - BD_0$	• Time between the actual beginning of degradation and the time when functional failure occurs
$\mathbf{PD_{MAX}}$	Estimated Max. Prognostic Distance	• Used in lieu of $\mathbf{PD_0}$ because BD and EOL are estimated

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Term	Name – Example Figure	Description/Explanation
	$PD_{MAX} = EOL_{MAX} - BD$	
PH	Prognostic horizon, For relative time (BD = 0) $PH_i = (TS_i - \text{BD}) + RUL_i$	• Estimated time after BD when functional failure occurs
$\mathbf{PH_\alpha}$	Subsequent PH estimates within $\alpha \leq$ $[abs\,(PD_{MAX} - PH)/PD_{MAX}]\,100$	• When the percent difference between $\mathbf{PD_{MAX}}$ and $\mathbf{PH}$ is within α percent
PITTFF	Predicted Initial Time-to-Functional-Failure used as an initial RUL estimate absent detection of degradation	• A specified value to be used for estimated RUL in the absence of detected degradation
RUL	Remaining useful life $RUL = EOL - TS$	• The distance between EOL (estimated) and the sample time
$\mathbf{RUL_\alpha}$	RUL for PH_α	• The value of RUL when $\mathbf{PH_\alpha}$ is achieved
SoH	State of Health	• Estimated health (in percent) of a prognostic target
$\mathbf{SoH_\alpha}$	SoH value for $\mathbf{PH_\alpha}$	• SoH value corresponding to $\mathbf{PH_\alpha}$
TB	Burst-mode period	• Time period during which data is sampled at a high-rate
TP	Sample period	• Time between samples
TS	Sample time	• Time when data is sampled (taken): includes count, record, etc.

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4. ARULE™ Installation

4.1 System Requirements

The following system requirements have been tested to ensure proper software installation and operation:

1. PC with Windows 10 Professional 64–bit operating system.
2. PC with at least 4GB of RAM installed.
3. Screen resolution of 1024x768.

Important Note: There is no specialized test equipment needed to run the software application, but the program does require some software dependencies from National Instruments. All required program dependencies are included and installed with the installation wizard. It is also recommended that the Windows PC has a minimum display resolution of 1920 x 1080 with the scale and layout settings set to 100%. The software is still operational if these specifications are not met, but the user may experience scaling issues.

4.2 Installing the Software for the First Time

Important Note: The dates and versions in these instructions and views are for illustrative purposes: they might not be the same as those for your particular install.

1. Download and extract the installation folder **ARULE_Installer.zip** from the download source as shown in Figure 33. Download source could be through Email, USB, Disk, Dropbox, etc.

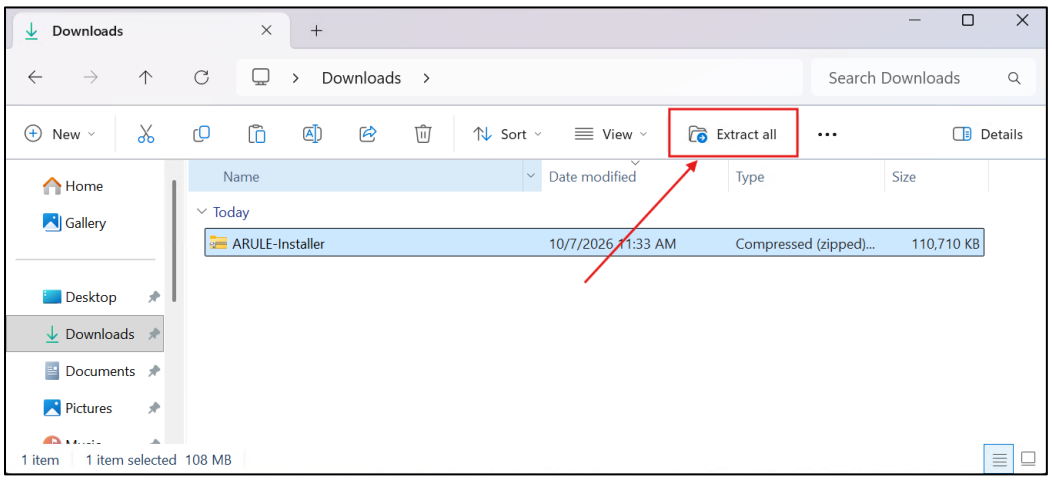


Figure 33 – ARULE™ compressed installation folder saved in Downloads

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2. Navigate to the following directory in the Windows File Explorer: **...\Downloads\ ARULE_Installer.**

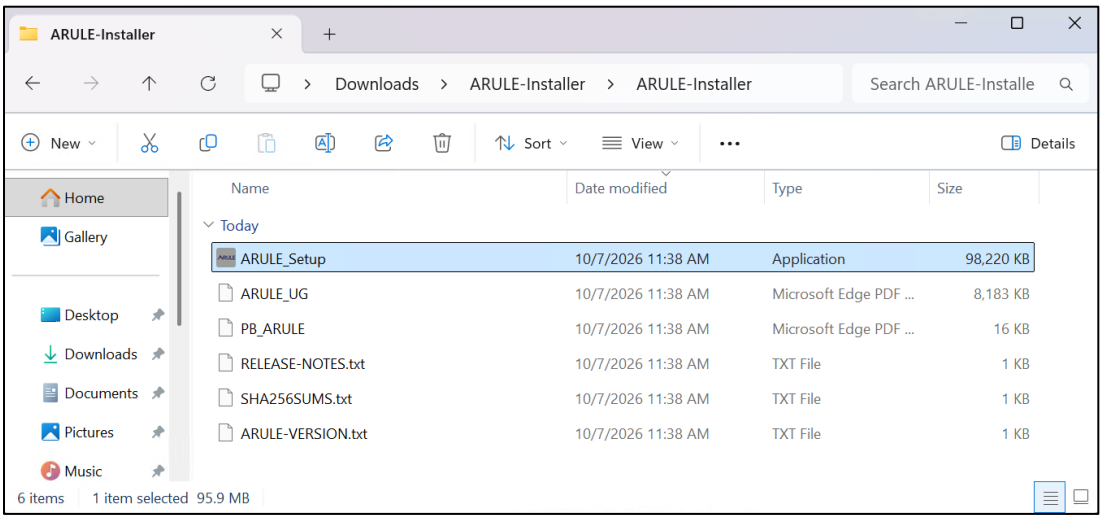


Figure 34 – Folder is located at ...\Downloads\ARULE_Installer

3. Click on ARULE_Setup.exe to run installer.
4. Verify that the installer initializes when you see the License Agreement in Figure 35. Choose "I Agree" if you accept the terms of the agreement to continue.

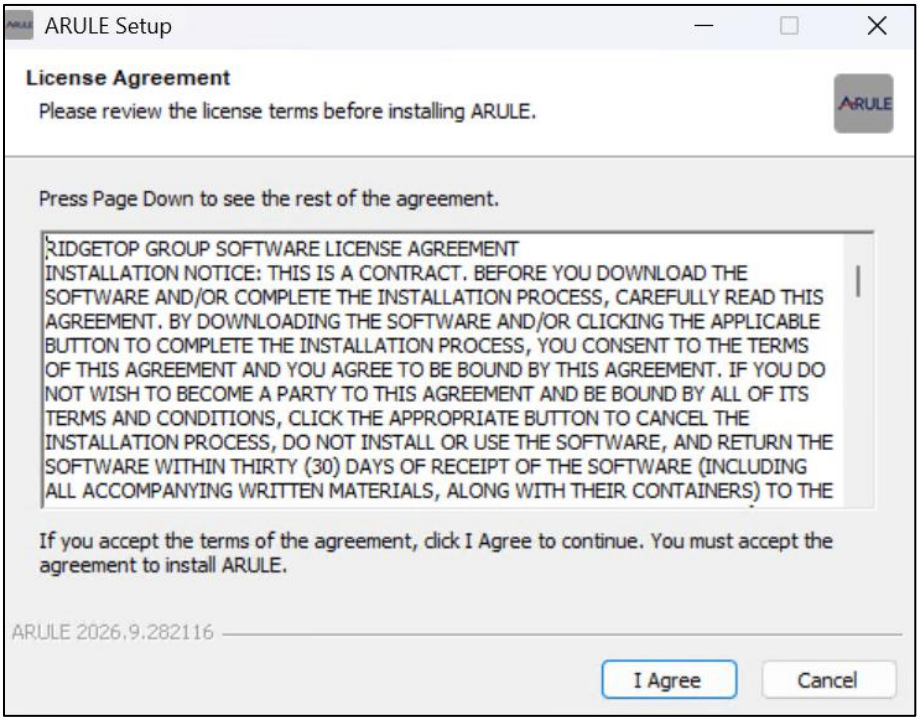


Figure 35 – ARULE™ installation wizard initializing

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5. You should see a modal dialog with a progress bar that shows the ARULE being installed (Figure 36).

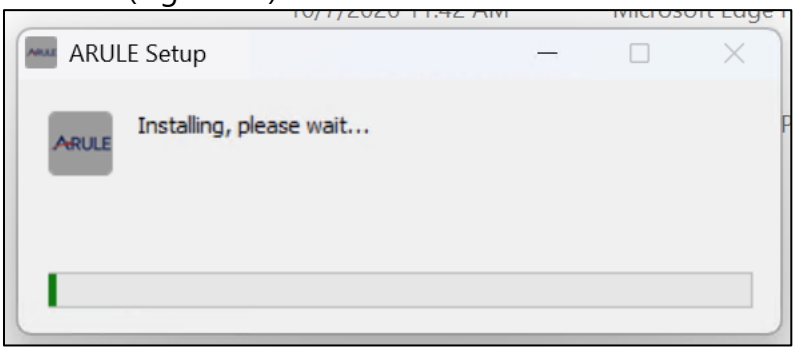


Figure 36 – Modal dialog of ARULE Installation Progress

6. When the installation is complete, the app will open and prompt user for email, serial number and activation code for ARULE (Figure 37).

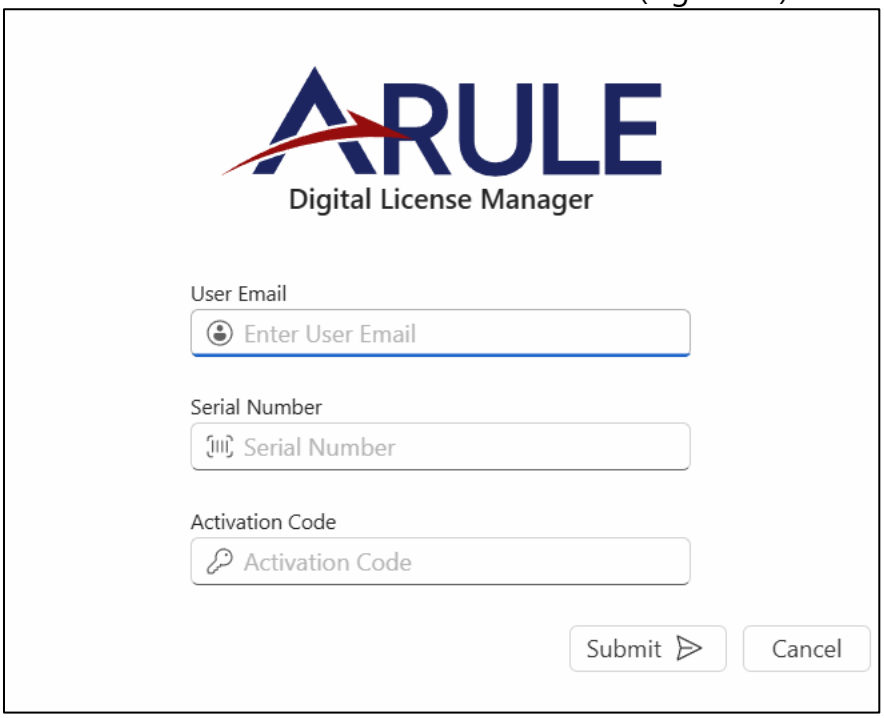


Figure 37 – Digital License Manager prompt.

7. After license is successfully activated, the following will be displayed (Figure 38), click OK to continue.

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Figure 38 – License Successfully Activated prompt, click OK to continue

8. You will then be prompted to setup an ARULE User directory (Figure 39). By default the ARULE directory will be setup in the user's Documents directory, but can be changed to different location by clicking on the "Change" button. Once the correct User Directory is selected, click "Set User Directory" to allow the application to setup the directories and files needed to use ARULE.

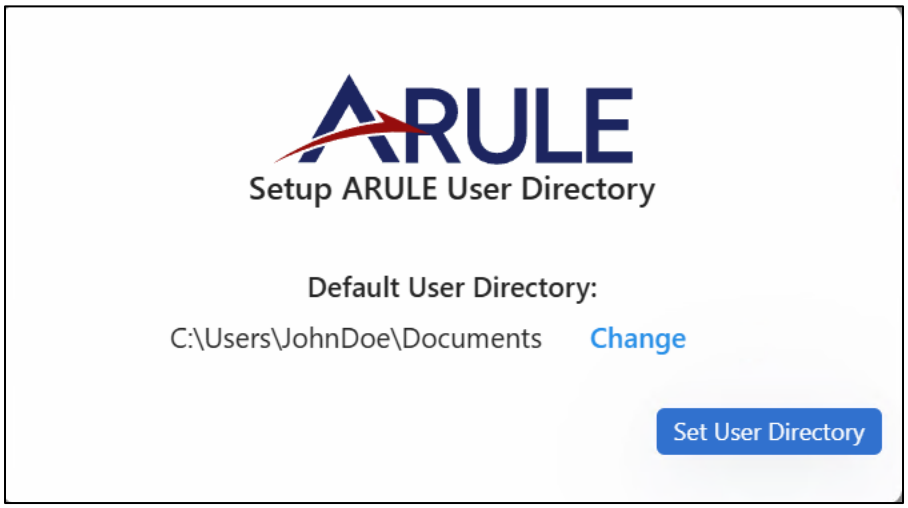


Figure 39 – Setup ARULE User Directory

9. After the ARULE directory is setup you should see the ARULE application ready to use (Figure 40). To see the ARULE directory (Figure 41), navigate to where you set the user directory and click on the ARULE folder.

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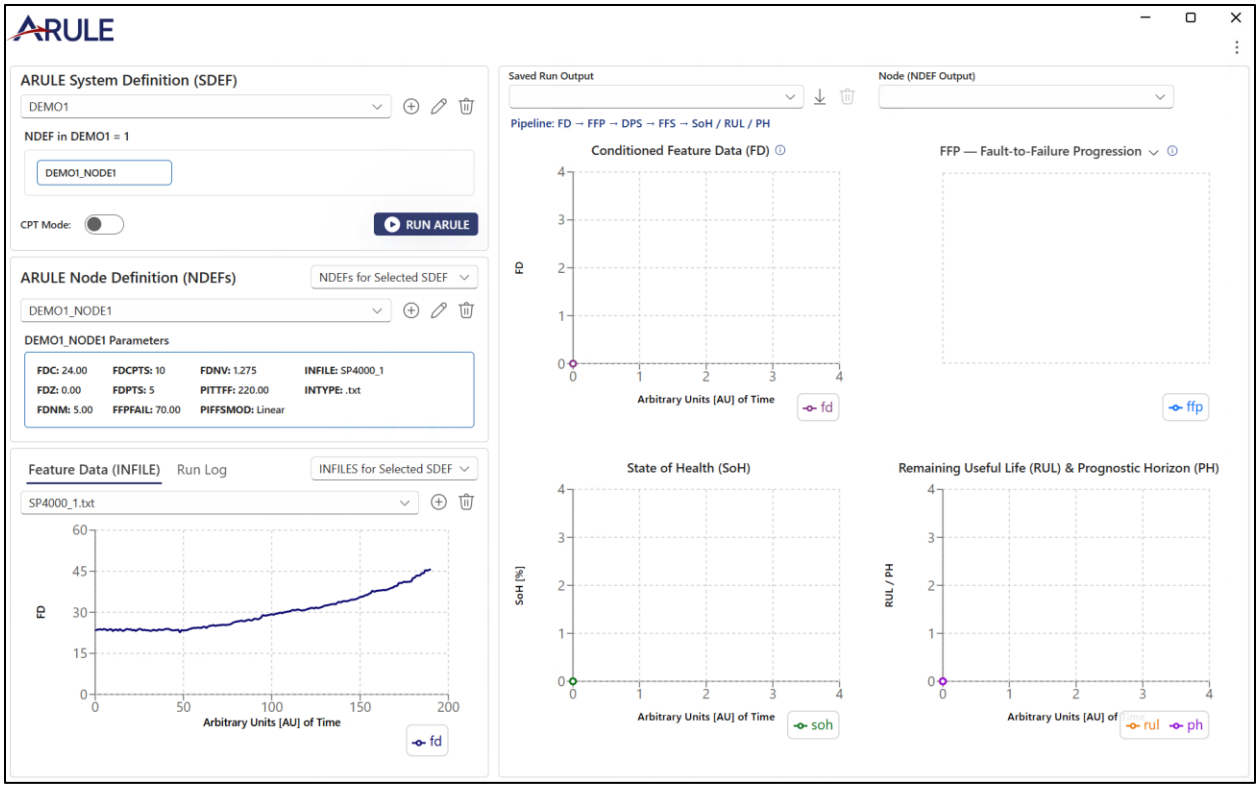


Figure 40 - ARULE Application after ARULE Directory Setup

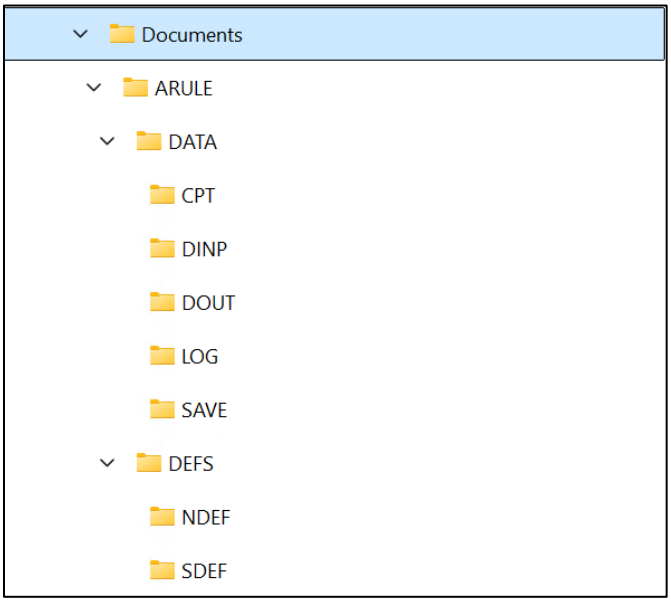


Figure 41 - ARULE User Directory

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5. ARULE™ User Interface

5.1 ARULE™ GUI Overview

The ARULE™ GUI comprises of the System Definition Input and Data Output, collectively in a single window. The programming logic follows a sequential flow of operations, where there are five main sections in the ARULE™ GUI. A visual for the single window setup is shown in Figure 42.

An outline of the 5 main program sections is as follows:

Single Window GUI Sections:

System Definition Input:

1. **ARULE System Definition (SDEF)** – A dropdown of available System Definition (SDEF) files.
2. **ARULE Node Definition Files (NDEF)** – A dropdown of Node Definition (NDEF) files for the selected SDEF file, or a dropdown option for all available Node Definition (NDEF) files.
3. **Feature Data (INFILE)** – A dropdown of the input files available to be used in Node Definition files. Chart displays the selected input file.
    a. **Run Log** – A text preview box for program messages, can be found by clicking the tab Run Log in the Feature Data (INFILE) Card.
4. **ARULE Run Arguments** – User specified program run arguments to process a specified number of data points (CPT Mode) or use the default option to process all input data points.

Data Output

5. **Data Output –** Two dropdown options that will display a scroll box containing a list of data outputs for either the CURRENT ARULE RUN or all previously SAVED ARULE RUNs.

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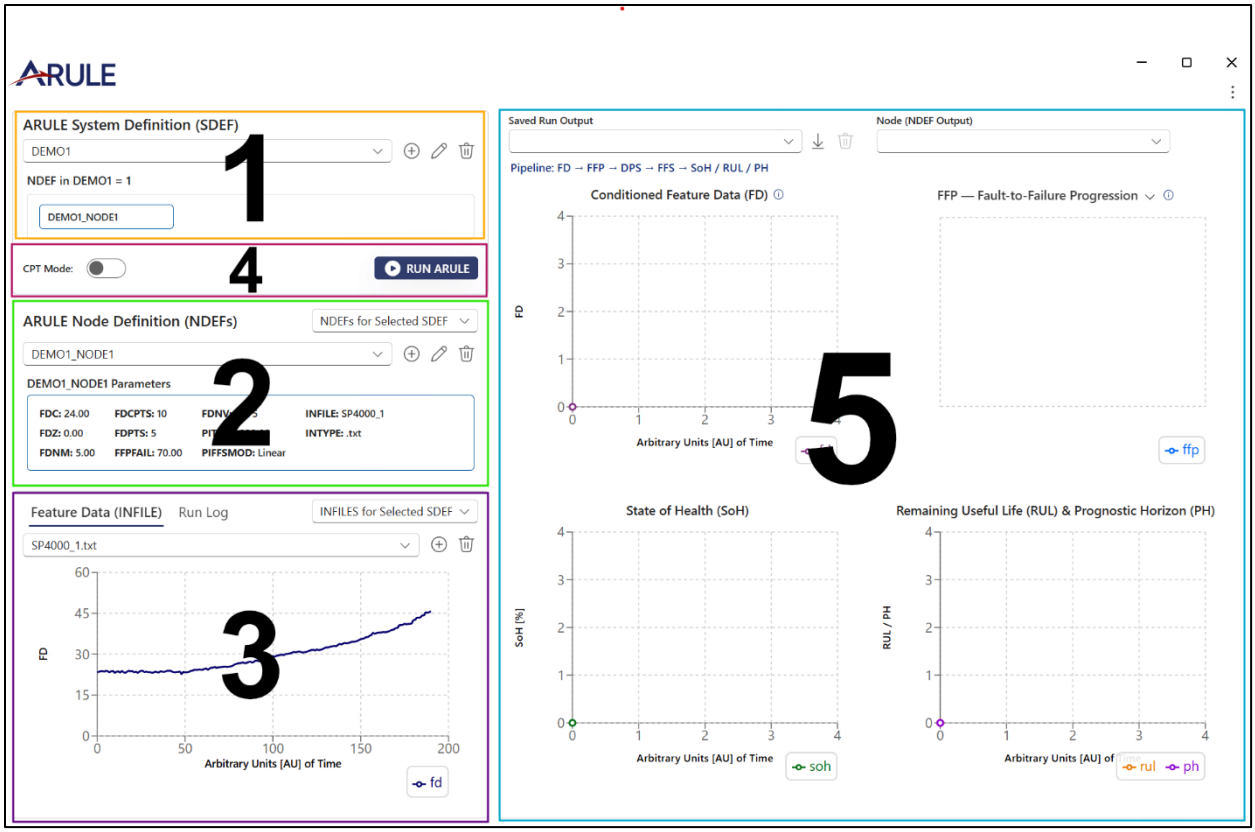


Figure 42 – Single Window GUI: System Definition Input and Data Output

5.2 System Definition Input

The System Definition Input section of the ARULE™ GUI, is where the user will select, create, and edit the System Definition files, Node Definition files, and specify the ARULE run arguments. The Feature Data (INFILE) section will provide a preview of the Condition–Based Data input plot that is linked to a data input file that is specified in the selected Node Definition file, which can be in either .csv or .txt format.

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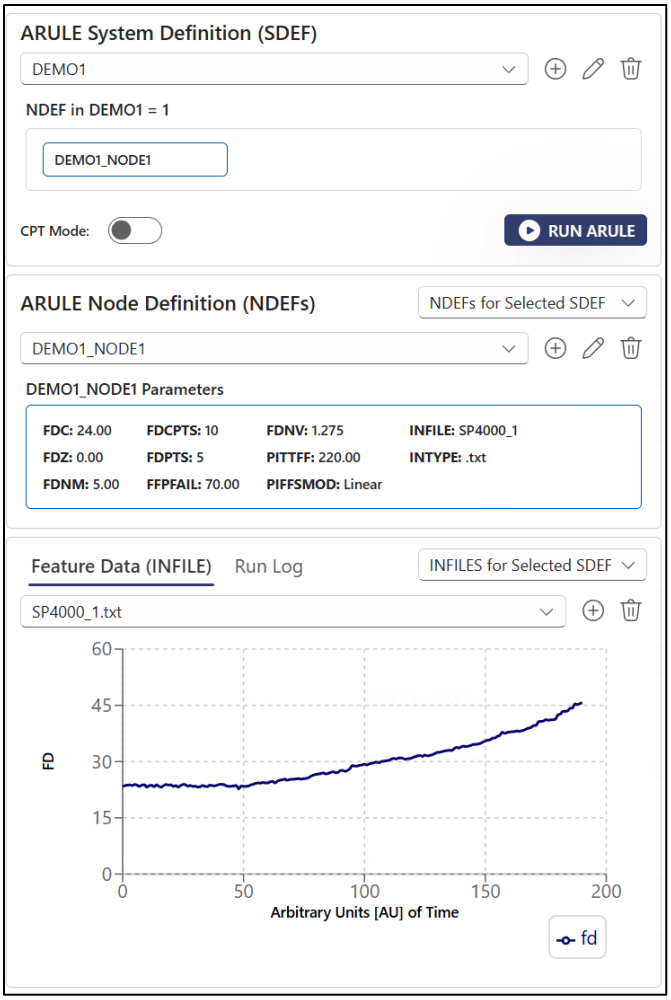


Figure 43 - Visual of the System Definition Input

5.2.1 System Definition (SDEF) Files

The **SDEF Dropdown** is where the user can define, edit, and delete System Definition (SDEF) files (Figure 44). An SDEF file defines a system of nodes that each have a specific node number ID (NDNUMID) and a corresponding Node Definition (NDEF) file name (NDFNAME). Each NDEF file contains key information and values for a particular system node. An SDEF file can contain up to twenty-five nodes (i.e. NDEFs) for a given system. An example SDEF filename that contains one system node is **DEMO1.txt**. (see Figure 43). Upon selecting the required SDEF file (in this case, **DEMO1**), the user can open it in the SDEF editor to edit or delete it. The user can also create a new SDEF file in the editor by clicking the plus button.

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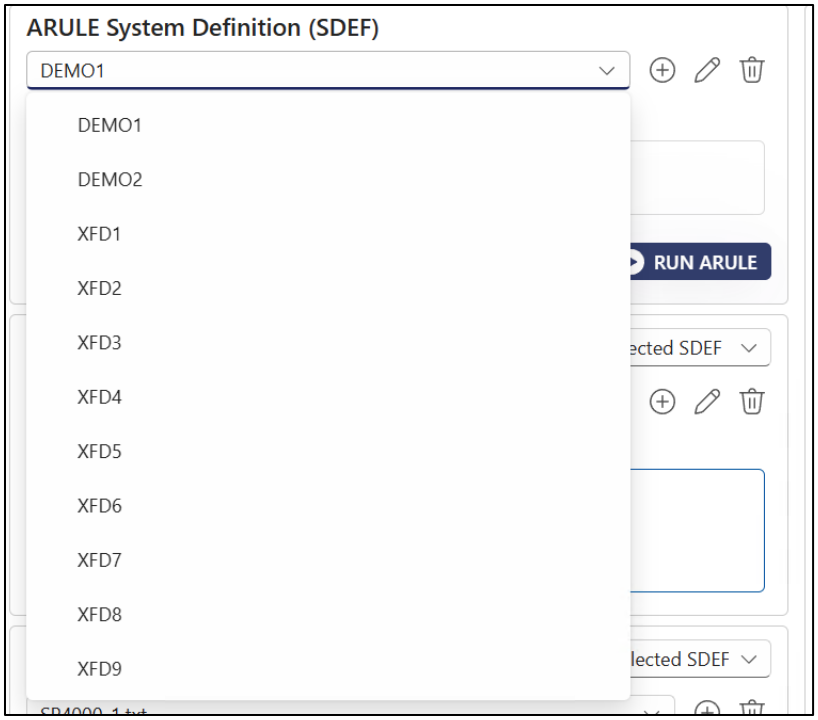


Figure 44 – Visual of the SDEF Dropdown List section for an SDEF file on the GUI

ARULE™ has the following prebuilt list of SDEF files in a scroll box in the SDEF List section. Each of these SDEF examples will be covered in Section 6 of this User Guide as they are based on real–world applications.

- DEMO1.txt
- DEMO2.txt
- XFD1.txt
- XFD2.txt
- XFD3.txt
- XFD4.txt
- …
- XFD9.txt

SDEF files can be created/edited. The first method is to click on the **Circle Plus** Button (to create a new SDEF file) or click on the **Pencil** Button (to edit an existing SDEF file) located on the right side of the **SDEF Dropdown**. A selected SDEF can also be deleted by clicking the **garbage can button** to the right of the **SDEF Dropdown**.

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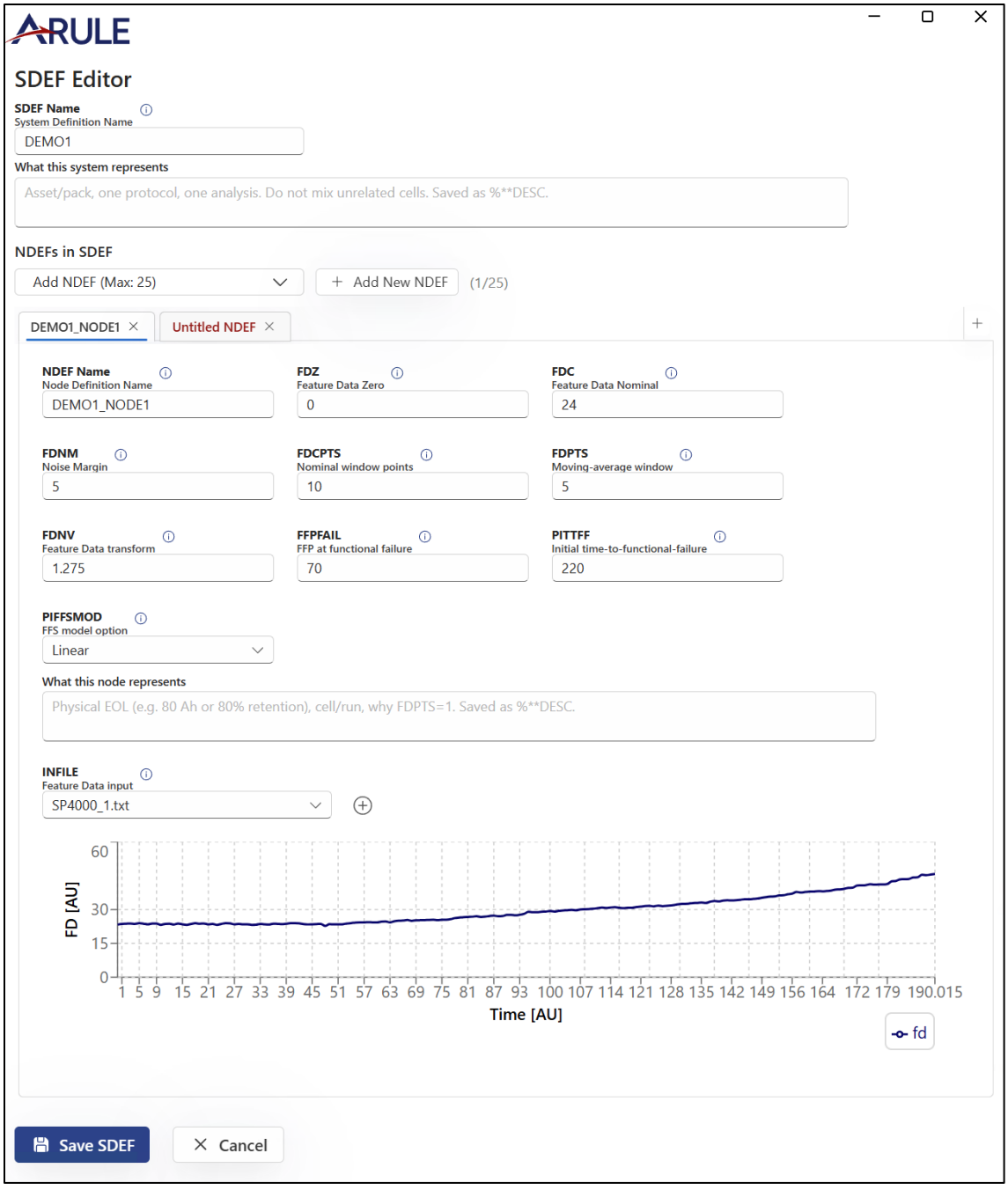


Figure 45 – Visual of the SDEF Editor

The SDEF Editor (Figure 45) allows user to set the following SDEF fields (see Figure 46):

1. Change the **System Definition Name** text field
2. Provide a description of the SDEF in the **What this system represents** text field.
3. Add NDEFs to an SDEF by selecting an NDEF from the **Add NDEF** dropdown or clicking **Add New NDEF** button and defining the NDEF in the NDEF section.

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4. Remove NDEFs from a SDEF by clicking on the **x** button next to the tab label for NDEF.
5. Users can also edit the selected NDEFs of the SDEF in the NDEF editor. Save NDEF button will be displayed if there are changes to the NDEF made (fields will be elaborated on in Section 5.2.3)
6. Users can Save Changes to SDEF by clicking **Save SDEF** button or can discard changes by clicking **Cancel**

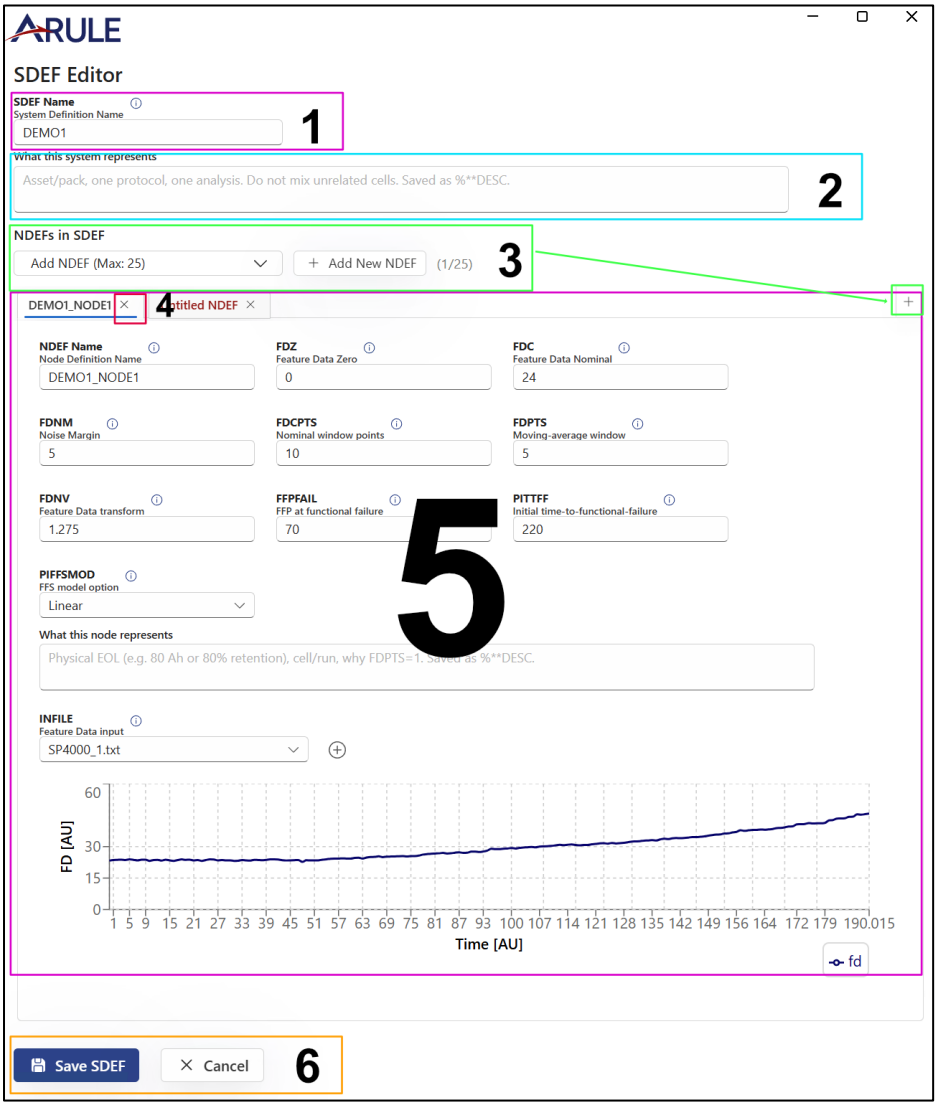


Figure 46 - ARULE Main GUI

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5.2.2 Node Definition (NDEF) Files

The Node Definition (NDEF) Files section of the GUI (Figure 47) allows users to view either the NDEFs for a selected NDEF or all NDEFs by clicking on the dropdown in the top right corner of the **ARULE Node Definition (NDEFs)** section. The parameters section displays the values for the NDEF.

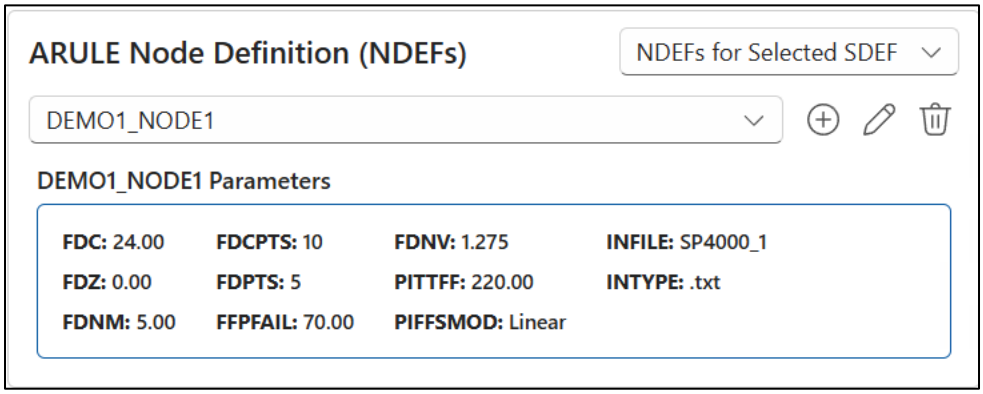


Figure 47 – Visual of the NDEF section in the main window

A user can select an NDEF by clicking on the dropdown button in the ARULE Node Definition section (see Figure 47).

- Clicking the Circle Plus button will allow a user to create a new NDEF in the NDEF Editor.
- Clicking the Pencil button will allow a user to edit the currently selected NDEF in the NDEF editor.
- Clicking the Trash button will delete the currently selected NDEF.

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5.2.3 NDEF Editor

A visual for the NDEF Editor is shown in Figure 48.

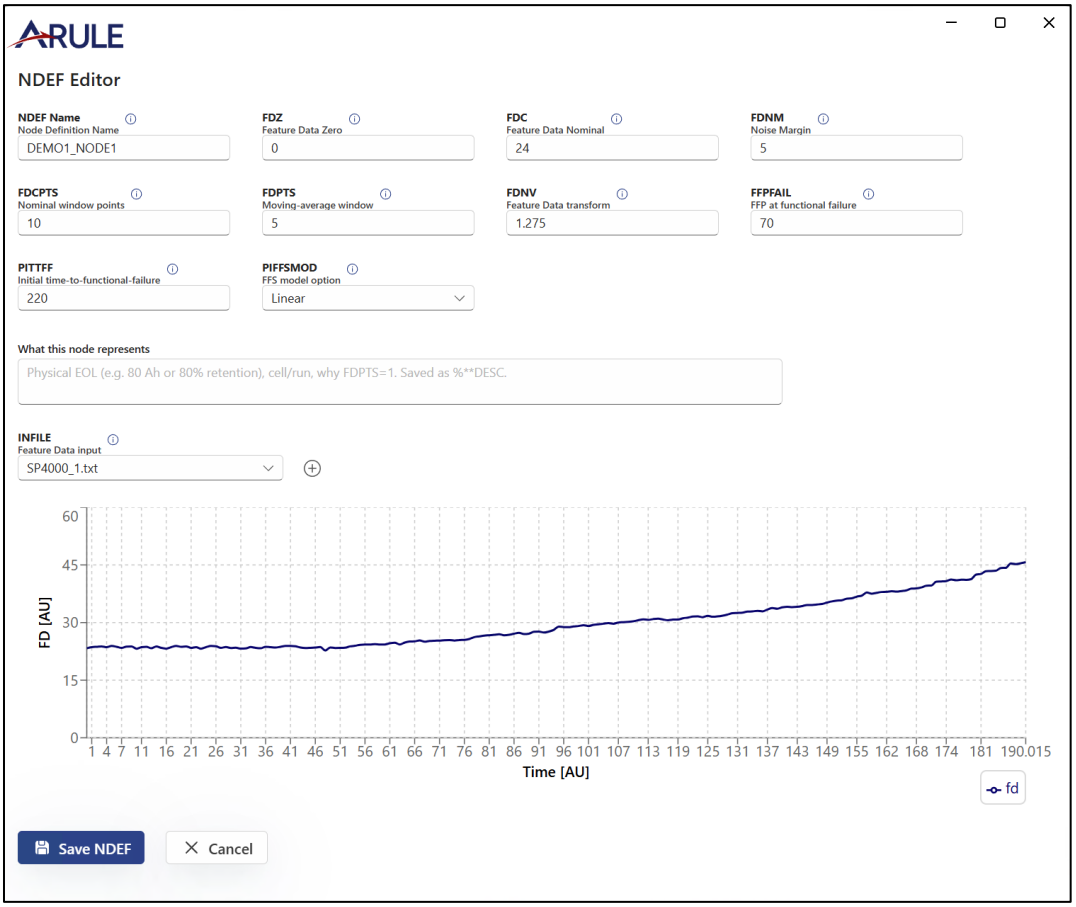


Figure 48 – NDEF Editor

The NDEF Editor displays the NDEF Name that was specified in the SDEF file. The user form also allows the user to select/specify the input data file using the **Circle Plus** button next to the INFILE dropdown and the Windows File Explorer. Once an input file is selected, the program then automatically sets the INFILE and INTYPE keywords in the NDEF file. For this example, the **DEMO1_NODE1** file (NDEF file) references **SP4000–1.txt**, which is in following default directory:

C:\Users*username*\My Documents\ARULE\DATA\DINP

Important Note: The input data file and type can be specified as either .csv or .txt; and, for usability, the prebuilt NDEF files in ARULE™ are named using the following convention: SDN_IFN(n), where SDN is the filename of the SDEF that uses the NDEF file, IFN is the filename of the input file specified in the NDEF, and (n) is used when two NDEF files use the same input file in a single SDEF. DEMO1_SP4000–1 is an example of the filename for a prebuilt NDEF. XFD1_TPWR.txt and XFD1_TPWRB.txt are NDEF examples that use the same input file for the XFD1 SDEF.

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NDEF Parameters:

- **NDEF Name** – Defines the name of the Node Definition File and the NDEFs identifier in SDEFs.
  - Any simple string identifier, no special characters.
- **FDZ** – Feature Data Zero Value as observed in your data input file. This is the zero (or absolute) reference when mapping Feature Data.
  - Any positive value and sets FD NOM in data output file.
  - 0.0 (Use FDC Value)
- **FDC** – Feature Data Nominal Value in the absence of degradation/damage. Sets the reference "healthy" amplitude for this feature.
  - Any positive value and is used to continuously calculate FD NOM based on FDCPTS.
  - 1.0 (default)
- **FDNM** – FD Noise Margin as a percent of the Nominal Value of FDC. A higher Noise Margin reduces sensitivity.
  - 0.0 to 25.0
  - 5.0 (default)
- **FDCPTS** – Number of data points to establish or refine FDC.
  - Integer value 0.0 to 25.0
  - 10.0 (default)
- **FDPTS** – Number of points to average for Feature Data. Number of Feature Data samples averaged before signal processing.
  - 1 to 5
  - 5 (default)
- **FDNV** – n or lambda Value for transforming FFP Signature data to DPS Signature data. Mapping / power option that shapes how Feature Data is shaped.
  - Any non–zero positive value
  - 1.0 (default)
- **FFPFAIL** – FFP value at which functional failure occurs. Sets the reference failure amplitude for FFP..
  - Any non–zero positive value
  - 70.0 (default)
- **PITTFF** – Initial time to functional failure: becomes RUL and PH in the absence of degradation.
  - Any non–zero positive value
  - 200.0 (default)

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- **PIFFSMOD** – Functional Failure Signature Model (FFSMOD) to use when processing input FFS data shown in Figure 22.
    - Linear (default)
    - 1: convex, 2: linear, 3: concave, 4: convex–concave, 5: concave–convex;
- **What this node represents** – Optional text field for description of the NDEF
- **INFILE** – Feature Data input file.
    - User can select an existing INFILE using the dropdown, and can see the data below.
    - Data file must be either a .txt or .csv file
    - User can import new data file by selecting the **Plus Circle** button to the right of the **INFILE** dropdown, and selecting the .txt or .csv in the file picker. Once imported, files should be selectable in dropdown.
- **Save NDEF Button** – if there are no errors detected, this button will allow user to save changes or create new NDEF.
- **Cancel Button** – clicking this will close the editor and discard any changes without saving.

5.2.4 Raw NDEF File in Notepad

An example of all NDEF parameters being displayed in a raw .txt file with Notepad is shown in Figure 49.

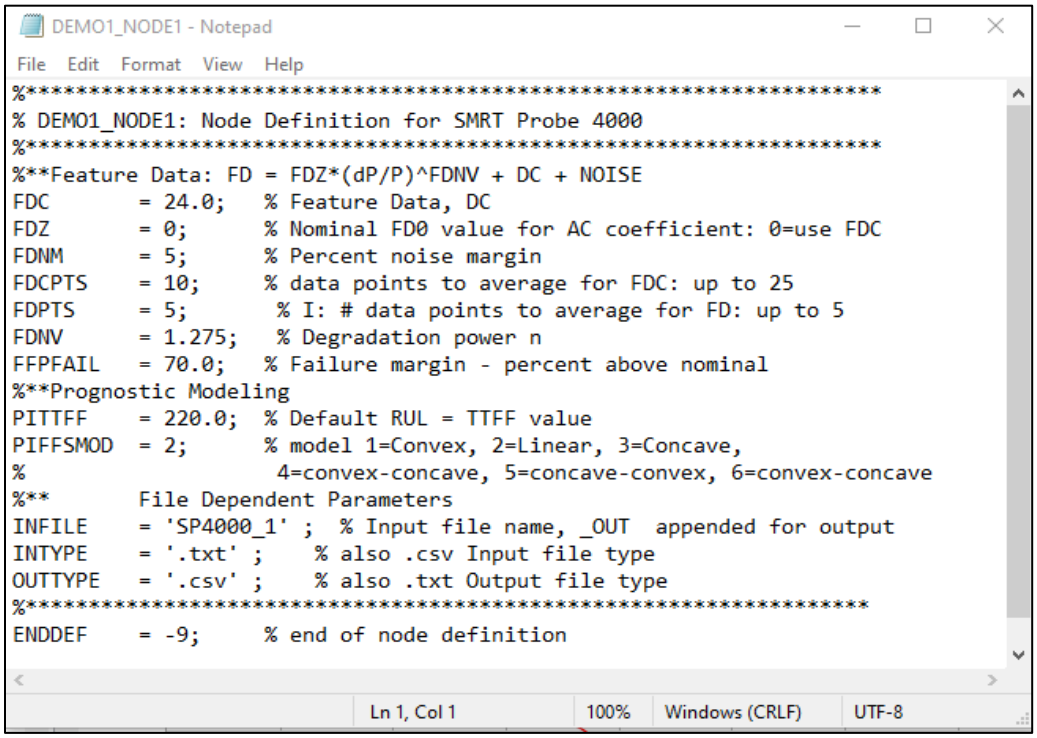


Figure 49 – Example of an NDEF being edited in raw .txt by Notepad

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Important Note: Ridgetop Group recommends that the SDEF and NDEF editors be used to edit and save NDEF and SDEF files.

5.2.5 Feature Data (INFILE) Section on Main Page

This section displays the Feature Data input plot for a selected NDEF file. Selecting the INFILE from the dropdown allows you to see the data in the chart below (see Figure 50).

- A user can either view **INFILES for Selected SDEF, INFILES for Selected NDEF** or **All INFILES** by selecting from the dropdown in the top right of the card.
- A user can add an **INFILE** by clicking the **Circle Plus** button to the right of the **INFILE** dropdown.
- A user can delete the selected **INFILE** by clicking the **Trash Can** button next to the **INFILE** dropdown.

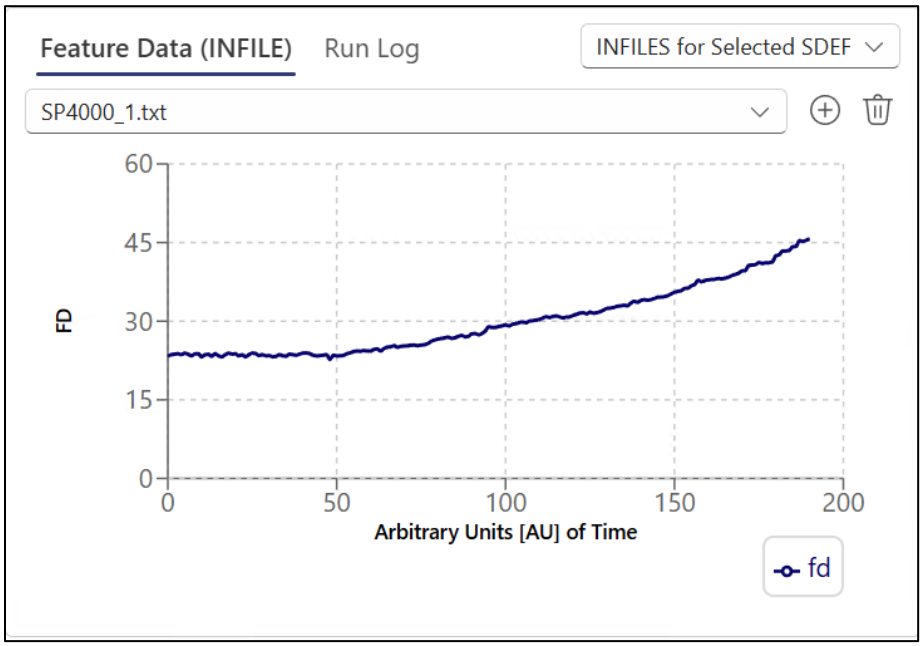


Figure 50 – Feature Data (INFILE) section of the main page

5.2.6 ARULE™ Run Arguments

This section of the ARULE™ software program will show a label for the SDEF file that will be processed by the ARULE™ APK, and allows the user to specify 2 input arguments before attempting a prognostic estimation run.

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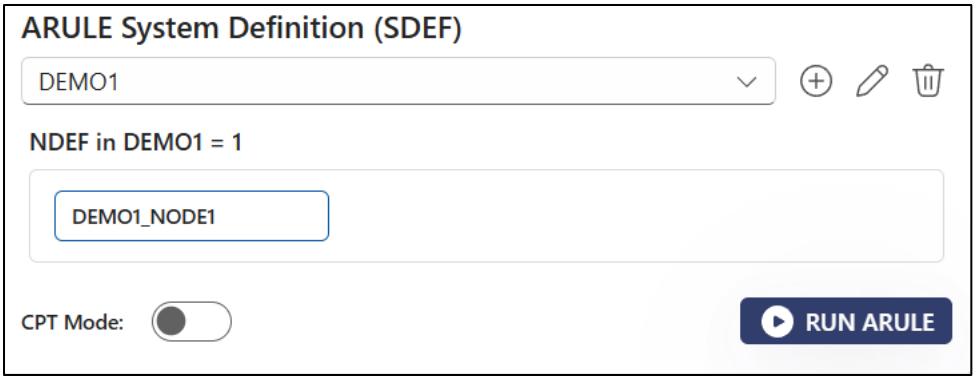


Figure 51 – ARULE™ Run Arguments section with CPT Mode Off (default)

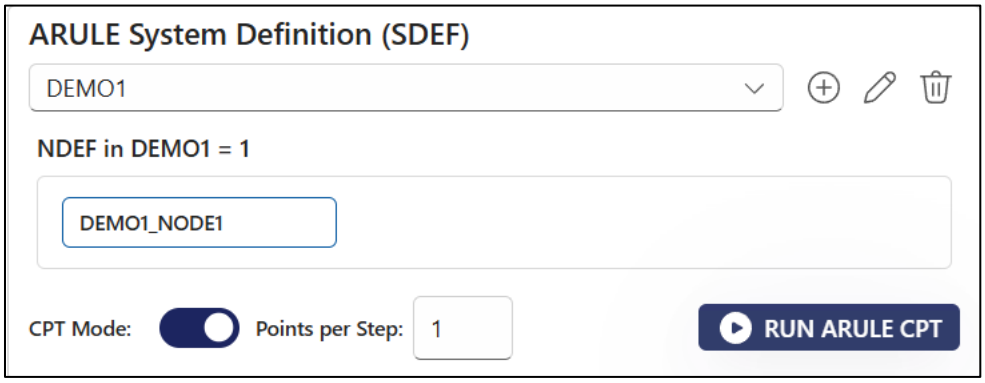


Figure 52 – ARULE™ Run Arguments section with CPT Mode On

- **Argument 1** – is the name of the SDEF file you selected, and is displayed as a label.
- **Argument 2** – is a Toggle Switch that enables Check Point Restart (CPT) mode to process a specified number of data points.
    - When the Toggle Switch is disabled, the program is in the Default Processing Mode and will process all data points for the input data files that were defined in the NDEF file.
    - When the Toggle Switch is enabled, the program is in Check Point Restart (CPT) Mode. In CPT mode, the user has the ability to process N# of data points between 1–10000 in the text input as shown in 52.

Enabling CPT Mode demonstrates the ability to call the ARULE™ APK multiple times to process the same data set at different instances in time. When CPT Mode is enabled, you may continue to process another set of input data points; or you may return to change the number of input data points to process; or you may choose to exit CPT Mode by processing a different SDEF and NDEF selection.

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System messages will also appear in the lower left corner of the GUI, and specific reason codes can be reviewed the log files here: **C:\ARULE™\DATA\LOG.**

Upon completing a successful prognostic estimation run with the ARULE™, there will be a **"UD_ARULE program is ended"** message shown in the **Run Log** section as in Figure 53 and the program outputs the key prognostic plots.

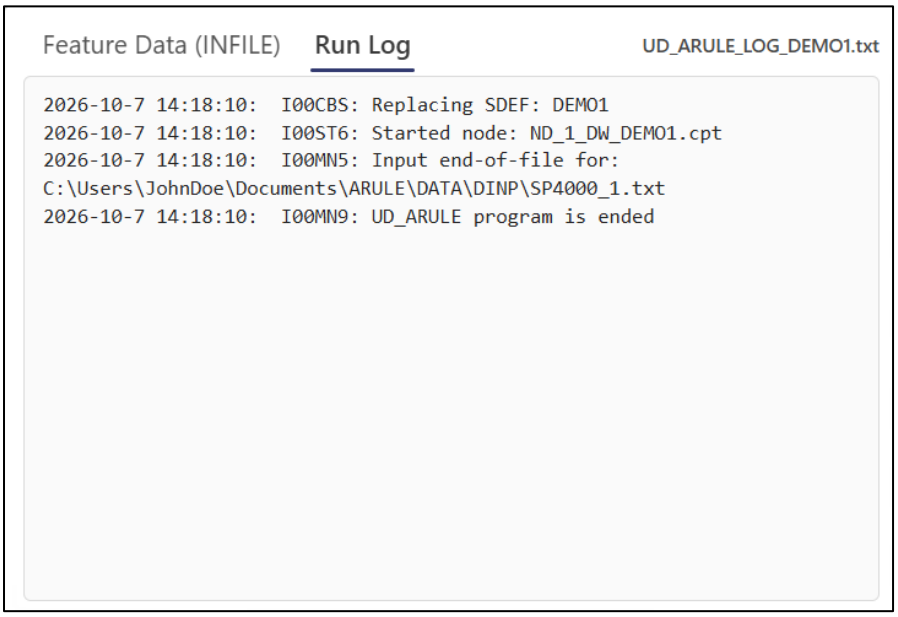


Figure 53 – Example of a Run Log: successful processing

5.3 Data Output

The Data Output section on the main window (Figure 54) shows a graphical representation of data output files for all processed data sets that were referenced in the NDEF file. When the data output files are selected using either the **CURRENT ARULE RUN or SAVED ARULE RUN** dropdown options, then four default data output graphs will populate for the selected data output file: 1) **FEATURE DATA**; 2) **FFP PROGRESSION**; 3) **SOH PLOT**; and 4) **RUL AND PH PLOT**. Note that the second graph has a drop-down option to toggle from the **FFP PROGRESSION** plot to the **DPS, FFS,** and **FFIN PROGRESSION** plots. These plots represent the different signature transforms within ARULE as covered in Section 2.5.

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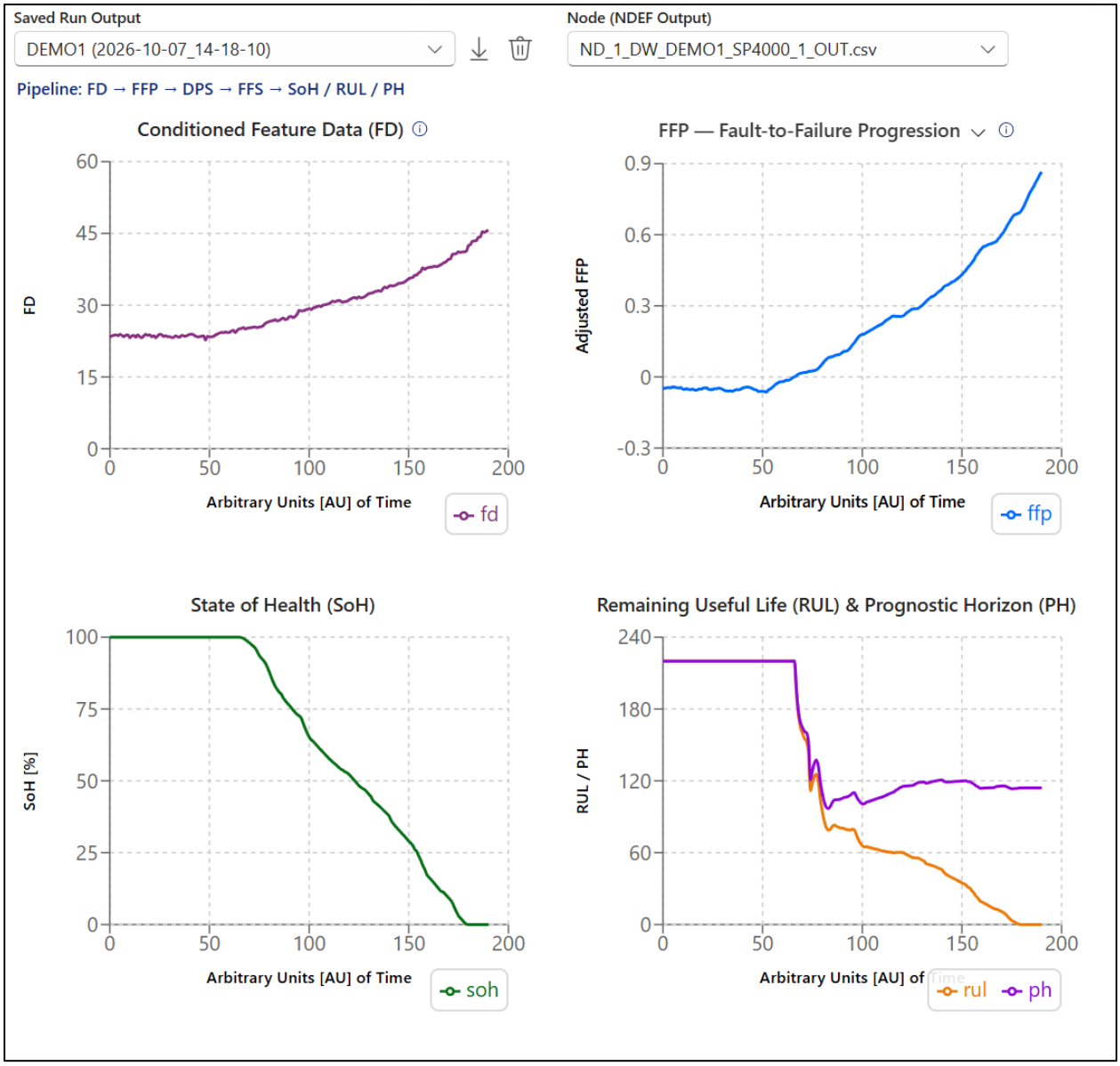


Figure 54 – Example of a Data Output section

5.3.1 Data Output Selection

The selection for data has two modes: Default and CPT Mode. You can toggle between these two modes by turning CPT Mode on and off.

Default Data Output Selection mode is used to view output from non-CPT simulations (Figure 55).



Figure 55 – Default Data Output Selection

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It consists of two dropdowns:

- **Saved Run Output –** Select the run you wish to view. This is denoted by the name of the SDEF and the time it was ran.
- **Node (NDEF Output) –** Select the node from the run selected in Saved Run Output to view the results of the ARULE run. These names are denoted by *ND_<<NODE NUMBER>>_DW_<<SDEF Name>>_<<INFILE Name>>_OUT.csv*.

These runs are automatically saved in the ARULE user directory under **ARULE\DATA\SAVE** folder. The **download arrow** button to the right of the **Saved Run Output** will let you save a compressed folder of the run. The **trash** button will allow you to delete the run.

CPT Output Selection mode is used to view output during a CPT run (Figure 56).

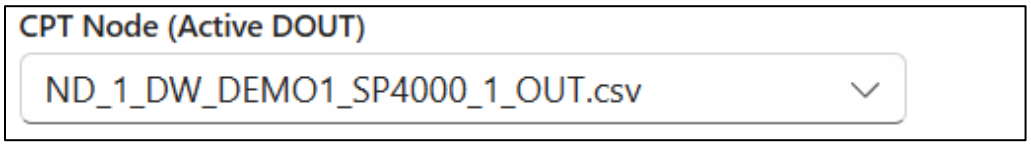


Figure 56 – CPT Data Output Selection

It consists of one dropdown:

- **CPT Node (Active DOUT) –** Select the node from the current run in CPT Mode that you wish to view. These names are denoted by *ND_<<NODE NUMBER>>_DW_<<SDEF Name>>_<<INFILE Name>>_OUT.csv.*.

These output files can be found in the ARULE user directory folder under **ARULE\DATA\DOUT** folder. This folder is reset at the beginning of a new CPT Run.

5.3.2 Data Outputs in .csv File

The Input-Output (IO) Bus of the ARULE API is the primary means by which an application program provides input data to ARULE and the means by which ARULE returns prognostic information to the application. RUL, SOH, and PH values are prognostic information useful in determining state of health and for predicting a future time of failure. The other values are provided for the following reasons: (1) support evaluation, including verification and validation, of ARULE results and performance; (2) support the creation of visualization aids, such as plots, by an application; and (3) support problem determination of incorrect parameter and other unexpected conditions (return and reason codes). The IO Bus is also used to provide return and reason codes for verification and validation of specified parameter values and for problem determination, accuracy evaluation and analysis, and performance evaluation by an application program. Table 4 lists the identifier (ID), a short description and other information for the values in the ARULE APK IO Bus.

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Table 4. Data Output Values

ID	DESCRIPTION	Notes
FLAG	Flag for internal program operation.	-
DT	Data Time	1, 6
DA	Data Amplitude	1
RUL	Remaining Useful Life	1
PH	Prognostic Horizon	1, 2
SOH	State of Health	1
BD	Begin of Degradation	1, 3
EOL	End of Life	1, 3
FDNOM	FD-nominal value (Set by FDZ or calculated by FDC & FDCPTS)	1
FD	Feature Data	1, 5
FFP	FFP signature data	1, 5
DPS	DPS signature data	1, 5
FFS	FFS signature data	-
FFIN	Functional Failure input data to ARULE prognosis	-
RC0	Highest Return Code for this data point	7
RS0	Reason for RC0	Table 5
RC1	First non-zero RC	7
RS1	Reason for indicated RC	Table 5
FLAG	Reserved flag for internal program operation.	-

Note 1: All values, except SOH, are arbitrary units of measure. It is the responsibility of the application to keep track of, for example, that a DT value of 8.5 means 8.5 hours, days, and so on.

Note 2: SOH values are percent values, for example, a SOH value of 92 means 92% healthy.

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Note 3: -99 value indicates not detected

Note 4: -99 value indicates not estimated

Note 5: Data after conditioning/smoothing and prior to next signature transform

Note 6: Time value for input data must be larger than previous time value

Note 7: Return code values are 0 (No errors/no exceptions); 4 (warning); 8 (error – ARULE is unable to continue processing data); 12 (severe error – possible programming error, ARULE is unable to continue processing)

Table 5. Reasons: Value, Descriptions, Action, and Remedy

RS	DESCRIPTION	ACTION	REMEDY
2	not assigned	none	none
4	FDNM is a negative value or greater than 25.0	Changed to 5.0	Specify a positive value between 0 and 25.0
6	not assigned	none	none
8	FDC/FDZ is a negative value	Changed to -FDZ	Specify a positive value
10	Not assigned	none	none
12	Calculated value of FDC is not within FDCVAR limits	Specified FDZ value is used	Change FDZ and/or FDZVAR as applicable
14	Not assigned	none	none
16	Not assigned	none	none
18	Not assigned	none	none
20	FDXPTS is a negative value or greater than 25	Changed to -FDXPTS and/or to 25	Specify a positive integer value up to 25
22	Not assigned	none	none
24	not assigned	none	none
26	Not assigned	none	none
28	not assigned	none	none
30	FFPFAIL is not greater than 0.0	Changed to 70.0	Specify a non-zero positive value
32	not assigned	none	none

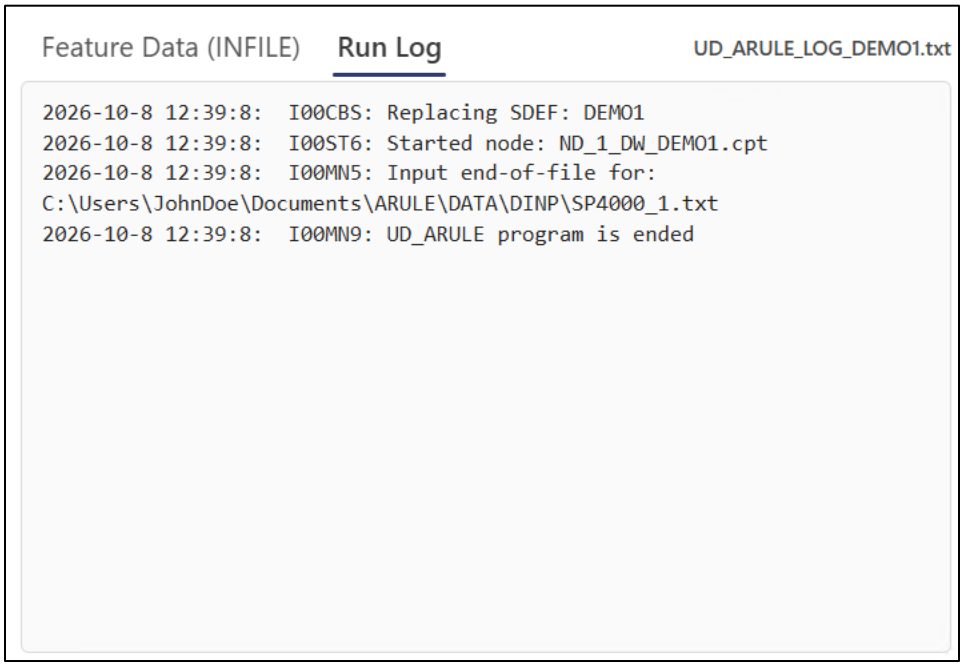
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34	not assigned	none	none
36	PITTFF is not greater than 0.0	Changed to 200.0	Specify a non-zero positive value
38	PIFFSMOD is not a recognized model number	Changed to 2	Specify a valid FFS-model number
40	Not assigned	none	none
99	Functional Failure has occurred	Data processing continues	Repair/replace the prognostic target
100	Value for input data time is not increasing	Data processing is ended	Correct the incorrect value of time
110	Value is not a number	Data processing is ended	Correct the incorrect value
900 9nn	Unexpected operating condition	Program operation is ended	Report the problem to your system administrator

Important Note: As required, additional reason codes might be added.

5.3.3 Run Log

This section of the GUI displays a text box that populates with any system messages, error codes, and reason codes that may occur during program operation.



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Figure 57 – View of Run Log

5.4 ARULE™ Global Features

5.4.1 Global Menu Options

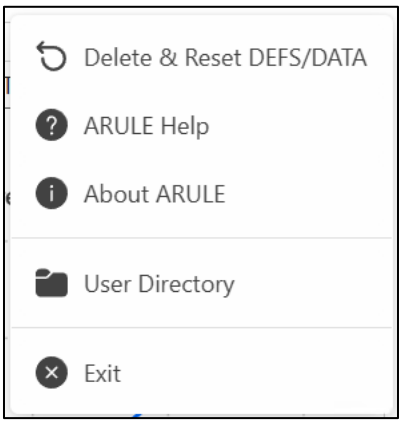


Figure 58 – View of Global Menu Options

There are the global menu options in the ARULE™ GUI as shown in Figure 58. An overview of each option is outlined as follows:

- **Delete & Reset DEFS/DATA**: This opens a popup (Figure 59), that allows you to delete and reset SDEFs, NDEFs, Input Data Files, Output Data or all the above. Select which option you want, and click the **Delete & Reset** Button. **This cannot be undone.** Click close to

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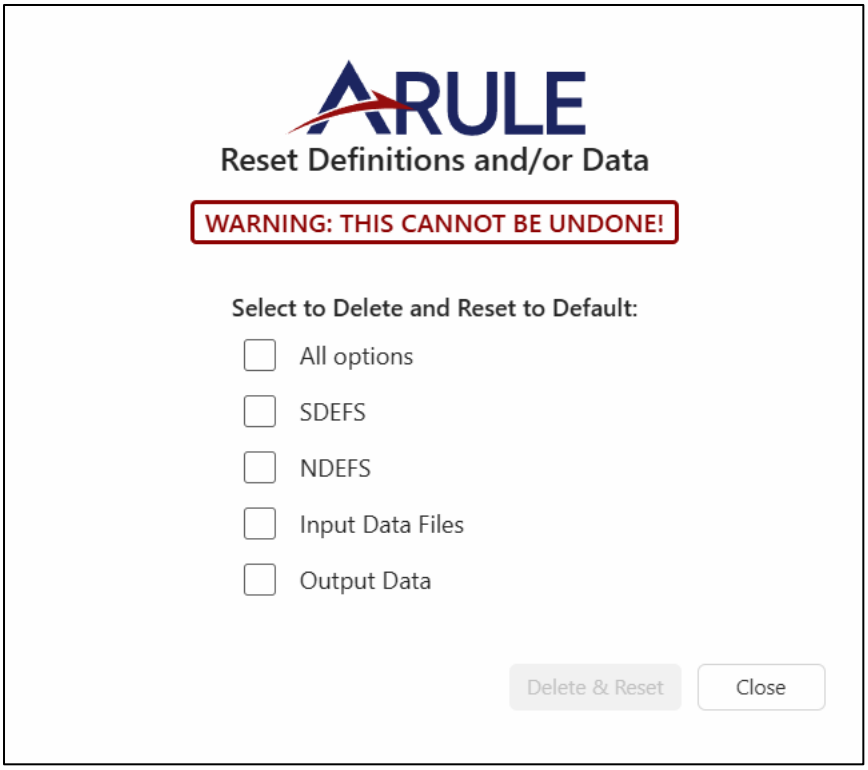


Figure 59 – Delete & Reset DEFS/DATA popup

- **ARULE Help**: This opens the help popup (Figure 60), which allows you to view the User Guide, Create a Support Ticket and provide you with email addresses for any other concerns.

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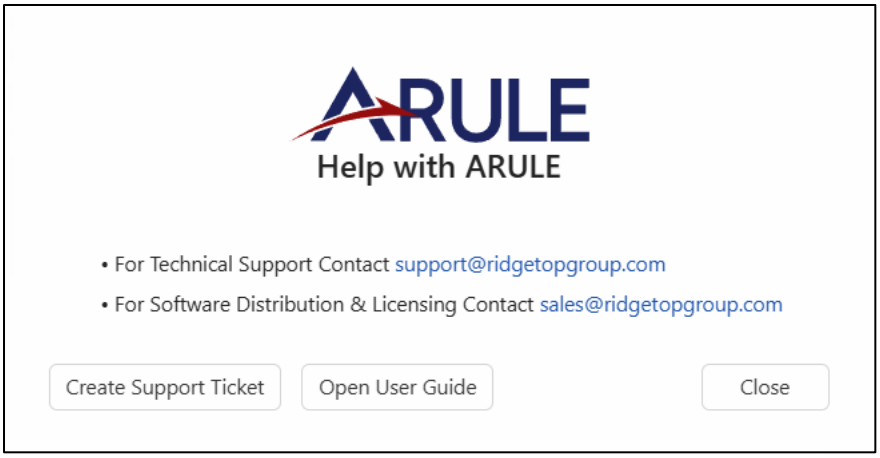


Figure 60 – Help Popup for ARULE

- **About ARULE**: This generates a popup window that tells you what version of ARULE™ is installed (Figure 61).

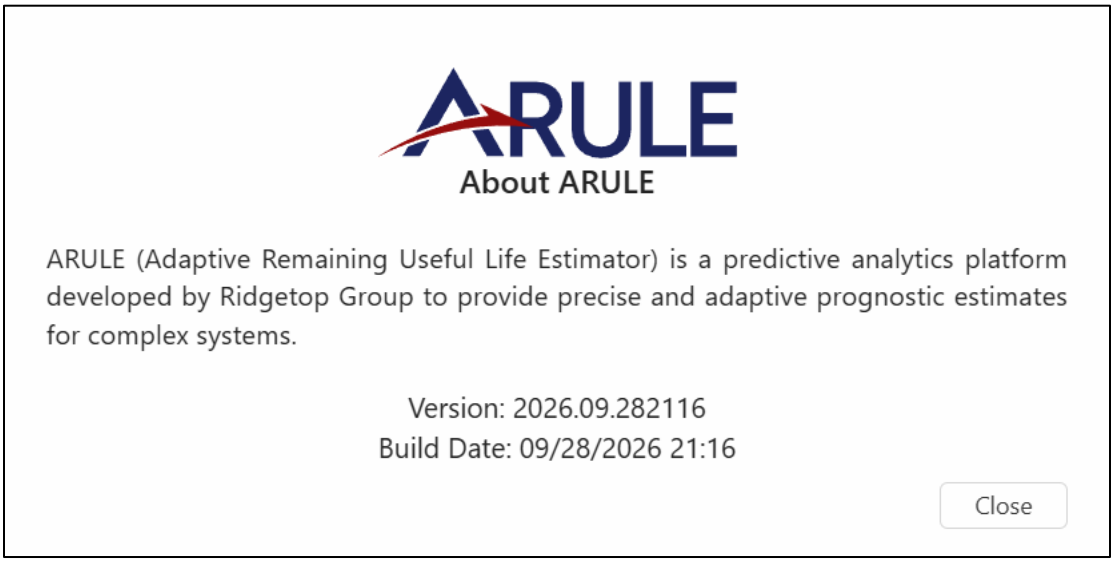


Figure 61 – View of About ARULE option when selected from the Menu

- **User Directory:** This opens the Setup ARULE User Directory popup which allows you to view the location of the current directory or to change the user directory to a different location.

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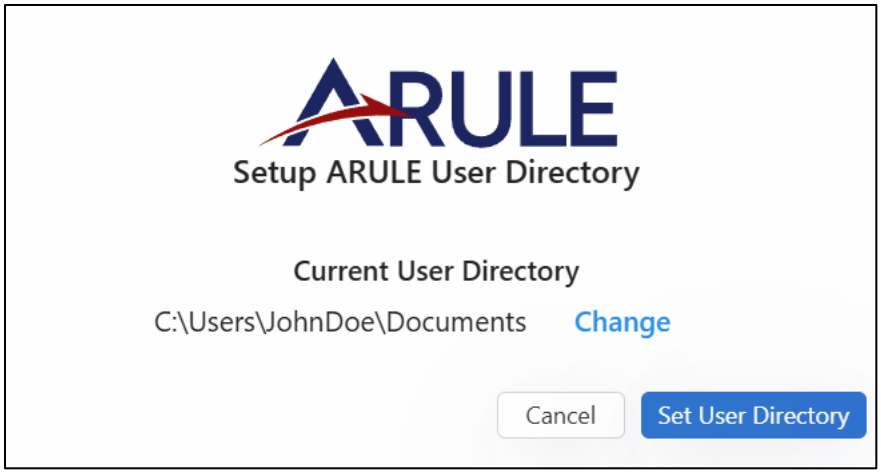


Figure 62 – User Directory Popup

- **Exit:** This prompts you to close and exit the program as shown in Figure 63.



Figure 63 – Prompt to exit the ARULE™ application

6. Real World Examples

Node definitions are used to process condition–based Feature Data (FD) and the major steps in creating and using an NDEF file in ARULE™ are the following:

1. Acquire a set of example FD that progresses from no degradation to functional failure.
2. Create an initial set of node definitions that define:
   a. A characteristic curve applicable to the example data.
   b. Initial points where degradation is detectable and where functional failure is likely.
3. Run ARULE™ to process the example FD.
4. Evaluate the results.
5. As necessary, update node definition(s) and repeat steps three and four.

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Step two in the above process comprises transforming condition–based data (CBD), also referenced as Feature Data (FD), into Fault to Failure Signature (FFS) data by selecting and defining NDEF keywords and values. The NDEF parameters then allow ARULE™ to transform FD signature data into FFP signature data, then into DPS data, and then into FFS data that is processed to produce prognostic information. The flow diagram shown in Figure 64 and the models shown in Table 6 summarizes the process of selecting applicable NDEF keywords and specifying values for those keywords. Further information is found in Section 2.

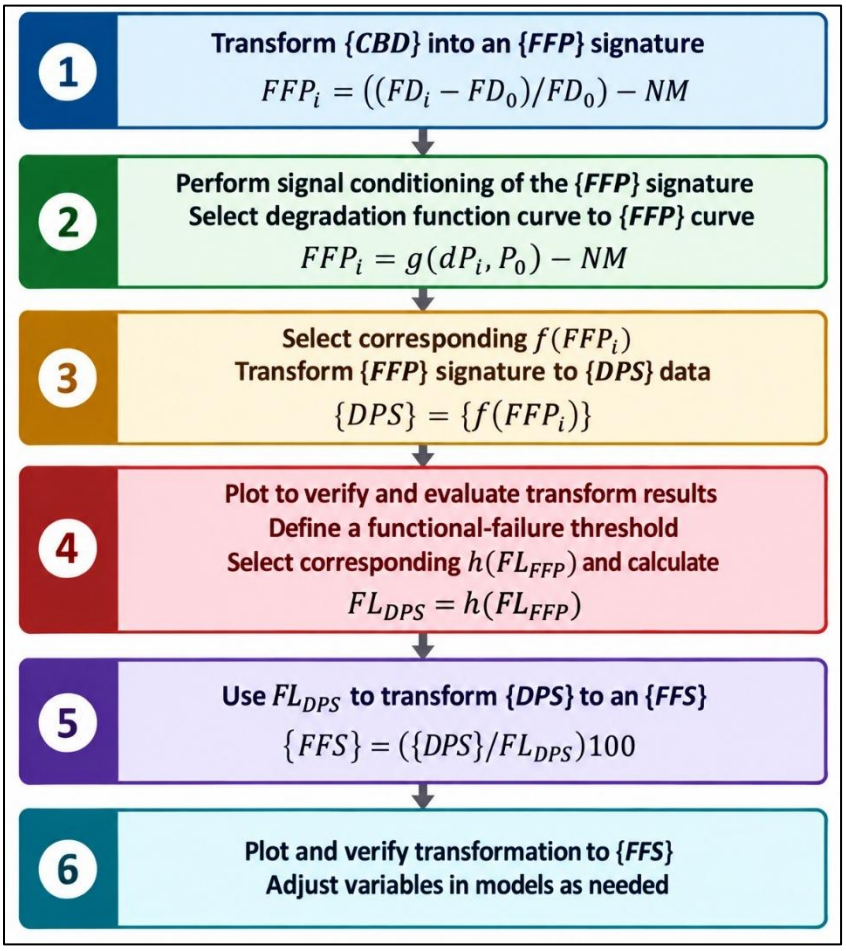


Figure 64 – Flow–diagram for transforming CBD signature data into FFS data

ARULE™ is packaged with sets of input data as well as SDEF and NDEF files that are used as examples for illustrative, instructional, and guidance purposes. The data used in those examples were originally collected from experiments, simulations, published works, internal and funded projects, and so on. Sometimes data from multiple collections and sources were combined to form more complete data sets; and sometimes dense data collections, such as those gathered from continuously running sensors, were transformed to less data collections. In prognostic applications, data should be collected periodically

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with high sampling frequencies. For example, the data plotted in Figure 84, page 90, comprises 109 data points extracted from an original data set of over 300,000,000 data points (500+ hours of about 160 data points per second).

Table 6. Table of FFP, DPS, and FFS Transformation Models

Increasing Signature Function	FFP Signature Model	FFP-to-DPS Transform Model	FFP-based FL to DPS-based FL Model
Set 1 Power	$(dP_i/P_0)^n$	$(FFP_i)^{1/n}$	$(FL_{FFP})^{1/n}$

6.1 DEMO1

The primary purpose of DEMO1 is to illustrate "batch–mode" processing of a single input data file to produce prognostic estimates of RUL, PH, and SoH. The input data file is in .txt format and represents the resonant frequency response from a PC power supply shown in Figure 65.

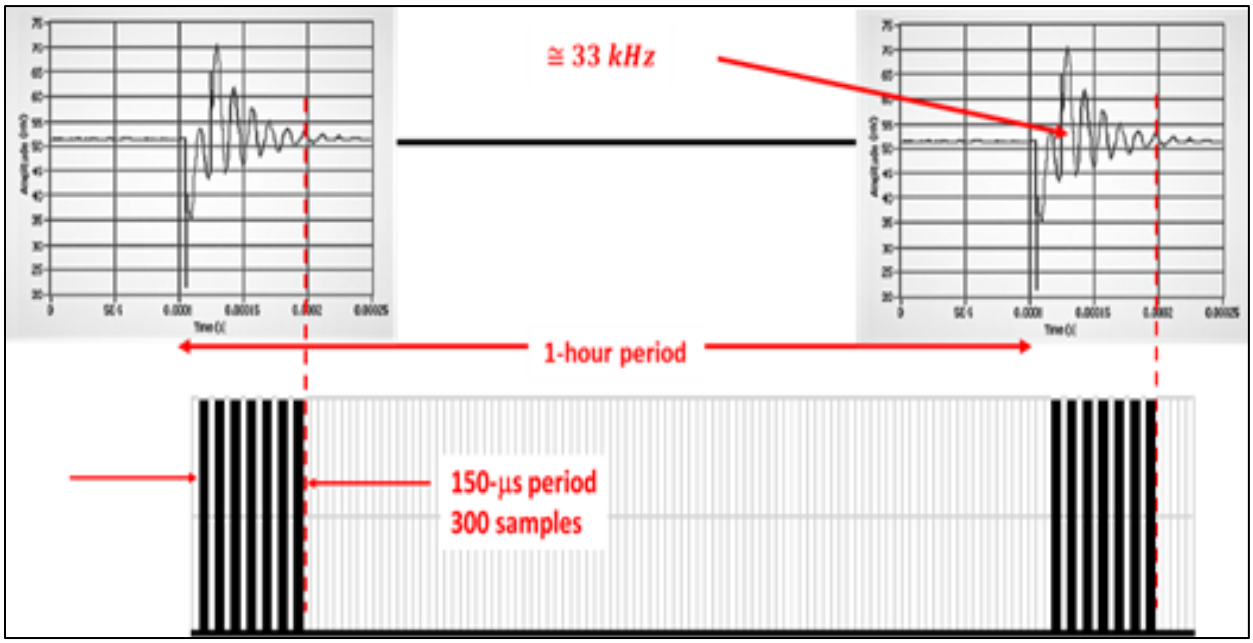


Figure 65 – Example of high–rate burst–mode sampling at a low–rate periodic–mode sampling

The data set was collected using Ridgetop's SMRT Probe 4000 sensing device which is a key component in the Sentinel Power family of prognostic solutions for power supply systems.

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6.1.1 DEMO1: Definitions

As shown in Figure 66, the SDEF file for this example points to a node definition number of 1, and an NDEF file named **DEMO1_NODE1**. Figure 67 shows the NDEF parameters and associated INFILE with DEMO1_NODE1.

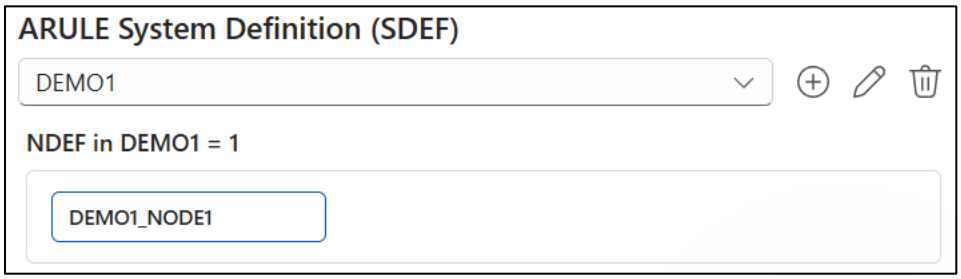


Figure 66 – DEMO1 SDEF

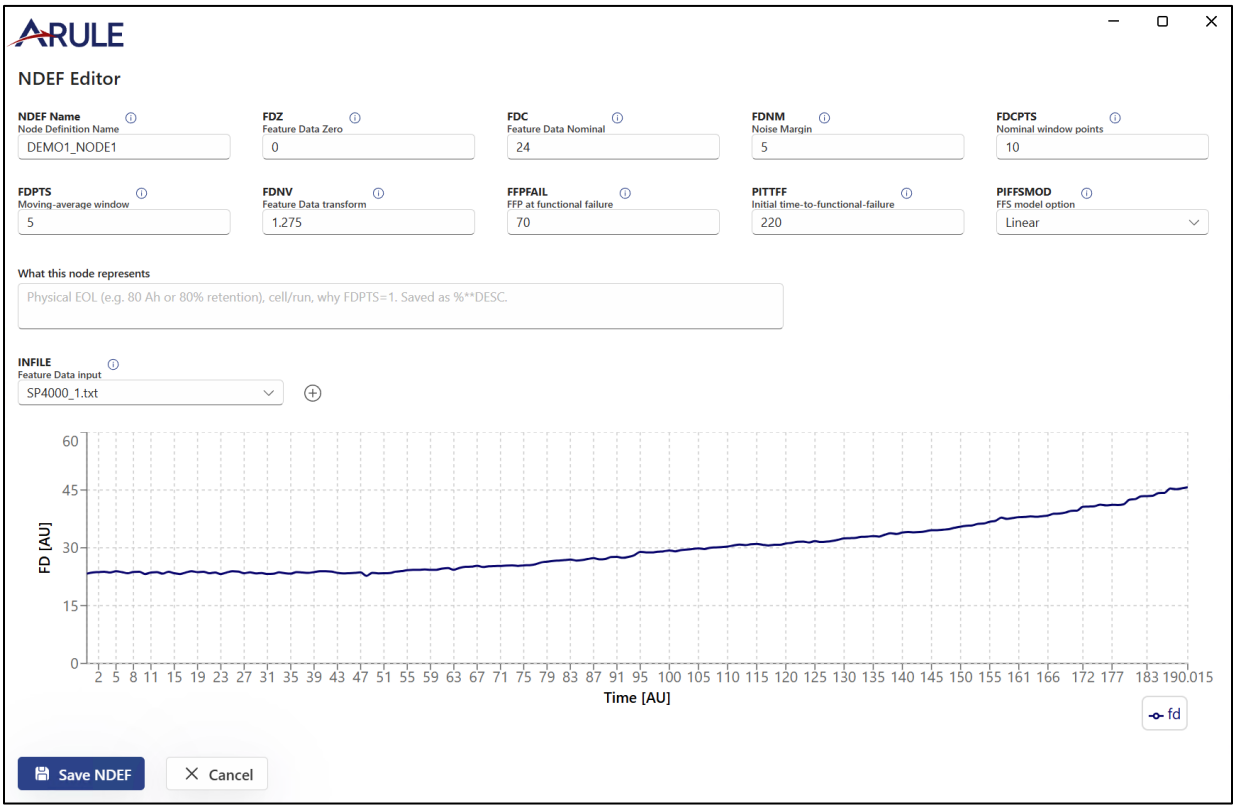


Figure 67 – DEMO1_NODE1 in NDEF Editor

6.1.2 DEMO1: Data Outputs and Prognostic Information

Figure 68 shows the plots of the prognostic information: FD, FFS Progression, RUL, PH, and SoH.

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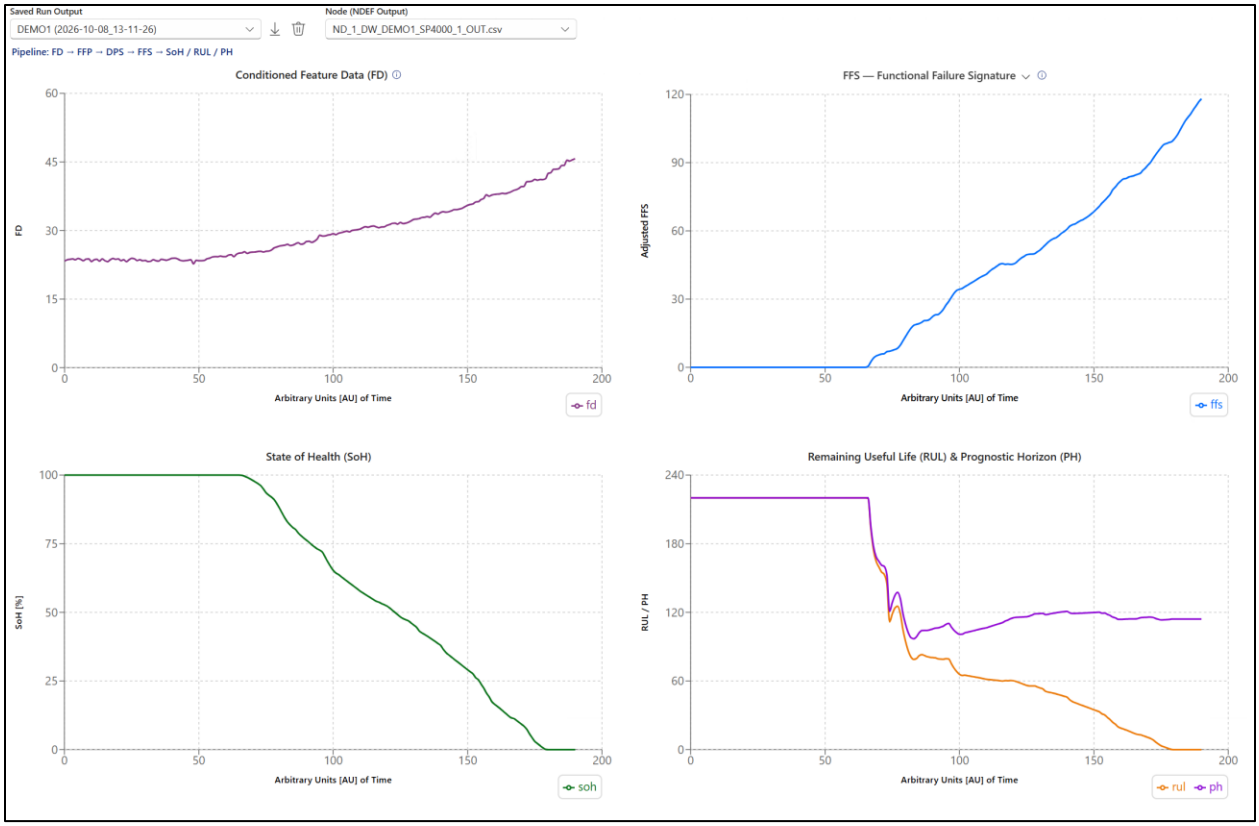


Figure 68 – Data Output section for DEMO1

6.1.3 DEMO1: Discussion

Power supplies supply power, voltage, and current, to a load and, as the load draws current, the magnitude of the output voltage changes. The magnitude of the change is dependent on multiple factors such as the magnitude of the current, the rate at which the current is drawn, the amount of capacitance and inductance used in the output filter circuit of the power supply. Capacitors have intrinsic series resistance that results in voltage changes called ripple voltage and a common design technique is to use many capacitors in parallel to reduce the total value of that intrinsic resistance. A common mode of failure is observed with the dielectric material of those capacitors: a capacitor fails short, high current results, and the capacitor connectors overheat and fail open – the net result is a loss of total capacitance, which causes the resonant frequency of the filter to increase.

A method to detect such failure is to attach a sensor, such as Ridgetop's SMRT Probe 4000, to the output of a power supply. When the sensor is commanded to sample, it introduces a small abrupt change in load that results in a damped–ringing response at the resonant frequency of the output filter of the power supply. The captured change in resonant frequency is shown in Figure 67. The plot is noisy because of thermal noise, switching noise, and distortion noise introduced by the feedback circuity of the power

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supply. Despite the noise, the results of processing the very irregular, noisy data are evaluated as excellent: refer back to Figure 68.

6.2 DEMO2

The primary purpose of DEMO2 is to illustrate "batch–mode" processing of three separate input data files of different types to produce prognostic estimates of RUL, PH, and SoH. The first and second input data sets are in .txt format with varying FDNV and FFPFAIL parameter values, and the third input data set is in .csv format. The data values are not identical, and are representative to the resonant frequency data analyzed in DEMO1.

6.2.1 DEMO2: Definitions

As shown in Figure 69, the SDEF file for this example points to three node definitions: **DEMO2_ NODE1**, **DEMO2_NODE2**, and **DEMO2_NODE3**. Figure 70, Figure 71 and Figure 72 shows the NDEF parameters, keywords, and the input data used for DEMO2: the three node definitions and the input data from SP4000_1.txt and SP4000_2.csv. These data are not the same, as evidenced by the plots in Figure 71 and in Figure 72. Note that data sets are not identical in values and FDNV & FFPFAIL parameters are varied in the case of the First and Second node.

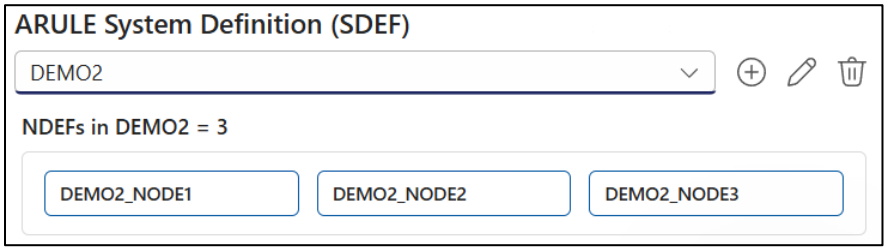


Figure 69 – DEMO2 System Definition (SDEF)

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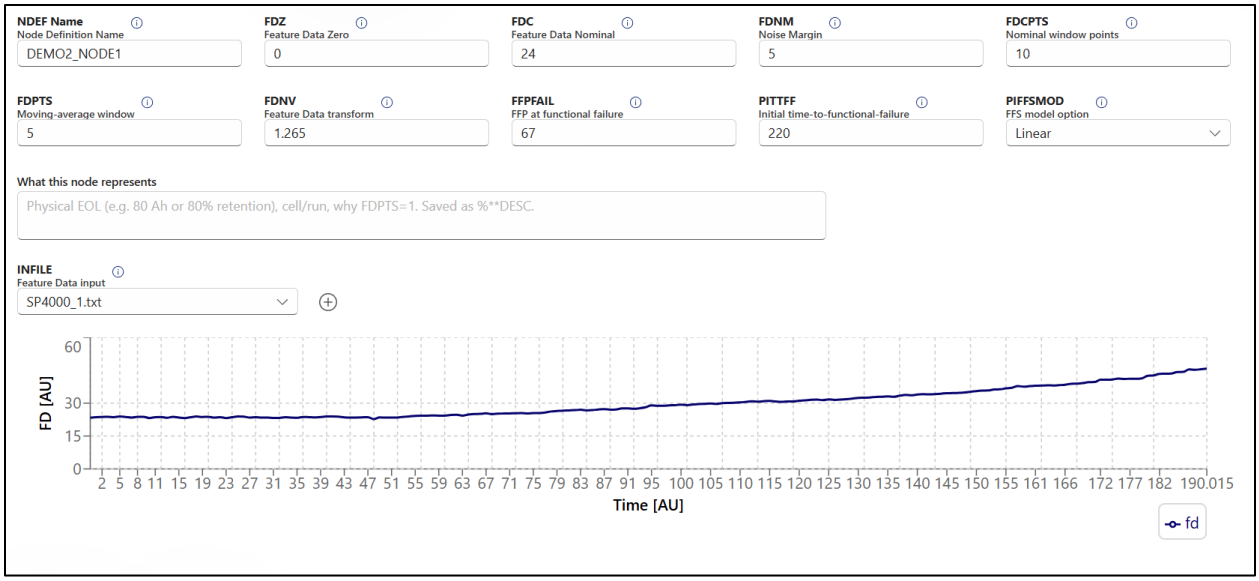


Figure 70 – DEMO2_NODE1 NDEF

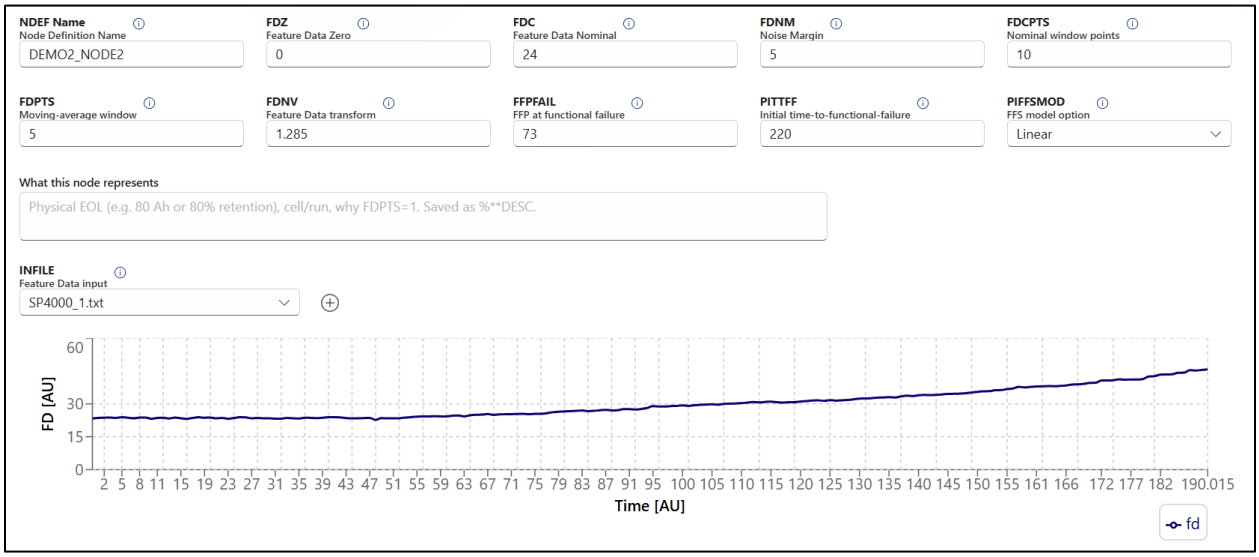


Figure 71 – DEMO2_NODE2 NDEF

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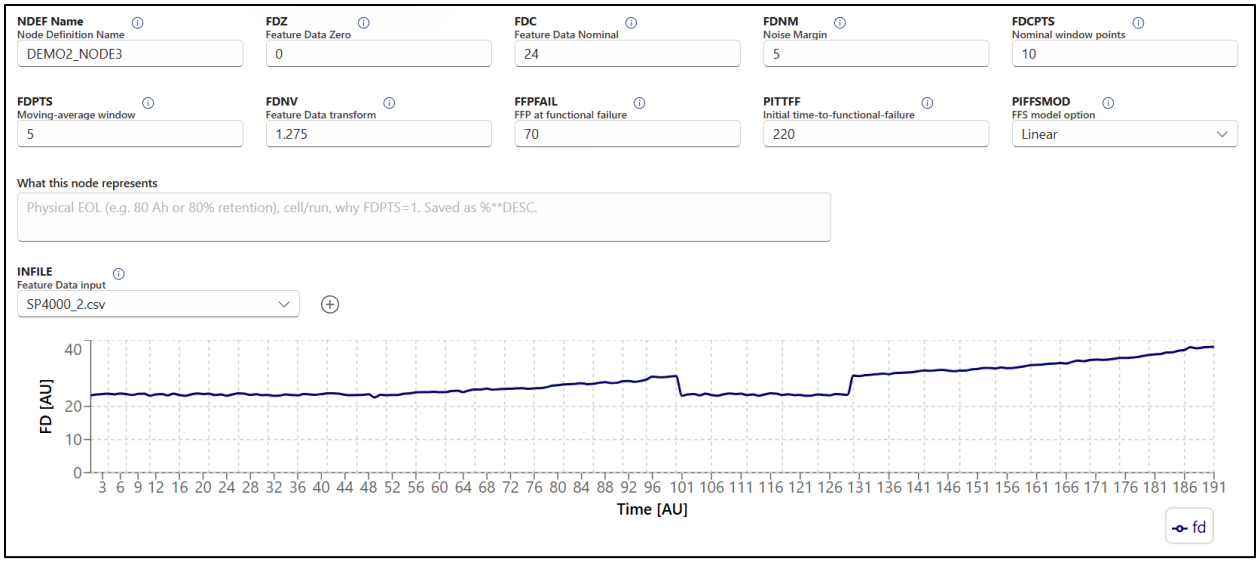


Figure 72 – DEMO2_NODE3 NDEF

6.2.2 DEMO2: Data Outputs and Prognostic Information

Figure 73, Figure 74, and Figure 75 show the data outputs and prognostic information for the First, Second, and Third Node respectively. These data outputs include the FD, FFS Progression, along with SoH, RUL, and PH. As expected, these plots are almost identical in the case of the First and Second node, but the Third node has different results.

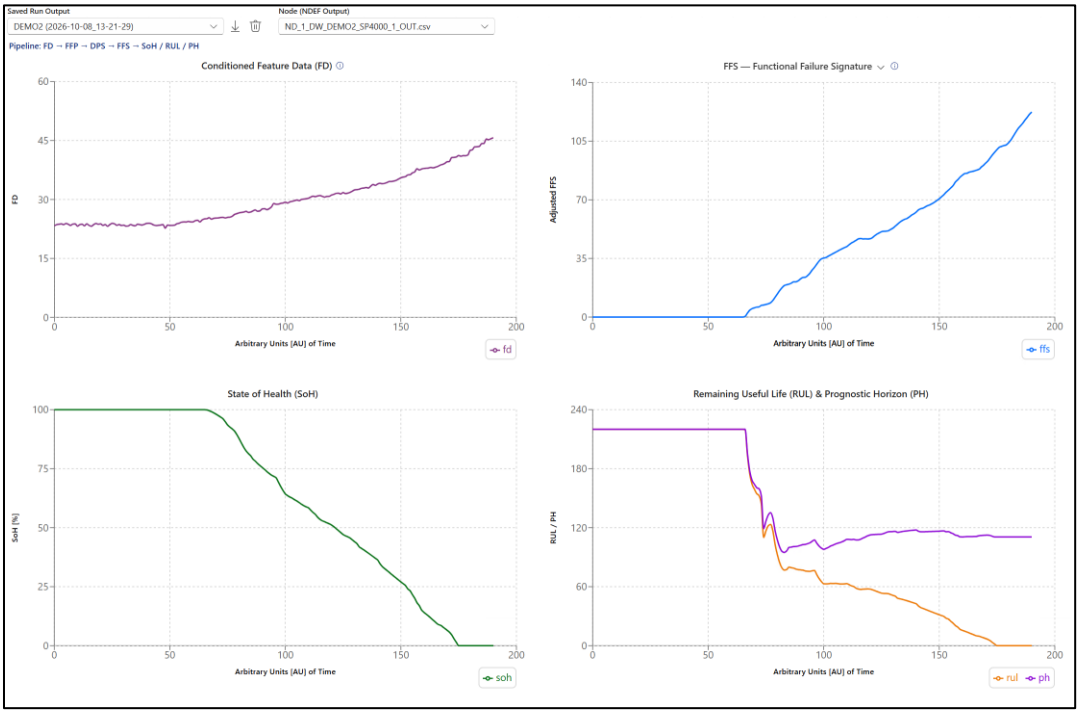


Figure 73 – Data Output for the First Node in DEMO2

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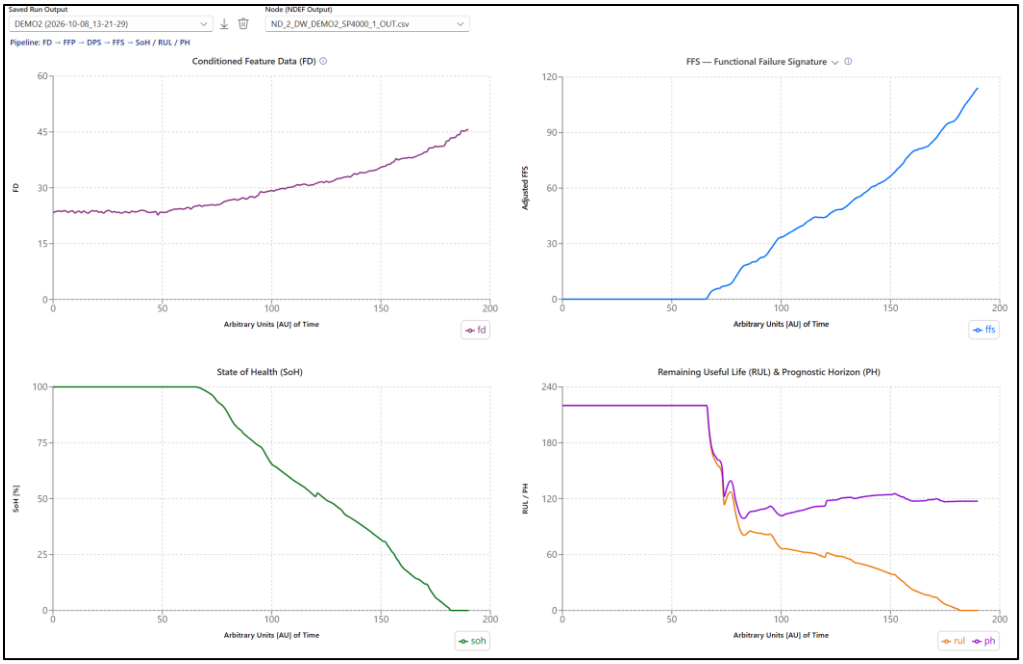


Figure 74 – Data Output for Second Node in DEMO2

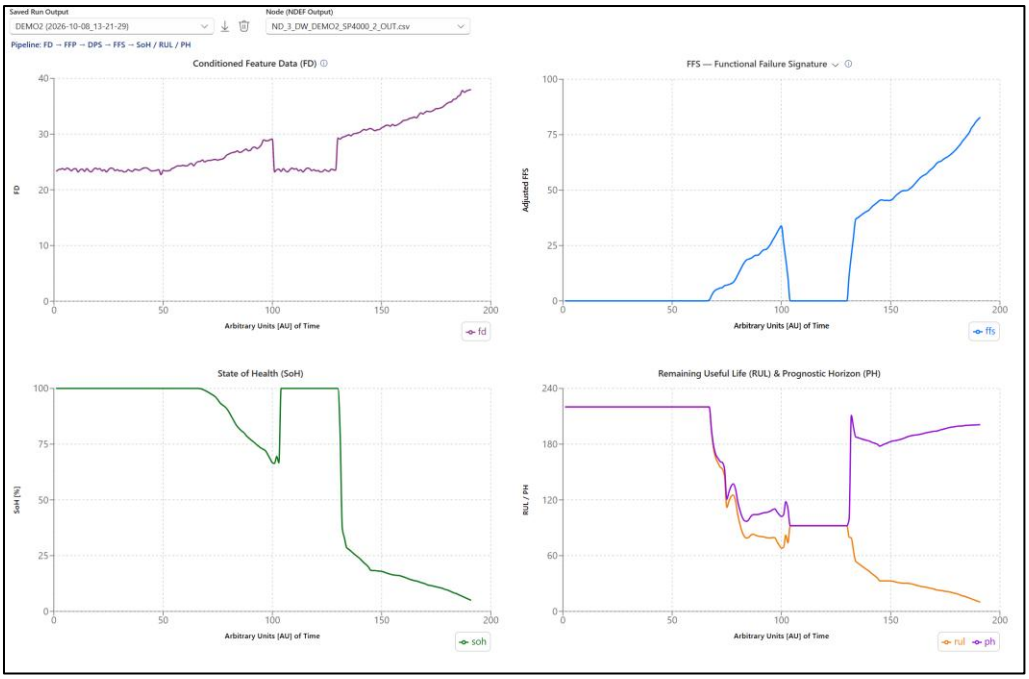


Figure 75 – Data Output for Third Node in DEMO2

6.2.3 DEMO2: Discussion

The first two data sets processed in this demonstration are equivalent to the DEMO1 data set, and therefore the same discussion notes can also be applied to this example. As previously noted, the primary purpose of this example was to demonstrate the capability of ARULE™ to process multiple NDEF files that point to two different types of input data

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sets. The first two input data sets were in .txt format, and the third input data set was in .csv format; and all output data sets are in .csv format.

A first glance of the plots of the output of all nodes might lead to a conclusion that the prognostic estimates are identical: they are not – because the two input files, SP4000–1.csv and SP4000–2.txt are not identical. Examples of non–identicality are shown in Figure 76

SP4000-1.csv			SP4000-2.txt		
Position	Time	Amplitude	Position	Time	Amplitude
1	1	23.3	1	0	23.3
26	26	23.913	26	25	23.9125

Figure 76 – Examples of differences between the input data in SP4000–1.csv and SP4000–2.csv

6.3 XFD1: Transmission Power

The first real world example is demonstrated with the SDEF titled XFD1. This example is related to the transmission of radio frequency (RF) being prone to loss of transmitted power. One method of monitoring this problem is to place a receiver at a known distance from the transmitter, receive transmitted signals, measure the received power, and use that measurement to detect loss.

6.3.1 XFD1: Definitions

As shown in Figure 77, the SDEF file for this example points to two node definitions. The NDEF files are named **XFD1_TPWR** and **XFD1_TPWRB**. Figure 78 and Figure 79 show the NDEF parameters and keywords used for the two nodes corresponding to XFD1, note that they both use same INFILE as well. There is only one difference between the two NDEF files: **XFD1_TPWR** has a PITTFF keyword parameter value of 200 as compared to a value of 60 for **XFD1_TPWRB**. By applying this small change to the initial estimate of RUL and PH, the convergence and accuracy of the ARULE™ outputs are drastically improved as detailed in the discussion section for this example.

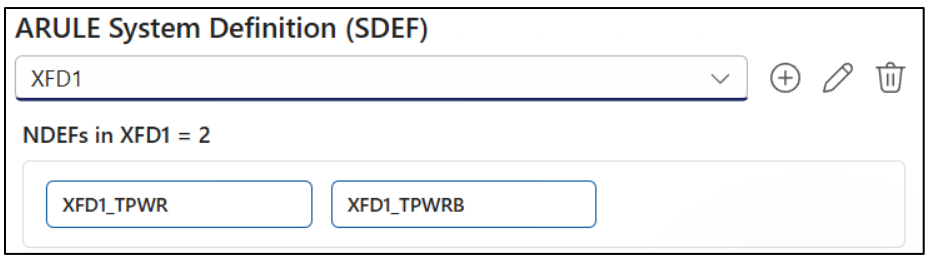


Figure 77 – XFD1 SDEF

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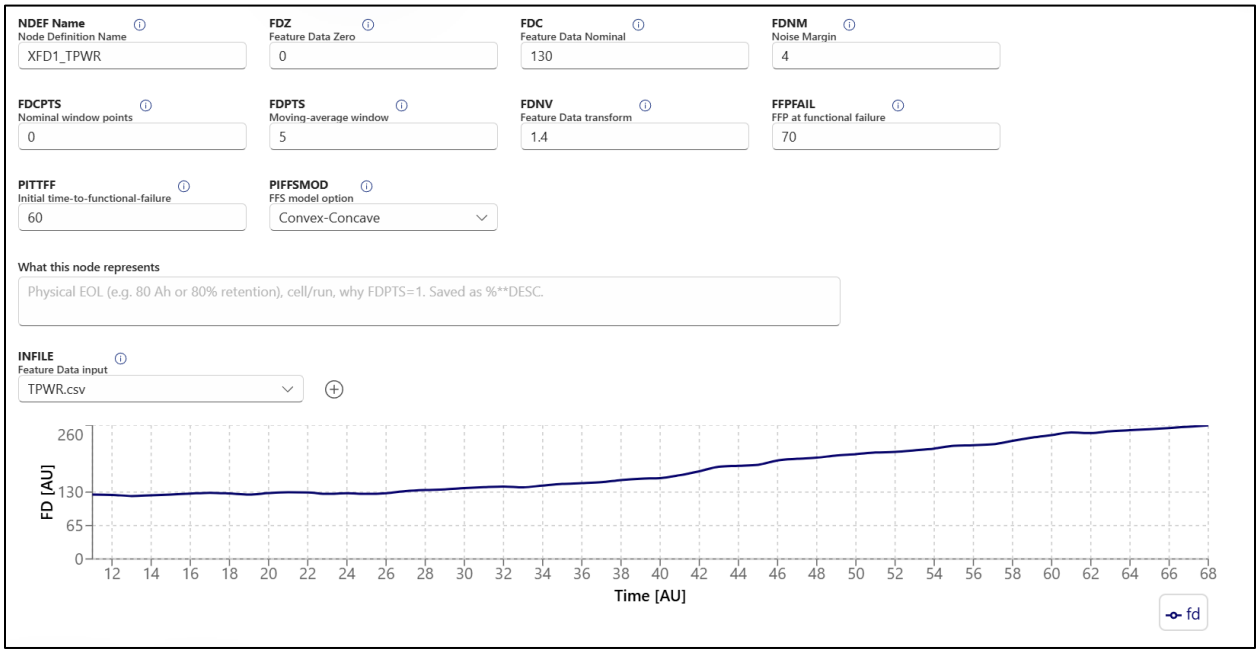


Figure 78 – XFD1_TPWR NDEF

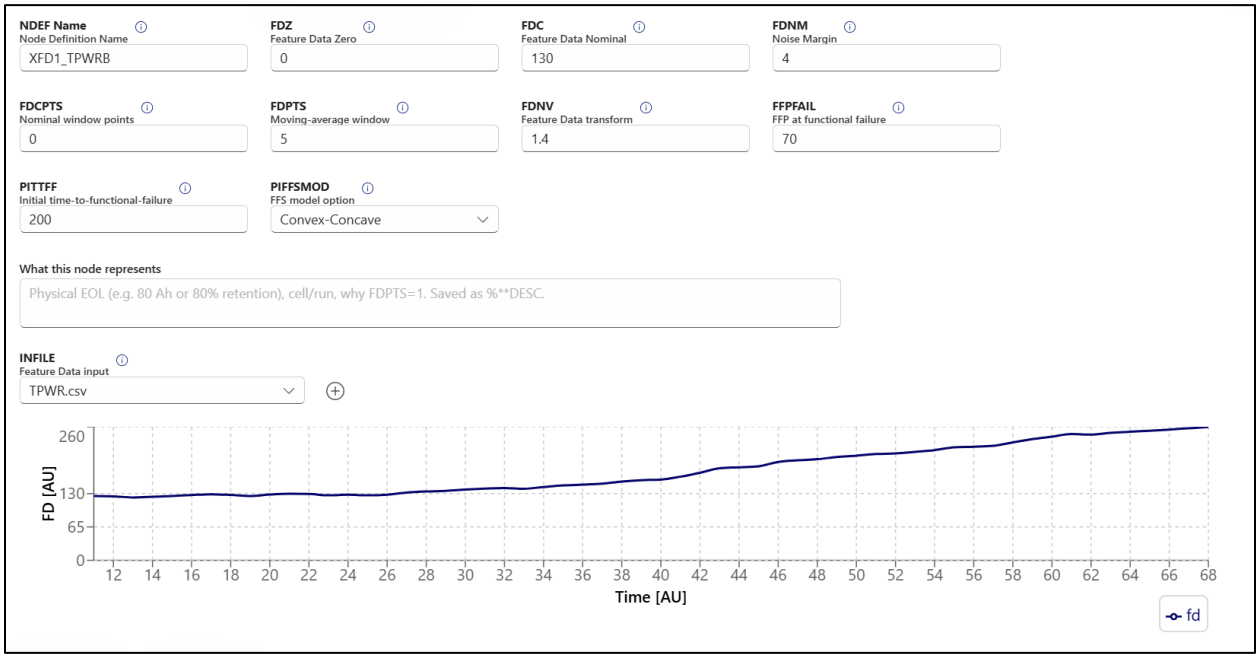


Figure 79 – XFD1_TPWRB NDEF

6.3.2 XFD1: Data Outputs and Prognostic Information

Figure 80 and Figure 81 show the Data Output for the two nodes in XFD1: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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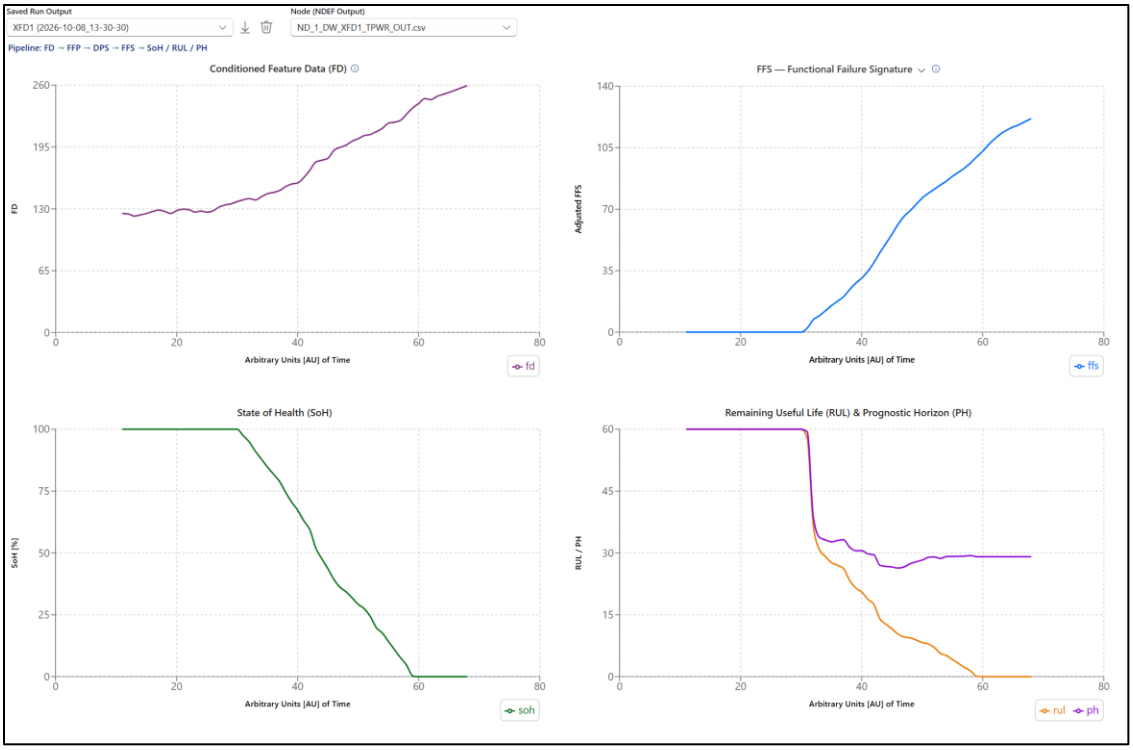


Figure 80 – Plots of the XFD1 Prognostic Information for the First Node

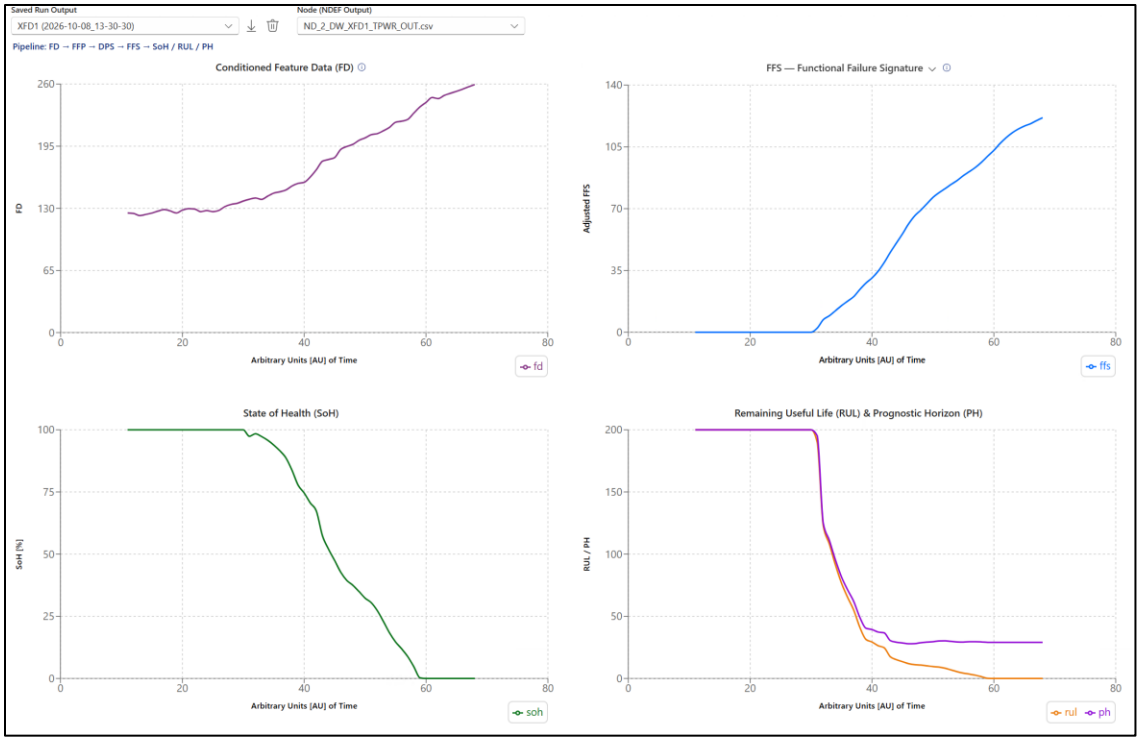


Figure 81 – XFD1 plots of the Prognostic Information for the Second Node

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6.3.3 XFD1: Discussion

Transmission of radio frequency (RF) is prone to loss of transmitted power: one method of monitoring is to place a receiver at a known distance from the transmitter, receive transmitted signals, measure the received power, and use that measurement to detect loss. The results of processing the very irregular, noisy data are evaluated as excellent.

Figure 80 and Figure 81 illustrate the effect of the initial estimate of how long it takes for functional failure to occur after the onset of degradation: ARULE™ converged from an initial–estimate error of 800% to less than 10% in eight (8) samples with PITTFF = 60 (Figure 80); ARULE™ converged from an initial–estimate error of 100% to less than 10% in only two (2) samples with PITTFF = 200 (Figure 81).

6.4 XFD2: Helicopter Gear Box – Pinion Gear

Ridgetop Group has developed an innovative sensing solution under the Sentinel Motion product line for detecting degradation in rotating–shaft equipment such as helicopter gear box systems, transmissions, and other mission critical equipment that rely on shock, vibration, and temperature monitoring. Sentinel Motion is based on the Internet–of–Things (IoT) where the system comprises a wireless network of RotoSense™ smart sensors, the Sentinel Gateway communications device, and the Sentinel MotionView software package for data acquisition, analysis, and sensor–gateway management. An example for how this sensing system was used to collect and process Condition–based Data (CBD) while working with NASA is shown in Figure 82 and this White Paper. Additional details pertaining to the full experiment can be reviewed in this NASA Spinoff Article, or on the Ridgetop Group website. This User Guide example illustrates how Sentinel Motion, RotoSense™, and the prognostic analysis from ARULE™ can be used to monitor and diagnose cracked gear teeth.

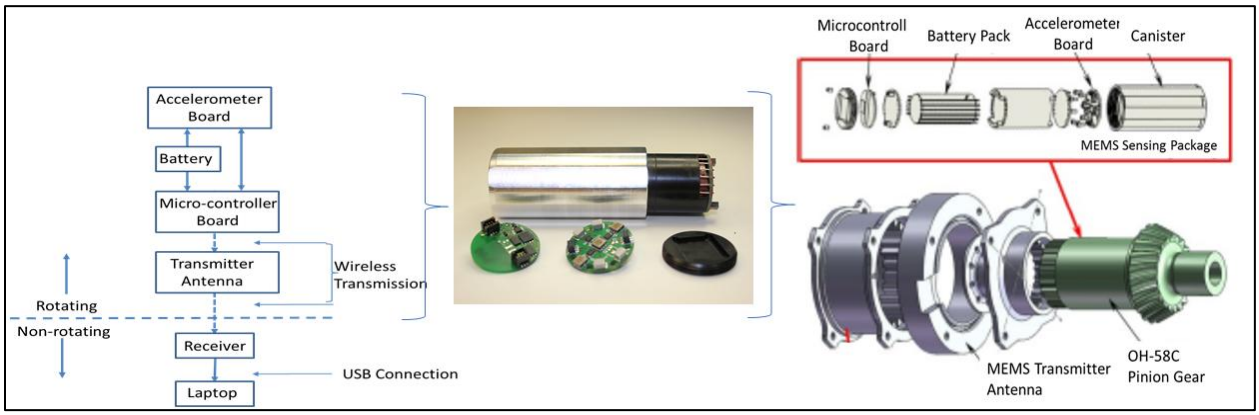


Figure 82 – RotoSense diagram, unit, and assembly in gearbox system.

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6.4.1 XFD2: Definitions

Figure 83 shows there are two nodes in XFD2: **XFD2_ROTOGB1** and **XFD2_ROTOGB2**. The NDEF files are identical: the difference is the second input data set was simulated to remove the instances of where the transmission was stopped to further damage the cracked gear tooth for this experiment. Figure 84 and Figure 85 show the NDEF parameters used for the two nodes corresponding to XFD2.

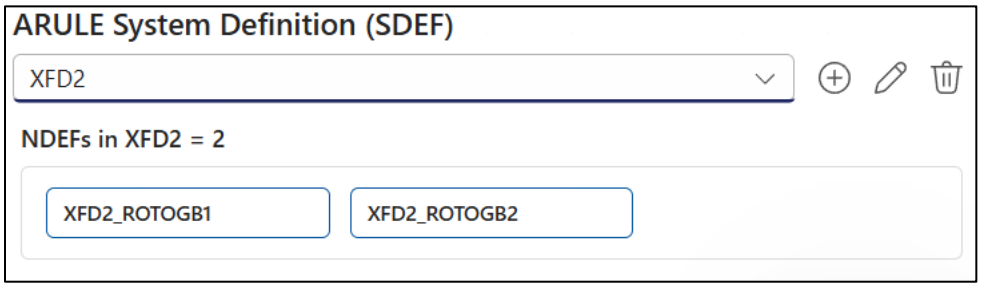


Figure 83 – XFD2 SDEF

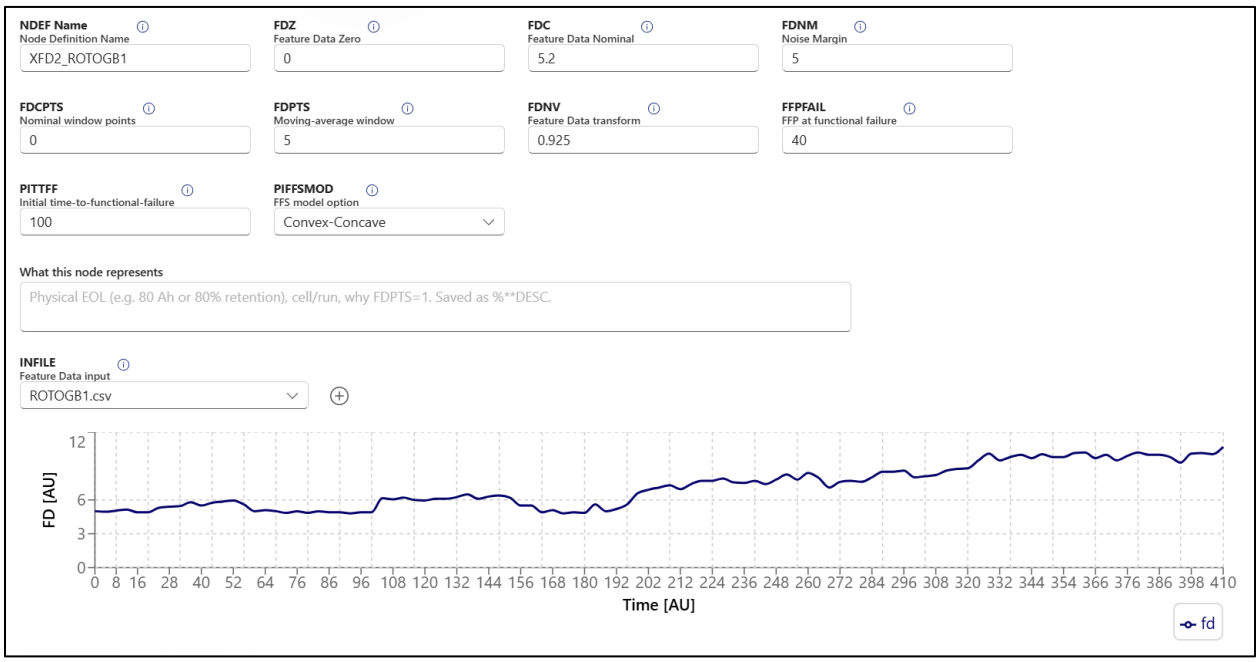


Figure 84 – XFD2_ROTOGB1 NDEF

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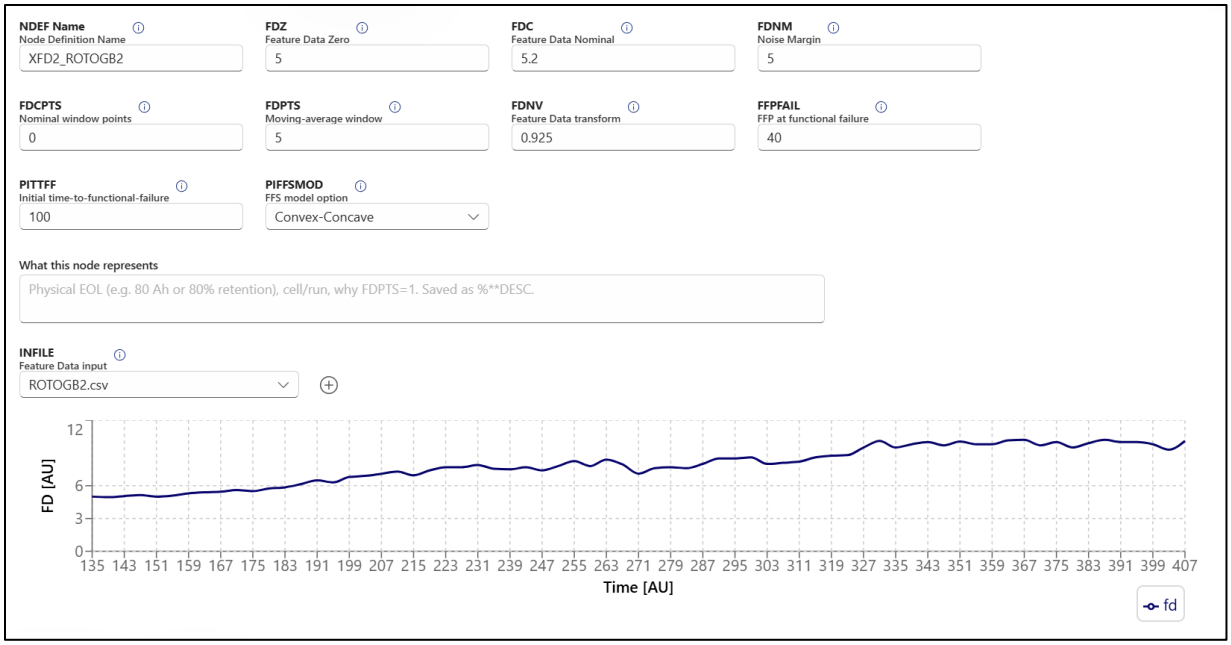


Figure 85 – XFD2_ROTOGB2 NDEF

6.4.2 XFD2: Data Outputs and Prognostic Information

Figure 86 and Figure 87 show the Data Output for the two nodes in XFD2: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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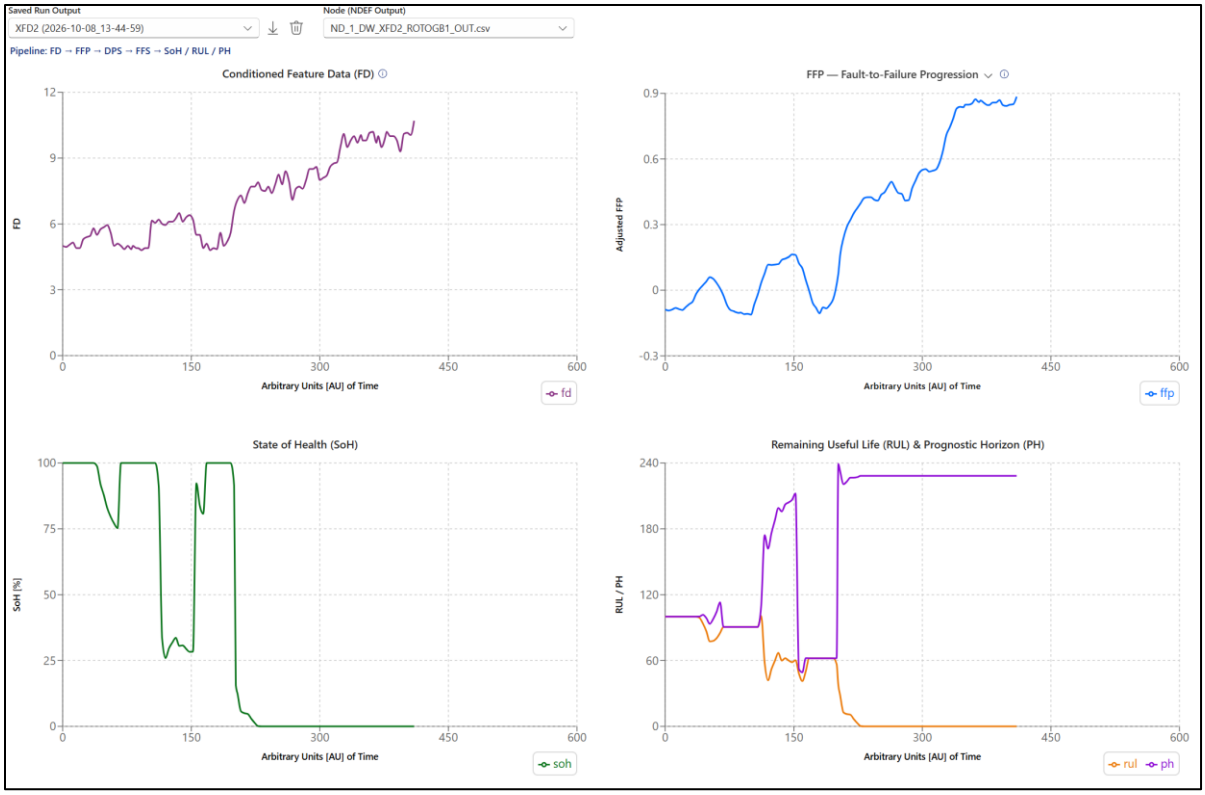


Figure 86 – Plots of the XFD2 Prognostic Information for the First Node

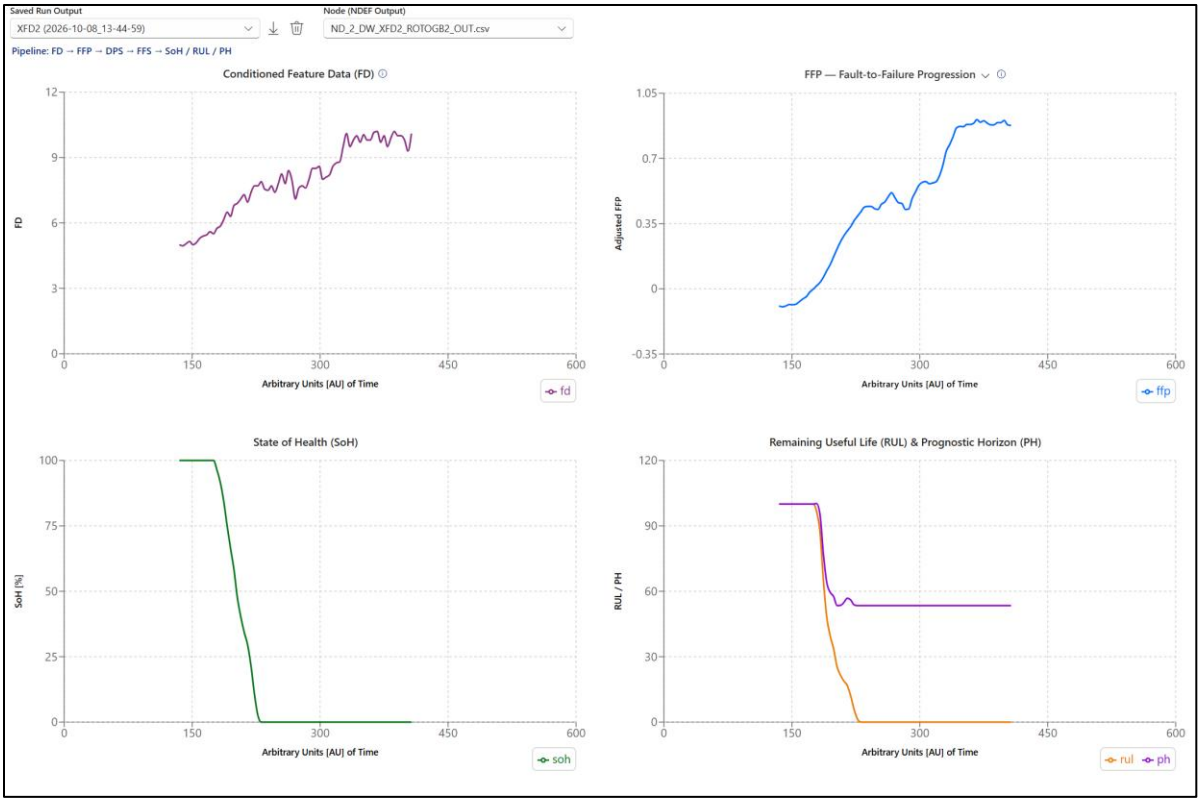


Figure 87 – Plots of the XFD2 Prognostic Information for the Second Node

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6.4.3 XFD2: Discussion

Node 1 Discussion: Figure 84 shows input data in **XFD2_ROTOGB1**. There are two features in the dataset that are the result of starting an experiment with a pinion gear that was slightly damaged; then in the First 5 and the First 16 hours of the test, the experiment was stopped, the pinion gear was removed, the defected gear tooth was further damaged (by filing), and then re–installed. After re–installation, the gear experienced a temporary reduction in force before the again showing evidence of increased damage: as evidenced by the prognostic information shown in Figure 87.

Node 2 Discussion: Figure 85 shows input data in **XFD2_ROTOGB2** which is the same dataset shown in Figure 84 with the exception being that the input data set for Node 2 was derived from the first data set to simulate not stopping the transmission and removing and replacing the pinion gear. This was done to illustrate the significance of spauling. There is one evident feature in Figure 84 that is the result of the pinion gear spauling twice: fracture followed by separation which resulted in lessening of binding and roughness between gear teeth (see Figure 88). About 200 hours later (time ~ 520 hours), there was a sharp drop in measured force as seen in Figure 89.

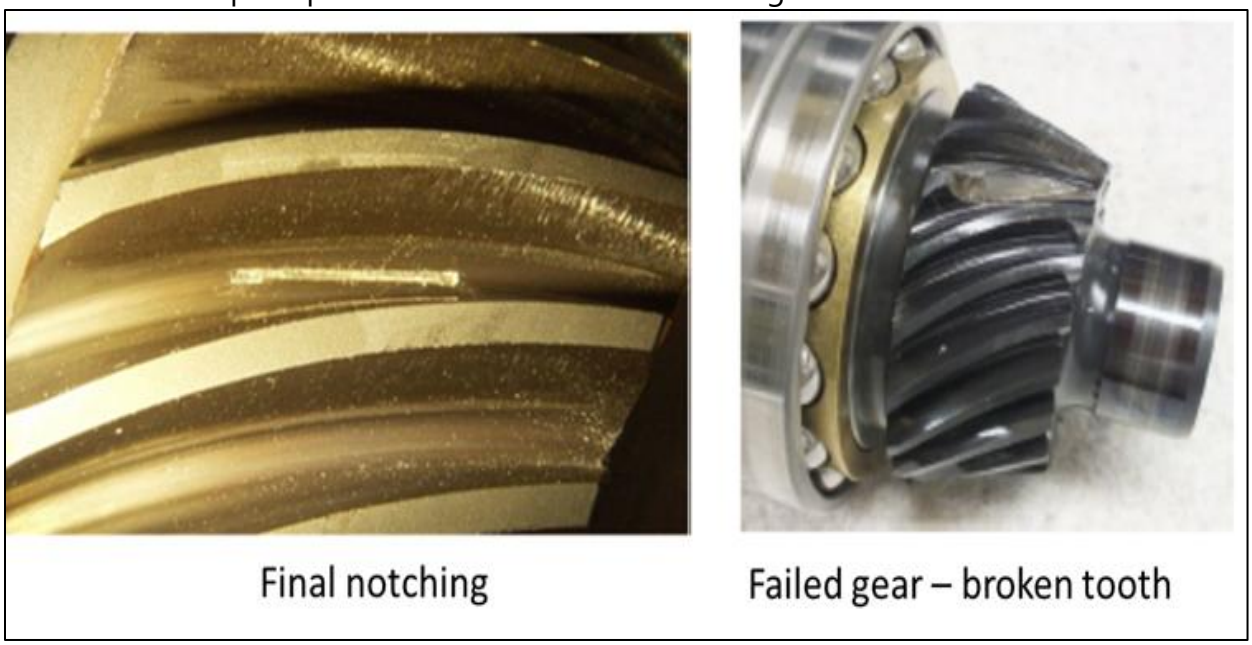


Figure 88 – Pinion gear after final notching @ 161 hours (left) and after the test was ended

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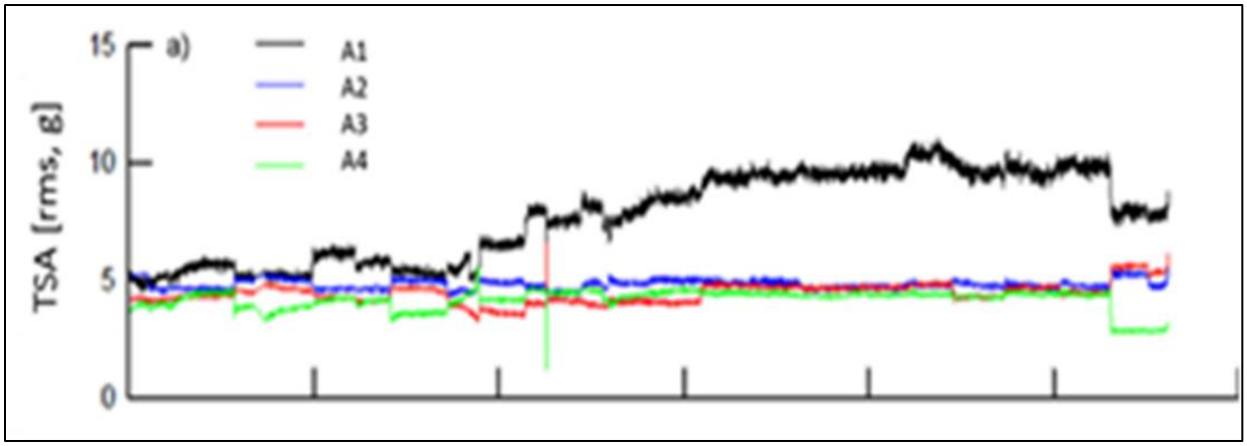


Figure 89 – Original data collection used to create input data for XFD2

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6.5 XFD3: Ripple Voltage, PC Power Supply, Full–PI filter

The CBD from power supplies comprises all manner of FD that can be extracted and used to support diagnostic and/or prognostic applications: Figure 90 illustrates six examples of FD – ripple voltage, thermal noise, damped–ringing response, harmonic distortion, droop, and switching spikes and glitches. Figure 90 is a plot of ripple–voltage data collected in response to increasing loss of filtering capacitance of a power supply in a personal computer (PC). This ARULE™ example has been proven through Ridgetop's Sentinel Power product line, which is a family of prognostic solutions for power supply systems.

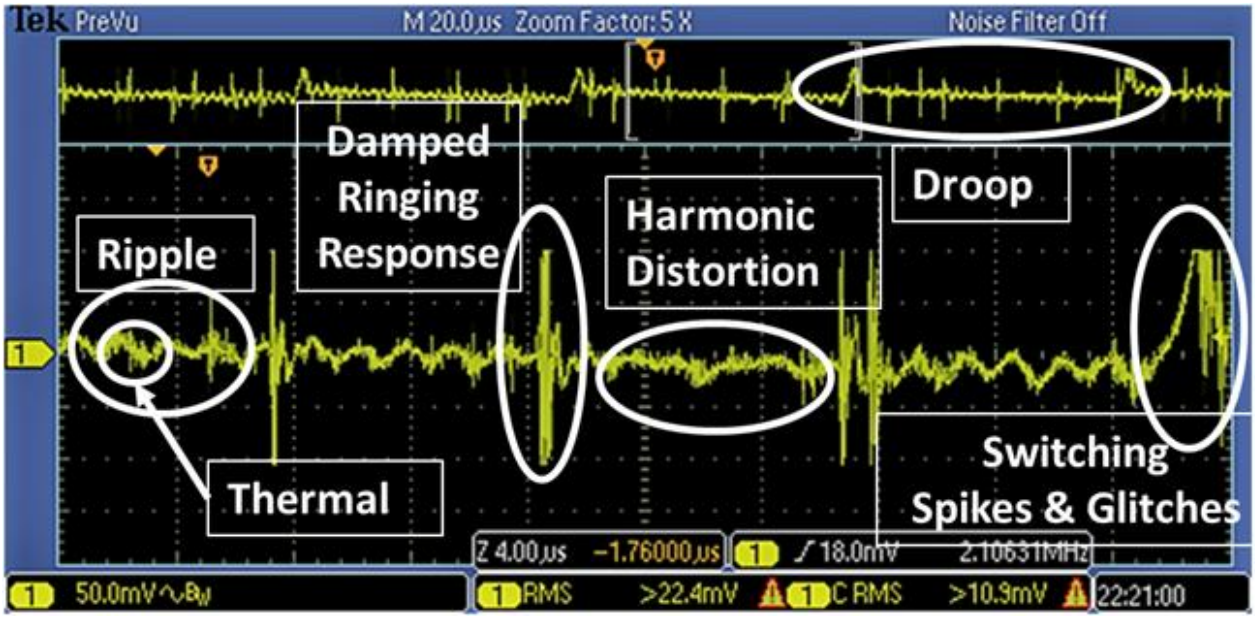


Figure 90 – Examples of Feature Data (FD) from power supply output

6.5.1 XFD3: Definitions

As shown in Figure 91, the SDEF file for this third example points to 2 node definitions, with NDEF files named **XFD3_VRPCAP** and **XFD3_VR2400**. Figure 92 and Figure 93 show the NDEF parameters, INFILES and keywords used for XFD3 for the First and Second Node, respectively.

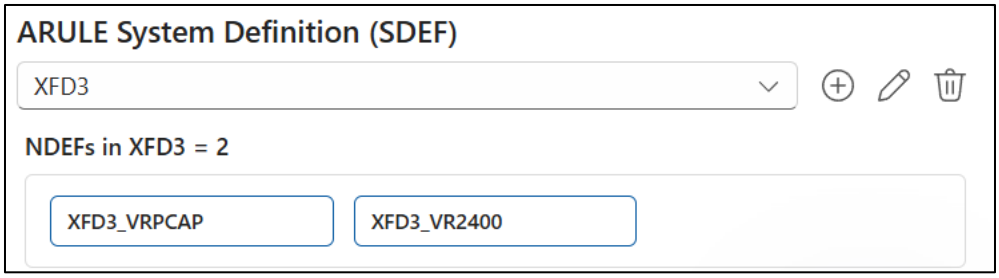


Figure 91 – XFD3 SDEF

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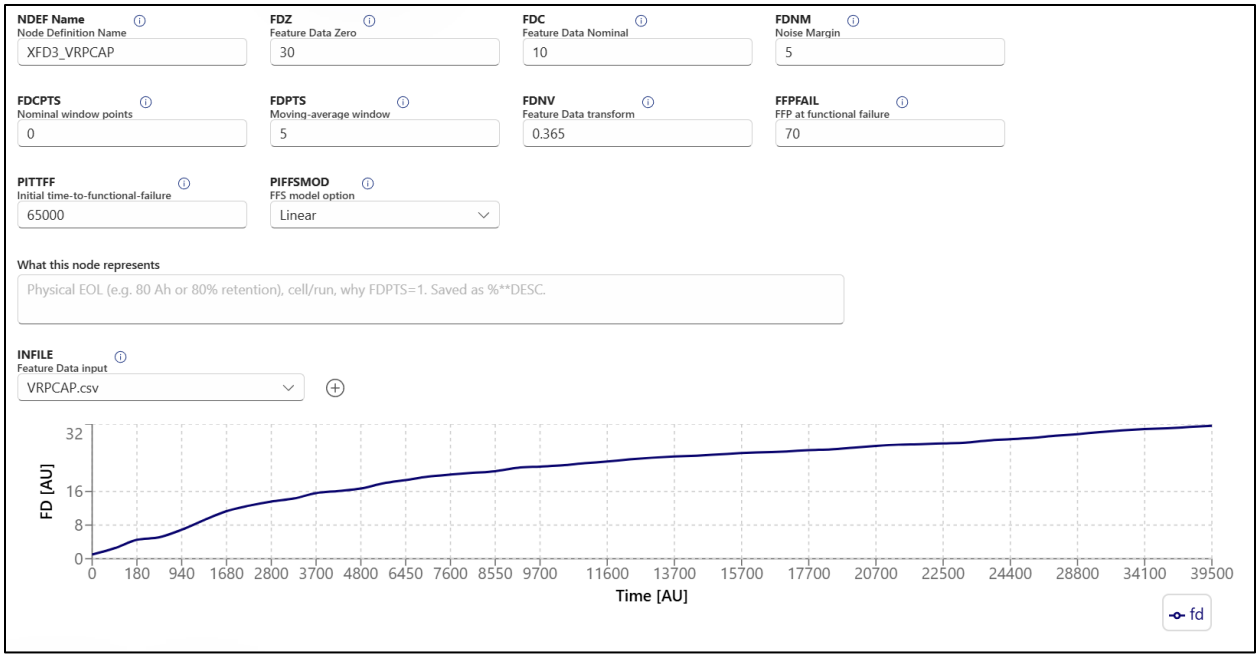


Figure 92 – XFD3_VRPCAP NDEF

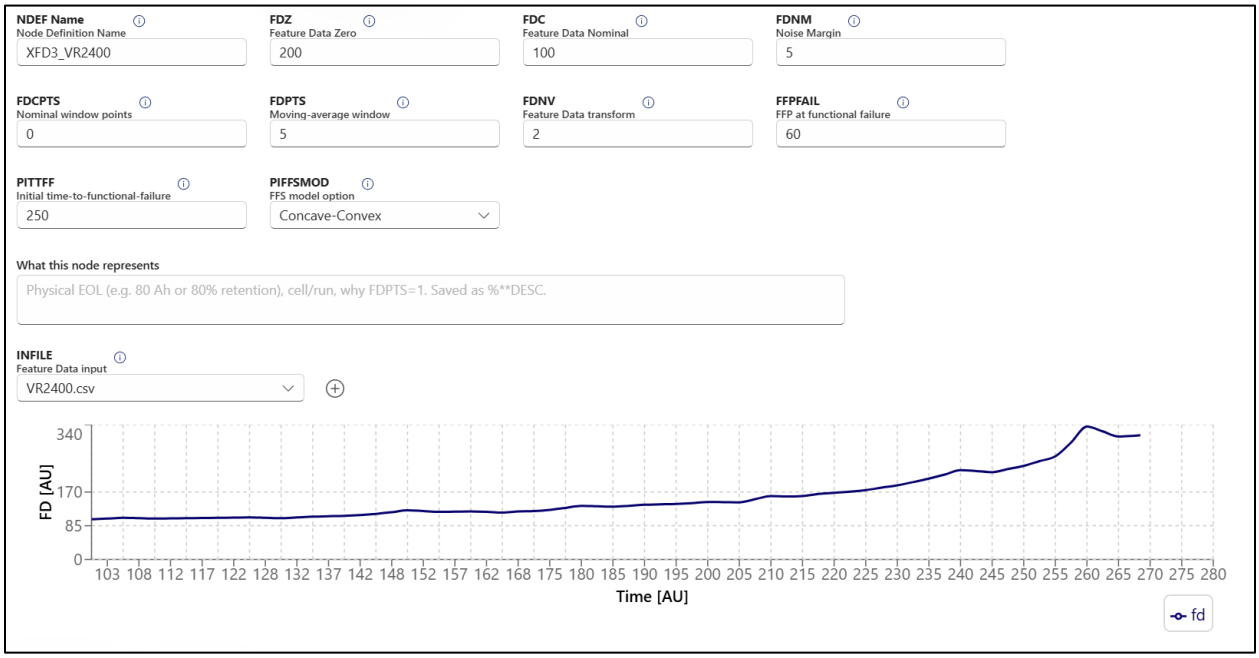


Figure 93 – XFD3_VR2400 NDEF

6.5.2 XFD3: Data Outputs and Prognostic Information

Figure 94 and Figure 95 show the Data Output for the two nodes in XFD3: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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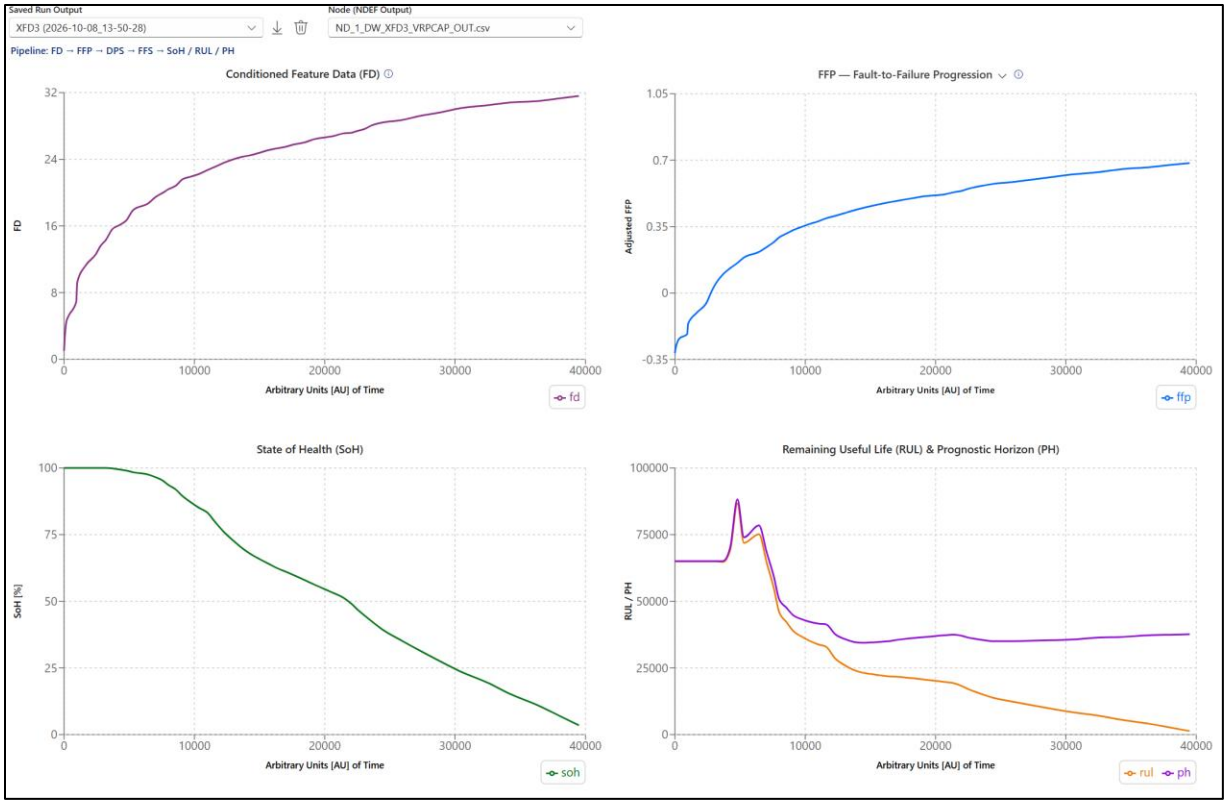


Figure 94 – Plots of the XFD3 Prognostic Information for the First Node

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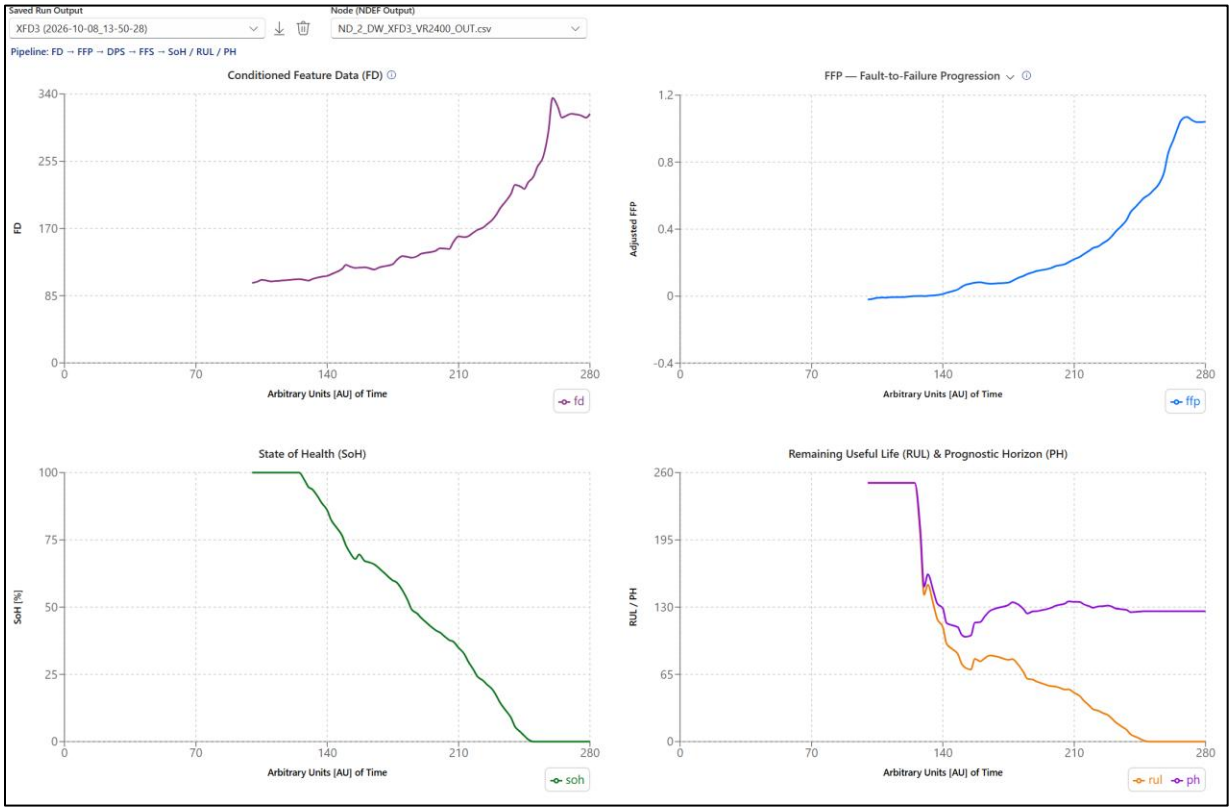


Figure 95 – Plots of the XFD3 Prognostic Information for the Second Node

6.5.3 XFD3: Discussion

This real–world example processes FD correlated to change in the amplitude of ripple voltage in the output of a power supply in response to loss of filtering capacitance. Analyses of this failure mode suggests a square root type of model, but experimentation suggests a higher model value yields more accurate prognostic information. That value offsets the significant step–like appearance in the ripple voltage during the initial onset of degradation, which is the result of stepped–fault injection to cause loss of filtering capacitance.

The two–stage ARULE™ APK is sufficiently robust to produce fairly accurate prognostic estimates without having to specify an exact model and parameter values as evidenced by comparing the output results shown in Figure 94 and Figure 95. Also note the comparison of the pseudo–ideal set of plots shown in Figure 29 on page 40.

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6.6 XFD4: Ripple Voltage, Helicopter Power Supply, Half–PI filter

As mentioned in XFD3, power supply systems have 6 different types of feature data that can be analyzed for degradation. This example focuses on the ripple–voltage type of Feature Data (FD). This type of FD from power supplies has two distinct forms of characteristic signatures when filtering capacitance is reduced: when the output filter of the supply is a full–pi circuit (capacitance, inductance, capacitance) or when the output filter of the supply is a half–pi circuit (inductance, capacitance) – step appearances occur because the filtering capacitance was reduced by electronically disconnecting parallel–connected capacitors one at a time. This ARULE™ example has also been proven through Ridgetop's Sentinel Power product line.

6.6.1 XFD4: Definitions

As shown in Figure 96, the SDEF file for this fourth example points to a node definition number of 1, and an NDEF file named **XFD4_BPS03**. Figure 97 shows the NDEF parameters, INFILE and keywords used for XFD4.

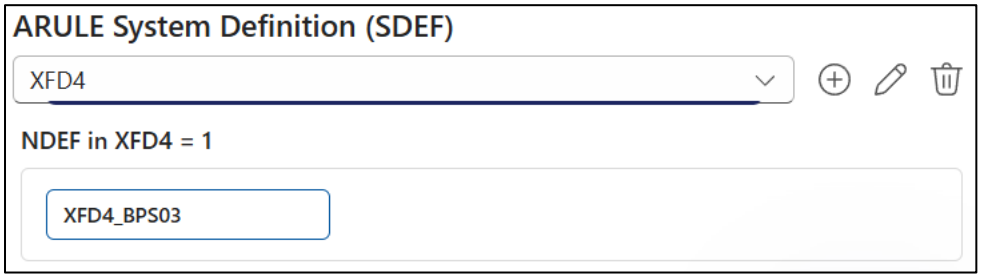


Figure 96 – XFD4 SDEF

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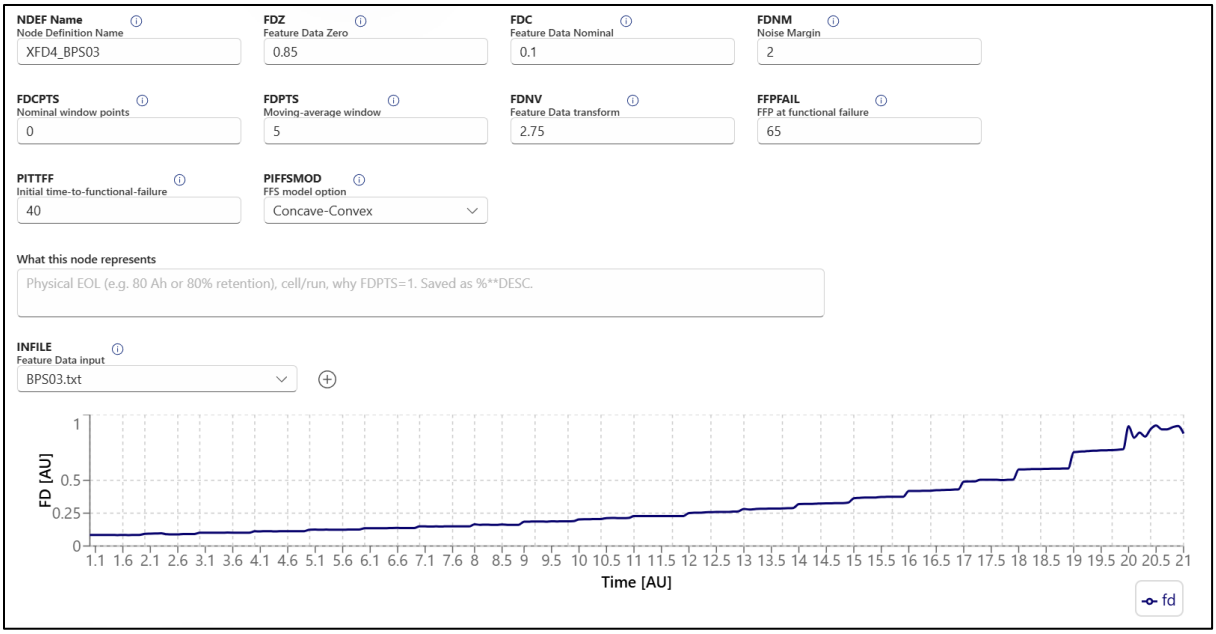


Figure 97 – XFD4 Node Definition (NDEF) User Form

6.6.2 XFD4: Data Outputs and Prognostic Information

Figure 98 shows the Data Output for the only node in XFD4: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

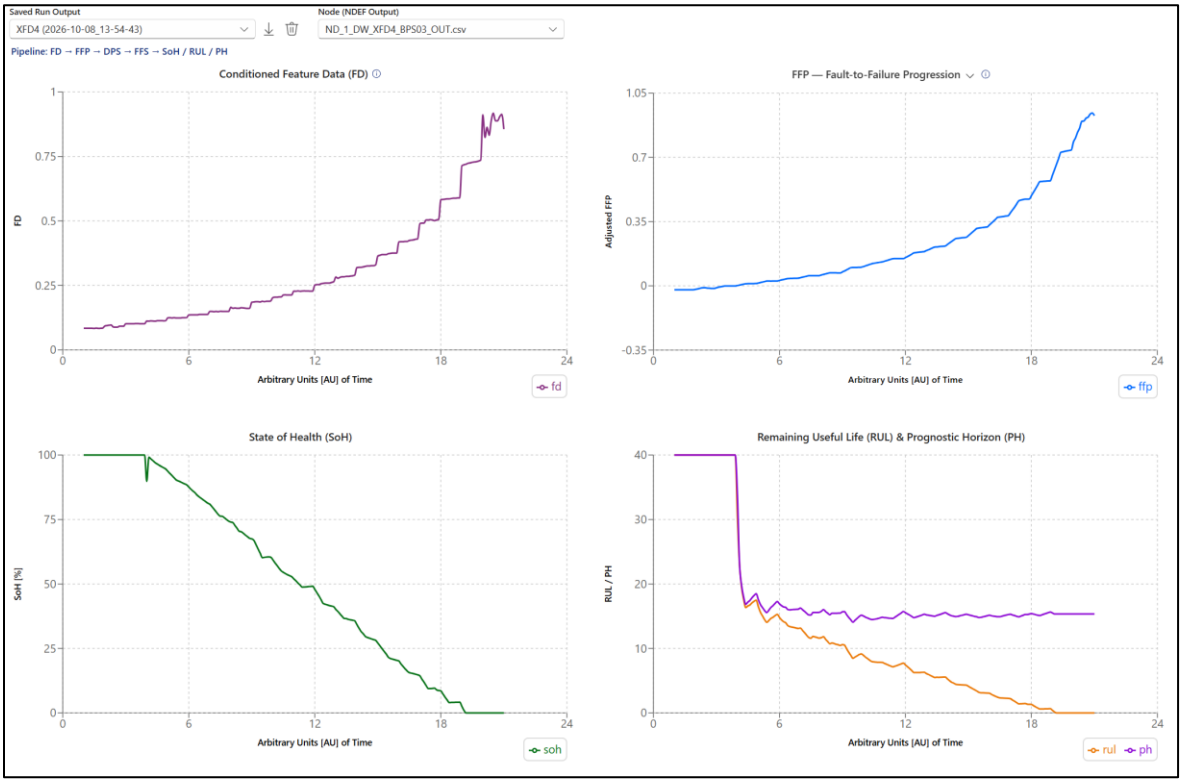


Figure 98 – Plots of the XFD4 Prognostic Information

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6.6.3 XFD4: Discussion

The results of processing the input data are evaluated as excellent. Note the difference in the characteristic curve of the input data in this example (Figure 97) compared to the previous example (Figure 92): the curves are different (convex versus concave) because of the difference in the configuration of the output filter (full PI versus half PI) and this example exhibits a pronounced "step" appearance because the loss of filtering capacitance was caused by abruptly disconnecting and connecting capacitors in the filter (a fault–injection technique).

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6.7 XFD5: Capacitor Leakage, Power Supply Filter

This ARULE™ example is another proven use case where Ridgetop has used the Sentinel Power product line to monitor and detect a common failure mode in power supply systems. Filtering capacitors, especially electrolytic type of capacitors, in power supplies are subject to a failure mode known as capacitor leakage. This failure mode is often the result of degradation of the dielectric material, and Ridgetop has developed the necessary hardware, firmware, and software solutions to prognostic enable several different types of power supply systems in the military, aerospace, and commercial market segments.

6.7.1 XFD5: Definitions

As shown in Figure 99, the SDEF file for this example points to 2 nodes with NDEF files namely **NODE_CLEAK** and **NODE_CLEAK2**. They are varied by changing the FDC parameter from 10 in **NODE_CLEAK** to 18 in **NODE_CLEAK2.** Figure 100 and Figure 101 show the NDEF parameters, INFILES and keywords used for XFD5. The previous examples had file names associated with specific SDEF files: such association is unnecessary and might be detrimental were an NDEF be used by more than one SDEF.

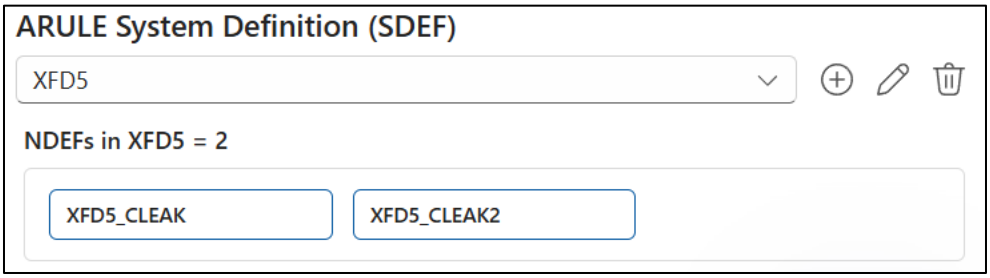


Figure 99 – XFD5 System Definition (SDEF)

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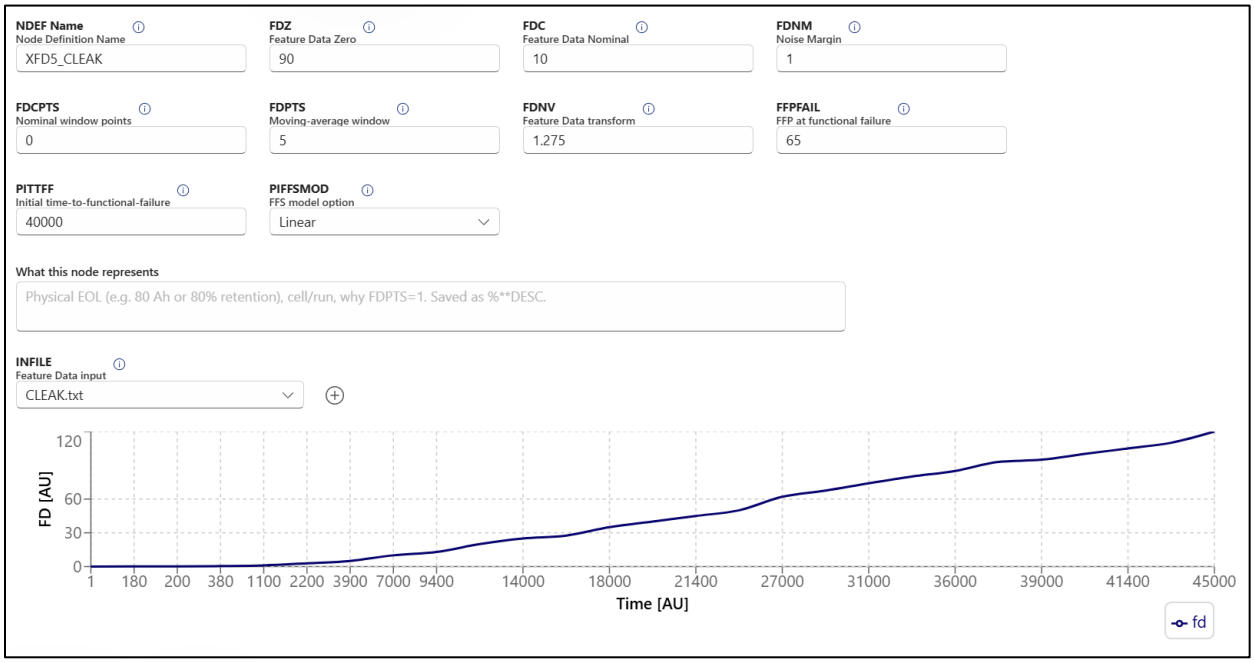


Figure 100 – XFD5_CLEAK NDEF

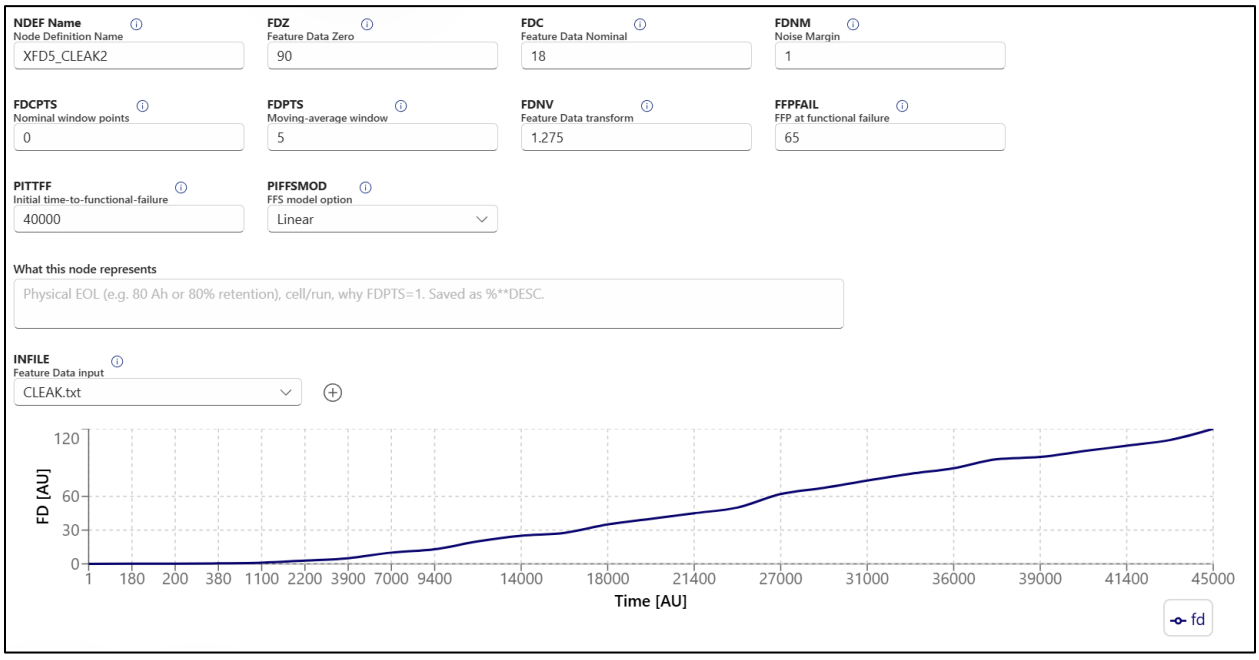


Figure 101 – XFD5_CLEAK2 NDEF

6.7.2 XFD5: Data Outputs and Prognostic Information

Figure 102 and Figure 103 show the Data Output for the two nodes in XFD5: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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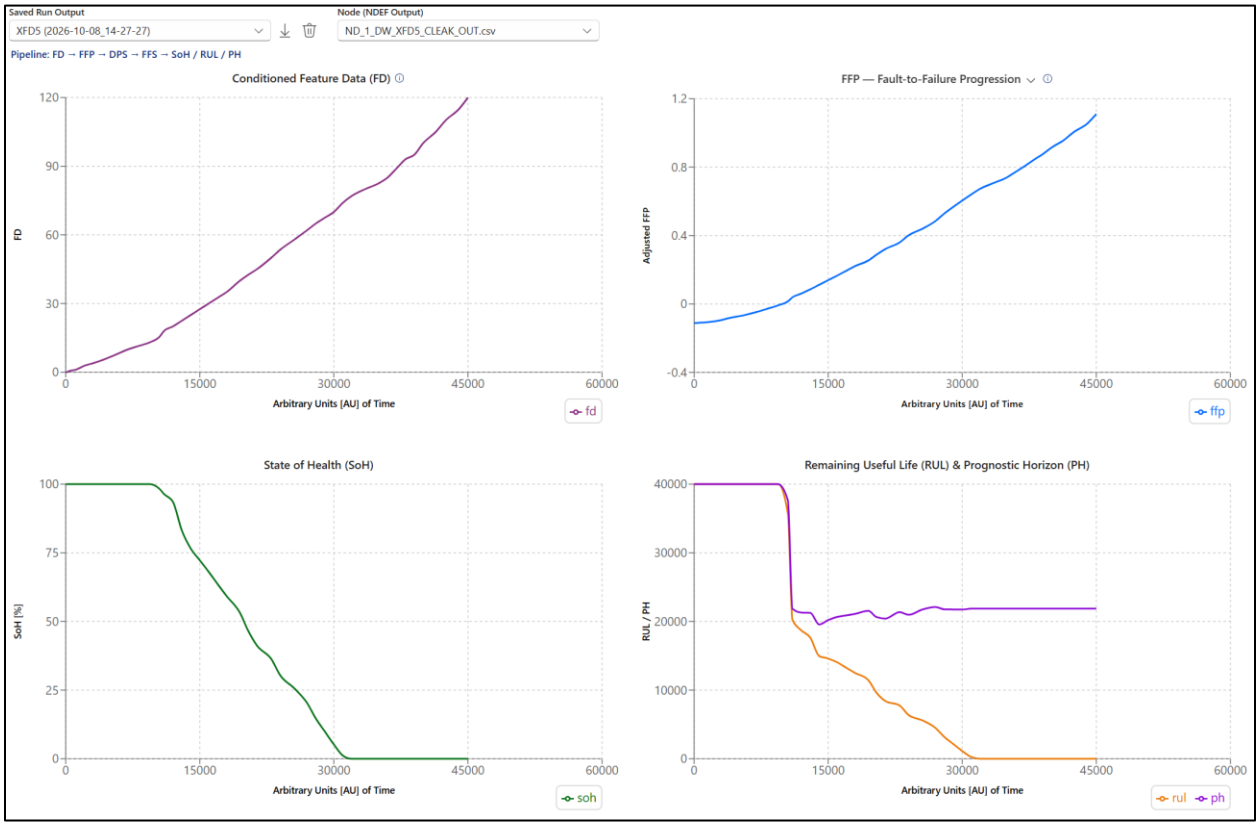


Figure 102 – Plots of the XFD5 Prognostic Information for the First Node

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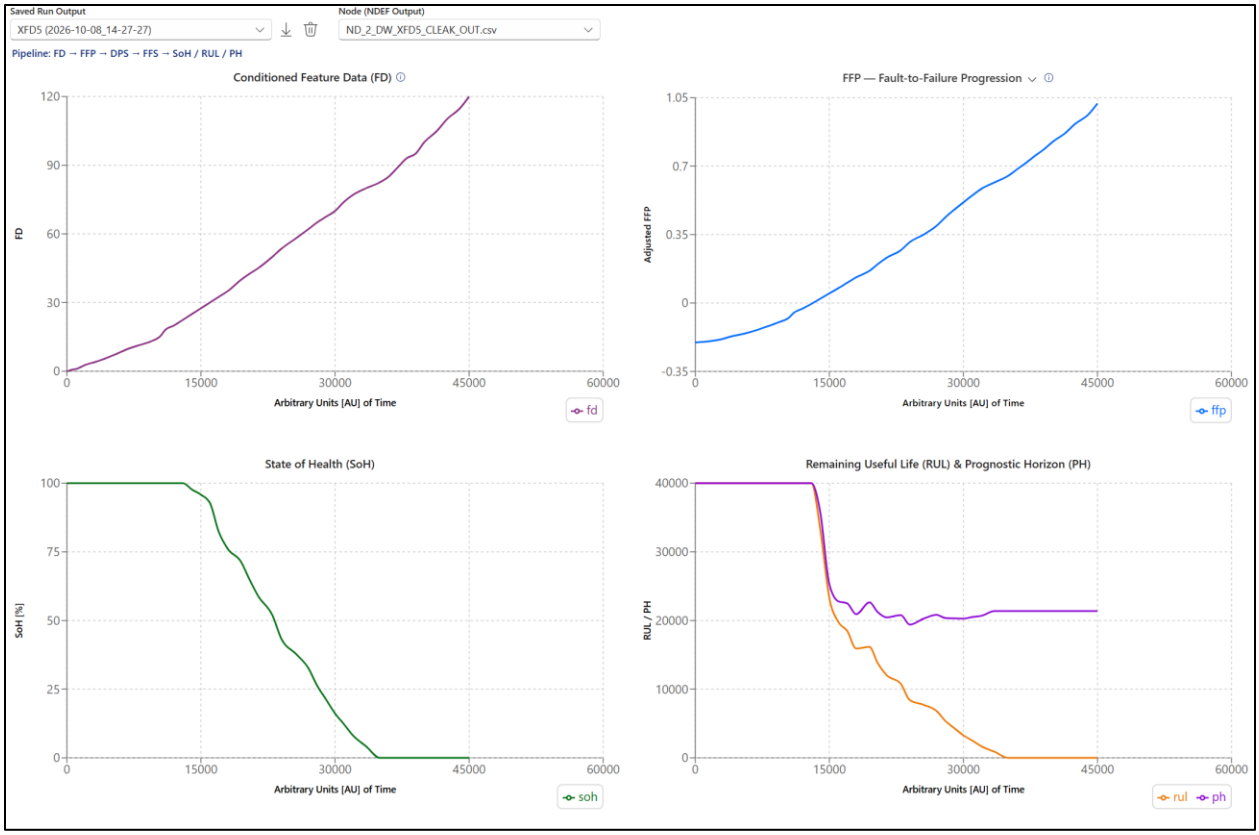


Figure 103 – Plots of the XFD5 Prognostic Information for the Second Node

6.7.3 XFD5: Discussion

Referring back to chart in Figure 100, the signature of the input data is fairly linear, so we do not need a model to linearize the input data. The results shown in Figure 102 and Figure 103 are very good, which confirms that a transformation model was not needed. FMEA (Failure Mode Effects Analysis) indicates that leakage current is typically below 1 mA to 5mA, and can be as high as 100 mA before evidence of physical failure.

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6.8 XFD6: Battery Internal Resistance

The next two ARULE™ examples, XFD6 and XFD7, are related to Ridgetop's prognostic solutions for batteries. Batteries are subject to a number of different failure modes including one called Internal Resistance/Internal Contact Resistance (IR/ICR). As the IR/ICR increases in batteries the output voltage of the battery is reduced as it supplies current to a load. This example demonstrates how the IR/ICR type of feature data (FD) can be analyzed with ARULE™, but this type of prognostic solution could also be implemented as an embedded process in a smart Battery Management System (BMS) design. This engineering service is offered through Ridgetop's CellSage product line, which also encompasses a suite of battery simulation and analysis software tools.

6.8.1 XFD6: Definitions

As shown in Figure 104, the SDEF file for this example points to one node and an NDEF file named XFD6_BATTIR. Figure 105 shows the NDEF parameters, INFILE and keywords used for XFD6.

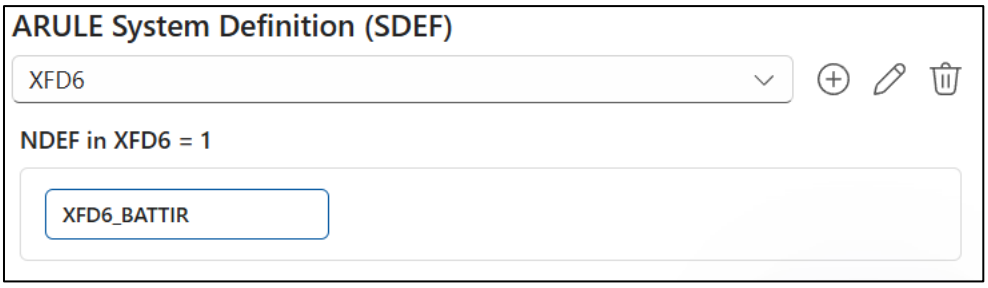


Figure 104 – XFD6 System Definition (SDEF) User Form

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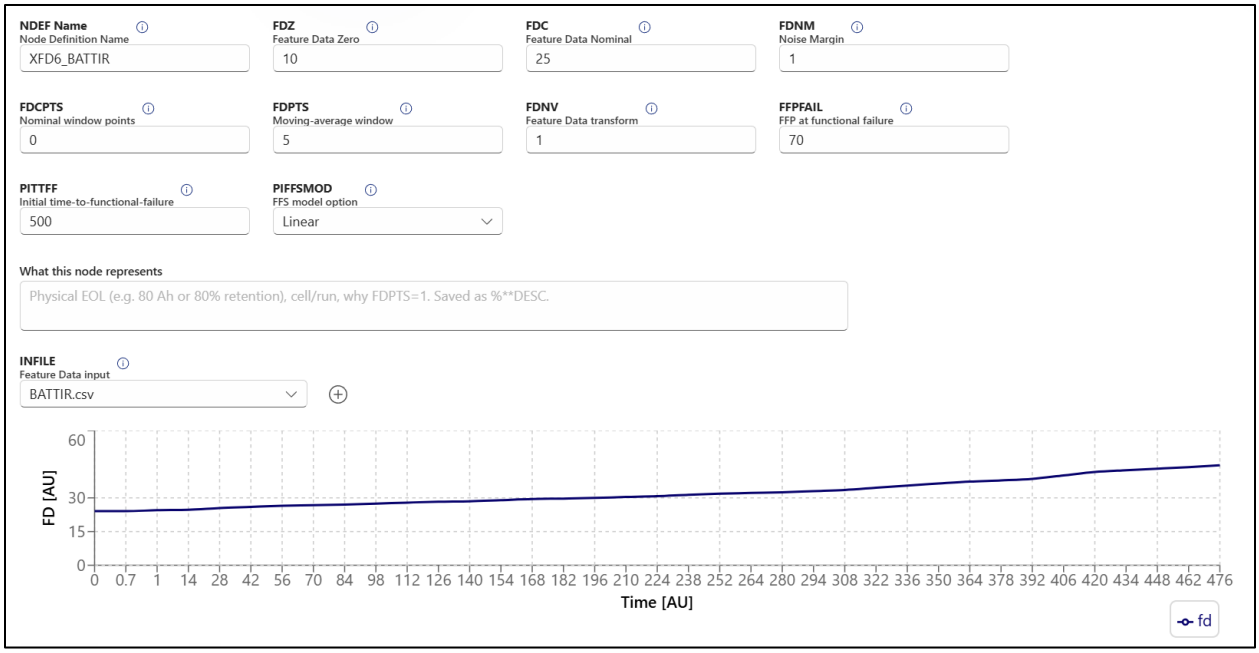


Figure 105 – XFD6_BATTIR NDEF

6.8.2 XFD6: Data Outputs and Prognostic Information

Figure 106 shows the Data Output for the only node in XFD6: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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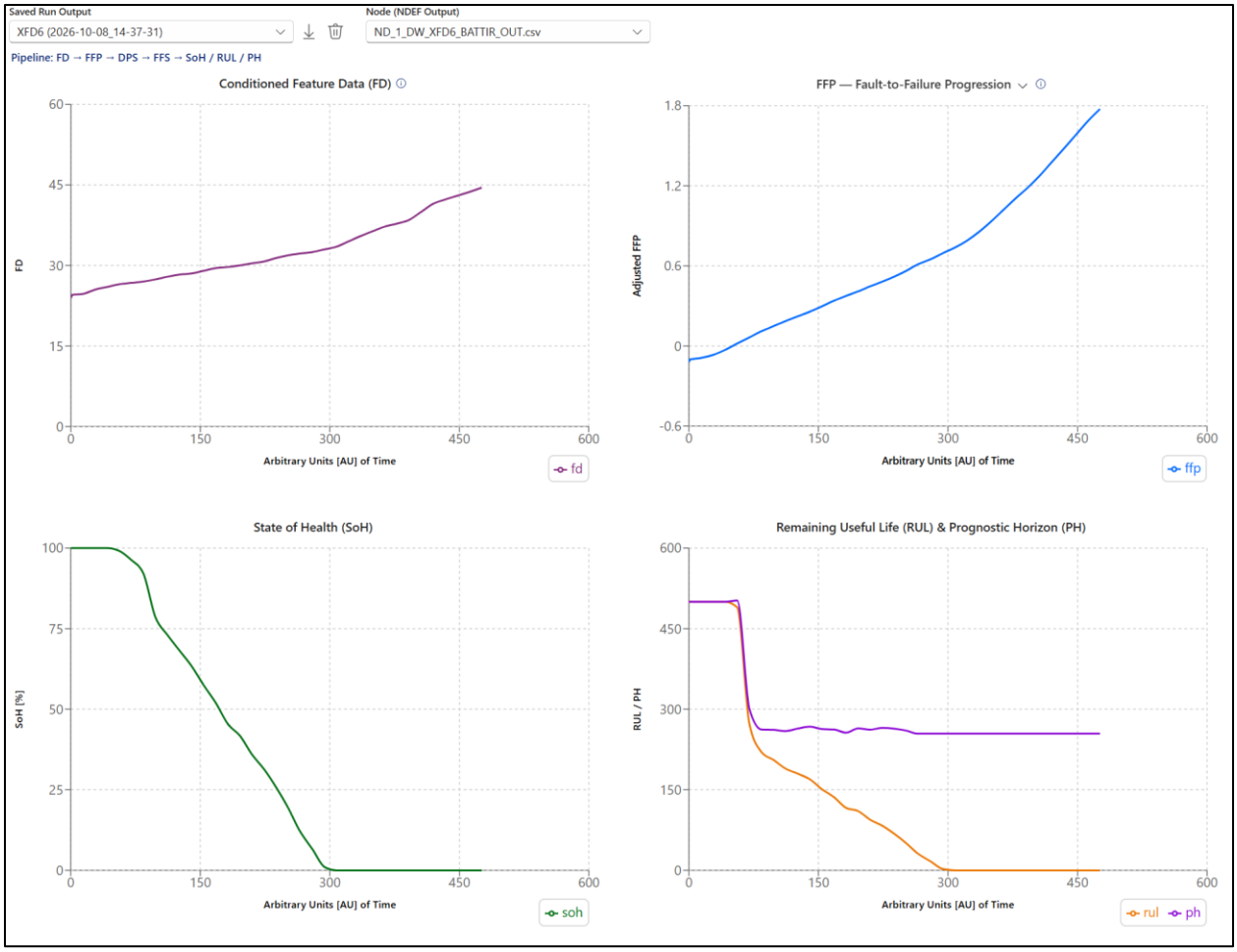


Figure 106 – Plots of the XFD6 Prognostic Information

6.8.3 XFD6: Discussion

Referring back to chart in Figure 105, the signature of the input data is fairly linear. The input signature exhibits an inflection point around day 80, which is reflected in the SoH estimates plotted in Figure 106 – despite that inflection, the estimates of RUL and PH produced by the ARULE™ APK are very, very accurate.

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6.9 XFD7: Battery Charging Capacity

As lithium–ion types of batteries degrade, in addition to increasing internal contact resistance, they also exhibit loss of charging capacity (see Figure 107). As covered in Ridgetop's CellSage product line, the loss of charging capacity is impacted by several different environmental and operational parameters. Such parameters include cell chemistry, temperature, state–of–charge, charging protocols, and application dependent duty cycles. Although each of these parameters have a known impact on battery ageing and degradation, the loss of charge capacity will typically follow an aging path that can be modeled by the single set of ARULE™ degradation signatures.

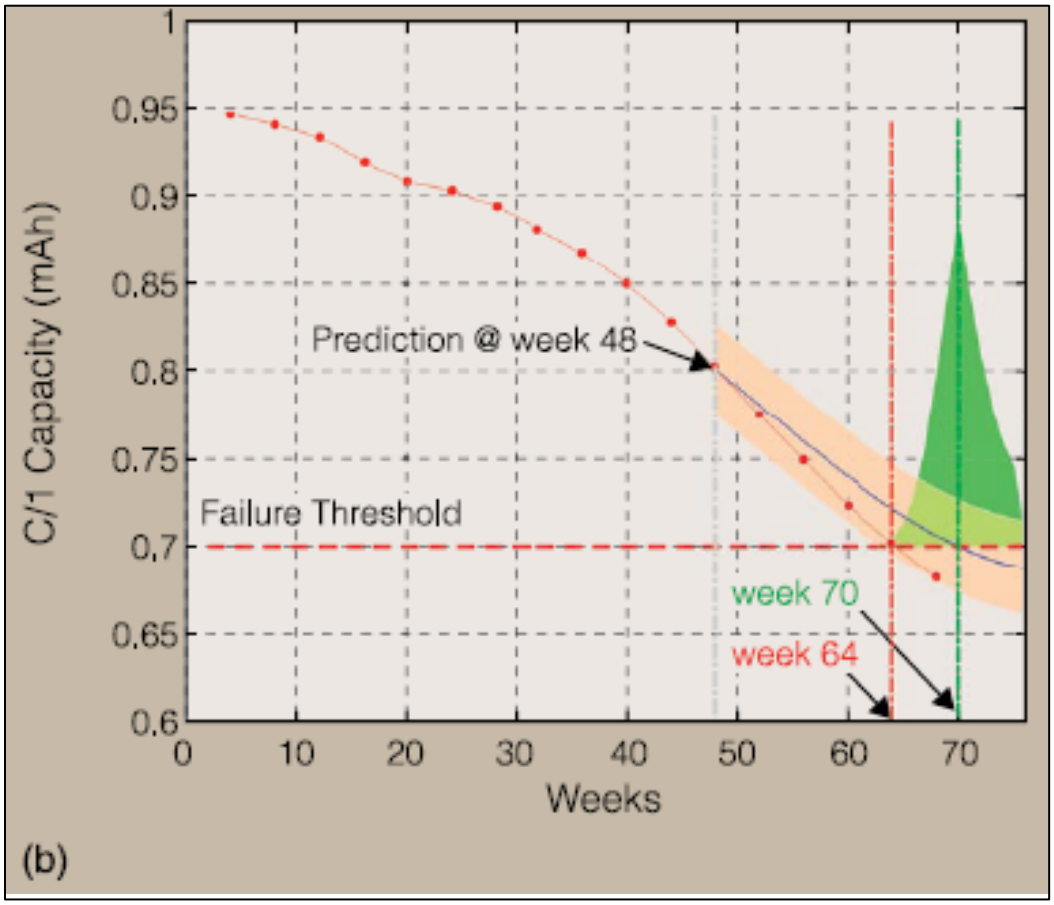


Figure 107 – Published data: battery capacity [Goebel, et.al, 2008, Prognostics in Battery Health Management, IEEE Instrumentation and Measurements Magazine]

This example will show how Ridgetop was able to successfully process capacity loss Feature Data (FD) from a design of experiments that were conducted through a NASA SBIR program as well as simulated battery degradation data from CellSage. There are three nodes in this example: the first node uses input data derived from experimental data shown in Figure 107; the second and third nodes use worst-case and best-case sets of

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simulated data output from CellSage as shown in Figure 108. The FD values are in mAh values in Figure 107 versus percent values in Figure 108.

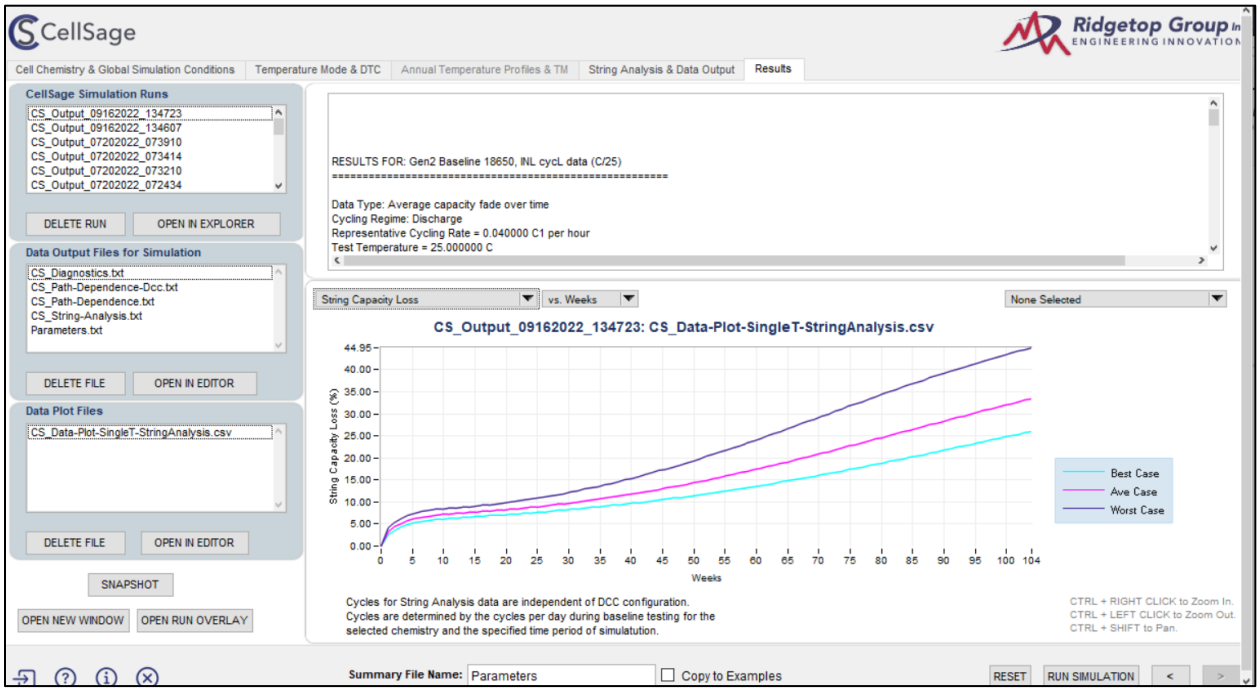


Figure 108 - CellSage screen shot showing loss of charge: annual temperature profile for Tucson, AZ for worst-case, average-case, and best-case simulation conditions

6.9.1 XFD7: Definitions

As shown in Figure 109, the SDEF file for this example points to 3 nodes with NDEF files named **XFD7_BATTCHG1**, **XFD7_BATTCHG2**, and **XFD7_BATTCHG3**. Figure 110, Figure 111 and Figure 112 show the NDEF parameters, input data, INFILES and keywords used for XFD7. The NDEF parameters are the same for all three cases except as follows:

1. The FDC value for Node 1 is specified as 0.07 (mAh) versus 10.0 (percent) for Node 2 and 3.
2. The FDZ value for Node 1 is specified as 0.20 (mAh) versus 23.0 (percent) for Node 2 and 3.

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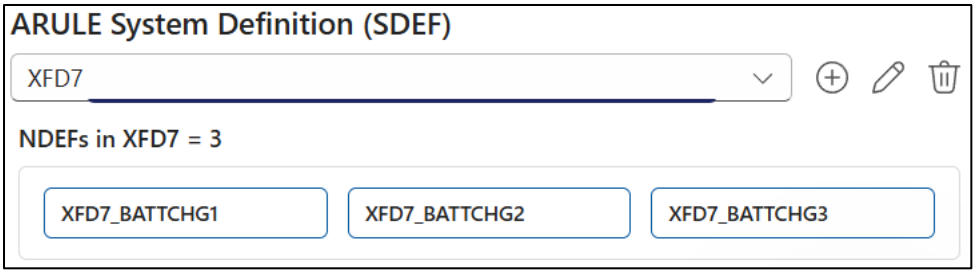


Figure 109 – XFD7 System Definition (SDEF)

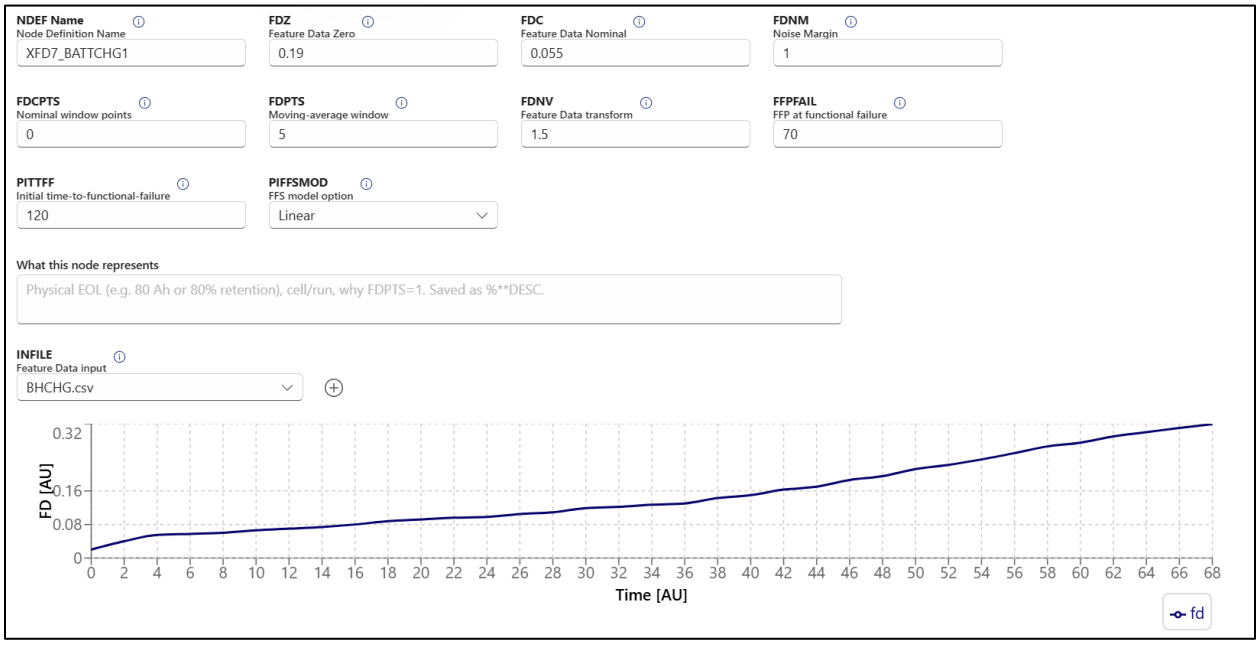


Figure 110 – XFD7_BATTCHG1 NDEF

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Figure 111 – XFD7_BATTCHG2 NDEF



Figure 112 – XFD7_BATTCHG3 NDEF

6.9.2 XFD7: Data Outputs and Prognostic Information

Figure 113, Figure 114, and Figure 115 show the Data Output for the three nodes in XFD7: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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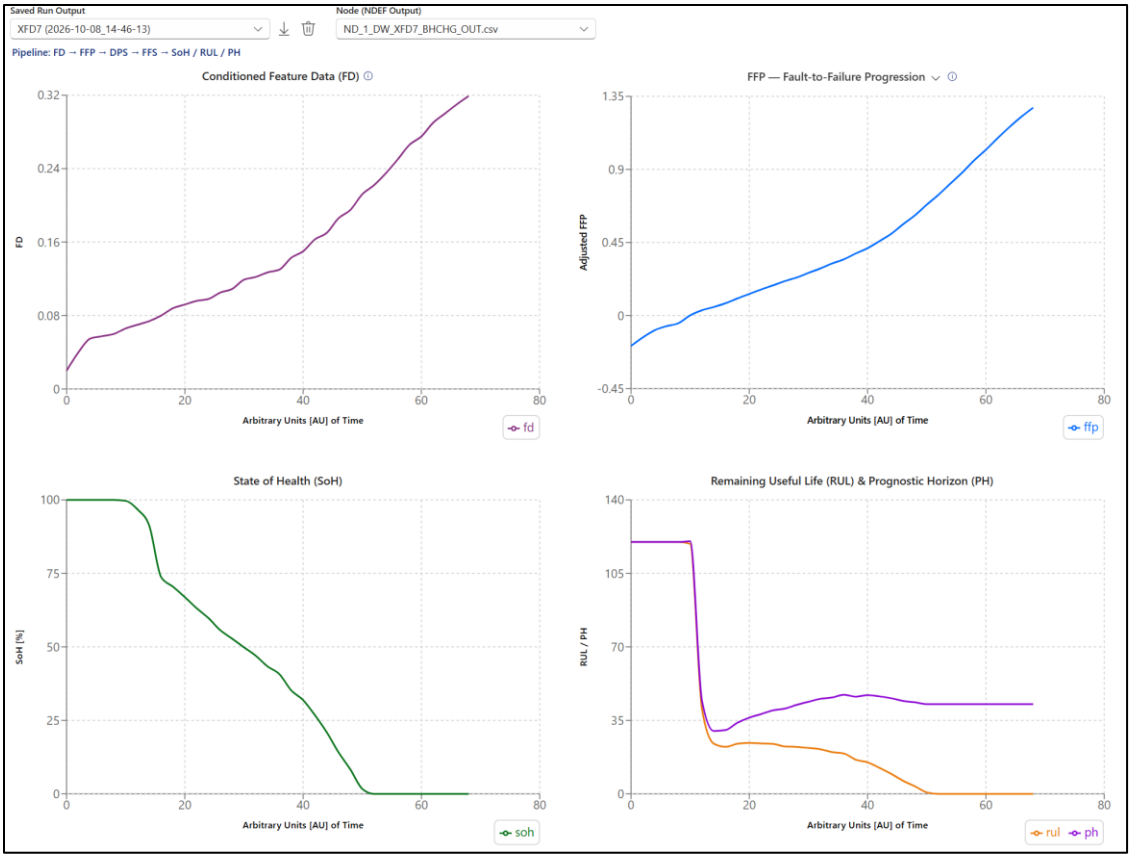


Figure 113 – Plots of the XFD7 Prognostic Information for the First Node.

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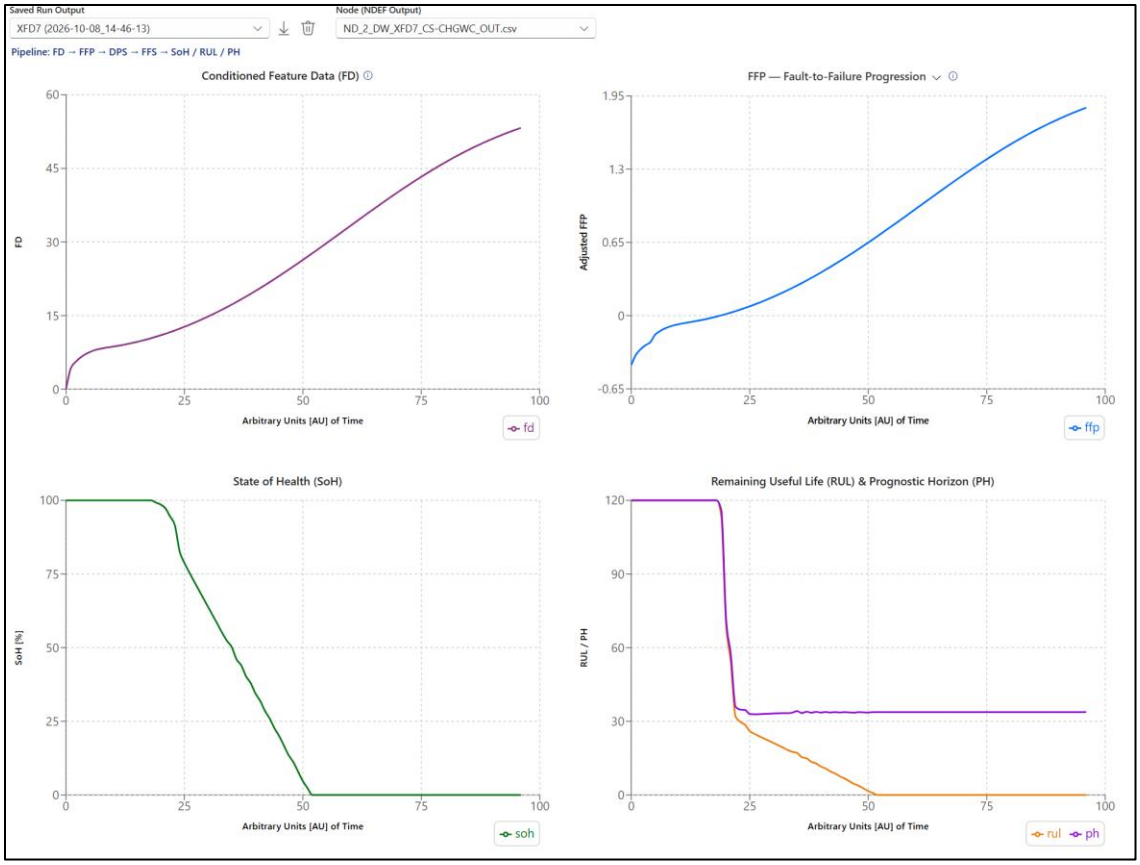


Figure 114 – Plots of the XFD7 Prognostic Information for the Second Node.

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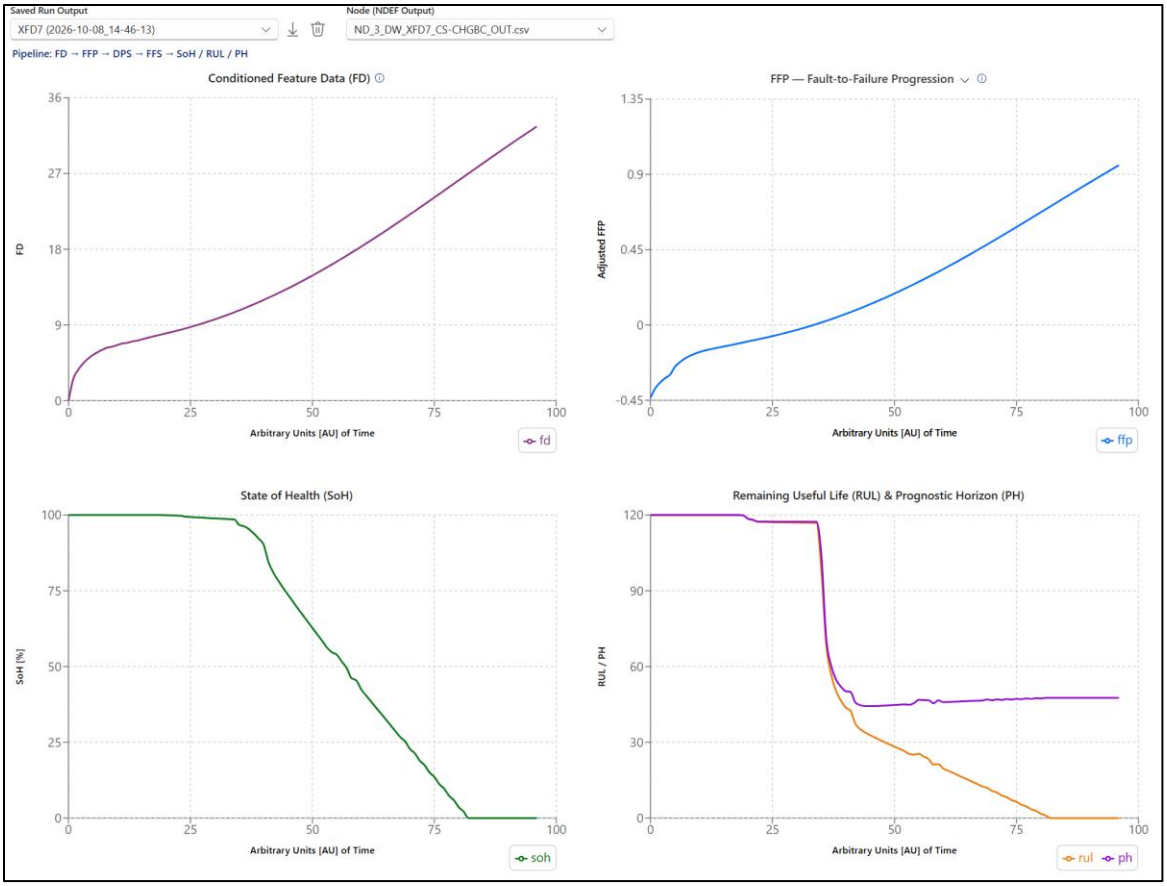


Figure 115 – Plots of the XFD7 Prognostic Information for the Third Node.

6.9.3 XFD7: Discussion

Referring back to charts in the following Figure 110, Figure 111, and Figure 112 the signature of the input data after the first inflection point resembles a square (power of n) function. Inspection of experimental data indicates the signature is somewhat flattened – indicating a power value closer to 1.5 rather than 2.0. Experimentation with parameters reveals that a wide range the square–root model values produce reasonable values of prognostic estimates: a final value of 1.25 is used to produce the very accurate estimates shown in Figure 113, Figure 114, and Figure 115. These types of prognostic solutions could also be implemented as an embedded process in a smart Battery Management System (BMS) design. This engineering service is offered through Ridgetop's CellSage product line, which also encompasses a suite of battery simulation and analysis software tools.

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6.10 XFD8: High Data Rate Cables

High Data Rate (HDR) cables, such as those with antennas on submarines, are attached to an antenna mast assembly at one end while at the other end they are attached to connectors. The connectors are cabled to transceivers inside the submarine (Figure 116 and Figure 117).

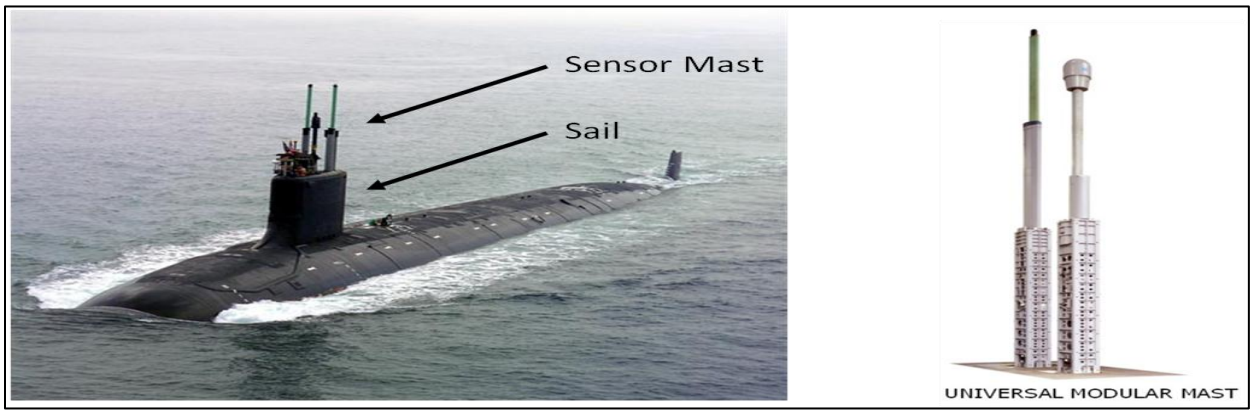


Figure 116 – Virginia class submarine with sensor mast attached to the sail

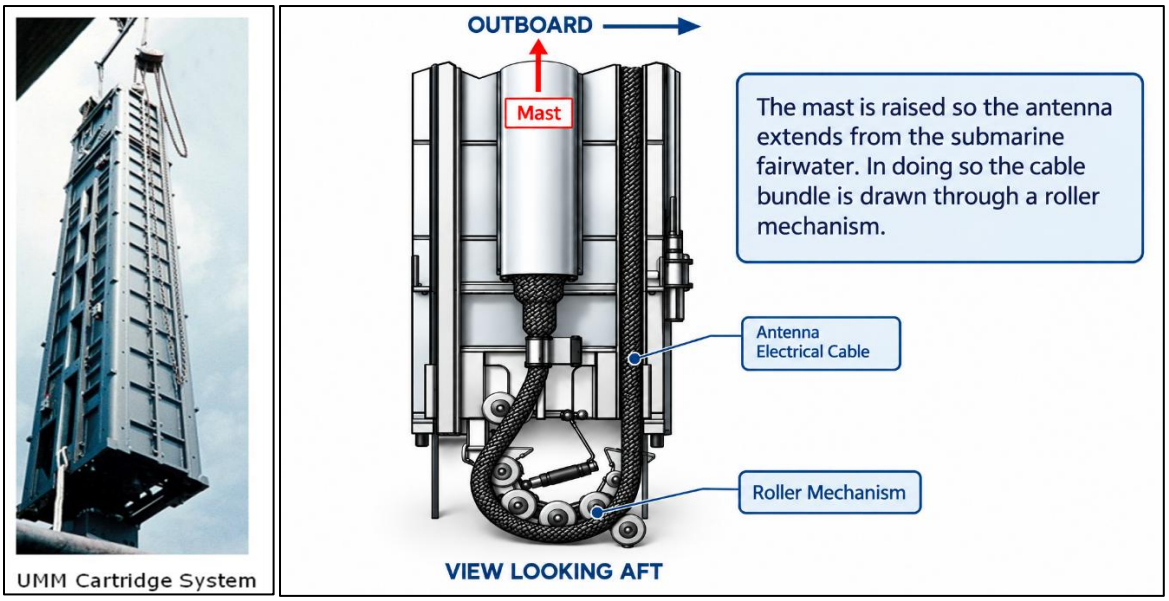


Figure 117 – Universal modular mast (UMM) cartridge and cabling system

Problem Description

As the antennas are raised and lowered, the HDR cables are reeled out and reeled in. During this process they can be subjected to crushing and crimping type of damage

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leading to loss of transmission and receiving power (Figure 118). Physical and traditional means of inspection HDR cables are problematic and impractical because the mast assembly is mounted inside the sail of a submarine.

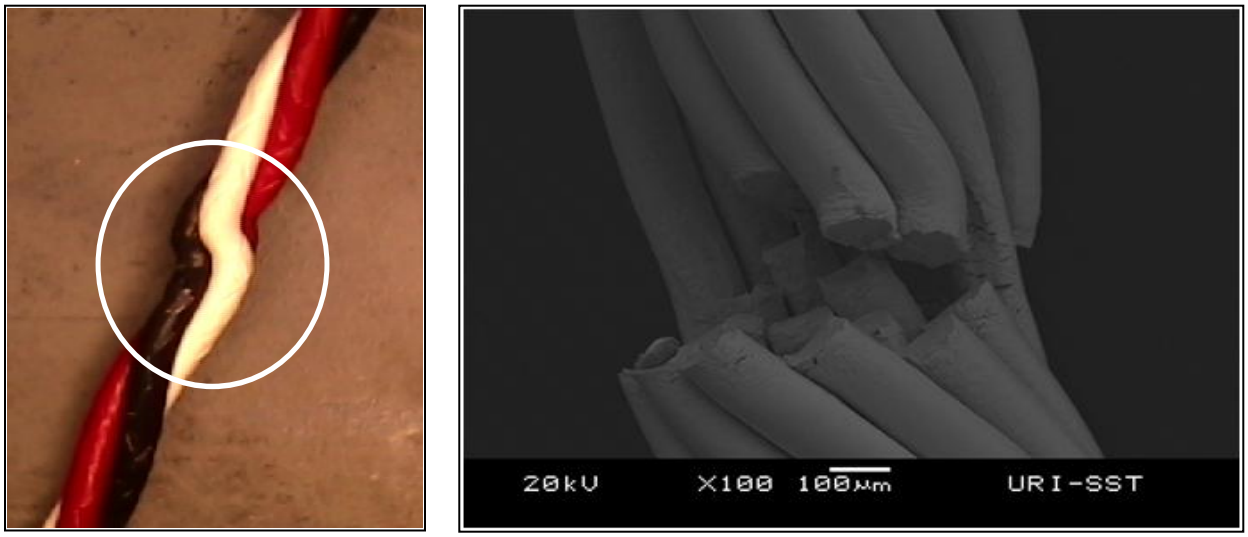


Figure 118 – Twisted triplet 24–AWG segment showing a kink; microscopic view of conductor failure

Solution Summary

Because high–frequency signals are transmitted and received by HDR cables, changes in distributed impedance can all also occur at different damage locations due to crimps, crushed damage dielectric material, distances, shielding, and wire conductors. Those impedance changes in transmission lines cause signal energy to be reflected and/or absorbed. A solution method to detect crushes and crimps is to do the following:

- Use a one–port network analysis method: Figure 119.
- Apply an RF signal and, capture its reflection, and process the signal with a smart processing unit (SPU) that produces CBD: Figure 120 and Figure 121.
- Send CBD output of the SPU to an ARULE™–based front–end application program as done in this example: Figure 122.

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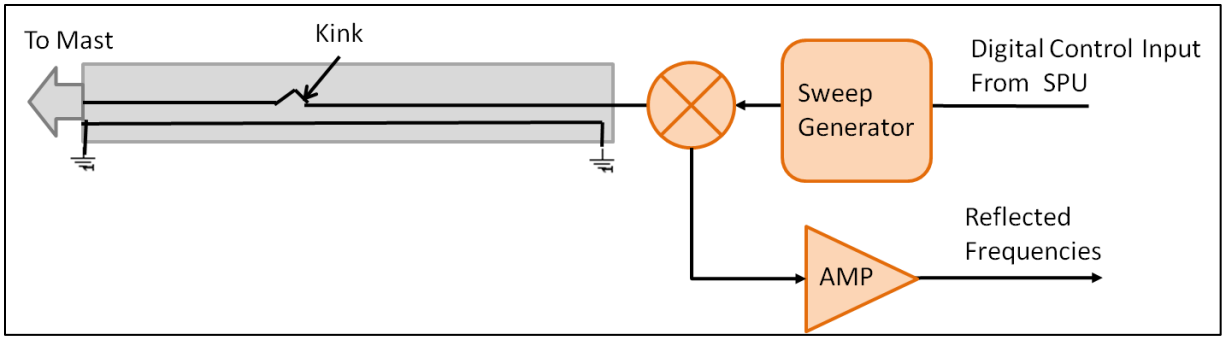


Figure 119 – SMRT Sweeping Probe block diagram

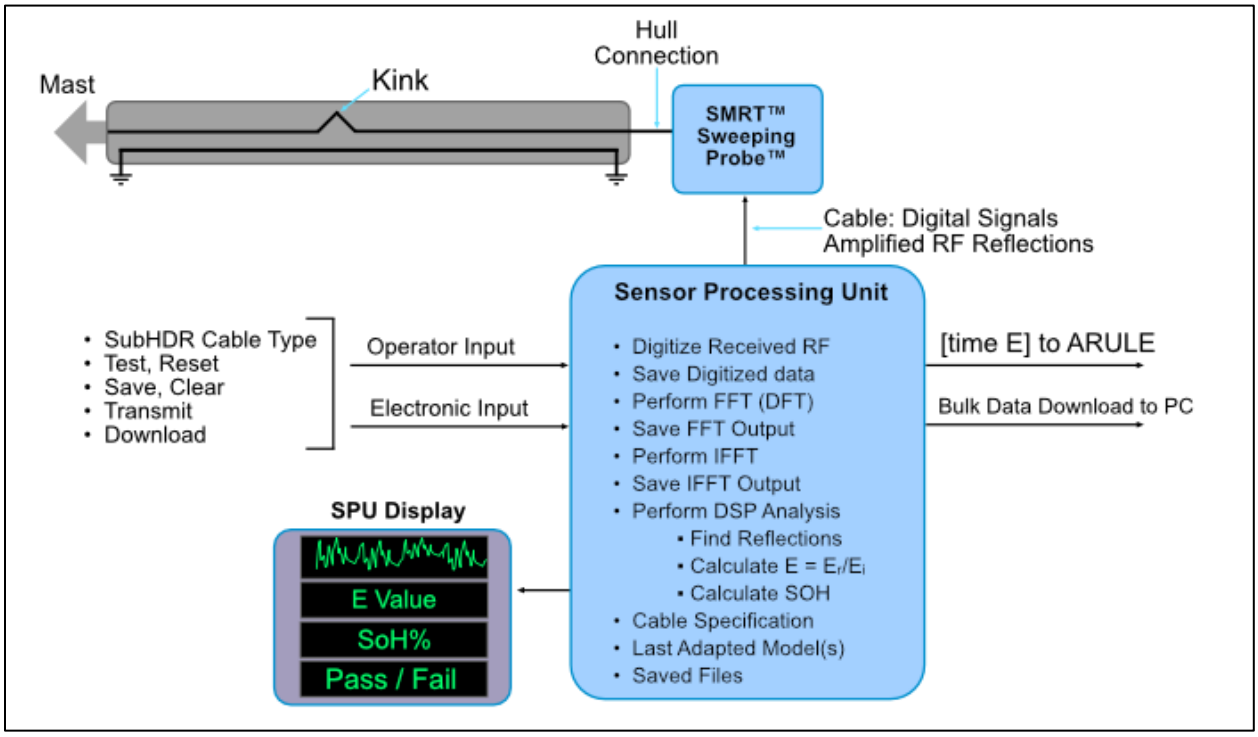


Figure 120 – SMRT Sweeping Probe and SPU.

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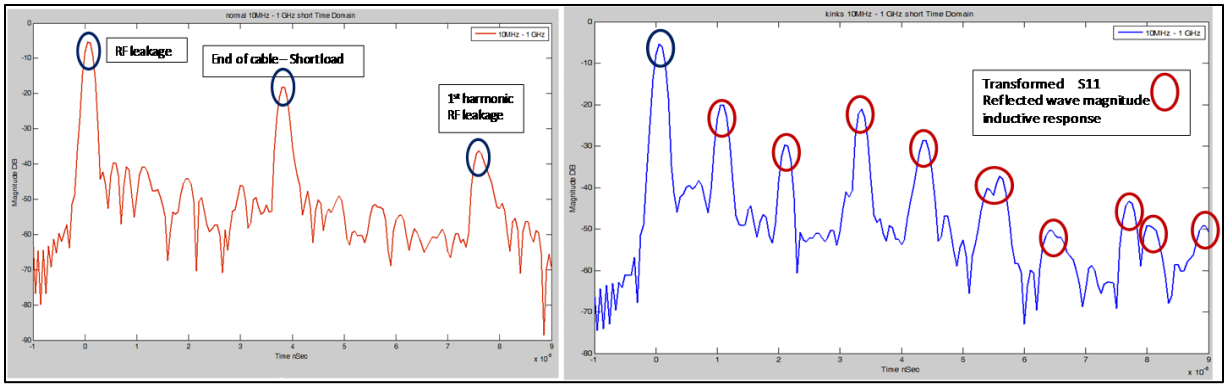


Figure 121 – CBD: Plots of the IFF transform – undamaged cable (left) and damaged cable (right)

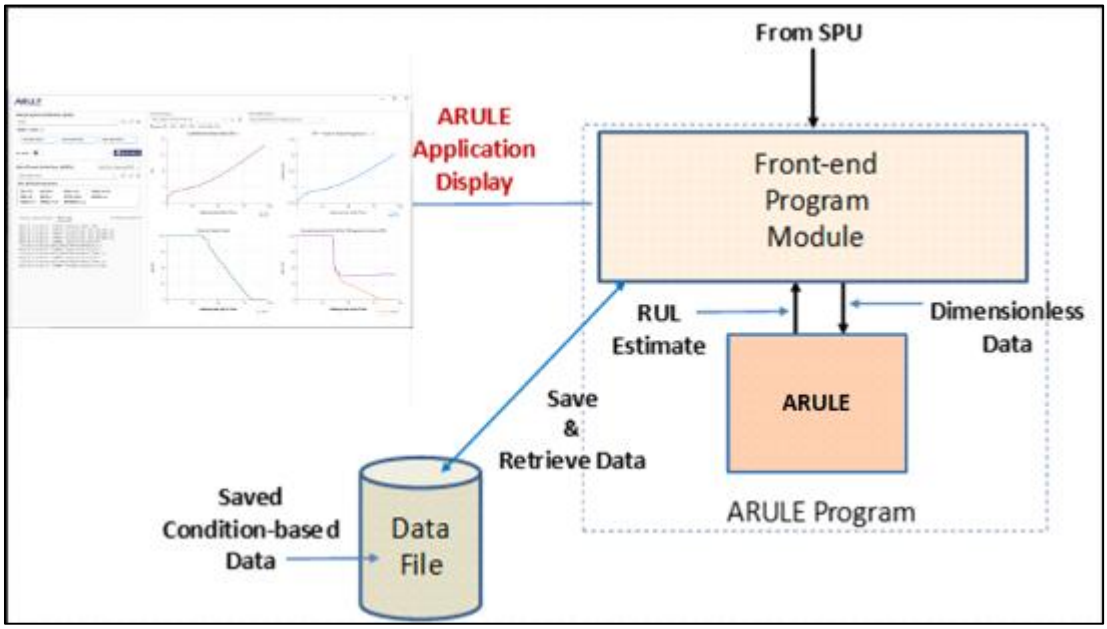
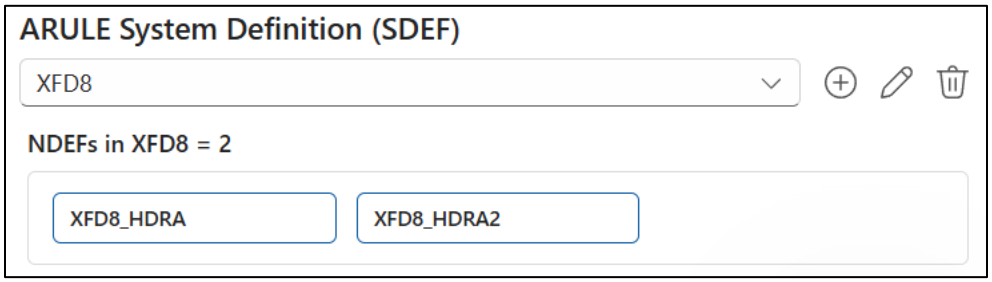


Figure 122 – ARULE™ Application

6.10.1 XFD8: Definitions

As shown in Figure 123, the SDEF file for this example points to 2 nodes with NDEF files named **XFD8_HDRA** and **XFD8_HDRA2**. Figure 124 and Figure 125 show the NDEF parameters, INFILEs and keywords used for XFD8.



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Figure 123 – XFD8 System Definition (SDEF) User Form

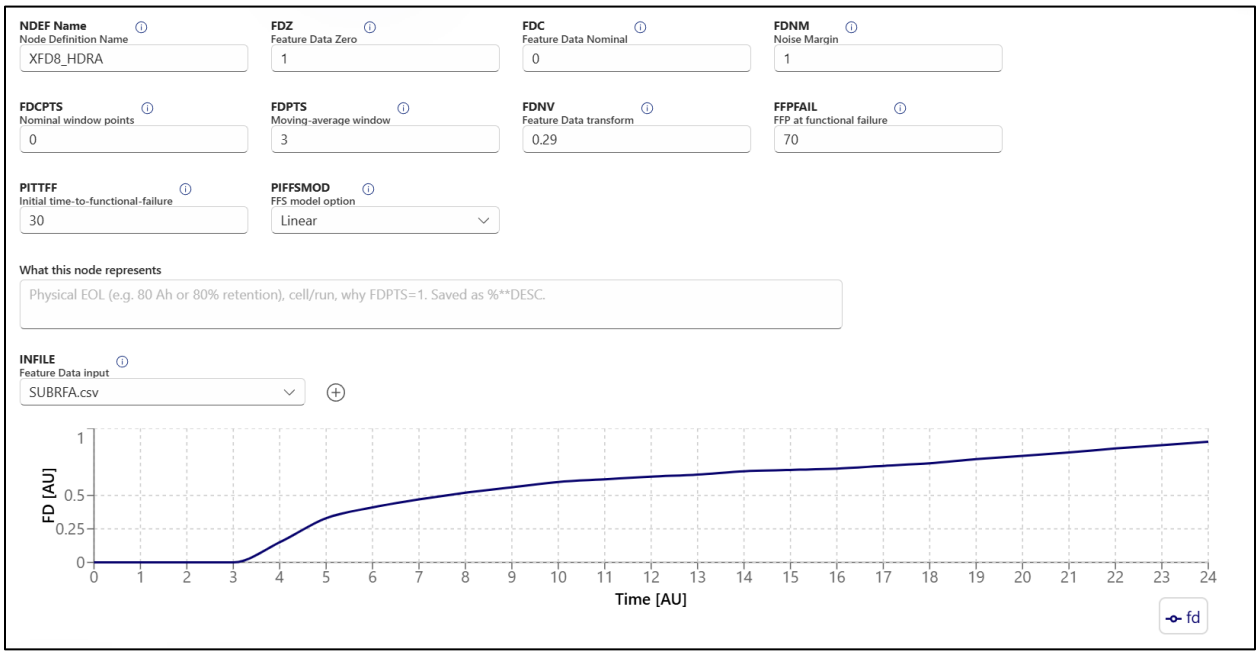


Figure 124 – XFD8_HDRA NDEF

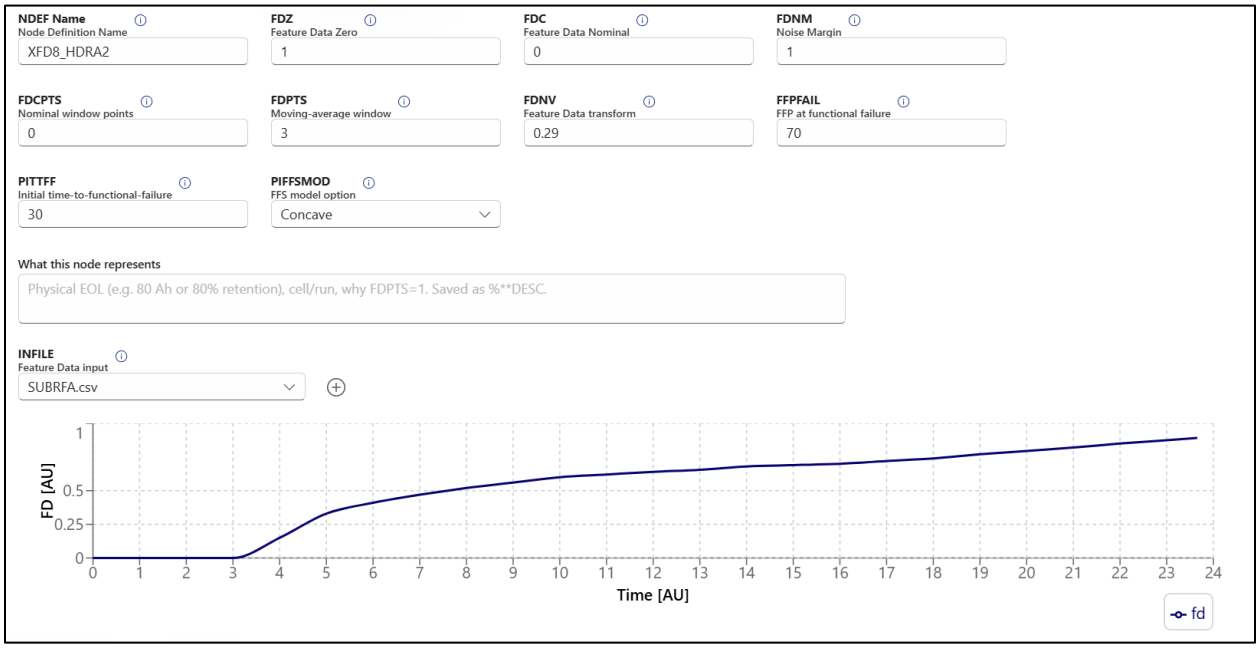


Figure 125 – XFD8_HDRA2 NDEF

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6.10.2 XFD8: Data Outputs and Prognostic Information

Figure 126 and Figure 127 show the Data Output for the two nodes in XFD8: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

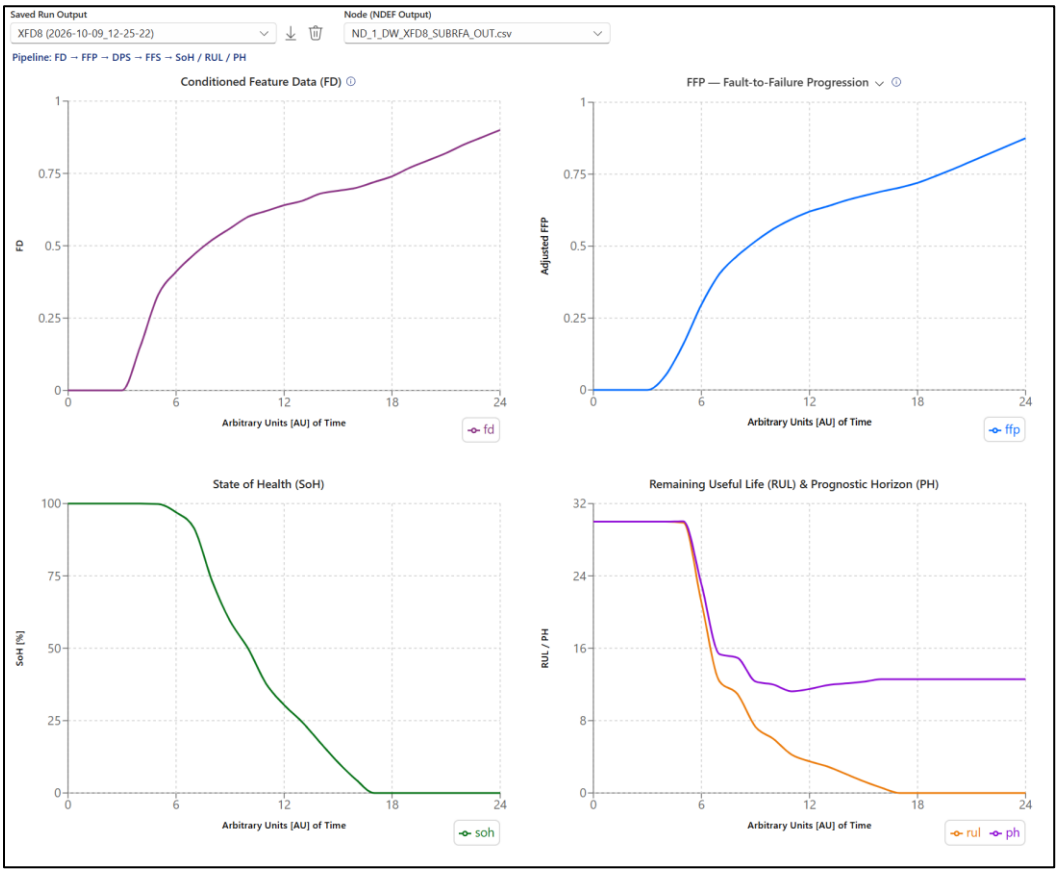


Figure 126 – Plots of the XFD8 Prognostic Information for the First Node

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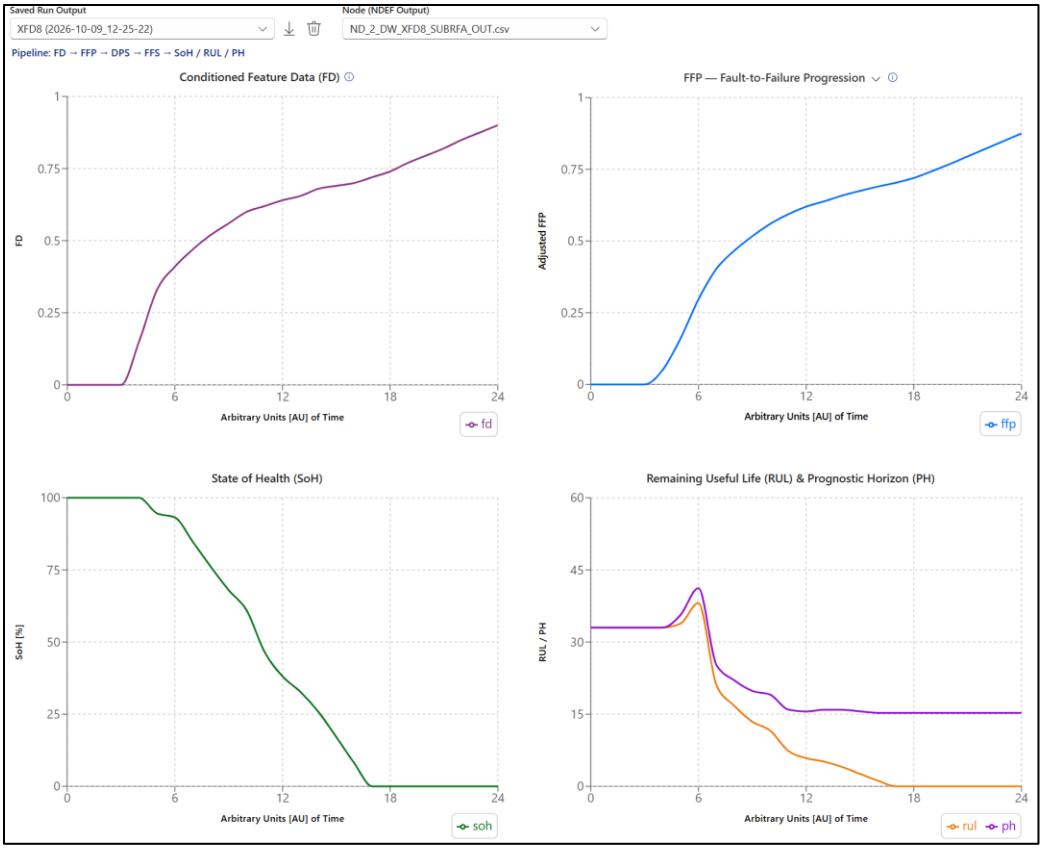


Figure 127 – Plots of the XFD8 Prognostic Information for the Second Node

6.10.3 XFD8: Discussion

Referring back to Figure 121, analysis of the reflected energy (left–hand plot) reveals the following:

1. In the absence of degradation in the form of damage to the wire (grounding shield, insulation material, or any of the twisted–wire conductors), three instances of high reflection occur: when the high–frequency energy is injected into the antenna, when the injected energy reaches the end of the wire; and when the energy is reflected back to the beginning of the antenna.
2. At every location of damage to the antenna, there will be change in the effective impedance because the distributed capacitance, inductance, and resistivity is changed,
3. When the magnitude of the reflected energy exceeds a predetermined level above the level for an undamaged state plus noise, the antenna is evaluated as being degraded.
4. As the level of damage increases, the magnitude of the reflected energy increases.

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5. The change in magnitude at that location is a FD having a signature as shown in Figure 126 and Figure 127.
6. A "weakest–link evaluation" can be used:
    - The SoH of the antenna is the lowest–valued SoH of all locations of damage.
    - The RUL of the antenna is the lowest–valued RUL of all locations of damage.

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6.11 XFD9: Electro–mechanical Actuators (EMAs)

This example provides a detailed overview for how Ridgetop has used ARULE™ to prognostic enable an electro–mechanical actuator system with multiple nodes. For a NASA–funded program, Physical Modeling and Diagnostic and Prognostic Reasoning (MAPR) program, Ridgetop Group designed and developed a number of solutions related to EMA devices that are widely used in different types of equipment, vehicles, and structures, to position and rotate surfaces. An EMA comprises devices, assemblies, and units: such as one or more power sources, positioning controllers, power converters, motors, and shafts. The Hardware Test Bed in Figure 128 comprises two line–replaceable units (LRUs): a switch–mode power supply (SMPS) with a sensor and an EMA motor with a phase–current sensor: the EMA motor comprises a brushless DC motor (BLDC) with a shaft to raise and lower a surface (Figure 128 and Figure 129). Likewise, the software in a host computer comprises two blocks (Figure 130 and Figure 131).

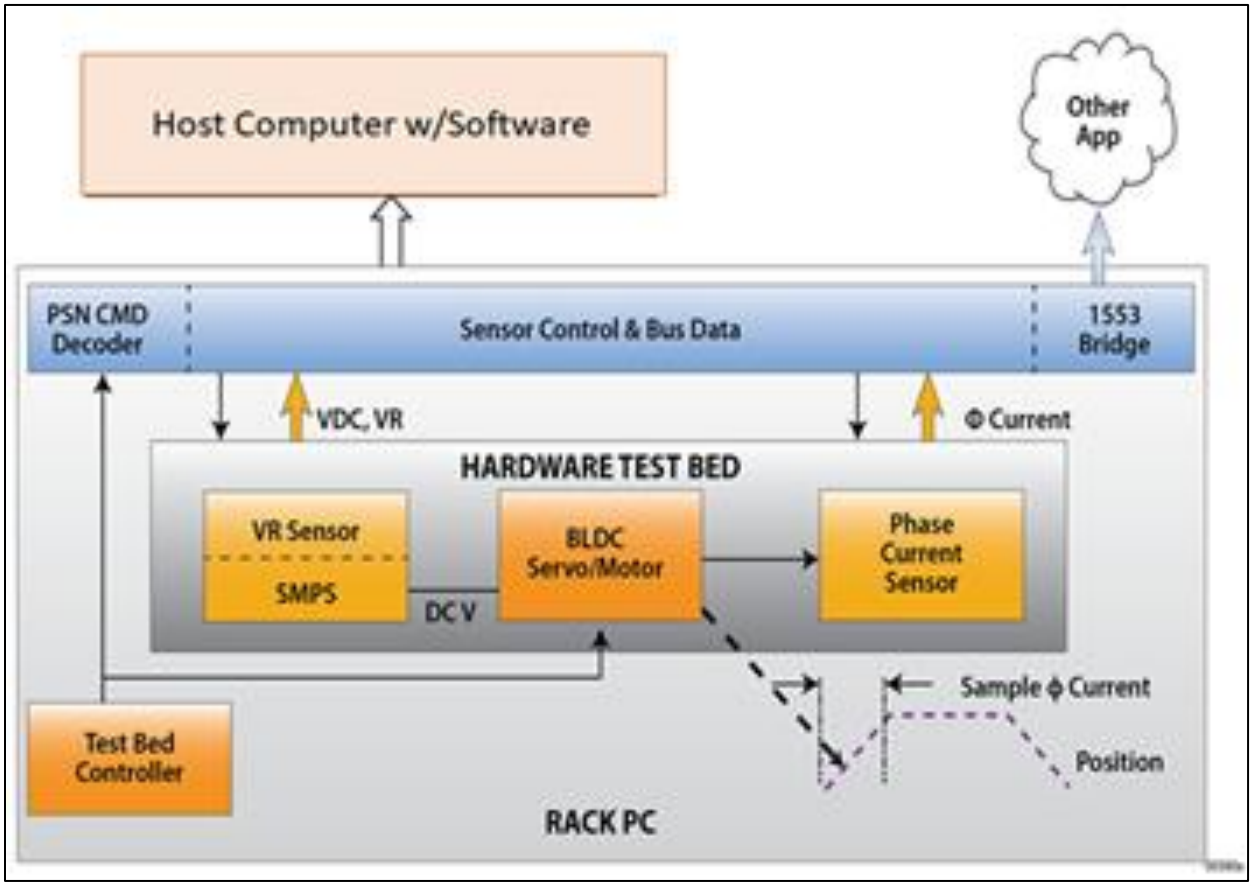


Figure 128 – EMA (controller and motor) test bed

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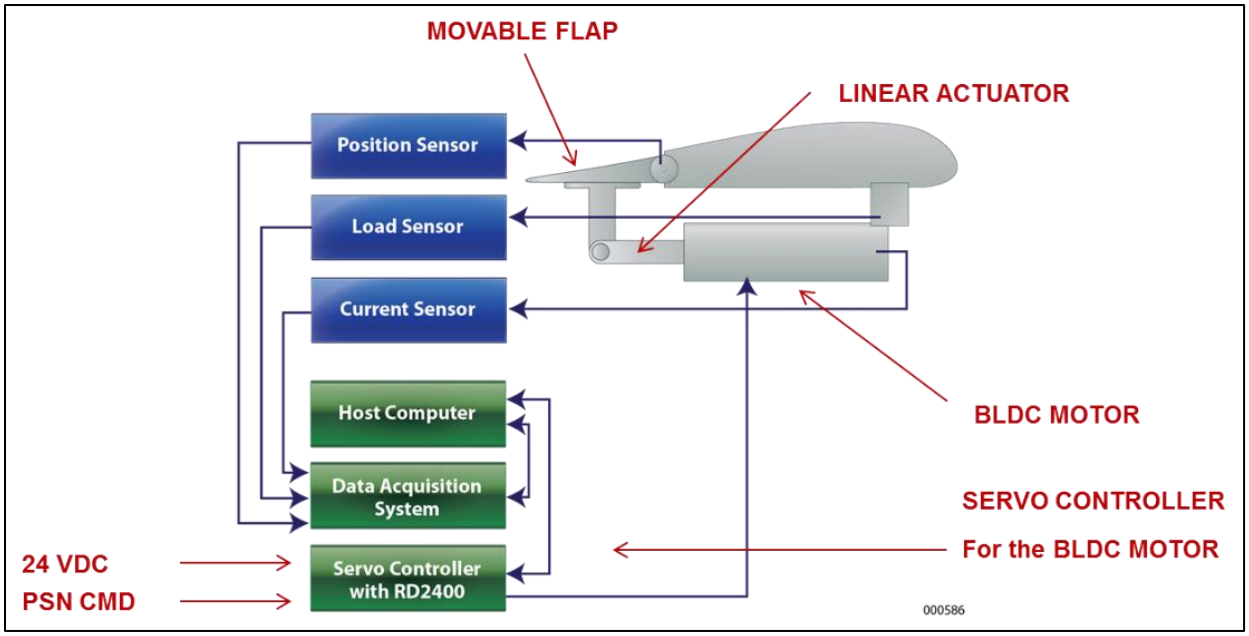


Figure 129 – Test Bed diagram

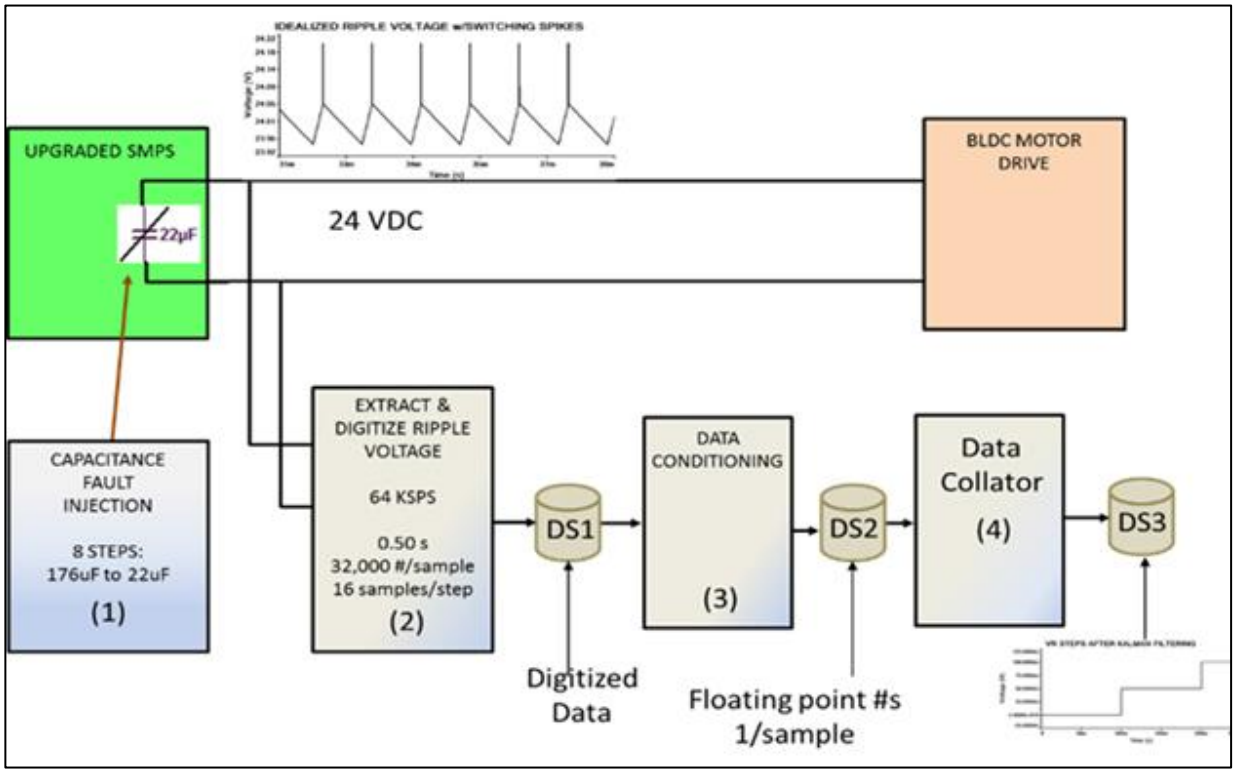


Figure 130 – Collect and process data from a switch–mode power supply: RD2400 in Figure 77

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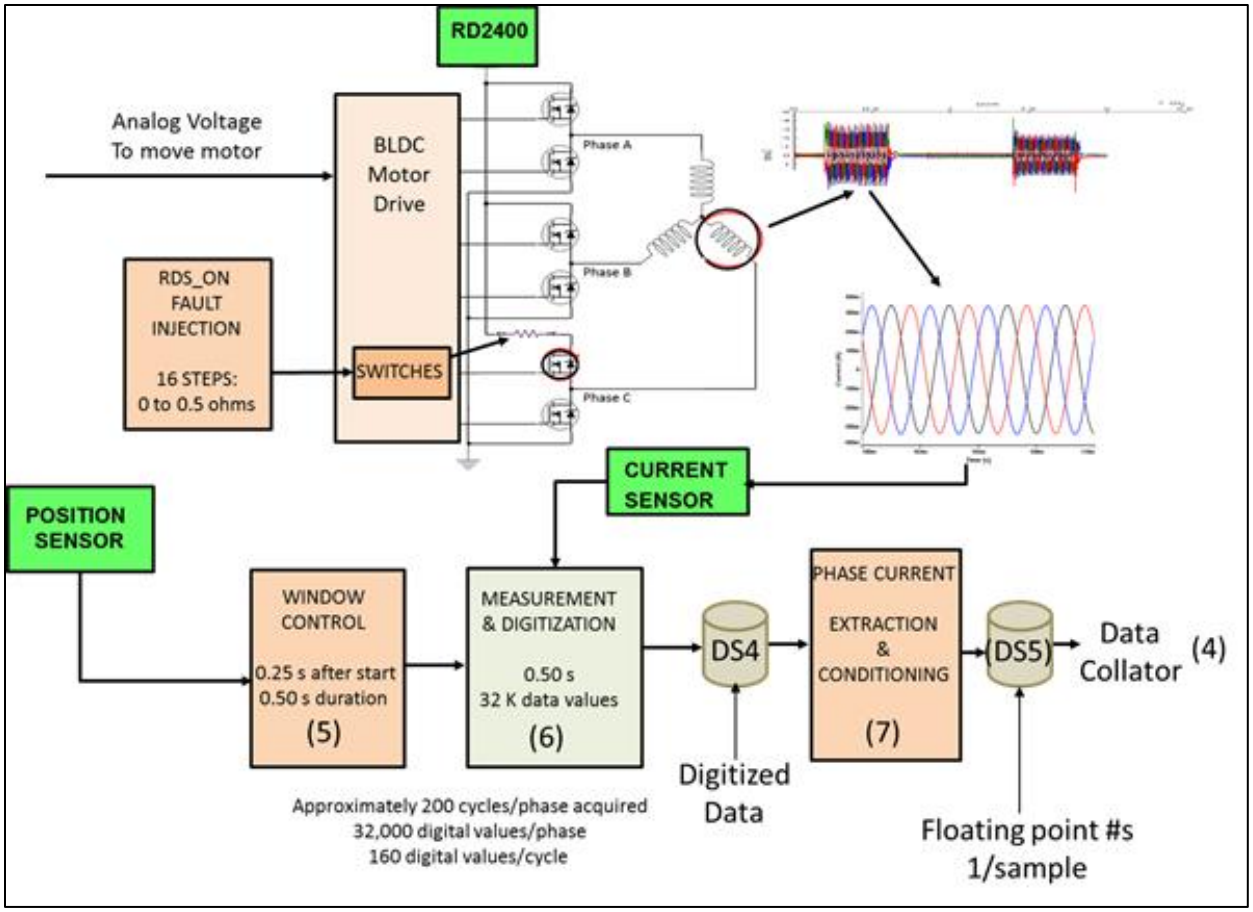


Figure 131 – Collect and process data from the EMA

The first LRU, a Ridgetop–designed and developed 5–V SMPS, had a smart sensor and a fault–injection controller that produced FD signature. Ridgetop–developed sampling, data management, and IEEE 1553 bus management collected that FD signature in a data file (DS3 in Figure 132). Figure 133 is a plot of the collected FD.

The second LRU, Ridgetop–designed and developed an EMA test bed that had four sensors (a positioning sensor and three AC phase–current sensors). The EMA test bed also had fault–injection hardware, firmware, and a gated–window software application to control data sampling. The positioning sensor is used with gating logic to create windows during which phase currents are sampled (see Figure 134).

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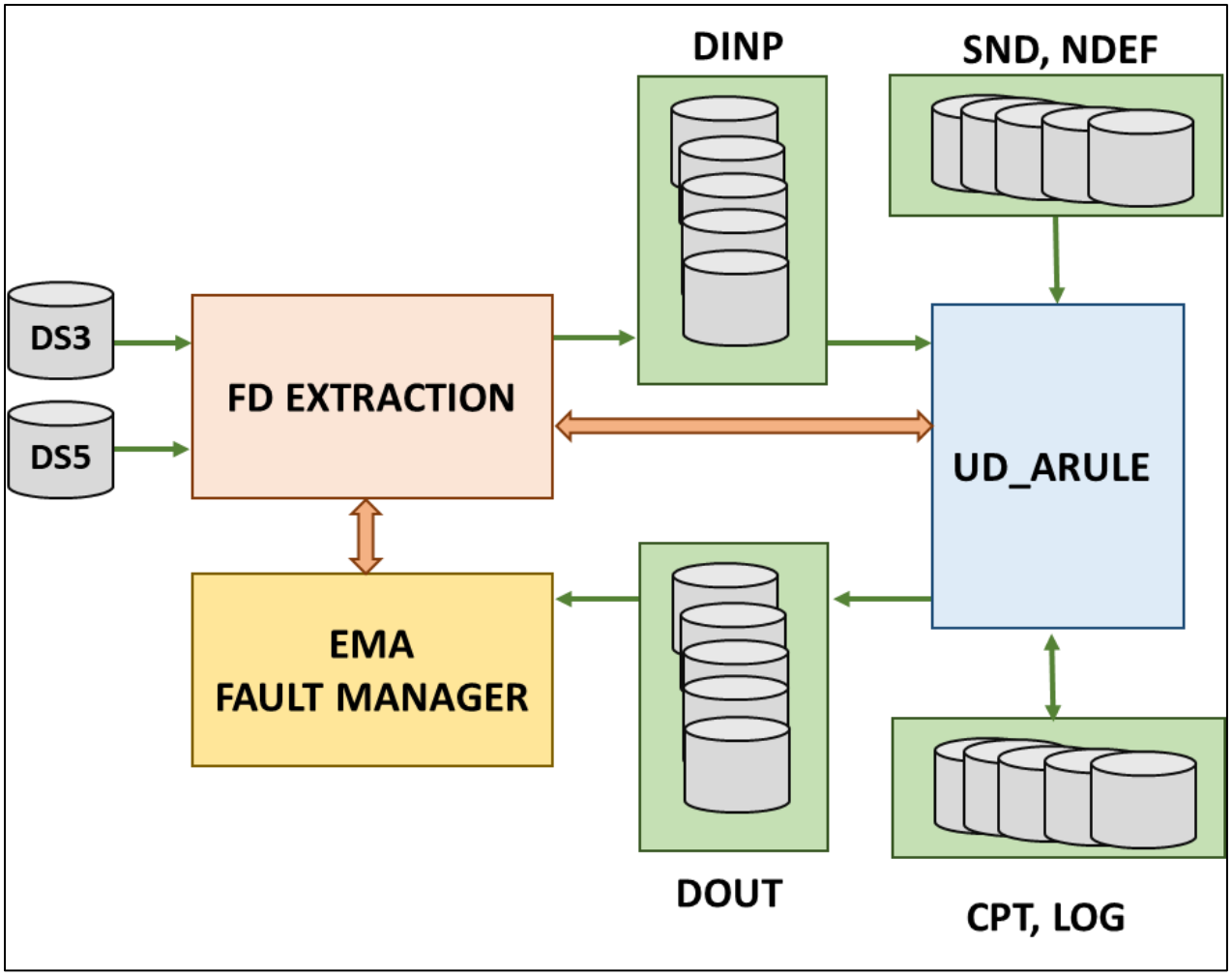


Figure 132 – Diagram: experimental solution

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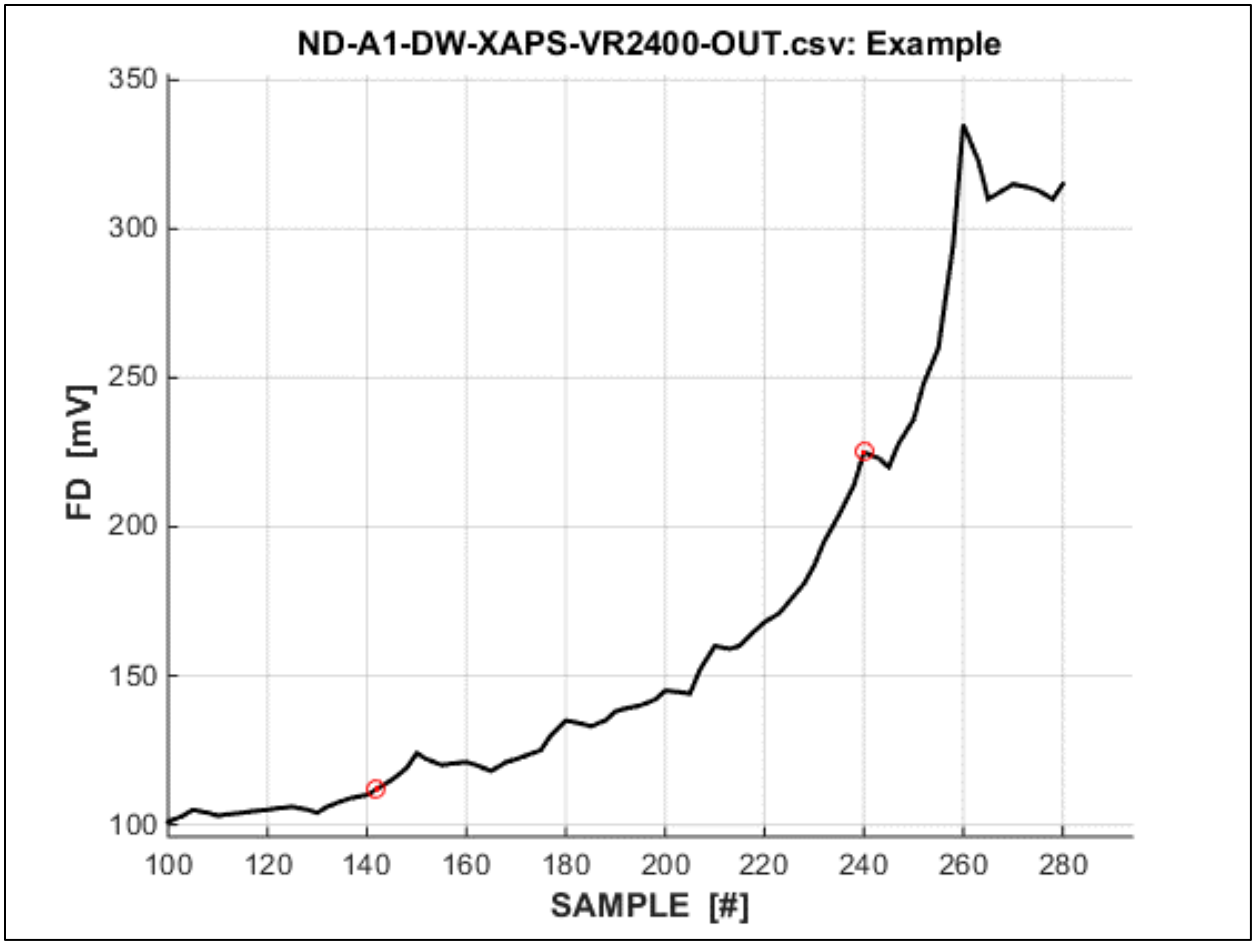


Figure 133 – Extracted FD from DS3: ripple voltage

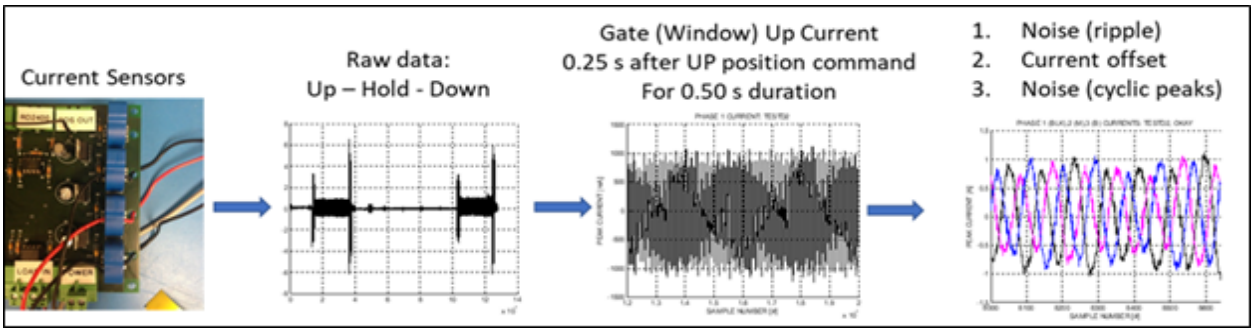


Figure 134 – Phase–current sampling: sensors, sensor output, sampled data, and examples of noise

The FD from sampling the phase currents exhibit all manner of noise: electrical (Figure 135), amplitude variability (Figure 136), current offset (Figure 137, unbalanced positive and negative current halves), and distortion (Figure 137): all of which required noise–mitigation and noise–canceling solutions).

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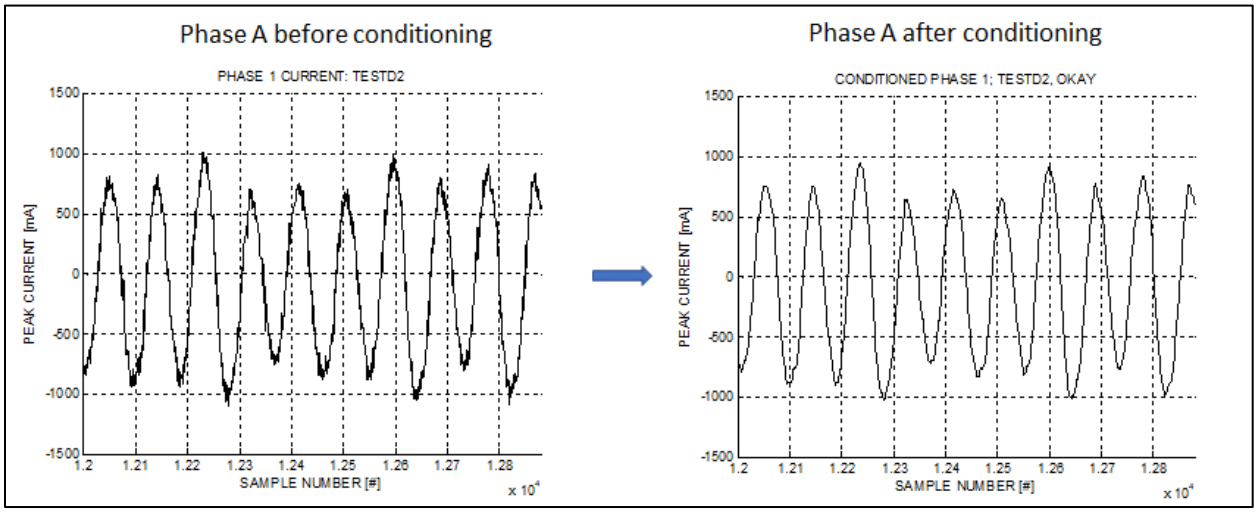


Figure 135 – Phase–current noise: electrical before (left) and after (right) conditioning

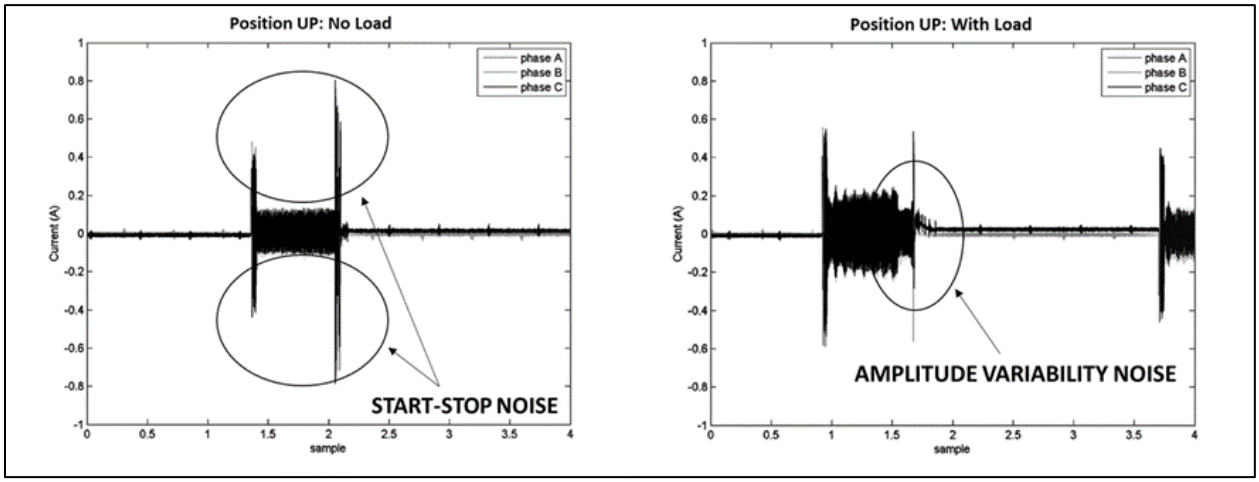


Figure 136 – Phase–current noise: start–stop (left) and load–variation (right)

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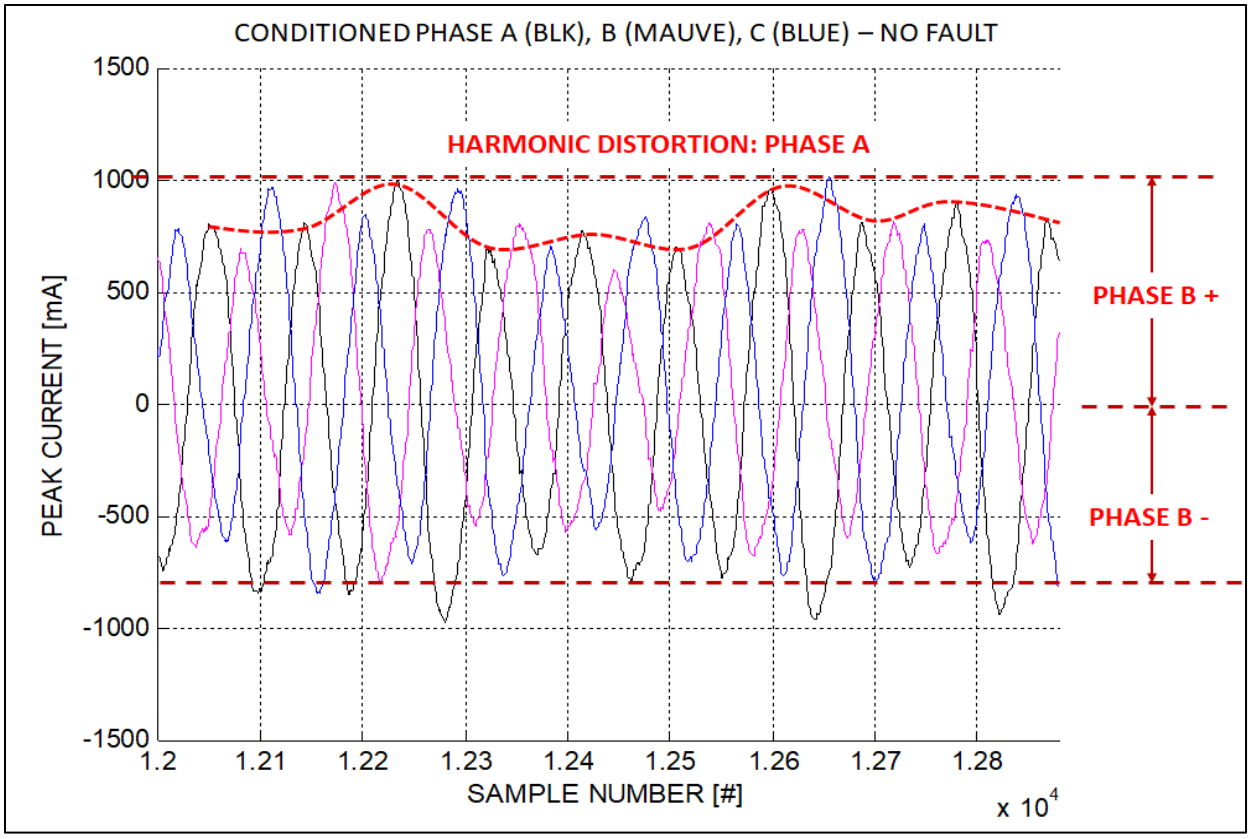


Figure 137 – Current offset and distortion

Experimentation, physics–of–operation and failure analyses, and physics–of–failure analyses revealed the following important characteristics of the EMA used in the test bed:

- The EMA motor uses a "floating" reference for the stator windings (Figure 138) so that, for example, an increase in impedance in one winding path causes all of the currents in all of the paths to change: zero sum of all of the currents into the floating node.
- The 120–degree phase shift is a primary cause of distortion: N reduction in phase–A current requires 2/3 N and 1/3 N increase in phase–B and phase–C currents.

Special methods for calculating rms and differential methods of comparison were required:

- To distinguish between failing nodes: switching–transistor (1 of 6), stator winding (1 of 3) and motor shaft (1 of 1).
- To support 10 degrees of freedom with only three sensors.
- To provide quantifiable computational values to support prognostics.

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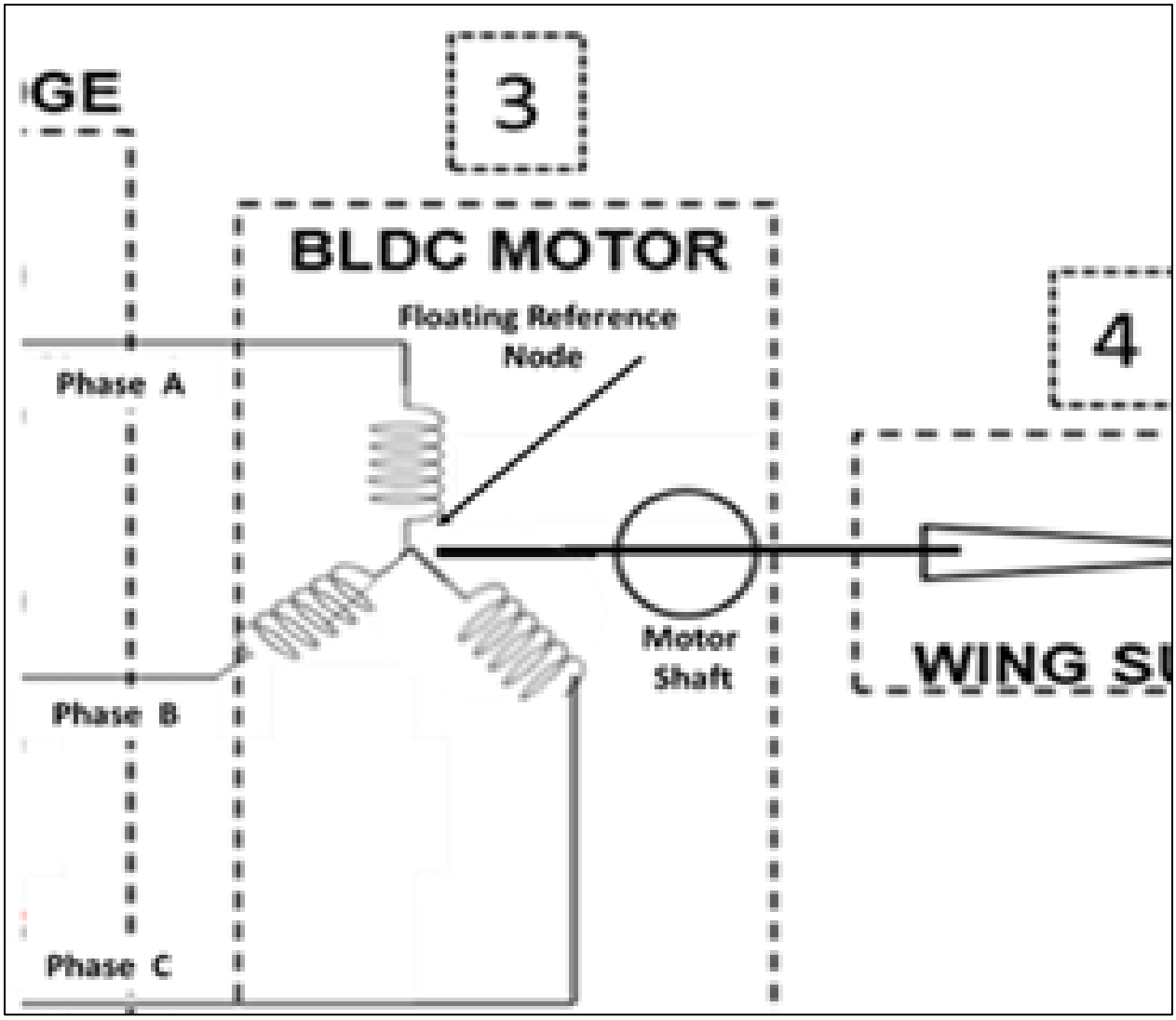


Figure 138 – Brushless DC motor with floating reference for the stator windings

The special methods for processing phase currents results in 10 sets of extracted FD signatures (refer back to Figure 132) as plotted in Figure 139, Figure 140, and Figure 141.

In Figure 139, the positive and negative halves of the phase–B current and phase–C current diverge in opposite directions while they diverge in the same (negative) direction for phase–A. All three of the phase currents diverging is indicative of degradation in a switching–transistor path: in this case it is the negative path for phase A in the H–bridge controller.

When there is (1) no divergence of the negative and positive halves of the phase currents, the fault tree is the following:

- There is no significant change in any of the three phase currents: no degradation.

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- There is a significant change in one of the phase currents: stator–winding fault (Figure 140).
- There is significant change in all three phase currents: motor–shaft fault (Figure 141).

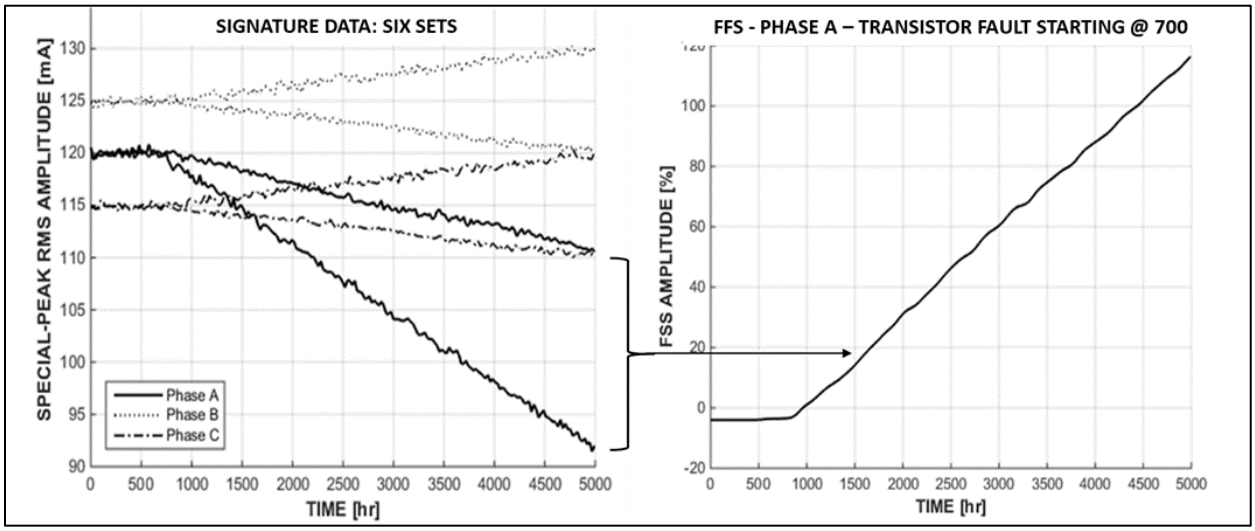


Figure 139 – Degradation in one–of–six transistor paths (phase A, negative path)

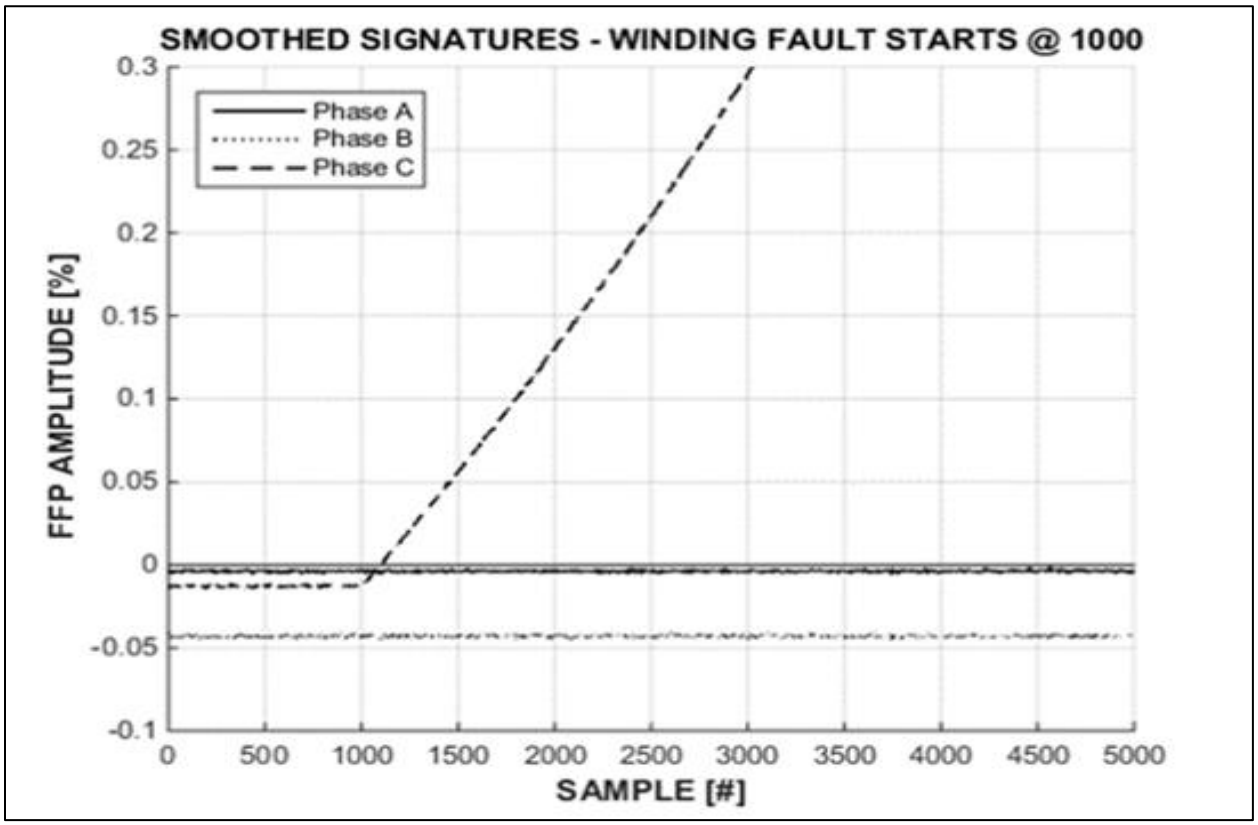


Figure 140 – Degradation in one–of–three stator windings

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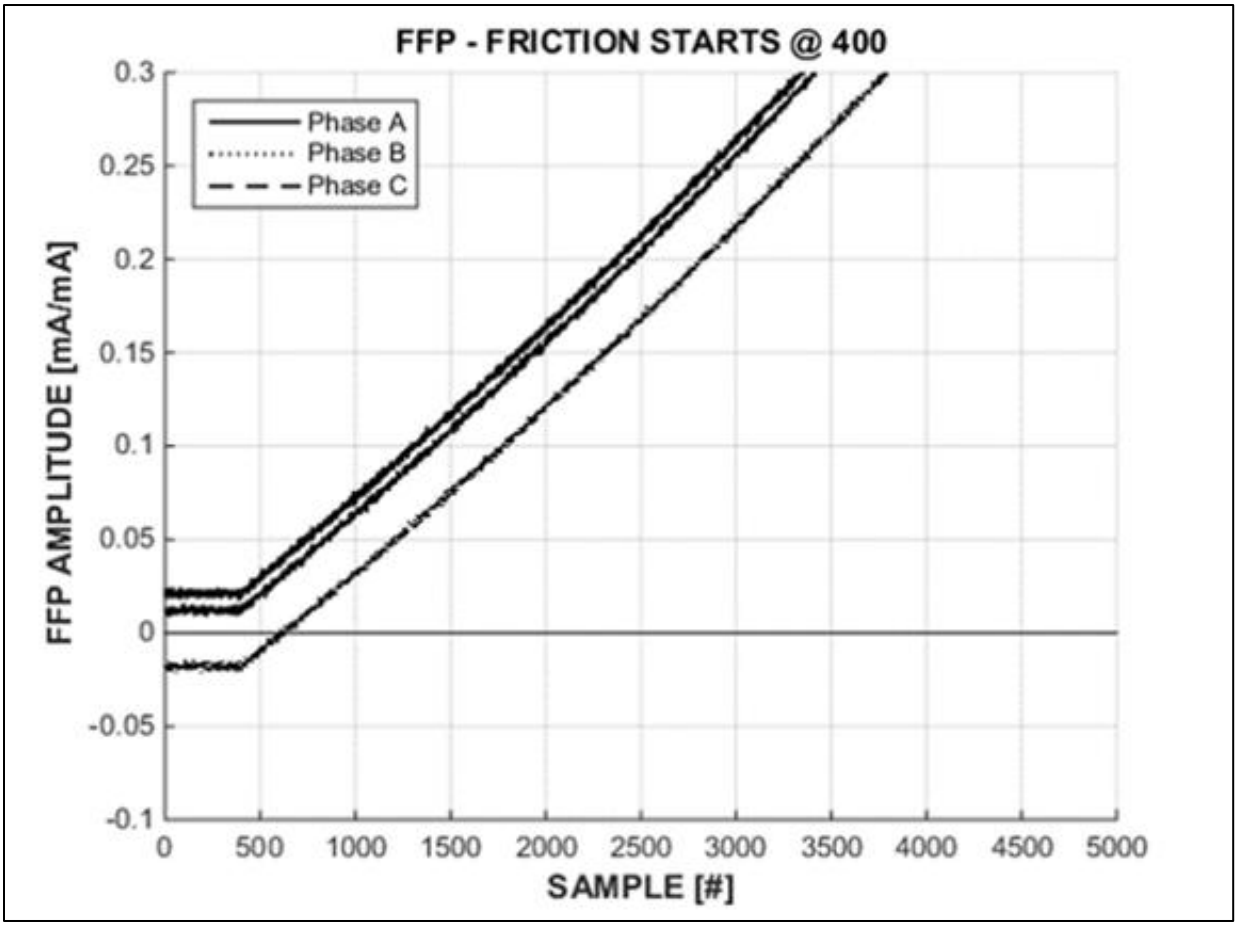


Figure 141 – Excessive friction/load on EMA rotating shaft

6.11.1 XFD9: Definitions

For illustrative and demonstration purposes, four examples of NDEFs for an EMA are created: XAPS for the SMPS, XAST for a switching transistor, XASW for a stator winding, and XAMS for a motor shaft: see Figure 142.

The input data plotted in Figure 139, Figure 140, and Figure 141 are simulated based on experimental results. The simulation results and experimental results were combined to create the example data used as input for this example.

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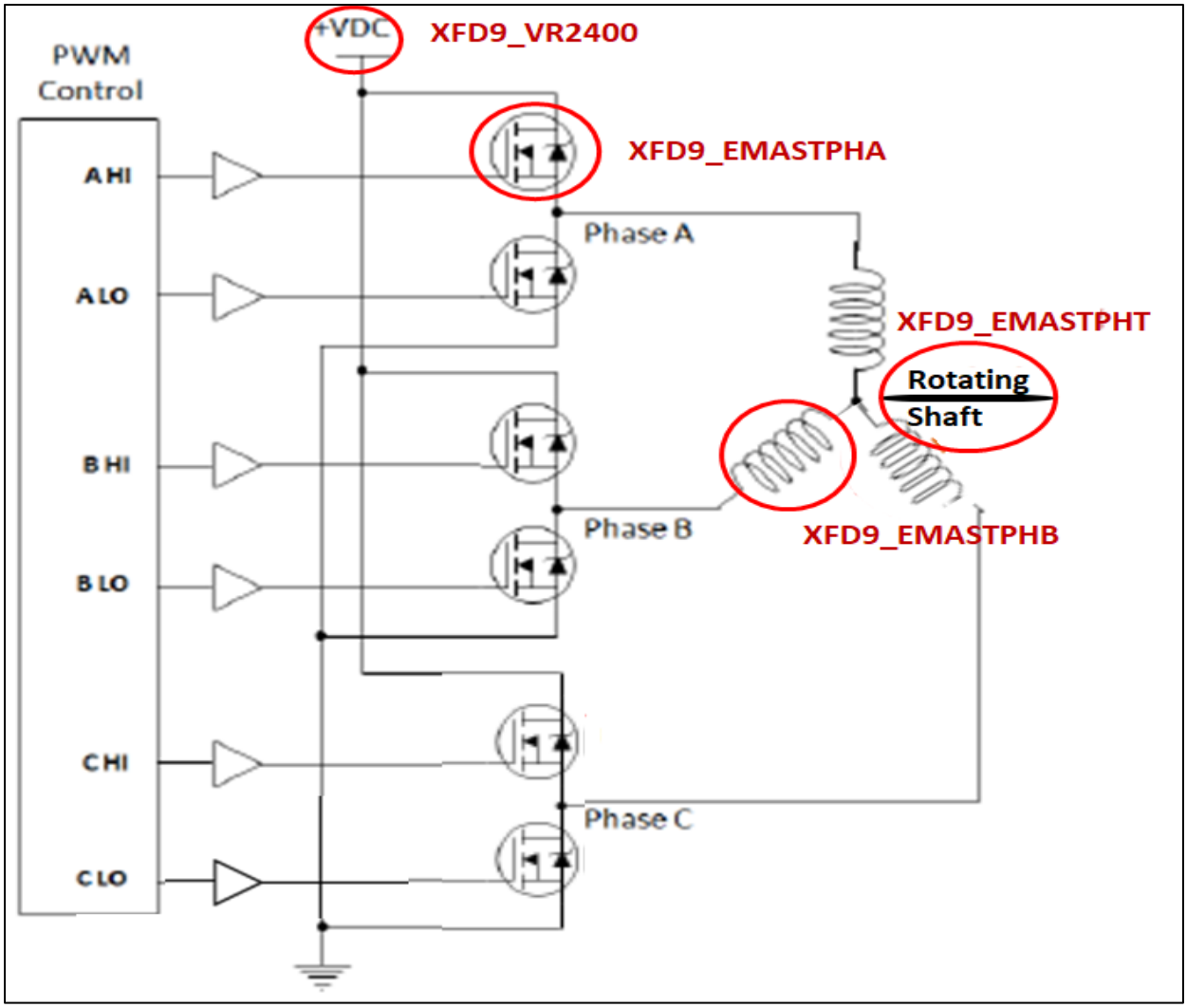


Figure 142 – EMA block diagram showing selected locations of four failures for XFD9

The full solution included a fault–tree analysis routine that compared prognostic estimates, relative magnitudes, and signature directions to determine whether or not degradation was detected and, if so, the likely source: power supply, phase–current branch (out of six), stator winding (out of three), and motor shaft. This example only presents failure for each of four major sources represented by the following NDEFs in XFD9 (see Figure 143):

- **XFD9_VR2400** SMPS for the EMA – Ripple Voltage
- **XFD9_EMASTPHA** Switching Transistor – Phase A
- **XFD9_EMASWPHB** Motor Stator Winding – Phase B
- **XFD9_EMARSPHT** Motor Rotating Shaft – Phase Total

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Figure 144, Figure 145, Figure 146, and Figure 147 show the NDEF parameters, INFILES and keywords used for XFD9.

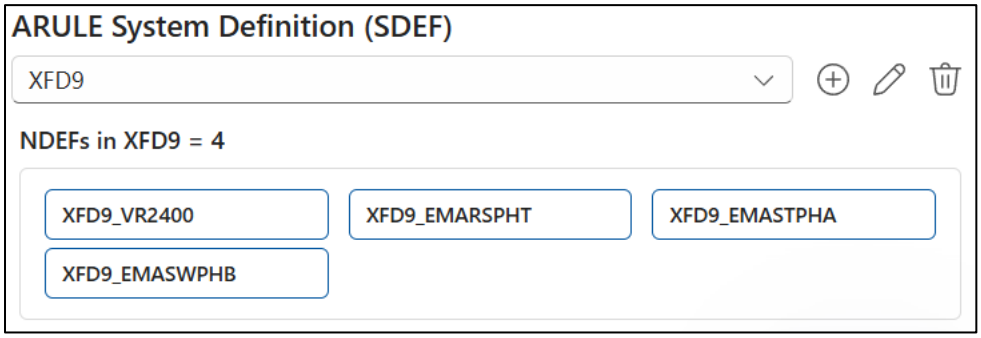


Figure 143 – XFD9 System Definition (SDEF) User Form

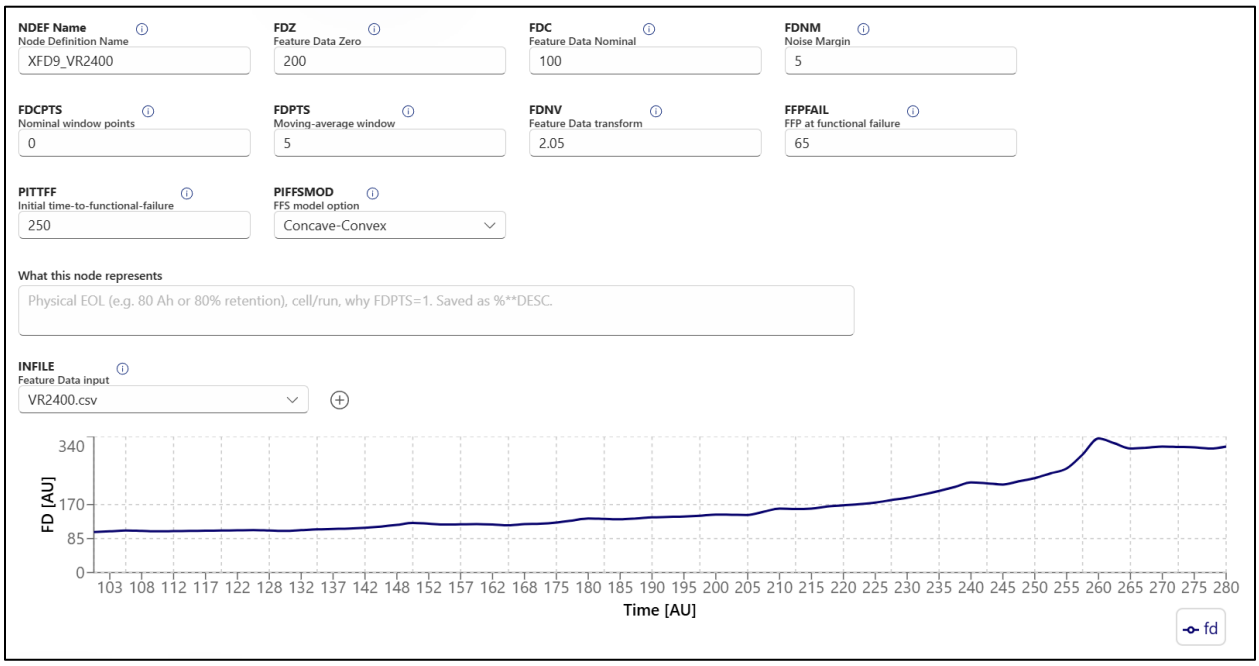


Figure 144 – XFD9_VR2400 NDEF

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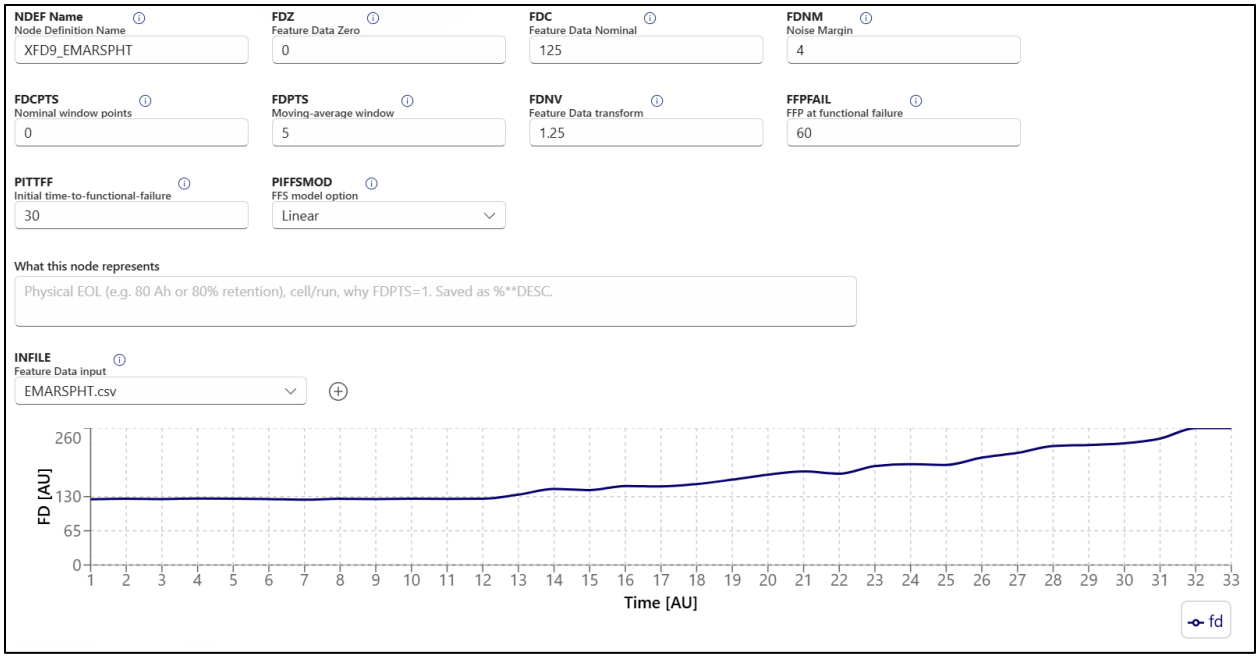


Figure 145 – XFD9_EMARSPHT NDEF

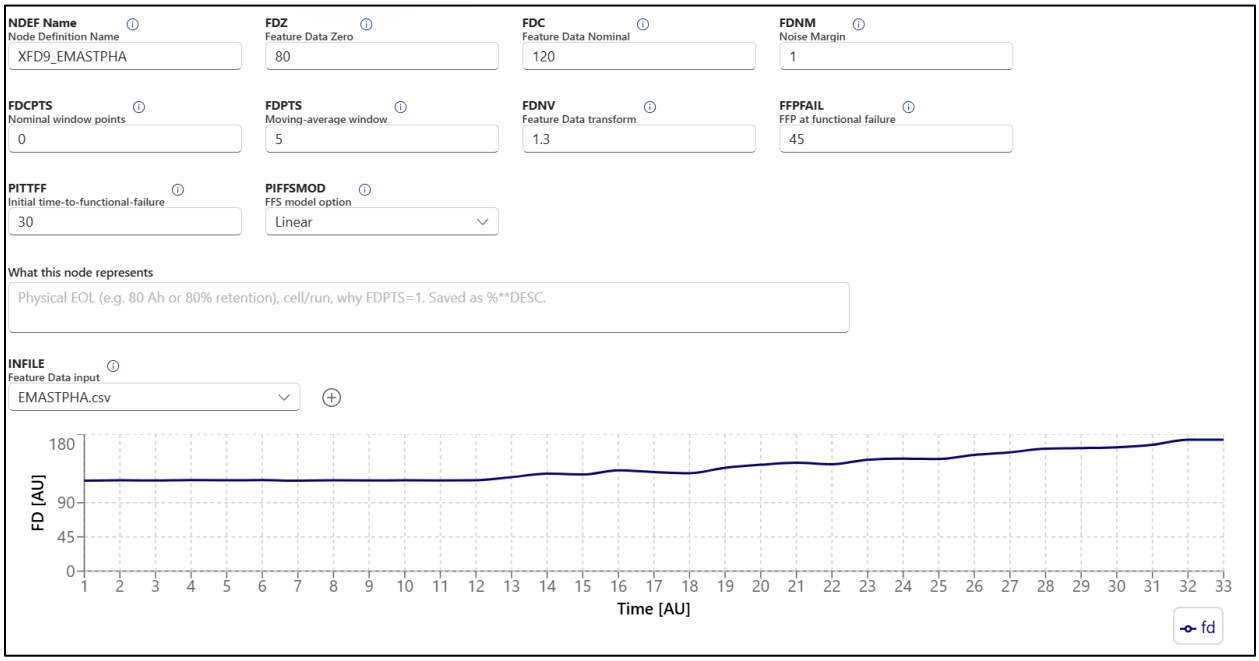


Figure 146 – XFD9_EMASTPHA NDEF

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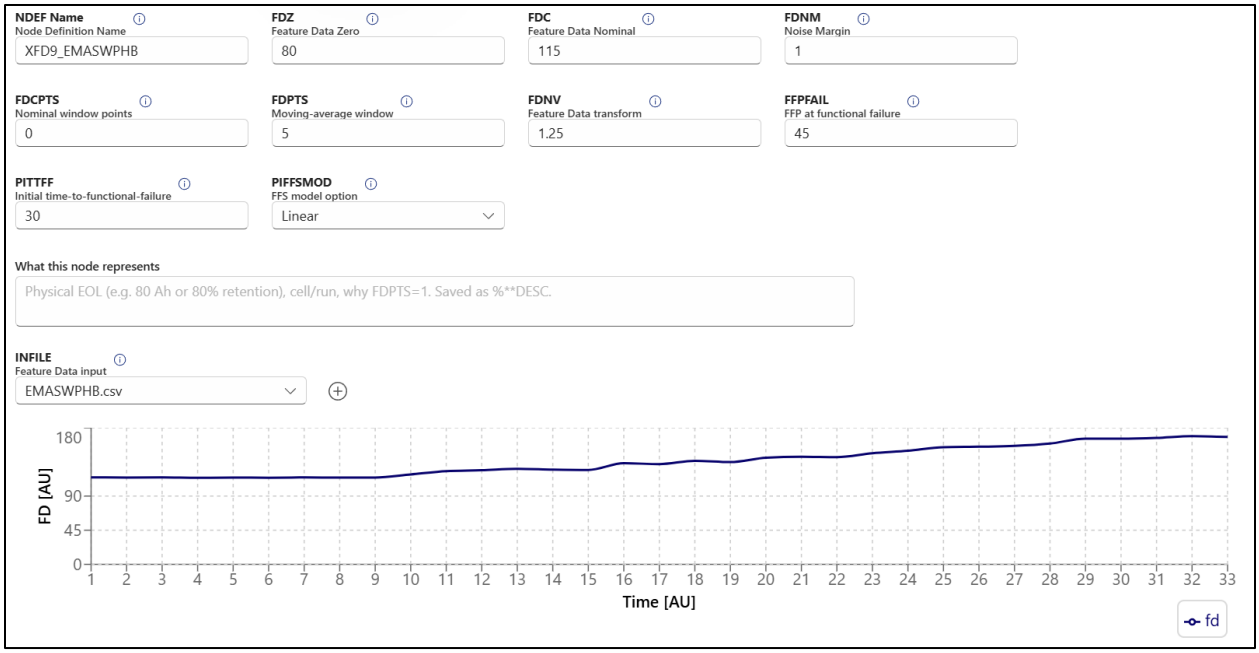


Figure 147 – XFD9_EMASWPHB NDEF

6.11.2 XFD9: Data Outputs and Prognostic Information

Figure 148, Figure 149, Figure 150, and Figure 151 show the Data Output for the four nodes in XFD9: This includes the FD, FFS Progression, along with SoH, RUL, and PH.

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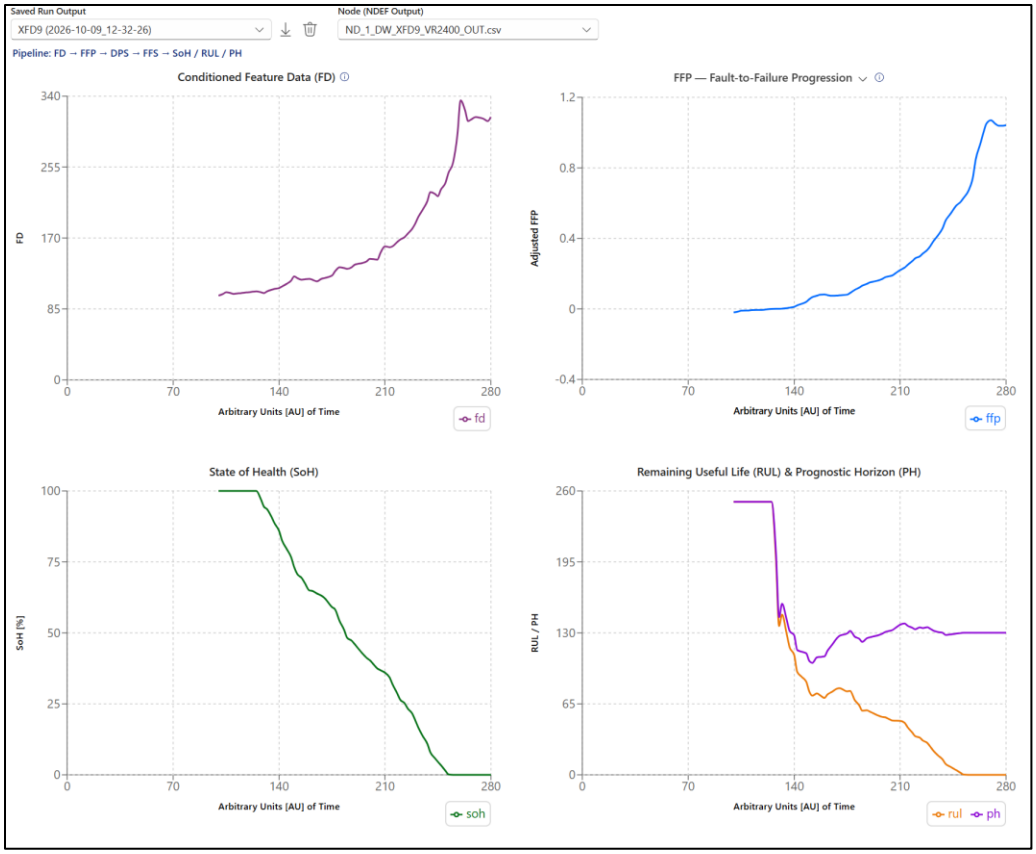


Figure 148 – Plots of the XFD9 Prognostic Information for the First Node

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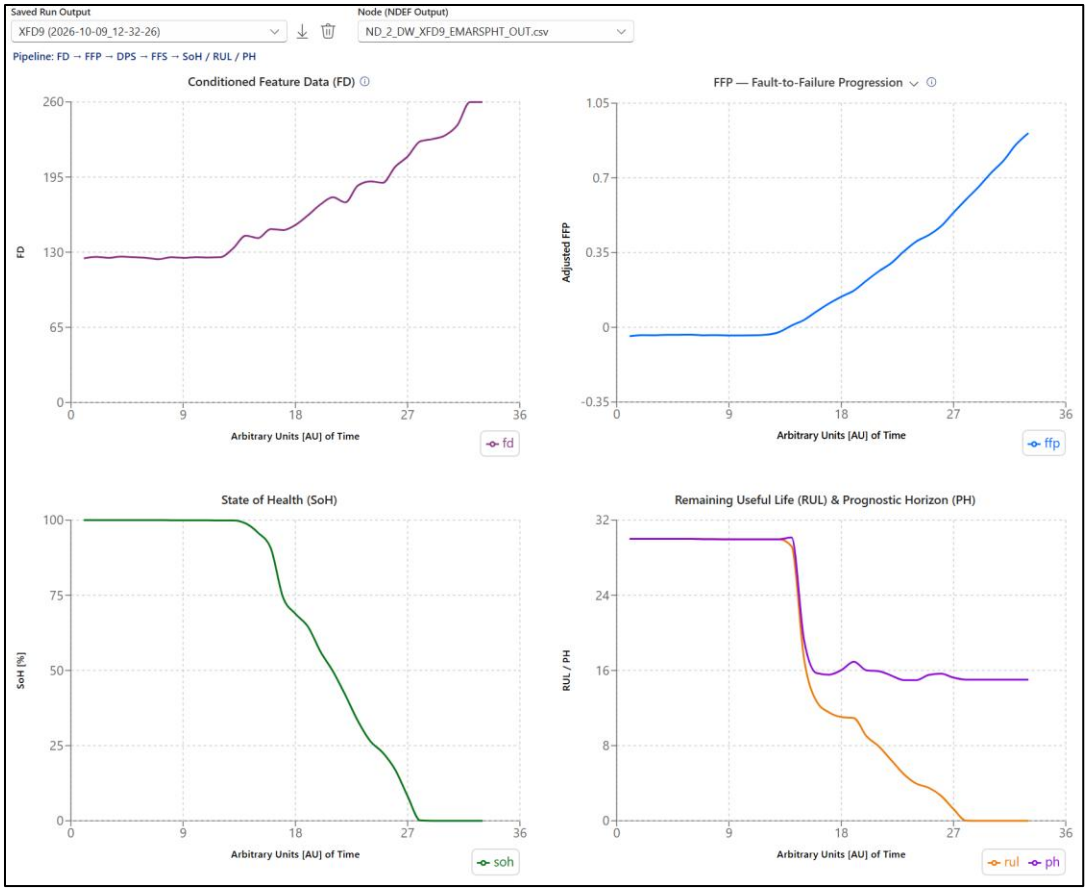


Figure 149 – Plots of the XFD9 Prognostic Information for the Second Node

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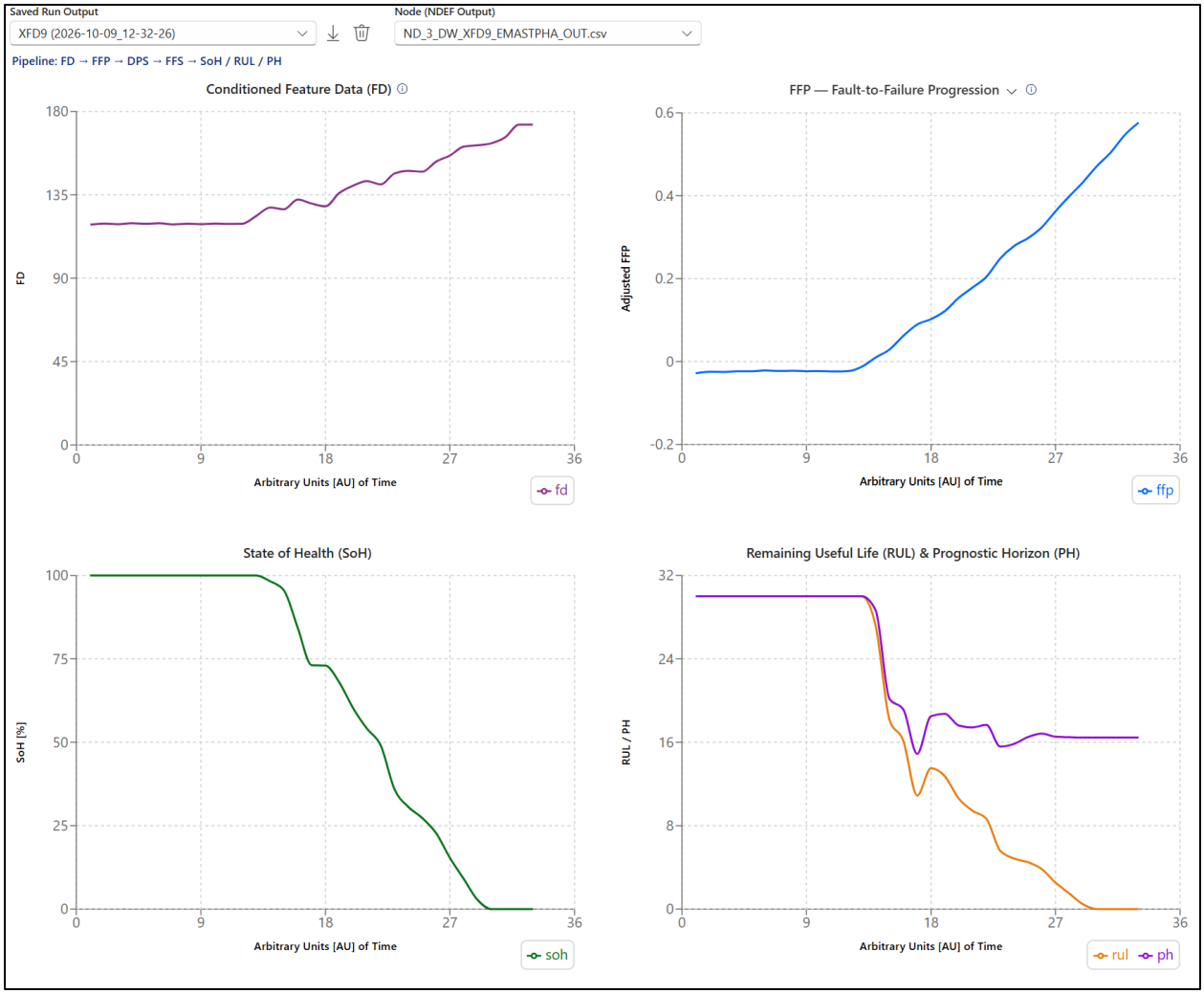


Figure 150 – Plots of the XFD9 Prognostic Information for the Third Node

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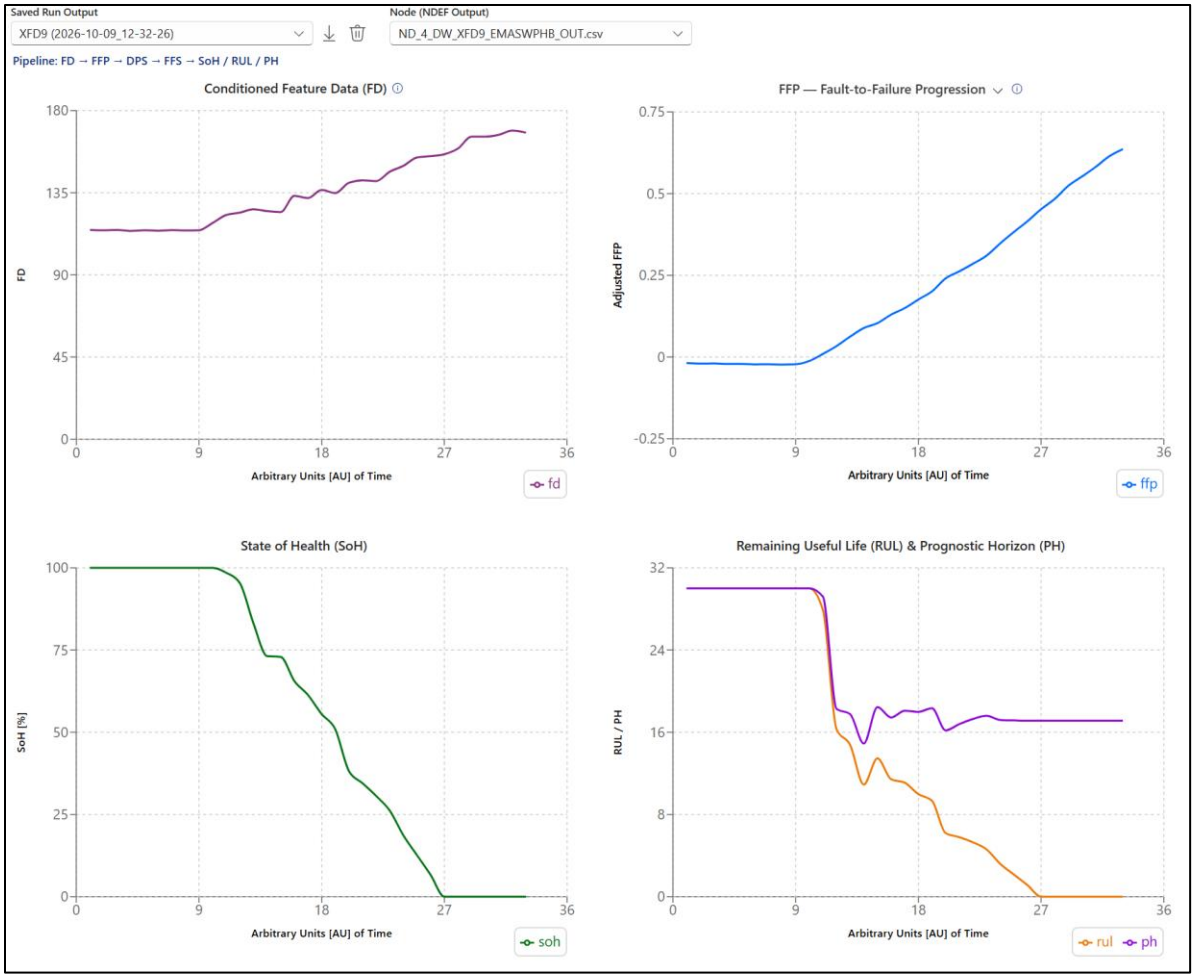


Figure 151 – Plots of the XFD9 Prognostic Information for the Fourth Node

6.11.3 XFD9: Discussion

XFD9_VR2400: A complete PHM solution would comprise 11 different sets of FD analyses: one for the power supply, one for each of the six switching transistor paths, one for each of the three stator windings, and one for the rotating shaft. This demonstration is for a failing case only. This demonstration is for the case when the power supply degrades (refer back to Figure 133, page 128).

XFD9_EMASTPHA: This demonstration is for the case where one of two sets of switching transistors for the Phase A current degrades. Referring back to Figure 139, page 122, the summation of the positive and negative halves for any undamaged switching path is essentially zero plus background noise. However, when the switching path is damaged, the positive and negative halves of the current change in the same direction – as shown for Phase A in the figure. If the change is in the positive direction, the positive half of the transistor pair is damaged. In this case, the change is in the negative direction because that was the location of the damaged switching transistor.

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XFD9_EMASWPHB: This demonstration is for the case where the stator winding for the Phase B current degrades. Referring back to Figure 140, page 120, when a stator winding is damaged, instead of a change in amplitude of both halves of all three currents, there is a change in both halves of only one phase and the change is in the same direction and of equal magnitude: in this changes, the damaged stator winding is for Phase B.

XFD9_EMARSPHT: Referring back to Figure 141, page 133, when there is excessive load (weight) and/or friction (binding) on the motor shaft, all halves of all three phases change in the same direction.

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7. How to Analyze Your Own Data with ARULE™

7.1 Overview and Prerequisites

This section of the user guide provides an application-independent procedure for analyzing custom Feature Data with ARULE™. It explains how to prepare and assess a degradation-related feature, create the required System Definition (SDEF) and Node Definition (NDEF) files, configure processing and functional-failure parameters, execute the analysis, and interpret and preserve the resulting prognostic outputs.

ARULE™ processes application-specific Feature Data rather than arbitrary raw sensor data. Before beginning, confirm that the selected feature represents degradation associated with a defined functional-failure condition and that its coordinate basis, engineering units, direction, expected variability, and sampling interval are understood.

Prerequisites. The user should have a suitable Feature Data file in .csv or .txt format, an engineering definition of functional failure, access to the ARULE™ GUI and working directories, and sufficient application knowledge to review the configuration and results. Parameters from an existing example must not be assumed to be valid for a new dataset.

7.2 Preparing and Importing Feature Data

Feature Data may be measured directly or derived from Condition-Based Data through filtering, feature extraction, normalization, fusion, or other application-specific processing. Select a feature that changes consistently with the failure mode of interest and can be related to a defined functional-failure condition.

Before importing the data, review the signature for direction, curvature, baseline stability, noise, outliers, gaps, and changes in operating conditions. Figure 152 illustrates representative increasing and decreasing degradation signatures. Use the figure as a visual reference only; the signature shape by itself is typically not sufficient to determine all NDEF parameter values.

- Confirm that the first column provides a monotonic coordinate such as time, cycle count, or sample index.
- Confirm that the second column contains the selected Feature Data value.
- Record the analysis-coordinate basis and corresponding engineering units. ARULE Remaining Useful Life (RUL) and Prognostic Horizon (PH) are interpreted using the defined progression-coordinate basis, whereas State of Health (SoH) is reported as a percentage.
- Verify that the dataset spans enough healthy and degrading behavior to support baseline, noise, transformation, and endpoint decisions.

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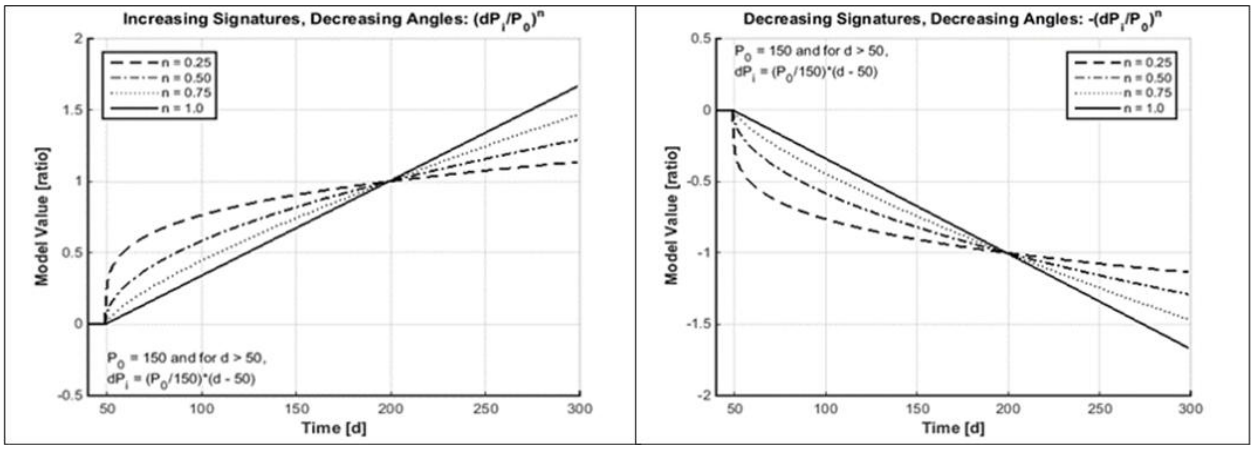


Figure 152 – Degradation signatures: increasing (left) and decreasing (right)

7.3 Creating a System Definition (SDEF)

Create a System Definition (SDEF) file to identify the monitored system and associate it with one or more Node Definition (NDEF) files. In the main window, select the control beside the SDEF list to open the SDEF editor, enter a unique and descriptive SDEF name, and add the NDEF files that belong to the analysis. Figure 153 shows the SDEF user form.

1. Confirm that each listed NDEF represents the intended node, component, or degradation feature.
2. Use stable names that identify the system, configuration, and dataset version.
3. Save the SDEF and verify that its NDEF associations are complete before execution.

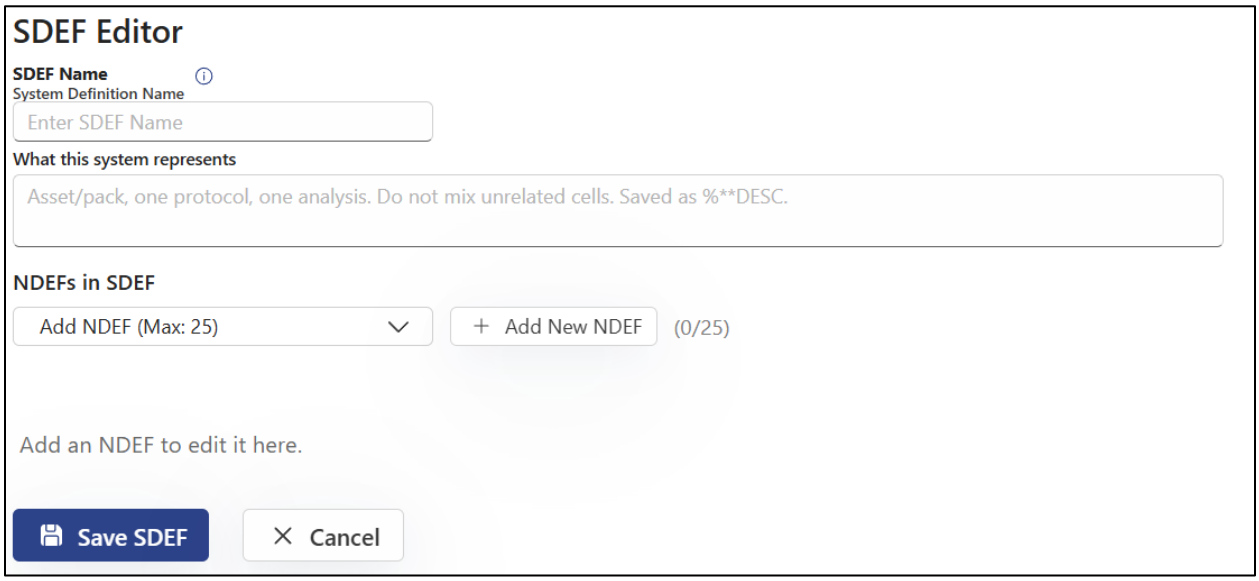


Figure 153 – Example of a new SDEF file: System Definition User Form

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7.4 Creating and Configuring a Node Definition (NDEF)

Create one NDEF for each node or Feature Data stream to be processed. If the required NDEF does not exist, open the Node Definition user form from the SDEF editor. Figure 154 shows the NDEF editor. Give the NDEF a descriptive name and maintain a clear one-to-one association between the NDEF, its input file, and its engineering configuration.

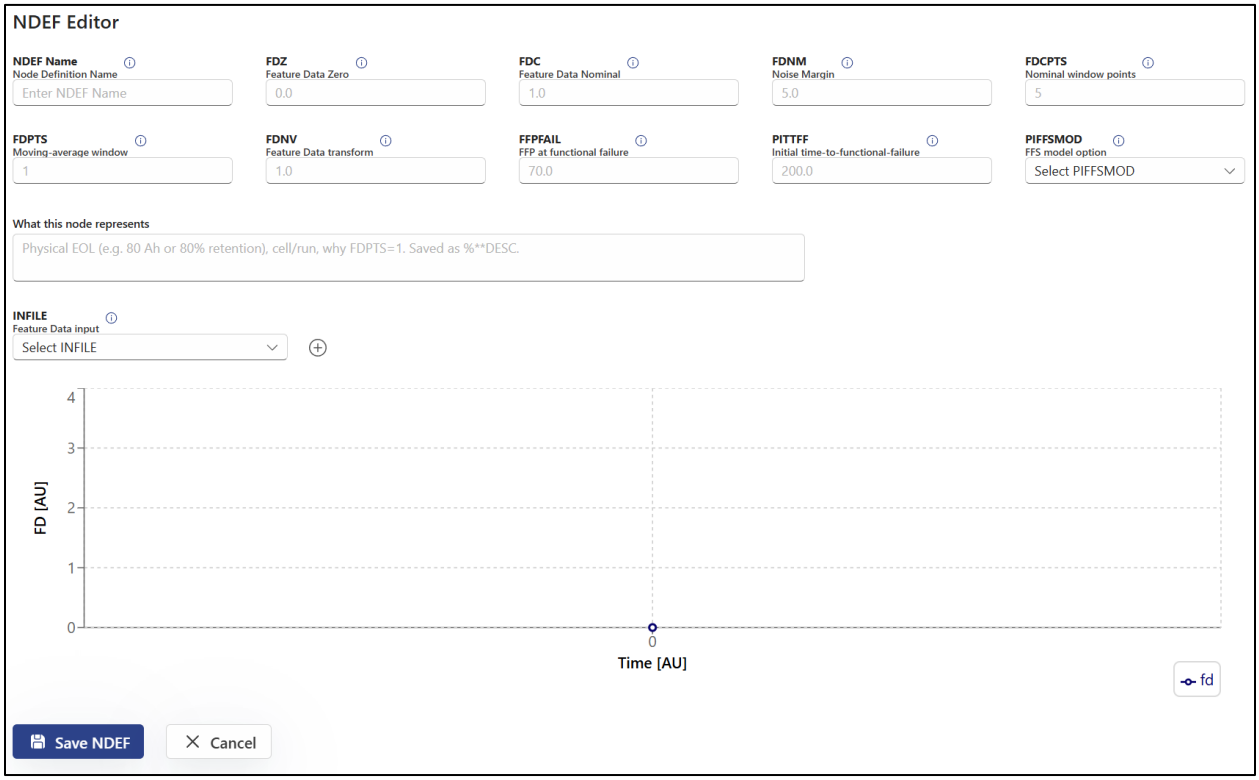


Figure 154 – Example of a new NDEF file and NDEF Editor

In the Node Definition user form, select **Import Data File** and choose the prepared input file for Feature Data in .txt. or .csv format. The default installation identifies **DATA\DINP** as the input-data directory. Figure 155 shows an example file selection.

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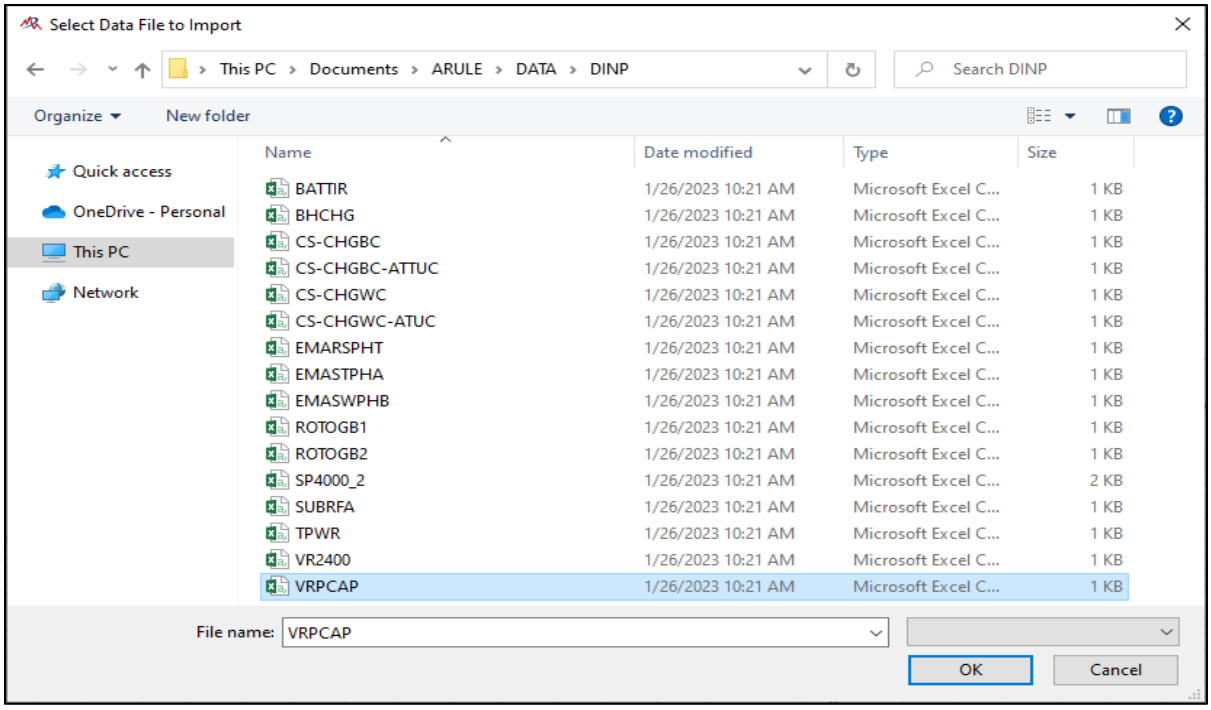


Figure 155 – DINP directory with VRPCAP.csv selected as the input data file

Input Format:

Use a supported text or CSV file containing two values per record: a coordinate in the first column and the Feature Data value in the second. The coordinate may represent time, cycles, sample index, or another monotonic basis. Figure 156 illustrates representative file layouts.

	A	B	C	D
1	0	1		
2	55	2.5		
3	180	4.5		
4	320	5.1		
5	940	6.9		
6	1010	9.2		
7	1680	11.3		
8	2440	12.6		
9	2800	13.6		
10	3200	14.3		
11	3700	15.6		
12	4250	16.1		

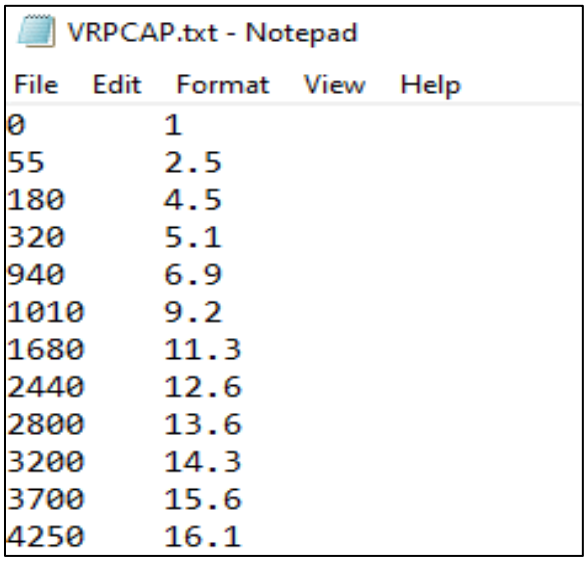


Figure 156 – Example of input data: .csv (left) and .txt (right)

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7.4.1 Define Functional Failure Before Entering Parameters

Define the physical or application-specific functional-failure condition before configuring the processing parameters. Functional failure is the point at which the monitored item no longer meets its required operating specification; it is not necessarily physical destruction or complete loss of operation.

Use the following sequence:

1. Identify the monitored feature, healthy baseline, coordinate basis, and engineering units.
2. Establish the physical or application-specific functional-failure criterion.
3. Choose normalization, noise-margin, averaging, and signature-transformation settings.
4. Map the failure criterion into the normalized FFP threshold used by **FFPFAIL**.
5. Configure the initial life assumption and final prognostic signature model.
6. Process representative data and verify that FFP, DPS, FFS, SOH, RUL, and PH behavior is consistent with the intended endpoint.

Caution: Do not enter a physical threshold percentage directly as **FFPFAIL** unless the normalization and transformation mapping demonstrates that the values are equivalent.

7.4.2 Select and Verify NDEF Parameters

Use the following questions to develop and review the NDEF configuration. Refer to the authoritative parameter definitions and valid ranges in Chapter 2.

Parameter	Configuration question
FDC	What Feature Data value represents the healthy or nominal condition?
FDZ	What normalization reference is appropriate for this feature and transformation?
FDNM	What level of natural feature variability should be ignored without masking meaningful degradation?
FDPTS	Is additional averaging of incoming Feature Data required, and what behavior could that averaging obscure?
FDCPTS	Should the nominal baseline be fixed or estimated from early observations?
FDNV	What transformation produces an appropriate degradation-progression signature? Verify the result rather than selecting from visual shape alone.
FFPFAIL	What normalized FFP threshold corresponds to the defined functional-failure condition?

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Parameter	Configuration question
PITTFF	What initial time-to-functional-failure assumption is appropriate in the selected coordinate basis?
PIFFSMOD	What final prognostic signature shape is supported by the transformed data and application model?

Important Note: Automatically populated values are starting values only. They are not universally valid settings for new Feature Data. Existing examples may be used to learn the interface, but their values must not be copied without an engineering basis.

Enter the reviewed values in the NDEF editor, save the NDEF, and reopen it to confirm that the file association and parameter values were retained. Figure 157 shows the parameter fields and a completed NDEF example; its values are illustrative, not a default configuration for unrelated data.



Figure 157 – Default: Values for Feature Data, Data Conditioning, Degradation signatures, and Prognostics Parameters and an example of a completed NDEF file

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7.5 Configuring Run Arguments and Executing ARULE

1. Return to the main window and select the SDEF to be processed. Review the displayed run arguments before starting the analysis. The SDEF identifies the associated NDEF files and therefore the input streams and configurations that ARULE™ will process.

- **Standard processing:** Process the configured input dataset using the selected SDEF and NDEF files.
- **Checkpoint/restart processing:** Use checkpoint files to continue or segment processing as supported by the installed implementation.

Use standard processing for a complete first-pass analysis unless a verified checkpoint/restart workflow is specifically required.

1. Confirm the selected SDEF, associated NDEF files, input paths, and run mode.
2. Start ARULE™ from the run controls.
3. Monitor the Run Log and verify that each configured NDEF completes successfully. Figure 158 shows an example success message.
4. If processing fails, stop and correct the input or definition-file issue before interpreting any partial output.

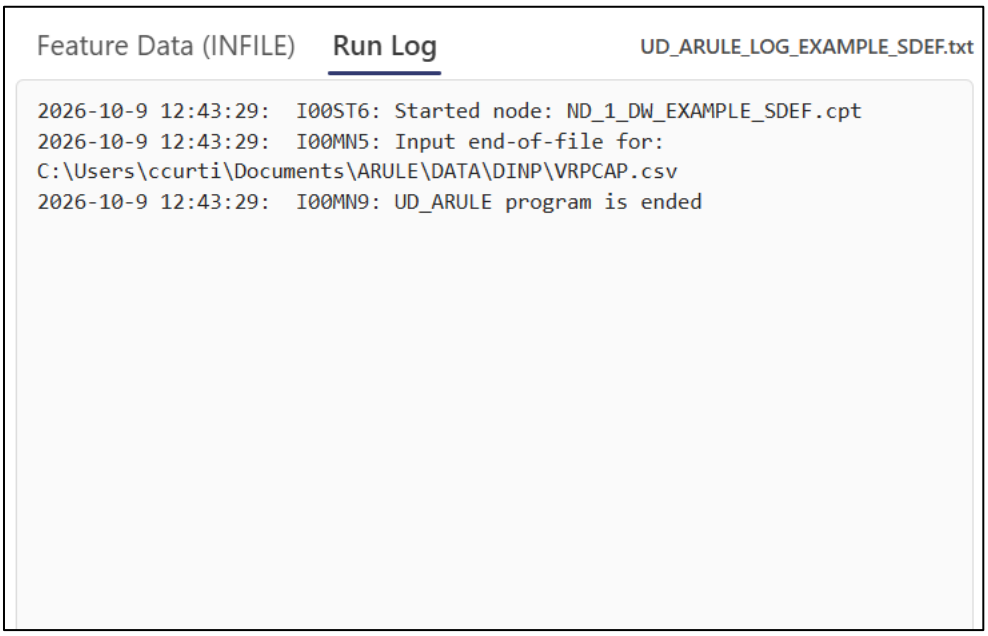


Figure 158 – Example message for successfully processing of the input data by ARULE™

Run-log review. Check for invalid records, path errors, missing NDEF associations, unsupported values, incomplete processing, and output-write failures. Preserve the log with the analysis record.

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7.6 Reviewing Prognostic Outputs

After successful processing, select the output associated with the NDEF under review. If the SDEF contains multiple NDEF files, review each output separately and confirm that the displayed dataset, configuration, and node name match. Figure 159 shows representative ARULE™ plots.

Interpret the outputs as a connected processing chain. A plausible RUL curve does not validate an unsuitable Feature Data input or an incorrect failure mapping.

- **Feature Data (FD):** The application-specific degradation feature supplied to ARULE™, not necessarily raw Condition-Based Data. Review its trend, noise, discontinuities, and coordinate basis.
- **Fault-to-Failure Progression (FFP):** The normalized progression produced from the configured baseline, normalization, and noise-margin settings.
- **Degradation Progression Signature (DPS):** The transformed progression used to reduce signature-model complexity. Review whether the configured transformation produces the intended behavior.
- **Functional Failure Signature (FFS):** The normalized signature referenced to the configured functional-failure endpoint. Verify that the endpoint corresponds to the engineering failure criterion.
- **State of Health (SOH):** The ARULE™ prognostic-health quantity calculated from the configured degradation and failure model. It is not automatically equivalent to a directly measured parameter such as battery capacity retention.
- **Remaining Useful Life (RUL):** The estimated distance from the current sample to functional failure. Its units follow the input coordinate basis unless a validated conversion is applied.
- **Prognostic Horizon (PH):** The estimated failure location or horizon represented in the same coordinate basis. Review convergence and stability in the context of data quality and configuration.

Do not accept results solely because the run completed. Compare the transformed signatures and outputs with known physics, historical observations, controlled test results, or other application evidence.

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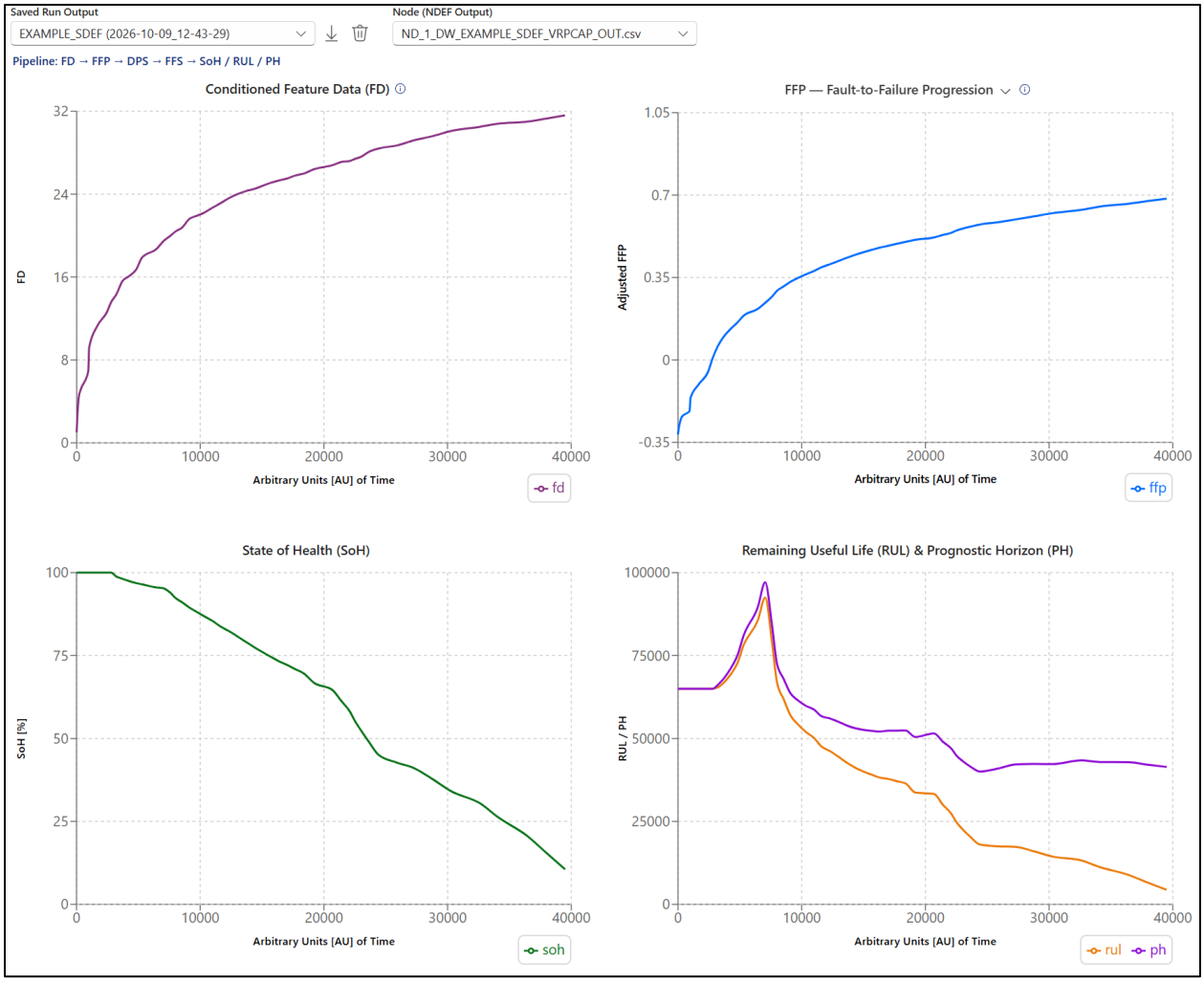


Figure 159 – Example Data Output plots: FD, FFS Progression, along with SoH, RUL, and PH

7.7 Saving and Exporting Results

Preserve enough information to reproduce the analysis. Save or archive the input Feature Data, SDEF, every associated NDEF, run arguments, software revision, run log, output data, plots, and a brief record of the engineering basis for each nondefault parameter.

- Use **DATA\DINP** only for input data unless the installed application explicitly documents another purpose.
- Treat **DATA\DOUT** as the documented data-output directory.
- Treat **DATA\SAVE** as the documented default save/download directory.

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7.8 Troubleshooting and Best Practices

Issue	Recommended checks
Input file is rejected	Check supported extension, delimiter, two-column structure, headers, blank records, missing values, numeric formatting, and monotonic coordinates.
Run cannot locate data or definitions	Check paths, file names, SDEF-to-NDEF associations, permissions, and whether files were moved after configuration.
SOH reaches zero too early	Review the physical failure definition, FDC, FDZ, FDNM, FDNV, FFPFAIL, data direction, outliers, and smoothing. Do not correct the symptom by changing one value without rechecking the full mapping.
RUL or PH is unrealistic or unstable	Confirm coordinate basis, PITTFF, PIFFSMOD, transformation behavior, sample spacing, noise, operating-condition changes, and whether the feature is prognostically suitable.
Results resemble an unrelated example	Confirm that example NDEF values were not copied into the new analysis without an engineering basis.
Logs or outputs are missing or incomplete	Confirm successful completion for every NDEF, output-write permissions, available storage, selected run, and the documented output and save locations.

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8. PHM Integration within CellNostix

8.1 Overview

The preceding sections of this User Guide describe the Adaptive Remaining Useful Life Estimator™ (ARULE™), its signature-processing methodology, prognostic outputs, System Definition (SDEF) and Node Definition (NDEF) configurations, and procedures for analyzing application-specific Feature Data (FD).

This section extends those concepts to the broader CellNostix™ Prognostics and Health Management (PHM) architecture and demonstrates how historical battery-degradation data originating from the CellNostix Data Acquisition (DAQ) application and CellSage™ modeling environment can be prepared, configured, and processed using ARULE.

Within CellNostix, ARULE provides the prognostic-analysis capability by processing qualified degradation-related Feature Data through the Adaptive Predictive Kernel (APK) to generate node-level prognostic information, including State of Health (SoH), Remaining Useful Life (RUL), Prognostic Horizon (PH), and associated execution/status information. These outputs support the broader health-assessment and decision-making processes implemented by the host application.

As illustrated in Figure 160, the CellNostix PHM architecture can be represented by five interconnected functional stages:

Sense and Acquire → Model and Compare → Analyze and Prognose → Advise → Act

- **Sense and Acquire** — Collect battery measurements, cycle information, test configurations, and operating conditions through the CellNostix DAQ application.
- **Model and Compare** — Use CellSage and other modeling capabilities to characterize expected battery behavior, support model-to-test comparisons, and prepare degradation-related features or Condition Indicators.
- **Analyze and Prognose** — Process measured, modeled, or derived Feature Data to evaluate degradation progression. ARULE provides the adaptive prognostic-estimation capability within this stage.
- **Advise** — Evaluate measured conditions and prognostic information using configured alert rules, thresholds, persistence criteria, and severity logic to generate actionable advisory information.
- **Act** — Support notifications, Health Management workflows, issue tracking, engineering review, and subsequent operational or maintenance decisions.

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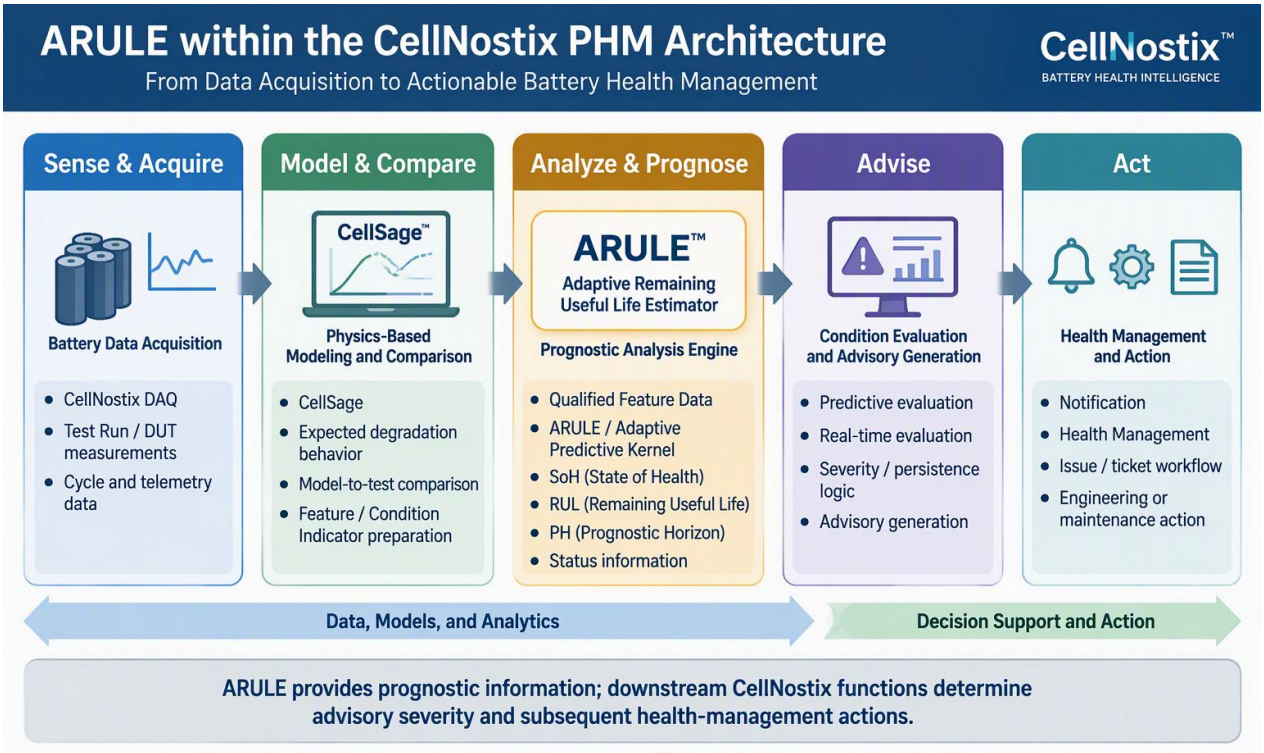


Figure 160. ARULE Within the CellNostix PHM Architecture

This architecture separates the responsibilities of data acquisition, modeling, feature preparation, prognostic estimation, and health-management decision-making. ARULE produces prognostic information but does not independently determine advisory severity, generate maintenance decisions, or define subsequent operational actions. Those responsibilities remain with the applicable CellNostix functions.

Two complementary modes of ARULE operation are supported within this framework:

1. Standalone engineering analysis and configuration development — Engineers prepare and evaluate historical or simulated Feature Data, develop SDEF/NDEF configurations, examine degradation-signature transformations, execute historical data replays, and interpret prognostic results.
2. Integrated prognostic execution within CellNostix — Qualified Feature Data and associated configurations may be supplied automatically to ARULE, enabling sequential prognostic estimation and the use of SoH, RUL, PH, and status information within automated health-monitoring and advisory workflows.

Figure 161 illustrates the relationship between these approaches. The upper portion represents the independent analysis of historical battery-degradation trajectories, including measured capacity loss from CellNostix DAQ and modeled Best, Average, and Worst capacity-loss trajectories from CellSage. The lower portion illustrates how an integrated CellNostix workflow may use an appropriately associated measured/model

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pair to derive a model-relative Condition Indicator, such as Capacity Residual Error (CRE), for subsequent ARULE prognostic estimation and advisory evaluation.

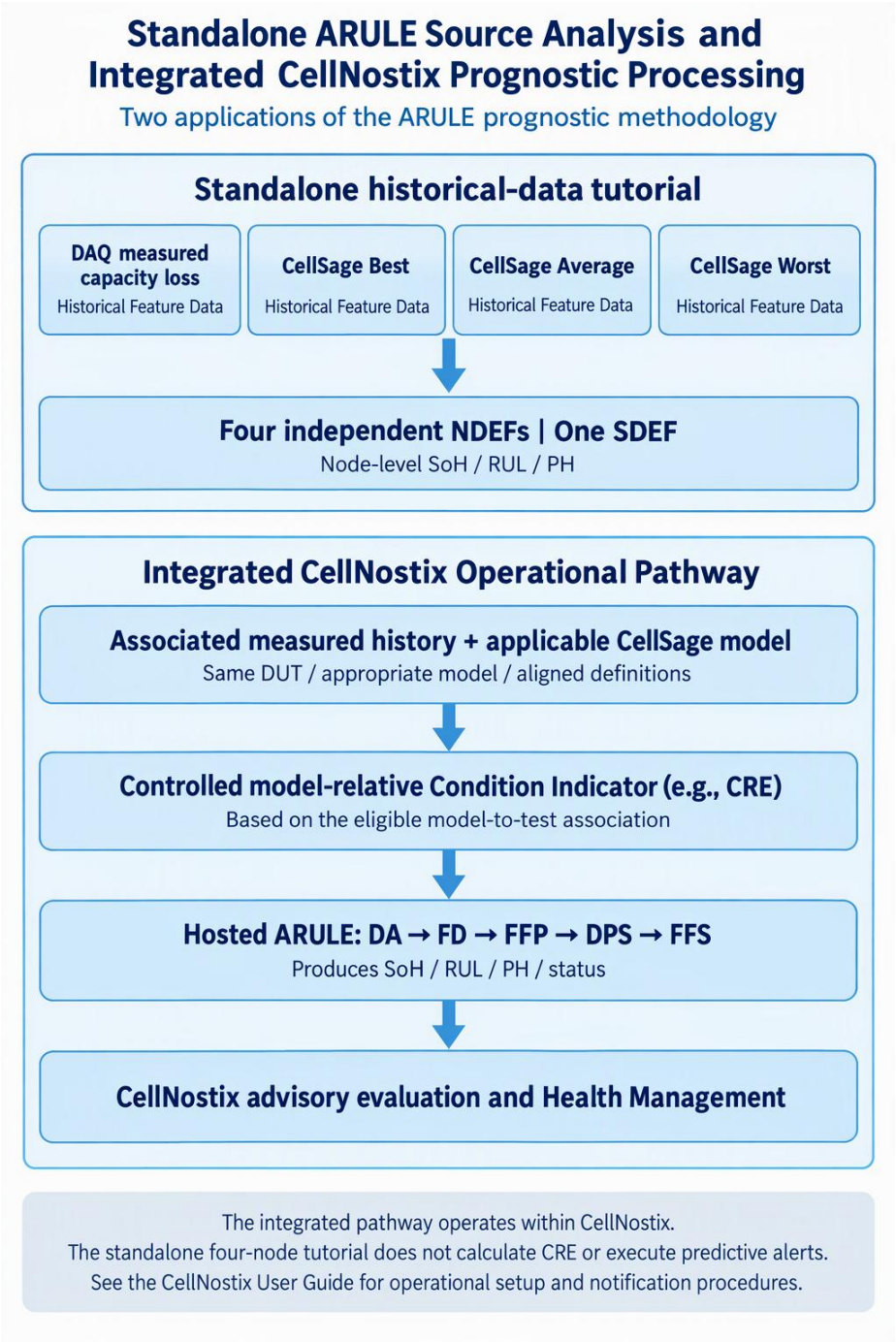


Figure 161. Standalone ARULE Source Analysis and Integrated CellNostix Prognostic Processing

The four-node worked example presented in this section focuses exclusively on standalone historical-data analysis. Each measured or modeled degradation trajectory is

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processed independently using its associated NDEF within a common SDEF. The example demonstrates Feature Data preparation, configuration, execution, result interpretation, and reproducibility; it does not establish a validated model-to-test relationship or execute the automated CRE-based predictive workflow illustrated conceptually in Figure 161.

The integrated workflow is included to demonstrate how the same ARULE prognostic methodology can support broader CellNostix applications. Detailed instructions for predictive-model association, Condition Indicator generation, advisory-rule configuration, alert thresholds, notifications, and Health Management functions are provided separately in the applicable CellNostix User Guide.

8.2 Purpose and Scope

The purpose of this worked example is to demonstrate how ARULE processes historical degradation-related Feature Data from measured and modeled sources using a common prognostic-analysis methodology. The example illustrates the preparation of input data, development of Node Definitions (NDEFs), configuration of a multi-node System Definition (SDEF), execution of the ARULE prognostic engine, and interpretation of the resulting degradation signatures and prognostic estimates.

The example uses four independently prepared battery-capacity degradation trajectories originating from two CellNostix applications: a historical cycle-life dataset acquired through CellNostix Data Acquisition (DAQ) and three simulated capacity-loss trajectories generated through CellSage String Analysis.

Each dataset is prepared as a two-column ARULE input file containing an ordered analysis coordinate and the corresponding Feature Data amplitude. For this example, all four nodes use cycle number as the analysis coordinate and capacity loss (%) as the Feature Data representation.

The four analysis nodes are organized within a common SDEF, as summarized below.

Node	Source system	Feature Data	Analysis coordinate
1	CellNostix DAQ	Measured Capacity Loss (%)	Cycle
2	CellSage String Analysis	Best-Case String Capacity Loss (%)	Cycle
3	CellSage String Analysis	Average-Case String Capacity Loss (%)	Cycle
4	CellSage String Analysis	Worst-Case String Capacity Loss (%)	Cycle

Although the four nodes share a common coordinate basis and capacity-loss representation, each NDEF is configured and processed independently. This demonstrates ARULE's ability to evaluate multiple degradation signatures within one

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system-level analysis while maintaining separate input associations, signature-processing configurations, and prognostic estimates.

The principal learning objectives of this example are to:

1. **Prepare historical Feature Data** — Identify the relevant degradation quantities from the original DAQ and CellSage exports and convert them into the two-column input format required by ARULE.
2. **Define the analysis configuration** — Establish the Feature Data meaning, coordinate basis, nominal behavior, normalization, and functional-failure mapping for each node.
3. **Configure a multi-node system** — Create or review four independent NDEFs and associate them with a common SDEF.
4. **Execute and interpret ARULE results** — Process the historical datasets and examine the resulting Feature Data transformations, State of Health (SoH), Remaining Useful Life (RUL), Prognostic Horizon (PH), and associated status information.
5. **Establish a reproducible analysis record** — Retain the source exports, prepared input files, configuration definitions, execution logs, and output data necessary to reconstruct and evaluate the analysis.

Figure 162 summarizes the relationship between the historical source datasets, prepared Feature Data inputs, individual NDEF configurations, and the common SDEF used for ARULE execution.

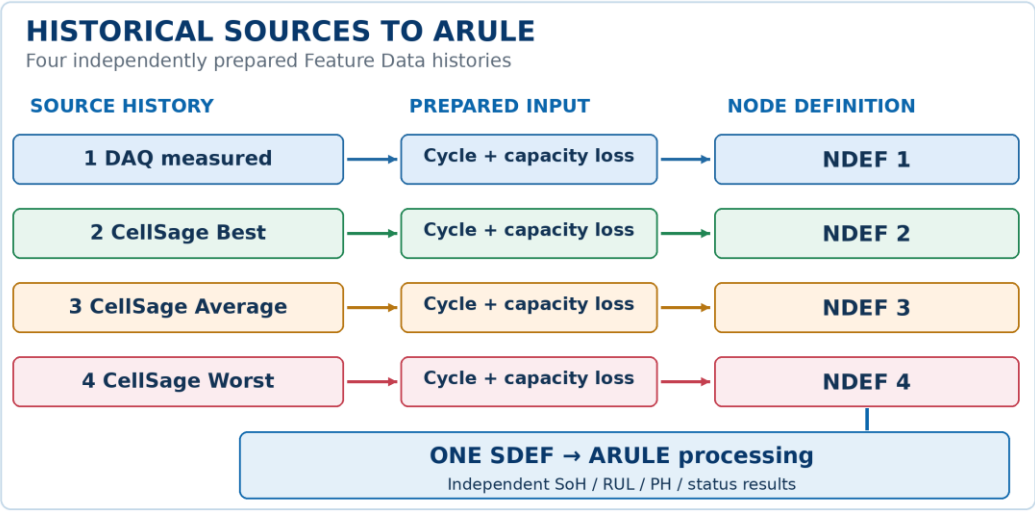


Figure 162. Historical Battery Source Data and Four-Node ARULE Analysis Workflow

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Scope of the Worked Example

The four degradation trajectories are processed as independent historical analysis nodes. Their inclusion within the same SDEF does not establish that the measured DAQ dataset and CellSage simulations represent the same battery, operating conditions, or a validated model-to-test comparison.

Accordingly, this example does not calculate Capacity Residual Error (CRE), evaluate CellSage prediction accuracy against the measured DAQ history, or demonstrate automated predictive-alert generation. Those capabilities are associated with the broader integrated CellNostix workflow introduced in Section 8.1 and discussed briefly in Section 8.7.

The focus of the following subsections is the practical application of ARULE to previously acquired or simulated degradation histories**,** including configuration, execution, scientific interpretation, and reproducibility. Successful execution demonstrates that the prepared histories can be processed under their respective configurations; it does not, by itself, establish independent prognostic accuracy or physical battery end-of-life.

8.3 Prerequisites

Before conducting an ARULE historical-data analysis, the user should establish the source-data identity, engineering meaning of each Feature Data representation, analysis coordinate, applicable configuration parameters, and software execution environment. These elements provide the basis for preparing the input data, interpreting the resulting prognostic estimates, and reproducing the analysis.

The four-node example uses historical battery-capacity data obtained from two CellNostix applications: the Data Acquisition (DAQ) application and the CellSage™ modeling environment. The following subsections describe the source datasets, their preparation, and the corresponding ARULE input requirements.

8.3.1 Execution and Software Baseline

The example uses the standalone ARULE application to process four historical degradation trajectories organized under a common System Definition (SDEF).

The retained example configuration is identified as:

CNX_4-Node_Battery_Capacity

For reproducibility, the analysis record should retain the following information:

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- **Source data:** Original DAQ and CellSage exports, including their associated Test Run or simulation identities.
- **Prepared inputs:** Four two-column ARULE input files containing the selected analysis coordinates and degradation-related feature amplitudes.
- **Configuration:** The SDEF, four associated NDEFs, and the applicable parameter settings.
- **Software baseline:** ARULE GUI and prognostic-engine version information, where available, together with relevant CellNostix and CellSage source-software identification.
- **Execution evidence:** Run settings, execution logs, node-level output files, and representative screenshots.
- **Data preparation:** Source-field selections, coordinate transformations, interpolation methods, and any other preprocessing applied before ARULE execution.

The reference package includes a recorded execution of the four-node configuration, with associated input files, output CSVs, and an execution log. The log records initialization and processing of the four source nodes through their respective input histories.

These artifacts support reconstruction and review of the example. The detailed execution evidence and reproducibility considerations are discussed in Section 8.6.

8.3.2 CellNostix DAQ Historical Capacity-Loss Data

The first dataset originates from a historical **CellNostix DAQ Cycle-Life Test Run**, which contains battery measurements, cycling information, test configuration, and exported data records.

The DAQ application provides a means of reviewing the historical battery response and obtaining the corresponding source data for further analysis. Figure 163 illustrates the DAQ output environment alongside the CellSage modeling output used in this example.

The measured degradation feature is obtained from the following DAQ cycle-summary export:

- **Cycle-Life_ECL_2025-02-01_CS.csv**

The source field capacity_fade is used to represent measured capacity loss over successive cycles. The corresponding ARULE input file is:

- **CNX_DAQ_CapLoss.csv**

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The prepared file contains cycle number in the first column and the exported capacity-fade value in the second column. The original DAQ export should be retained because it establishes the source of the measured Feature Data and preserves the capacity-reference calculation implemented by the DAQ application. The capacity_fade values should not be recalculated using an assumed Beginning-of-Life (BOL) reference unless a separate engineering procedure explicitly requires that transformation.

The measured capacity-loss history contains point-to-point variability, including some negative values during the earlier cycles. These observations are retained in the prepared input rather than being artificially clamped, smoothed, or forced into a monotonically increasing degradation trajectory. This preserves the recorded feature behavior and allows ARULE's response to the measured data to be evaluated using the selected NDEF configuration. The significance of this variability is examined in Section 8.5.

8.3.3 CellSage Best, Average, and Worst Capacity-Loss Data

The remaining three datasets originate from a CellSage String Analysis simulation, which generates modeled battery-capacity degradation trajectories over the selected simulation conditions and operating horizon.

The source report contains three modeled String Capacity Loss representations:

- **Best Case** — the modeled trajectory identified as having the least capacity loss among the reported cases.
- **Average Case** — the intermediate modeled capacity-loss trajectory.
- **Worst Case** — the modeled trajectory identified as having the greatest capacity loss among the reported cases.

These labels describe the CellSage model scenarios and should not automatically be interpreted as statistical confidence intervals or measured battery histories.

The source export used for this example is:

- **StringAnalysis-SingleT.csv**

Although the CellSage graphical interface displays the simulated capacity-loss trajectories against aging time in weeks, the exported report also provides a cycle-coordinate field and the corresponding capacity-loss values. To prepare the ARULE inputs, the selected Best, Average, and Worst capacity-loss trajectories were extracted from the source report and linearly interpolated onto integer cycle coordinates from 1 through 400.

This produces three independent, cycle-based Feature Data histories that can be processed using separate ARULE NDEF configurations.

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The prepared files are:

- CNX_CS_Best_CapLoss.csv
- CNX_CS_Avg_CapLoss.csv
- CNX_CS_Worst_CapLoss.csv

The original CellSage report remains the source record for the model-generated trajectories. The selected source columns, cycle-coordinate values, and interpolation procedure should be retained so the prepared ARULE inputs can be reconstructed. These modeled histories are processed independently of the DAQ measured dataset. Their inclusion in the same SDEF does not establish that they represent the same battery or operating conditions.

8.3.4 Feature Data Definition and Analysis Coordinate

ARULE requires an ordered sequence of degradation-related observations, with each observation associated with an analysis coordinate and a corresponding feature amplitude.

For the standalone ARULE input format used in this example:

- **Column 1 — Analysis Coordinate:** The independent variable describing the progression of the observations.
- **Column 2 — Feature Amplitude:** The degradation-related quantity imported for subsequent ARULE conditioning and prognostic processing.

In this example, all four datasets use:

- **Analysis Coordinate:** Cycle Number
- **Feature Amplitude:** Capacity Loss (%)

The historical degradation signatures can therefore be represented conceptually as:

- **Capacity Loss = f(Cycle Number)**

Although both measured and modeled datasets use capacity loss as their feature representation, their underlying sources and engineering interpretations remain distinct.

The analysis coordinate also determines the interpretation of ARULE's coordinate-based prognostic outputs. Because the first column represents cycle number, Remaining Useful Life (RUL) is interpreted in cycles rather than elapsed time.

The current standalone ARULE graphical interface may display the legacy x-axis label *"Arbitrary Units [AU] of Time."* For the datasets presented here, that label should be interpreted according to the actual input-coordinate definition, which is cycle number.

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During execution, ARULE imports the supplied amplitude as Data Amplitude (DA) and applies the applicable NDEF conditioning parameters to produce Feature Data (FD) for subsequent signature processing. These quantities may be numerically equivalent when the configuration applies no multi-point averaging, but they remain distinct processing representations. Additional information about the analysis coordinate, Feature Data conditioning, and signature-processing sequence is provided in Sections 1, 2, and 7.

8.3.5 Prepared ARULE Input Files

The four prepared input files used in this example are summarized as follows:

Node	Prepared input file	Source-column / processing rule
1	CNX_DAQ_CapLoss.csv	DAQ Cycle number + exported capacity_fade (%)
2	CNX_CS_Best_CapLoss.csv	CellSage Cycle number + interpolated String Cap Loss (%) Best
3	CNX_CS_Avg_CapLoss.csv	CellSage Cycle + interpolated String Cap Loss (%) Avg
4	CNX_CS_Worst_CapLoss.csv	CellSage Cycle + interpolated String Cap Loss (%) Worst

Each prepared file contains 400 numeric observations corresponding to cycles 1 through 400, organized as two columns without a header row.

The original DAQ and CellSage exports are retained as source records, while the prepared two-column CSV files serve as the corresponding ARULE analysis inputs.

The distinction between these file types is important: source exports establish data provenance, while prepared input files establish the exact observations supplied to ARULE.

Figure 163 illustrates the relationship between the two source applications, their respective outputs, and the four prepared Feature Data files.

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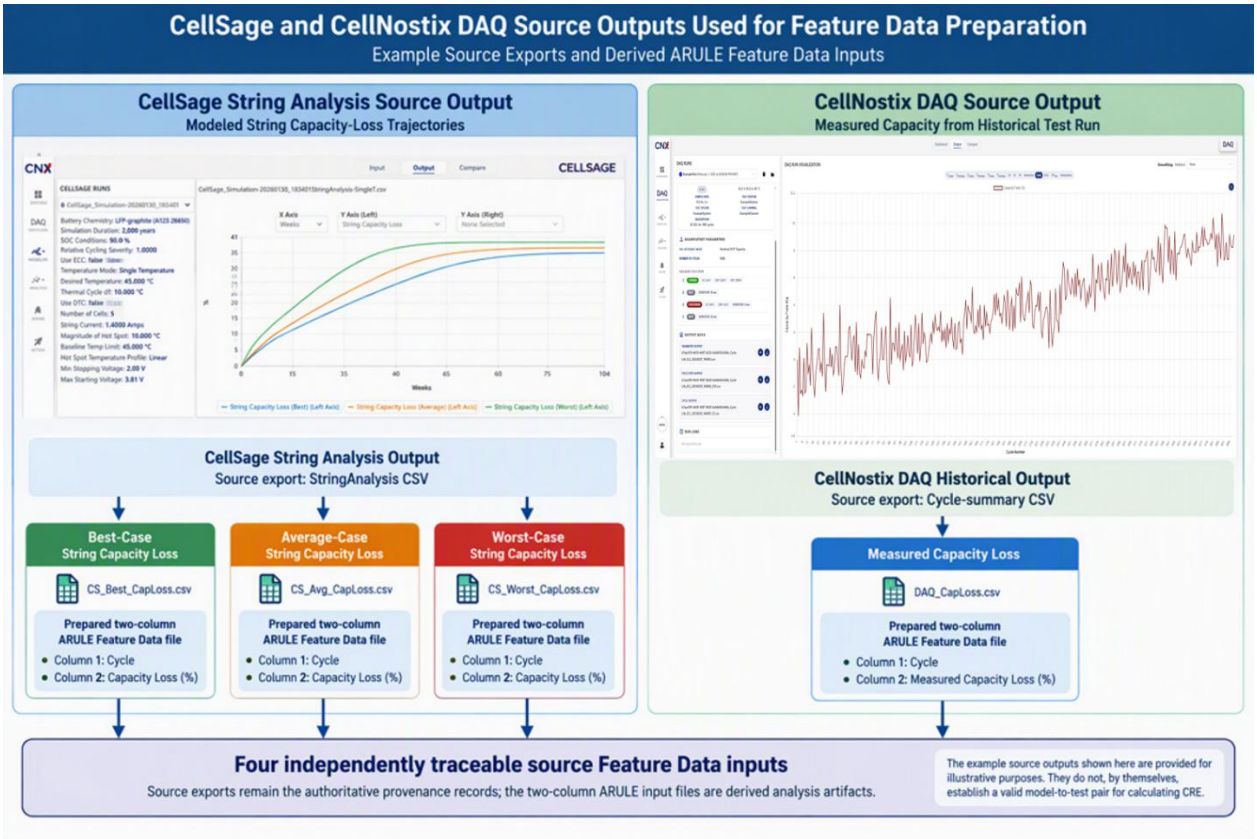


Figure 163. CellSage and CellNostix DAQ Source Outputs Used for Feature Data Preparation

With the four input files prepared, the next step is to review the associated NDEF configurations, organize them within the common SDEF, and execute the historical-data analysis as described in Section 8.4.

8.4 Multi-Node Battery-Capacity Source and Tutorial Analysis

The engineering question is how ARULE processes each of four historical degradation-related Feature Data series under its own configured nominal reference, transformation parameters, functional-failure definition, and prognostic initialization. The purpose is source-level prognostic evaluation, not measurement-versus-model accuracy testing.

The common SDEF is a container for four independent analysis nodes. Each NDEF associates a prepared input file with the parameters that control signature conditioning, normalization, and prognostic estimation. Use the same ordered cycle basis for all nodes, but do not infer that one node's failure mapping or estimated RUL is physically interchangeable with another node's result.

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8.4.1 Engineering Objective and Configuration Structure

The primary engineering objective is to evaluate how ARULE processes measured and modeled battery-capacity degradation histories using a consistent prognostic methodology while maintaining independent configuration and estimation for each analysis node.

The four-node example consists of:

- **Node 1 — DAQ Measured Capacity Loss:** Historical capacity-loss data acquired through the CellNostix DAQ application.
- **Node 2 — CellSage Best-Case Capacity Loss:** Modeled degradation trajectory representing the CellSage Best-Case scenario.
- **Node 3 — CellSage Average-Case Capacity Loss:** Modeled degradation trajectory representing the CellSage Average-Case scenario.
- **Node 4 — CellSage Worst-Case Capacity Loss:** Modeled degradation trajectory representing the CellSage Worst-Case scenario.

Each node processes an independent Feature Data sequence using the associated NDEF settings. Although all four datasets use cycle number as the analysis coordinate and capacity loss as the feature representation, their input histories, nominal-reference behavior, and resulting prognostic estimates are evaluated separately.

The common System Definition used in this example is:

- **CNX_4-Node_Battery_Capacity**

The SDEF organizes the four NDEFs into a single analysis configuration. It does not establish a physical relationship between the DAQ Test Run and the CellSage simulations or imply that the modeled trajectories represent predictions for the measured battery.

This example therefore demonstrates multi-node historical-data processing, rather than model-to-test validation.

Node	Source	Feature Data	Coordinate	Class	Engineering Meaning
1	CellNostix DAQ	Measured Capacity Loss	Cycle	Source	Observed historical degradation
2	CellSage String Analysis	Best-Case String Capacity Loss	Cycle	Source	Modeled Best-Case degradation
3	CellSage String Analysis	Average-Case String Capacity Loss	Cycle	Source	Modeled Average-Case degradation

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4	CellSage String Analysis	Worst-Case String Capacity Loss	Cycle	Source	Modeled Worst-Case degradation

8.4.2 Review the Four Source NDEF Configurations

Before executing ARULE, review the NDEF configuration associated with each source dataset. Each NDEF defines the input-data association and the parameters used for Feature Data conditioning, normalization, degradation-signature transformation, functional-failure mapping, and prognostic initialization.

The retained configuration parameters for the four-node example are summarized below.

Parameter	Node 1 DAQ (1)	Node 2 Best (2)	Node 3 Average (3)	Node 4 Worst (4)
FDC	1.00	1.00	1.00	1.00
FDZ	20.00	20.00	20.00	20.00
FDNM	5.00	5.00	5.00	5.00
FDCPTS	0	5	5	5
FDPTS	1	1	1	1
FDNV	1.000	1.000	1.000	1.000
FFPFAIL	100.00	100.00	100.00	100.00
PITTFF	1000.00	1000.00	1000.00	1000.00
PIFFSMOD	2 (Linear)	2 (Linear)	2 (Linear)	2 (Linear)

These values represent the recorded tutorial configuration and should not be interpreted as universally applicable parameter settings for battery prognostics.

Several configuration characteristics are important to understanding the example:

Feature Data conditioning (FDPTS):

All four nodes use FDPTS = 1, which does not apply multi-point averaging to the imported Data Amplitude (DA) sequence. Consequently, the conditioned Feature Data (FD) remain numerically equivalent to the imported amplitudes for the retained records. This preserves the point-to-point variation present in the measured DAQ history.

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Nominal-reference calculation (FDC and FDCPTS):

The DAQ node uses FDCPTS = 0, while the three CellSage nodes use FDCPTS = 5. Although all four configurations specify FDC = 1.00, the active nominal-reference behavior depends on the applicable parameter settings and ARULE implementation. Identical configured FDC values do not necessarily establish identical effective nominal references.

Normalization and degradation transformation (FDZ, FDNM, and FDNV):

These parameters influence how Feature Data amplitudes are transformed into the normalized degradation representations used by ARULE. Their engineering interpretation should be considered together with the selected nominal reference and feature direction.

Functional-failure mapping (FFPFAIL):

The configured FFPFAIL value establishes the failure endpoint within the applicable ARULE signature representation. It should not be interpreted directly as a physical capacity-loss percentage without evaluating the associated normalization and transformation.

Prognostic initialization (PITTFF and PIFFSMOD):

All four nodes use PITTFF = 1000 and the linear failure-signature model, PIFFSMOD = 2. These are the retained initialization and model-selection settings for this example, not independently established estimates of the actual failure life of the measured or modeled battery trajectories.

For the DAQ node, the retained transformation implies an input Feature Data endpoint of approximately 21.05 capacity-loss percentage points when FFP reaches 1. This is a configured feature-domain endpoint rather than an independently established physical battery end-of-life criterion.

The numerical relationship is demonstrated in Section 8.5.2. General guidance for selecting and evaluating NDEF parameters is provided in Section 7.

8.4.3 Assemble and Review the Four-Node SDEF

After reviewing the four NDEFs, confirm that they are correctly associated with the common System Definition.

The SDEF used in this example is identified as:

- **CNX_4-Node_Battery_Capacity**

Its four node associations are:

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```
NDNUMID = 1;
NDFNAME = 'CNX_DAQ-CapLoss';

NDNUMID = 2;
NDFNAME = 'CNX_CS-Best';

NDNUMID = 3;
NDFNAME = 'CNX_CS-Avg';

NDNUMID = 4;
NDFNAME = 'CNX_CS-Worst';

ENDDEF = -9;
```

Each NDNUMID identifies an analysis node, while the corresponding NDFNAME identifies the Node Definition containing its input association and processing parameters.

Figure 164 illustrates the four-node SDEF structure and the relationships between the NDEFs and prepared Feature Data files.

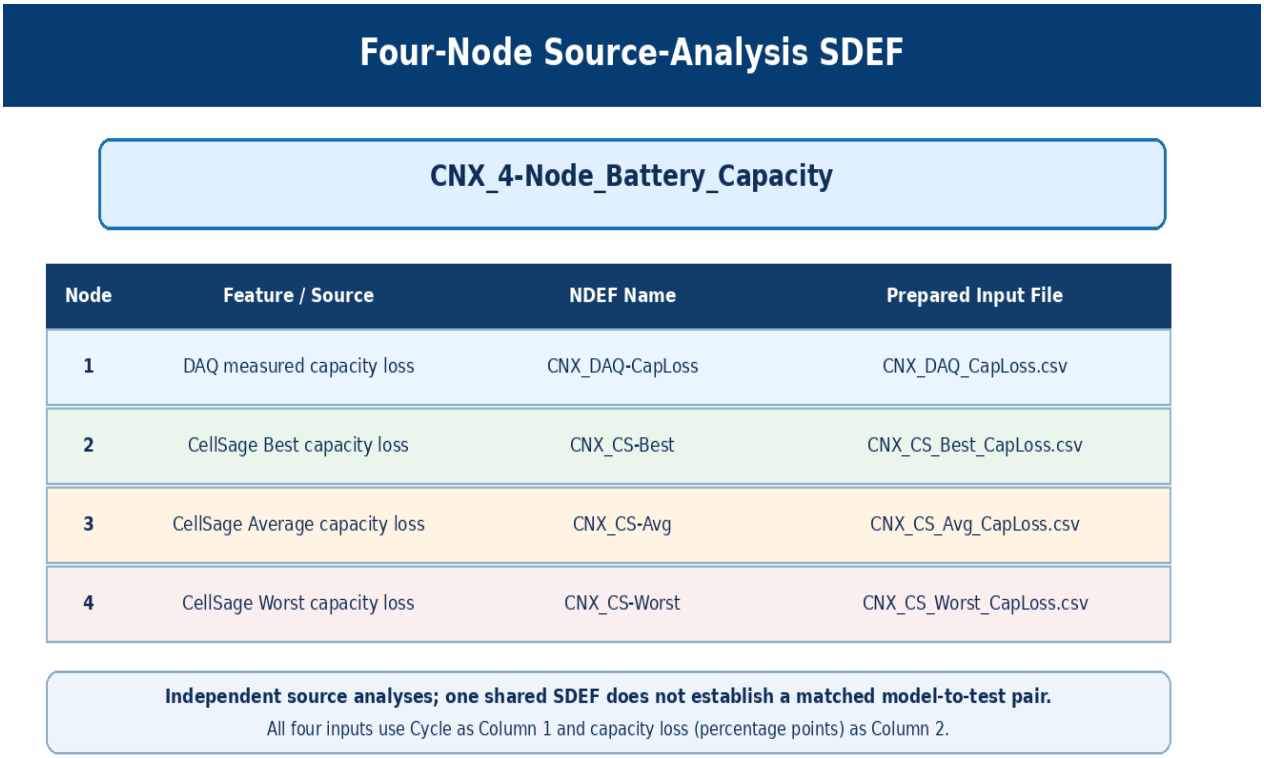


Figure 164. Four-Node Source-Analysis SDEF and Prepared Input Associations

Before proceeding, confirm that:

1. The SDEF contains exactly four node associations.
2. Each NDEF is assigned to the intended node number.

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3. The NDEF identifiers match the corresponding configuration files.
4. Each NDEF references the correct prepared Feature Data file.
5. The selected configuration parameters match the retained analysis record.

These checks help prevent configuration or input-association errors before execution.

8.4.4 Load and Inspect Each Input

With the SDEF configured, open the standalone ARULE application and select the four-node System Definition.

Use the SDEF/NDEF Editor to inspect each node's input association, configuration parameters, and Feature Data plot.

The input associations used in this example are:

Node	NDEF	Prepared Input File
1	CNX_DAQ-CapLoss	CNX_DAQ_CapLoss.csv
2	CNX_CS-Best	CNX_CS_Best_CapLoss.csv
3	CNX_CS-Avg	CNX_CS_Avg_CapLoss.csv
4	CNX_CS-Worst	CNX_CS_Worst_CapLoss.csv

For each node, confirm that the input data have been loaded correctly and that the displayed Feature Data signature corresponds to the selected source.

The following characteristics should be observed.

Node 1 — DAQ Measured Capacity Loss

The DAQ Feature Data plot should display the historical capacity-loss measurements over the 400-cycle analysis interval. Unlike the modeled trajectories, the measured series exhibits appreciable point-to-point variability superimposed on an overall increasing degradation trend.

Some observations may fall below zero because the exported capacity-fade calculation uses a capacity reference that can be exceeded by individual measurements. These observations should be retained unless a separate engineering evaluation establishes an appropriate preprocessing procedure.

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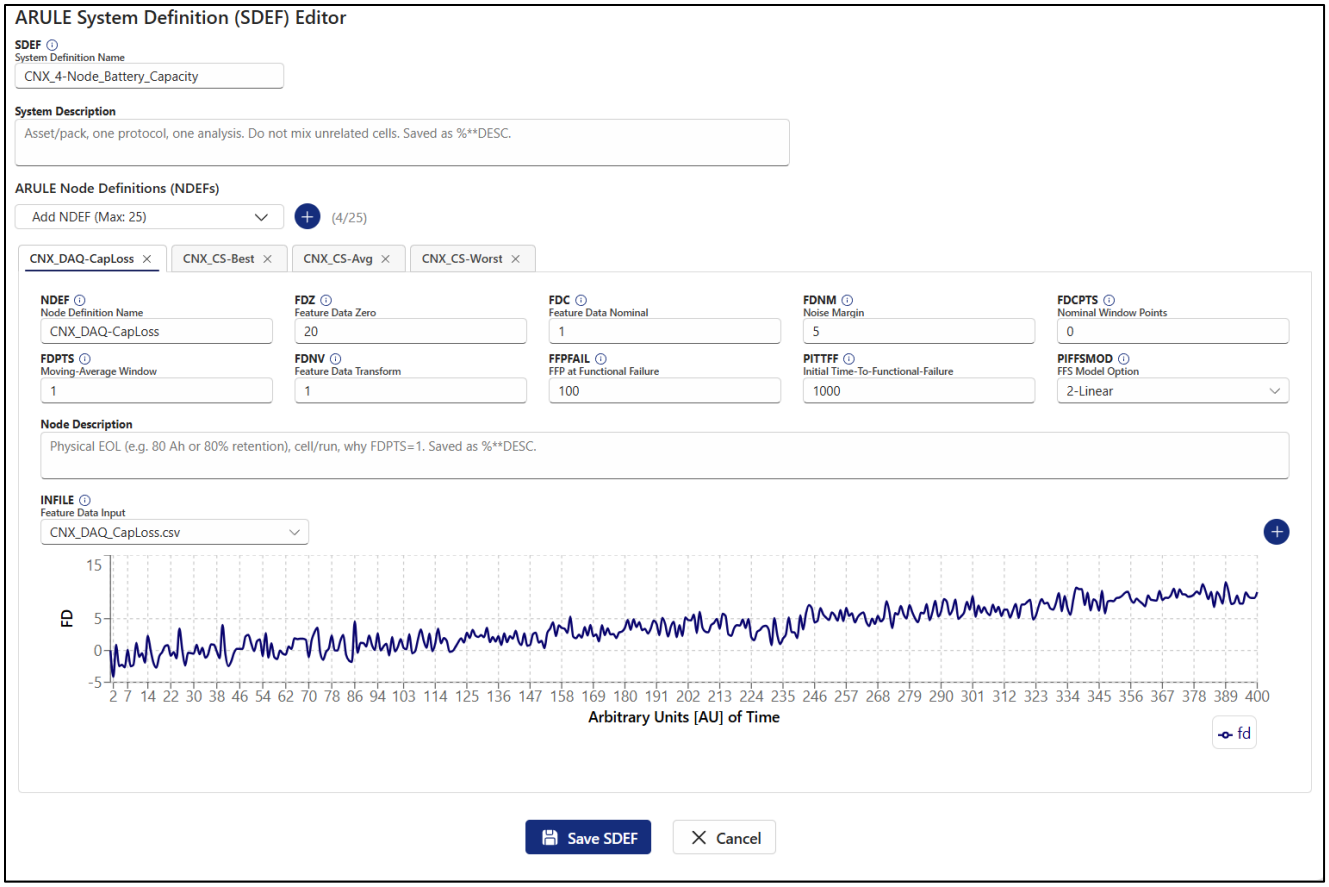


Figure 165. Node 1 — DAQ Node Definition and Historical Capacity-Loss Feature Data

Node 2 — CellSage Best-Case Capacity Loss

The Best-Case dataset should display a smooth, increasing modeled capacity-loss trajectory over the cycle-coordinate interval. This scenario exhibits the least capacity loss among the three CellSage model histories used in the example.

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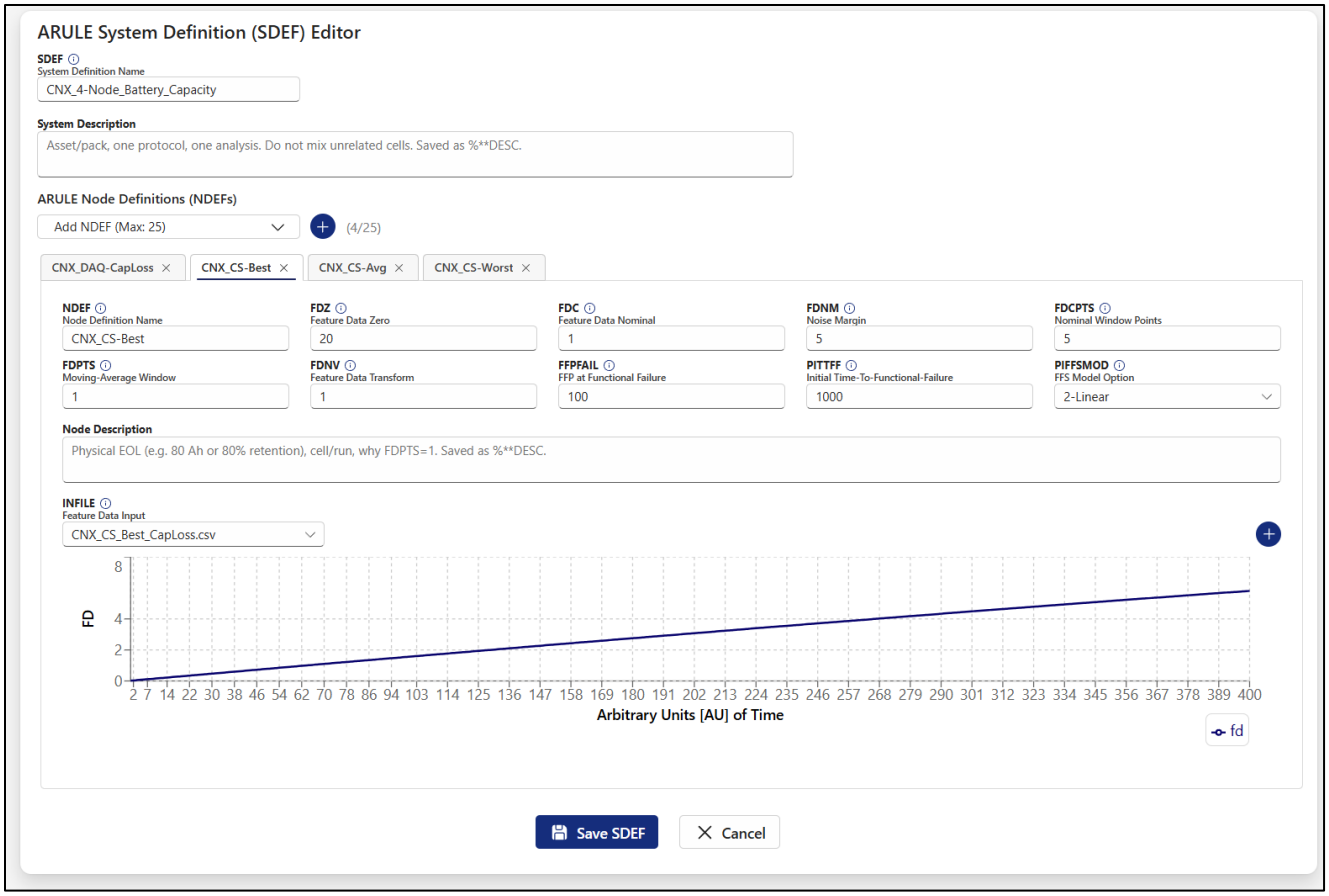


Figure 166. Node 2 — CellSage Best-Case Node Definition and Feature Data

Node 3 — CellSage Average-Case Capacity Loss

The Average-Case dataset should display an intermediate modeled degradation trajectory, with greater capacity loss than the Best-Case representation and less than the Worst-Case representation over the illustrated interval.

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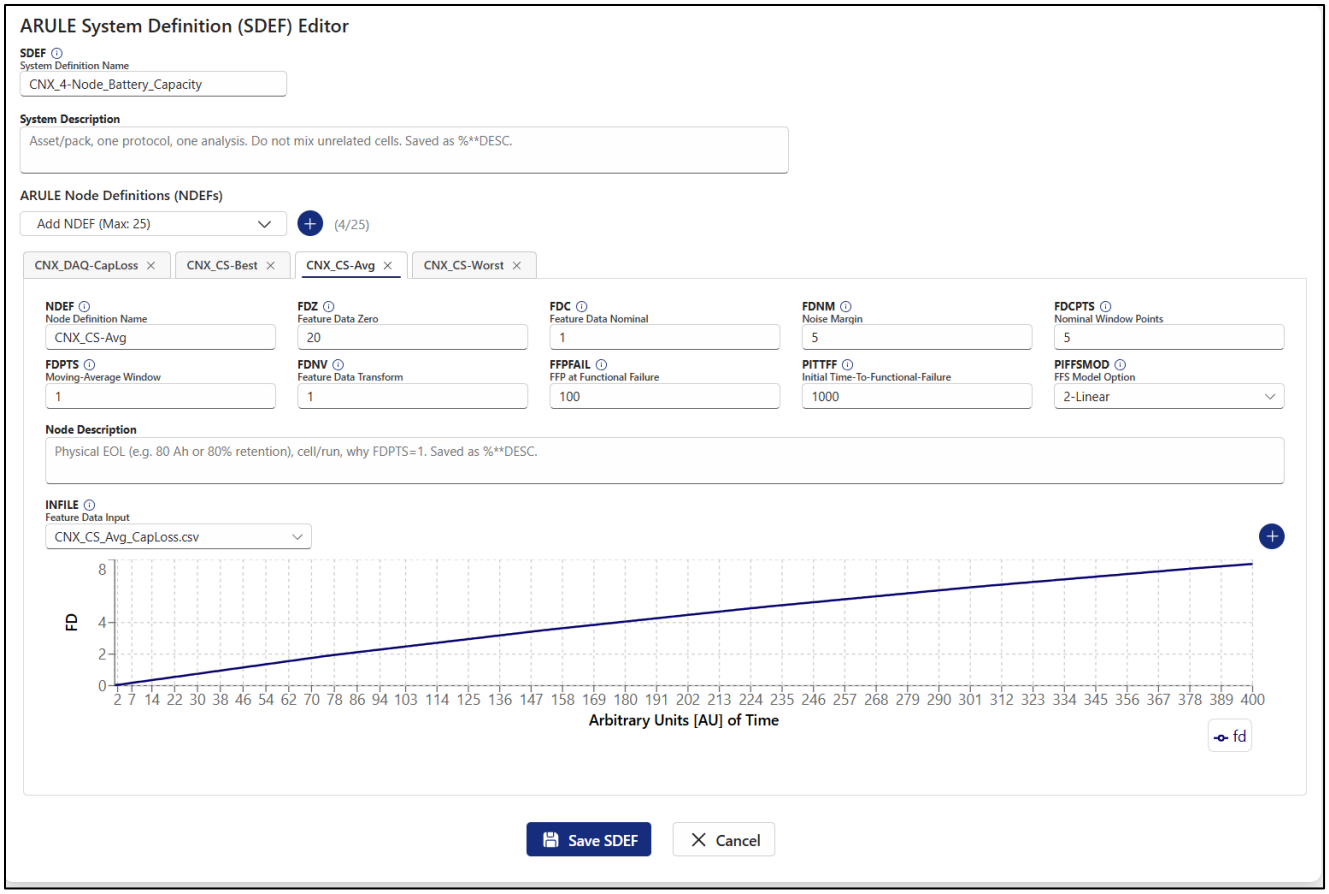


Figure 167. Node 3 — CellSage Average-Case Node Definition and Feature Data

Node 4 — CellSage Worst-Case Capacity Loss

The Worst-Case dataset should display the greatest modeled capacity loss among the three CellSage scenarios. Its degradation progression is therefore more advanced over the same cycle-coordinate interval.

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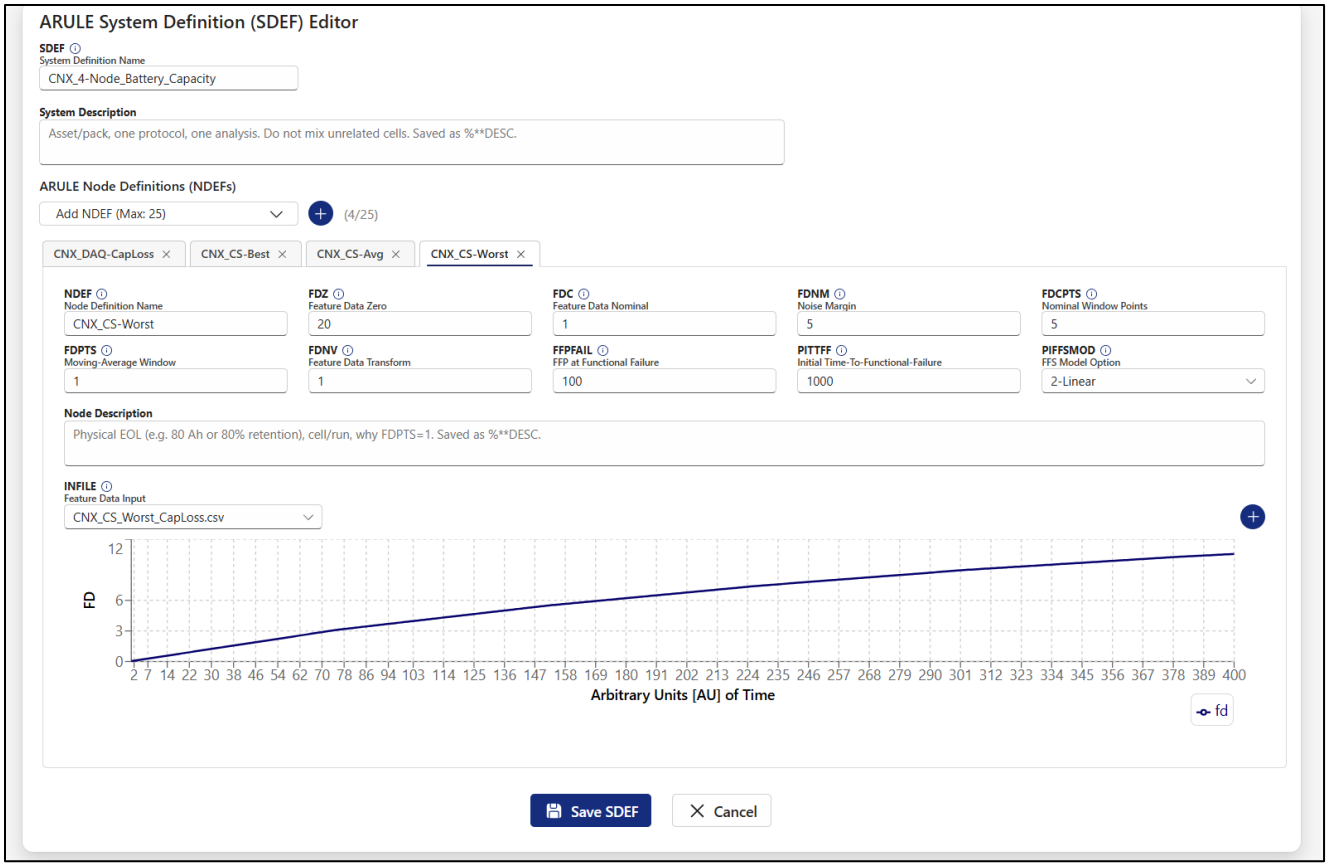


Figure 168. Node 4 — CellSage Worst-Case Node Definition and Feature Data

The four input plots demonstrate that ARULE can process different degradation histories using a common analysis-coordinate basis while maintaining independent node-level configurations.

At this stage, the objective is to verify input association and Feature Data representation, not to determine whether the resulting prognostic estimates are accurate.

8.4.5 Execute the Four-Node Analysis

After verifying the SDEF, NDEF parameters, and Feature Data input associations, execute the configured historical-data analysis using the standalone ARULE application.

During execution, ARULE processes the observations associated with each node through the configured signature-processing and adaptive prognostic-estimation sequence.

For each node, the processing chain includes:

- **DA → FD → FFP → DPS → FFS → Prognostic Estimation**

The resulting outputs include State of Health (SoH), Remaining Useful Life (RUL), Prognostic Horizon (PH), and associated status information.

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The retained execution record for CNX_4-Node_Battery_Capacity documents startup of Nodes 1–4, processing of their associated input histories through end-of-file, and termination of the UD_ARULE program.

These entries provide evidence that the four-node configuration executed and processed the supplied historical data. The associated output CSV files provide the numerical record required to examine the resulting degradation signatures and prognostic estimates.

After execution:

1. Review the execution log for node initialization, input processing, and any reported errors or warnings.
2. Confirm that output data were generated for each of the four nodes.
3. Select each node in the ARULE output interface.
4. Review the Feature Data and transformed degradation signatures.
5. Examine the resulting SoH, RUL, PH, and status information.
6. Compare the observed results with the retained output CSV files.

The numerical results and their engineering interpretation are presented in Section 8.5.

Successful program execution demonstrates that the configured analysis processed the supplied input histories. It does not, by itself, establish the validity of the functional-failure criteria, the physical meaning of the prognostic estimates, or independent prediction accuracy.

8.4.6 Save the Analysis Record

After completing the historical-data execution, retain the files and configuration information necessary to reconstruct the analysis.

The retained analysis package should include:

- Original CellNostix DAQ and CellSage source exports.
- Four prepared ARULE Feature Data CSV files.
- The four NDEF configuration files.
- The common SDEF file.
- ARULE software and engine identification, where available.
- The execution log.
- The four node-level output CSV files.
- Representative input, configuration, and output screenshots.

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- Relevant data-preparation procedures and engineering assumptions.

The original source exports should remain distinguishable from the prepared ARULE input files, and any changes to the data or configuration should be documented.

Section 8.6 describes the reproducibility and evidence-retention requirements in greater detail, including the comparison of corresponding source-node outputs from the retained execution records.

The completed four-node analysis provides the basis for evaluating how the different measured and modeled degradation signatures influence ARULE's signature-processing behavior and prognostic estimates, as discussed in Section 8.5.

8.5 Reviewing and Interpreting ARULE Results

Following execution of the four-node System Definition, ARULE generates individual output records containing the processed Feature Data, intermediate degradation-signature representations, prognostic estimates, and associated status information for each node.

This section examines those outputs to demonstrate how the ARULE signature-processing and adaptive prognostic-estimation methodology responds to different historical degradation trajectories.

The analysis focuses on two complementary cases: the variability present in measured battery-capacity degradation and the comparatively smooth degradation behavior generated by the CellSage simulations.

Figures 169 through 172 illustrate the ARULE output displays for the four analysis nodes.

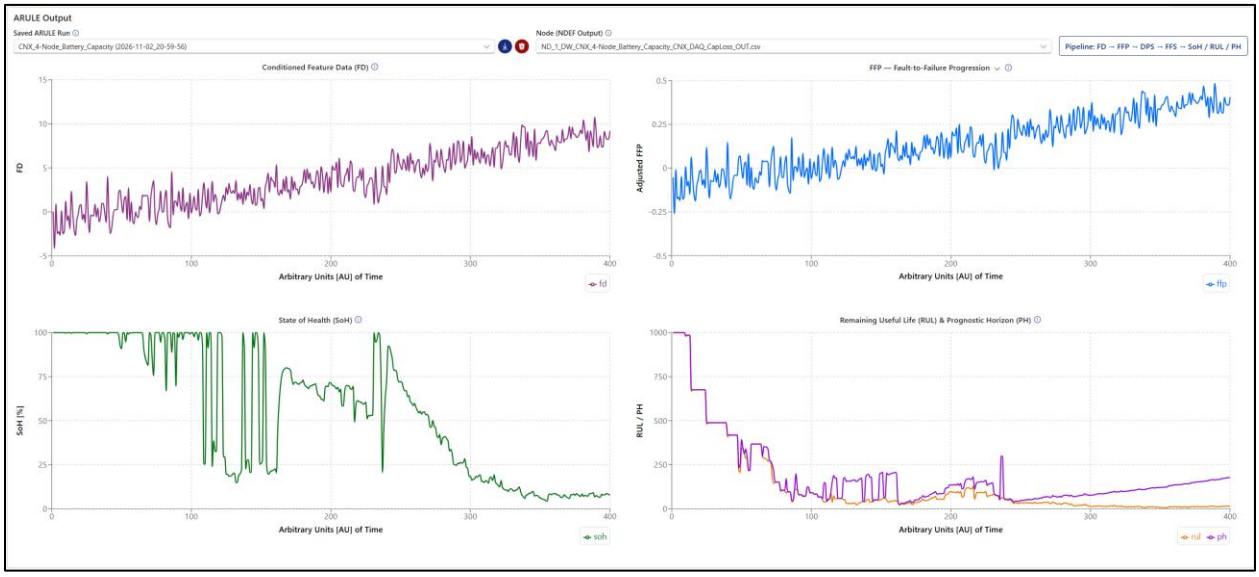


Figure 169. Node 1: DAQ Measured Capacity-Loss ARULE Outputs

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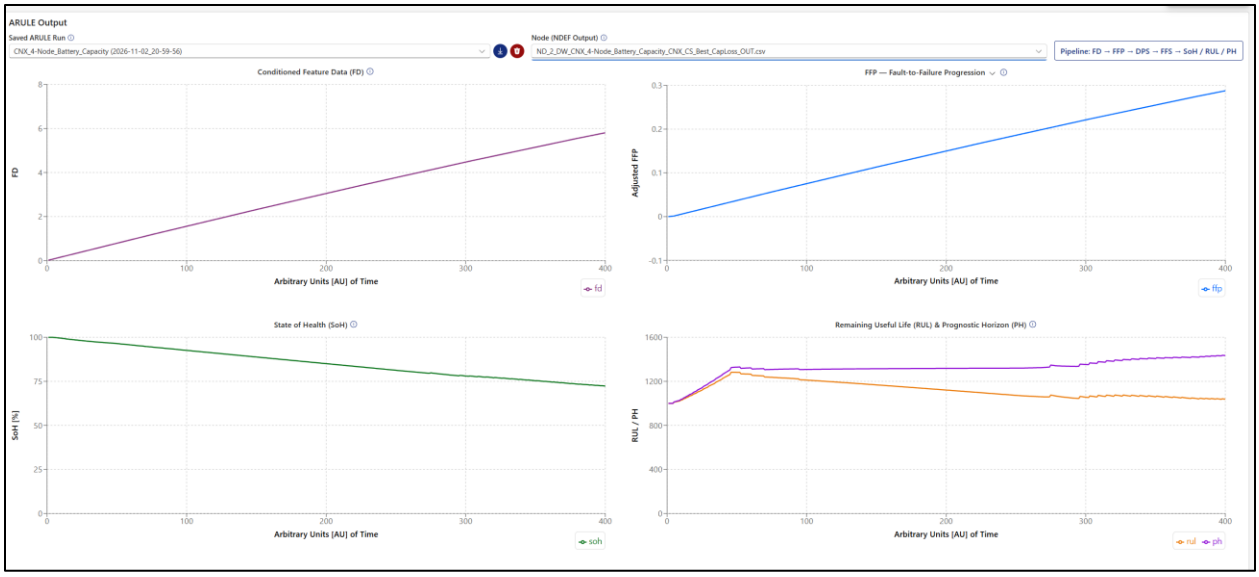


Figure 170. Node 2: CellSage Best-Case ARULE Outputs

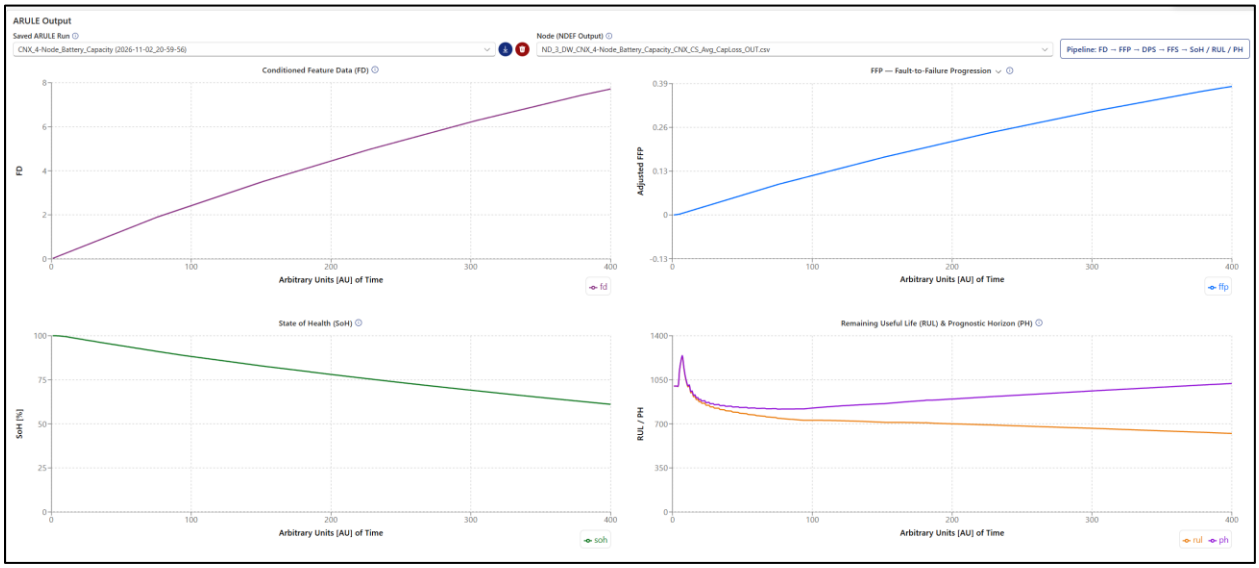


Figure 171. Node 3: CellSage Average-Case ARULE Outputs

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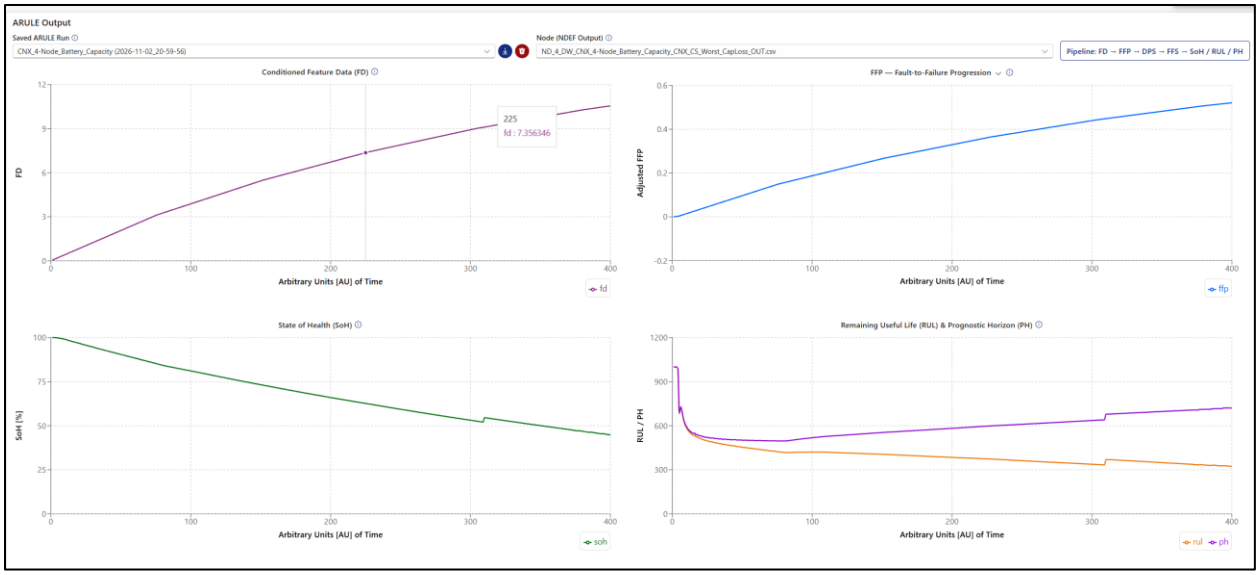


Figure 172. Node 4: CellSage Worst-Case ARULE Outputs

These figures provide a visual representation of the processed Feature Data, degradation progression, and evolving prognostic estimates. The corresponding output CSV files provide the numerical records used for the evaluations presented below.

8.5.1 ARULE Signature Processing and Prognostic Outputs

As described in Section 2, ARULE applies a sequence of transformations to the supplied degradation-related Feature Data before performing adaptive prognostic estimation.

The processing sequence is:

Data Amplitude (DA) → Feature Data (FD) → Fault-to-Failure Progression (FFP) → Degradation Progression Signature (DPS) → Functional Failure Signature (FFS) → Prognostic Estimation

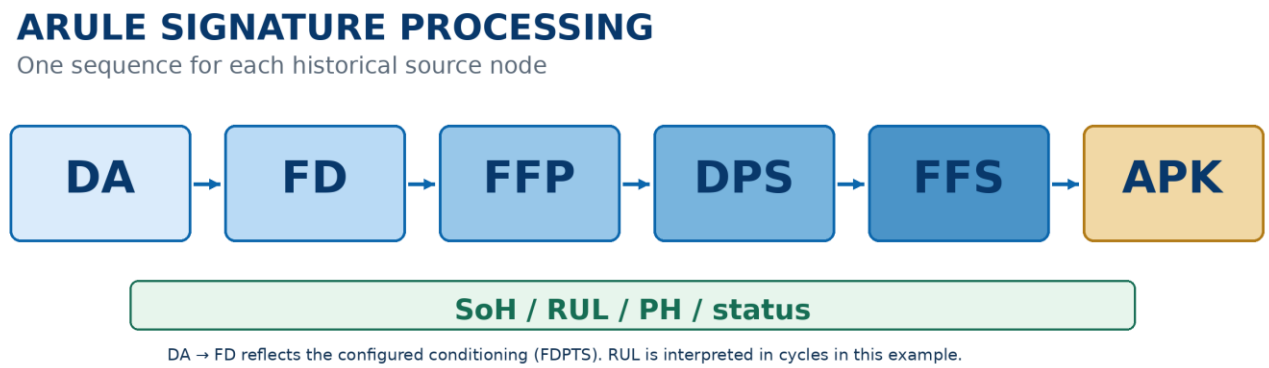


Figure 173. ARULE Signature-Processing and Prognostic Estimation for Historical Feature Data

Each stage serves a distinct purpose within the prognostic-analysis process.

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- **Data Amplitude (DA)** represents the imported amplitude associated with each observation in the input dataset.
- **Feature Data (FD)** represents the resulting feature after applying the applicable NDEF conditioning parameters. Depending on the configuration, DA and FD may be numerically equivalent or may differ because of averaging or other supported conditioning operations.
- **Fault-to-Failure Progression (FFP)** represents the transformed feature relative to the applicable nominal reference, noise margin, and normalization settings.
- **Degradation Progression Signature (DPS)** represents the transformed degradation progression used by the subsequent ARULE processing methodology.
- **Functional Failure Signature (FFS)** expresses progression within the configured failure-signature representation and provides information used by the prognostic-estimation process.

The Adaptive Predictive Kernel (APK) uses the degradation information and estimator state to generate prognostic outputs, including:

- **State of Health (SoH):** An ARULE health estimate associated with the configured degradation-to-functional-failure progression.
- **Remaining Useful Life (RUL):** The estimated remaining distance along the analysis coordinate to the configured functional-failure condition.
- **Prognostic Horizon (PH):** The corresponding prognostic-horizon quantity defined by the ARULE methodology.
- **Status Information:** Additional execution and estimator information used to support interpretation and troubleshooting.

These quantities are related but are not interchangeable. In particular, ARULE SoH is not necessarily equivalent to measured battery-capacity retention, and the RUL estimate is dependent on the selected Feature Data, functional-failure mapping, and prognostic configuration.

Because the analysis coordinate in this example is cycle number, RUL and PH are interpreted using the cycle-coordinate basis. The formal prognostic-output definitions, including PH, are provided in Section 3.

8.5.2 Numerical Example of the ARULE Feature Transformation

The DAQ measured-capacity node provides a numerical example of how the selected Node Definition (NDEF) parameters influence the transformation of Feature Data into ARULE's degradation-signature representations.

The retained Node 1 — DAQ Measured Capacity Loss configuration includes the following parameters:

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- FDC: 1.00
- FDZ: 20.00
- FDNM: 5.00%
- FDCPTS: 0
- FDPTS: 1
- FDNV: 1.000
- FFPFAIL: 100.00
- PITTFF: 1000.00
- PIFFSMOD: 2 (Linear)

At cycle 400, the recorded DAQ output contains the following values:

- Data Amplitude (DA): 9.180
- Feature Data (FD): 9.180
- Fault-to-Failure Progression (FFP): 0.4065
- Degradation Progression Signature (DPS): 0.4065
- Functional Failure Signature (FFS): 40.65

Feature Data Transformation

For the retained DAQ configuration, the recorded FFP is consistent with the following transformation:

- **FFP = [FD − FDC(1 + FDNM/100)] / FDZ**

Substituting the configured parameters and recorded Feature Data value:

- **FFP = [9.180 − 1.00(1 + 5/100)] / 20.00**

The nominal reference, including the configured 5% noise margin, is:

- **1.00(1 + 5/100) = 1.05**

Therefore:

- **FFP = (9.180 − 1.05) / 20.00**
- **FFP = 8.130 / 20.00 = 0.4065**

With FDNV = 1.000, the recorded Degradation Progression Signature is numerically equivalent to FFP for this cycle-400 record:

- **DPS = 0.4065**

The corresponding Functional Failure Signature is:

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- **FFS = 100 × DPS**
- **FFS = 100 × 0.4065 = 40.65**

These calculations reproduce the relationships observed in the retained DAQ output at cycle 400 and illustrate how the selected nominal reference, noise margin, and normalization settings influence the degradation-signature representation generated from the input Feature Data.

This numerical reconstruction describes the specific retained DAQ configuration and cycle-400 record. It should not be generalized to all ARULE applications without confirming the applicable ARULE processing methodology and NDEF behavior.

In particular, the three CellSage nodes use FDCPTS = 5, whereas the DAQ node uses FDCPTS = 0. Consequently, the active nominal-reference behavior and resulting transformations for the CellSage nodes should be interpreted according to their respective NDEF settings rather than assumed to follow the DAQ transformation identically.

Interpretation of the Configured Functional-Failure Endpoint

The same recorded-case transformation can be used to examine the feature-domain endpoint implied by the retained DAQ configuration.

For the demonstrated mapping, the normalized functional-failure condition corresponds to:

- **FFP = 1**

Rearranging the preceding relationship for the Feature Data value gives:

- **FD(fail) = FDC(1 + FDNM/100) + FDZ × FFP(fail)**

Substituting the retained DAQ parameters:

- **FD(fail) = 1.00(1 + 5/100) + 20.00(1)**
- **FD(fail) = 1.05 + 20.00**
- **FD(fail) = 21.05**

Under this recorded-case transformation, the configured functional-failure endpoint therefore corresponds to approximately 21.05 capacity-loss percentage points.

This value is a consequence of the retained NDEF normalization and functional-failure mapping. It should not be interpreted as independent evidence that the tested battery physically reaches end of life at precisely 21.05% capacity loss.

If the intended application requirement defines functional failure at exactly 20% capacity loss, equivalent to 80% capacity retention under a consistent beginning-of-life reference, the NDEF configuration and its associated engineering rationale should be

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reviewed to establish the appropriate mapping. This document does not make that engineering disposition.

Any resulting change to the NDEF would constitute a revised configuration and would require a separate execution and corresponding analysis record.

The numerical example therefore illustrates the relationship among the imported Feature Data, the retained NDEF parameters, the transformed degradation signatures, and the configured feature-domain functional-failure endpoint. The following subsections examine how these representations relate to the recorded ARULE State of Health (SoH), Remaining Useful Life (RUL), and Prognostic Horizon (PH) estimates.

8.5.3 Recorded Four-Node Results

The following table summarizes the final recorded outputs at cycle 400 for each of the four analysis nodes.

The values are taken from the retained node-level output CSV files associated with the four-node execution.

Node / Source	FD (%)	FFS	SoH (%)	RUL (cycles)	PH (cycles)
1 — DAQ	9.180	40.65	7.90	13.98	176.98
2 — CellSage Best	5.803	28.77	72.33	1,037.94	1,434.94
3 — CellSage Average	7.713	38.17	61.10	623.61	1,020.61
4 — CellSage Worst	10.548	52.09	44.78	322.71	720.71

Several observations can be made from these results.

First, the three modeled CellSage trajectories exhibit the expected ordinal progression within the selected reference configurations. The Best-Case trajectory has the lowest capacity loss and the largest reported RUL, while the Worst-Case trajectory has the greatest modeled capacity loss and the smallest RUL. The Average-Case results fall between these two cases.

Second, the measured DAQ history produces a substantially different prognostic result. Although the final measured capacity-loss feature is approximately 9.18%, ARULE reports an SoH of approximately 7.90% and RUL of approximately 13.98 cycles.

This difference reinforces an important distinction:

- **ARULE SoH and RUL are calculated prognostic quantities resulting from the complete degradation history, transformation parameters, estimator behavior, and configured functional-failure condition. They are not direct**

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restatements of measured capacity retention or a simple extrapolation of the final Feature Data value.

Third, PH and RUL remain distinct outputs. For example, the DAQ node reports RUL of approximately 13.98 cycles and PH of approximately 176.98 cycles, while the CellSage Best-Case node reports RUL of approximately 1,037.94 cycles and PH of approximately 1,434.94 cycles.

These quantities should be interpreted using their respective definitions from Section 3. Their numerical differences alone do not establish an alternative PH calculation or a physical time-to-failure relationship.

The recorded results demonstrate different estimator responses under the four retained configurations. Their scientific interpretation requires additional examination of the underlying Feature Data and signature-processing behavior.

8.5.4 Scientific Interpretation of DAQ Measured Results

The DAQ measured-capacity dataset provides an important illustration of how Feature Data variability can influence ARULE's degradation-signature processing and prognostic estimates.

Unlike the comparatively smooth CellSage trajectories, the measured DAQ history contains substantial point-to-point variation superimposed on an overall increasing capacity-loss trend.

Across the 400 recorded observations, the Feature Data range from approximately −4.15 to 10.74 capacity-loss percentage points.

The presence of negative values or temporary decreases in the measured capacity-loss feature should not automatically be interpreted as physical capacity recovery. Such behavior may arise from measurement variability, test-condition changes, the underlying capacity-reference calculation, or other factors that require examination of the acquisition record.

The retained DAQ configuration uses FDPTS = 1, meaning that no multi-point averaging is applied to the imported observations through that setting. Consequently, short-term variations in the source Feature Data remain visible in the transformed degradation-signature sequence.

The nominal/noise-margin boundary for the recorded DAQ transformation is FDC × (1 + FDNM/100) = 1.00 × 1.05 = 1.05. A short sequence of observations near cycles 143–145 illustrates how changes in the Feature Data relative to this boundary affect the reported ARULE outputs.

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Cycle	FD (%)	FFP	DPS	FFS	SoH (%)
143	1.46	0.0205	0.0205	2.05	20.81
144	0.86	-0.0095	0.0000	0.00	100.00
145	2.68	0.0815	0.0815	8.15	91.45

At cycle 143, the Feature Data exceed the configured boundary, producing a positive FFP and FFS. The corresponding ARULE SoH estimate is approximately 20.81%.

At cycle 144, the recorded FD decreases below the boundary. The resulting FFP becomes negative, while DPS and FFS are reported as zero. The SoH output returns to 100%.

At cycle 145, the Feature Data again exceed the configured boundary, producing positive DPS and FFS values and a different SoH estimate.

These consecutive changes do not demonstrate that the battery physically recovered to full health and then immediately degraded again. Instead, they illustrate how the retained ARULE configuration responds to changes in the supplied Feature Data and

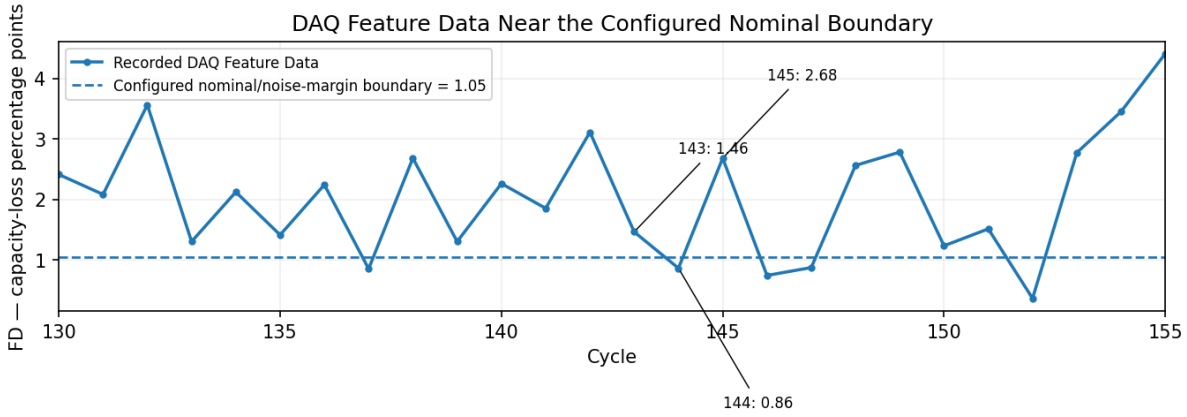


Figure 174. DAQ Feature Data Near the Configured Nominal Boundary (Cycles 130–155)

Figure 174 illustrates the measured Feature Data near the configured nominal/noise-margin boundary, including the observations at cycles 143–145.

Across the complete DAQ execution, 88 of the 400 recorded observations have both FFS = 0 and SoH = 100% under the retained configuration.

This recurring behavior emphasizes the importance of distinguishing between the measured capacity-loss feature, the transformed degradation signature, and the resulting prognostic state estimate.

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It also indicates that the selected Feature Data conditioning and NDEF parameters warrant further engineering assessment before the resulting SoH and RUL estimates are interpreted as physically meaningful battery-life predictions.

A separate sensitivity study could examine the effects of FDPTS, nominal-reference selection, FDNM, prognostic initialization, and other applicable configuration parameters while preserving the original historical source data.

Such a study should be documented as a separate engineering evaluation rather than modifying the retained reference configuration solely to produce smoother or more intuitive plots.

8.5.5 Interpretation of CellSage Modeled Results

The three CellSage datasets provide a complementary demonstration of ARULE behavior when processing smooth, model-generated degradation trajectories.

The Best, Average, and Worst capacity-loss histories represent different modeled degradation progressions over the same cycle-coordinate interval.

At cycle 400, the source Feature Data exhibit the following progression:

- **Best Case: 5.803% → Average Case: 7.713% → Worst Case: 10.548%**

The corresponding FFS values increase from 28.77 to 38.17 and 52.09, while the reported SoH and RUL estimates decrease.

This ordinal relationship demonstrates how the selected degradation-signature progression influences the resulting prognostic estimates under the retained configurations.

Compared with the measured DAQ history, the modeled CellSage trajectories are smoother and exhibit more regular degradation progression. Their resulting ARULE output histories are correspondingly more regular.

However, smoothness and consistent ordering should not be interpreted as evidence of forecast accuracy.

The modeled Best, Average, and Worst trajectories are independently generated source representations. They are not established as statistical confidence intervals or as predictions for the specific battery used in the DAQ historical Test Run.

Furthermore, the modeled NDEFs use FDCPTS = 5, and the relationship between the configured nominal settings and the active nominal-reference calculations should be considered when interpreting differences in their transformed signatures.

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The example therefore demonstrates ARULE's response to different modeled degradation histories, rather than validating the accuracy of CellSage or establishing which modeled trajectory best represents the measured battery.

8.5.6 Limitations and Forecast-Performance Evaluation

The four-node example demonstrates historical-data preparation, configuration, execution, degradation-signature processing, and recorded prognostic-estimation behavior.

It also provides an opportunity to evaluate the effects of different degradation representations on ARULE outputs.

However, several distinct levels of technical assessment should be recognized.

- **Execution:** The supplied datasets were processed under their retained NDEF/SDEF configurations, and corresponding output records were generated.
- **Reproducibility:** The retained configurations, inputs, output files, and execution records provide a basis for reconstructing and comparing the analysis.
- **Engineering interpretation:** The observed output behavior can be examined in relation to the source Feature Data, transformation settings, and selected functional-failure definitions.
- **Forecast-performance validation:** Independent evaluation of prediction accuracy requires additional information and a controlled assessment methodology.

For example, retrospective forecast evaluation would require selecting a prediction origin, restricting the available input history to observations preceding that origin, and comparing the resulting RUL estimate against an independently established later functional-failure coordinate.

Appropriate error metrics, acceptance criteria, and persistence requirements would also need to be defined. The current example does not establish that the reported cycle-400 RUL values accurately predict physical battery failure. Likewise, the DAQ and CellSage datasets are not established as a matched measured/model pair, so the example does not provide a basis for validating CellSage predictions against the measured Test Run. The results should therefore be interpreted as demonstrations of ARULE's historical-data processing and estimator behavior**,** not as independently validated battery prognostic predictions or evidence of readiness for unattended operational deployment.

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8.6 Reproducibility and Evidence Retention

A reproducible ARULE analysis requires more than screenshots of the resulting graphs. The original source data, prepared inputs, configuration definitions, execution records, and output files should be retained together so the analysis can be reconstructed and evaluated.

For the four-node historical-data example, the retained evidence package includes the following classes of material.

Evidence class	Required material	Purpose
Source exports	DAQ cycle-summary CSV; CellSage String Analysis CSV	Preserve measured and modeled origin
Prepared inputs	Four two-column ARULE CSVs	Define cycle and feature amplitude
Configuration	Four NDEFs; four-node SDEF	Record engine settings and node association
Execution	Engine/GUI identification; run settings; run log	Identify runtime and processing history
Results	Four output CSVs; selected screenshots	Support reproducible analysis and interpretation
Engineering record	Source-field map; interpolation method; failure-map rationale	Explain preparation and configuration

The retained execution log for CNX_4-Node_Battery_Capacity records startup of Nodes 1–4, input processing through the end of the four associated CSV files, and termination of the UD_ARULE program.

Together with the retained configuration and output records, these entries provide supporting evidence that the four historical histories were processed under the recorded configuration.

Reproducibility Comparison

A further comparison was performed between the four source-node outputs from the dedicated four-node execution and their corresponding outputs from an earlier execution that also included a fifth demonstration node.

The comparison included:

- Four corresponding source nodes.
- 400 observations per node.

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- 19 numeric output fields per observation.
- Exact numeric comparison of the parsed CSV values.

No numeric differences were identified across the compared output records.

This represents agreement across **30,400 compared numeric values**:

- 4 x 400 x 19 = 30,400

The result provides supporting evidence that the four source-node outputs were reproduced between the retained executions under the corresponding inputs and configurations.

It also shows that removal of the separate fifth demonstration node did not change the compared outputs of the four source nodes in these recorded executions.

This finding supports reproducibility of the specific executed analysis. It does not independently demonstrate prognostic accuracy, physical model validity, general software qualification, or satisfaction of a formal verification requirement.

The execution records and comparison results should be retained with the tutorial evidence package for subsequent engineering review.

8.7 Relationship to CellNostix Integrated Prognostics

The historical-data example presented in this section demonstrates how degradation-related Feature Data can be prepared, configured, and analyzed using ARULE.

Within an integrated CellNostix application, the same underlying prognostic methodology may be used in a more automated workflow. Qualified Feature Data and associated configurations can be supplied to ARULE by the host application, allowing new observations to be processed and updated SoH, RUL, PH, and status information to be returned for subsequent health assessment.

As illustrated conceptually in Figure 161, a future integrated application may generate model-relative Feature Data by comparing measured degradation with an appropriately associated CellSage model. Such a workflow requires controlled source association, compatible measured and modeled quantities, defined comparison logic, and appropriate ARULE configuration.

The resulting prognostic information may be used by CellNostix advisory functions to evaluate predictive conditions and support notifications, engineering review, and Health Management activities.

These integrated capabilities are distinct from the four-node historical replay demonstrated here. Detailed procedures for model association, Condition Indicator

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preparation, advisory-rule configuration, alert thresholds, severity and persistence settings, notification delivery, and Health Management are provided in the applicable CellNostix User Guide.

8.8 Summary and Limitations

This section demonstrated the application of ARULE to four historical battery-capacity degradation trajectories originating from measured DAQ data and CellSage model-generated outputs.

The worked example illustrated how to:

1. Prepare degradation-related Feature Data using a common cycle-coordinate basis.
2. Configure individual NDEFs for measured and modeled degradation histories.
3. Organize the four NDEFs within a common SDEF.
4. Execute ARULE and examine the resulting degradation-signature and prognostic outputs.
5. Interpret the effects of Feature Data behavior and configuration on SoH, RUL, and PH.
6. Retain the source, configuration, execution, and output records required for reproducibility.

The measured DAQ case demonstrated the importance of Feature Data variability, nominal-reference behavior, and conditioning parameters when interpreting degradation signatures and prognostic estimates. The CellSage cases demonstrated how different modeled degradation progressions influence ARULE outputs under their respective configurations.

The numerical transformation example and recorded output comparisons provided additional insight into the relationship between NDEF parameters, Feature Data amplitudes, transformed degradation signatures, and the resulting prognostic estimates.

The retained four-node execution and reproducibility comparison provide supporting evidence of historical-data processing under the documented configurations.

The example does not independently establish physical battery functional failure, validated CellSage model accuracy, ARULE out-of-sample prediction accuracy, equivalence between ARULE SoH and measured capacity retention, or suitability of the tutorial configuration for operational deployment.

Nevertheless, the procedures demonstrate how the ARULE standalone application can be used as an engineering environment for developing, examining, and documenting application-specific prognostic analyses.

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These capabilities provide a foundation for subsequent integration within automated CellNostix prognostic workflows, where qualified Feature Data, controlled configurations, and host-level advisory functions can support more comprehensive PHM applications.

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9. Appendix

9.1 Frequently Asked Questions (FAQ)

1. Are there any prerequisites for installing and running the ARULE™ standalone application?

Answer:
The software dependencies required by the supported ARULE installation are included with the ARULE installation package and are installed through the installation wizard.

Users should also confirm that the computer meets the system requirements identified in Section 4 and that they have the permissions necessary to install ARULE, activate the software license, and access the configured ARULE user directory.

2. What type of data does ARULE™ require?

Answer:
ARULE operates on degradation-related **Feature Data (FD)** rather than assuming that arbitrary raw sensor measurements are directly suitable for prognostic analysis.

Feature Data may be:

- directly measured;
- calculated from Condition-Based Data (CBD);
- derived through signal processing;
- obtained from a physics-based, empirical, statistical, or data-driven model;
- generated from model-to-test comparisons; or
- represented by another validated Condition Indicator related to degradation.

The selected Feature Data should have an established engineering relationship to the degradation mechanism or functional-failure condition being evaluated.

3. Can ARULE™ analyze any type of Feature Data?

Answer:
ARULE is designed to be **application-independent**, meaning that the same underlying

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prognostic methodology can be applied across different asset classes and degradation mechanisms.

However, application-independent does not mean that every dataset is automatically suitable for prognostics. The selected Feature Data should exhibit behavior that can be meaningfully related to degradation progression and a defined functional-failure condition.

Before configuring ARULE, the user should understand the Feature Data source, physical or engineering meaning, nominal behavior, degradation direction, variability, analysis coordinate, and failure criterion.

Section 7 provides the recommended workflow for analyzing user-defined Feature Data.

4. What does the first column, or analysis coordinate, of an ARULE™ input file represent?

Answer:
The analysis coordinate represents the ordered progression of the Feature Data observations and functions as the independent variable, or x-axis, of the Feature Data signature.

Depending on the application, the coordinate may represent:

- elapsed time;
- operating hours;
- charge/discharge cycles;
- accumulated usage;
- mission count;
- distance or revolutions;
- sequential sample or inspection number; or
- another ordered application-specific quantity.

The analysis coordinate should not be confused with the engineering domain from which the Feature Data were derived.

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For example, a vibration Feature Data value may be calculated from frequency-domain data while successive Feature Data observations are ordered by operating hours. In that case, frequency describes the Feature Data amplitude, while operating hours define the ARULE progression coordinate.

5. What determines the units of Remaining Useful Life (RUL)?

Answer:
RUL is interpreted in the same coordinate basis used to order the Feature Data.

For example:

- if the analysis coordinate represents hours, RUL is interpreted in hours;
- if the coordinate represents cycles, RUL is interpreted in cycles;
- if the coordinate represents missions, RUL is interpreted in missions; and
- if the coordinate represents sequential observations or another arbitrary progression coordinate, RUL remains expressed in that coordinate basis unless an external conversion has been established.

The physical units of the Feature Data amplitude do not determine the units of RUL.

6. What is the difference between Data Amplitude (DA) and Feature Data (FD)?

Answer:
In the standalone ARULE workflow, **Data Amplitude (DA)** represents the amplitude imported from the applicable input-data field.

ARULE may then apply configured conditioning, such as averaging or other preprocessing, to the imported DA values to produce the **Feature Data (FD)** used by the signature-processing chain.

The applicable processing sequence is:

DA → FD → FFP → DPS → FFS → Prognostic Estimation

When no multi-point averaging or other applicable conditioning changes the imported value, DA and FD may be numerically equivalent. They should nevertheless be treated as distinct quantities because different configurations may produce different FD values from the same imported DA sequence.

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7. What are SDEF and NDEF files?

Answer:
ARULE uses configurable definition structures to describe how an analysis is organized and processed.

A **System Definition (SDEF)** identifies the analysis system and the Node Definitions associated with that system.

A **Node Definition (NDEF)** defines the Feature Data input association and the parameters required to condition, normalize, transform, and prognostically process a particular analysis node.

An SDEF may reference one or more NDEFs, allowing multiple monitored nodes or degradation features to be organized within a single system-level analysis.

Together, the SDEF, NDEF, input data, ARULE software version, output files, and run logs form important parts of the reproducible ARULE analysis record.

8. What does ARULE™ mean by functional failure?

Answer:
Functional failure is the defined condition at which the monitored system, subsystem, component, or analysis node no longer satisfies the applicable performance or operating criterion used for the analysis.

Functional failure is not necessarily the same as catastrophic physical failure.

For example, a component may still physically operate but no longer meet a required capacity, vibration, electrical, thermal, mission, or performance specification.

ARULE estimates progression toward the **configured functional-failure condition**. The selected physical or engineering failure criterion must therefore be mapped appropriately into the Feature Data and ARULE signature-processing configuration.

A physical threshold should not automatically be entered directly as an ARULE normalized failure parameter without first verifying the applicable mapping.

9. What are the primary prognostic outputs produced by ARULE™?

Answer:
The primary prognostic outputs are:

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- **Remaining Useful Life (RUL)** — the estimated remaining distance along the analysis coordinate from the current observation to the predicted functional-failure condition;
- **State of Health (SoH)** — an ARULE health estimate associated with progression through the configured degradation-to-functional-failure interval; and
- **Prognostic Horizon (PH)** — the applicable prognostic-horizon quantity generated by the ARULE methodology.

ARULE also generates intermediate signature-processing values, status information, return codes, and reason codes that can be used to evaluate execution behavior, troubleshoot configurations, and interpret the resulting prognostic estimates.

The detailed output definitions are described in Sections 2, 3, and 5.

10. How do I know whether a Feature Data signature is suitable for ARULE™?

Answer:
A Feature Data signature should have an engineering basis for relating changes in the feature to increasing degradation or progression toward functional failure.

Useful evaluation considerations include:

- whether the feature responds consistently to the degradation mechanism of interest;
- whether nominal and degrading behavior can be distinguished;
- whether the degradation direction is understood;
- whether noise and operating-condition effects are manageable;
- whether the analysis coordinate is meaningful and correctly ordered;
- whether a defensible functional-failure endpoint can be established; and
- whether the transformed FFP, DPS, and FFS behavior is technically reasonable.

A successful ARULE execution does not by itself prove that the selected Feature Data are prognostically valid.

Section 7 provides additional guidance for evaluating user-defined Feature Data.

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11. Does ARULE™ determine when an alert, maintenance action, or operational response should occur?

Answer:
Not by itself.

ARULE produces prognostic information such as SoH, RUL, PH, and applicable status information. A larger PHM or health-management application may combine those results with:

- alert thresholds;
- persistence rules;
- severity criteria;
- mission context;
- safety constraints;
- maintenance policies; or
- other application-specific decision logic.

Within CellNostix™, for example, ARULE provides the prognostic-analysis capability within a broader workflow that also includes data acquisition, modeling, Feature Data preparation, predictive-alert logic, advisory evaluation, and health-management functions.

12. What is the difference between the standalone ARULE™ application and integrated ARULE execution?

Answer:
The standalone ARULE application provides an engineering environment for:

- loading and reviewing Feature Data;
- creating and modifying SDEF and NDEF configurations;
- evaluating functional-failure mappings;
- replaying historical, experimental, or simulated datasets;
- reviewing signature-processing behavior;

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- evaluating SoH, RUL, and PH estimates; and
- reviewing output files and run logs.

Integrated execution uses the same underlying prognostic methodology within a larger software environment. In this case, another application may automatically prepare and qualify Feature Data, invoke the ARULE prognostic engine, and consume the returned prognostic information.

Section 8 describes the relationship between standalone ARULE engineering analysis and integrated execution within CellNostix™.

13. What information should be retained to reproduce an ARULE™ analysis?

Answer:
A reproducible ARULE analysis should retain, as applicable:

- the input Feature Data file;
- the SDEF;
- each applicable NDEF;
- the ARULE software or engine version;
- execution or run settings;
- output data files;
- run logs;
- checkpoint or restart information, when used;
- Feature Data and coordinate definitions;
- functional-failure definition and mapping basis; and
- any relevant engineering assumptions used to establish the configuration.

Because ARULE processes data sequentially, execution state and observation order may also be important when comparing repeated runs or checkpoint/restart behavior.

14. Where should I start if I want to analyze my own data with ARULE™?

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Answer:
Begin with **Section 7 — How to Analyze Your Own Data with ARULE™**.

Before configuring ARULE, identify:

1. the system, subsystem, component, or analysis node being evaluated;
2. the degradation mechanism or failure mode of interest;
3. the source Condition-Based Data;
4. the Feature Data to be used for prognostics;
5. the ordered analysis coordinate;
6. the nominal or healthy Feature Data behavior;
7. the expected degradation direction;
8. the intended functional-failure condition; and
9. the engineering basis for the associated ARULE configuration.

The SDEF and NDEF parameters should then be selected based on that engineering definition rather than copied directly from an unrelated example.

A successful software execution does not by itself demonstrate that the Feature Data, failure mapping, or resulting prognostic estimates are technically valid.

15. Does Ridgetop Group provide ARULE™ training, application support, and technical assistance?

Answer:
Yes. Ridgetop Group provides remote and on-site training, technical support, and engineering assistance for ARULE™, PHM, CBM, IVHM, and related Ridgetop technologies.

Training and engineering support may include:

- ARULE software operation;
- Feature Data preparation and evaluation;
- SDEF and NDEF configuration;

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- functional-failure mapping;
- prognostic-analysis workflows;
- troubleshooting and result interpretation;
- application-specific PHM development; and
- integration of ARULE into larger monitoring or health-management systems.

Technical-support contact information is provided at the beginning of this User Guide.

When requesting assistance with a specific ARULE issue, users should provide sufficient information to reproduce the condition where practical, including the ARULE software version, applicable SDEF and NDEF, input-data identity, execution settings, relevant logs, error or status messages, and a description of the expected and observed behavior.

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